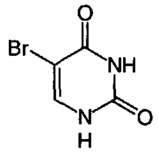
bromouracil *symbol*: BrUra; 5-bromouracil; 5-bromo-2,4(1*H*,3*H*)-pyrimidine-dione; a synthetic analogue of thymine with mutagenic activity. It is incorporated into DNA as **bromodeoxyuridine**, which replaces thymidine and induces transitions of G–C base-pairing to A–T pairing. It can be used as a density marker for DNA. The free compound is an inhibitor of dihydrouracil dehydrogenase.



bromouridine *symbol*: B *or* BrUrd; 5-bromouridine; 5-bromouracil riboside; a synthetic analogue of ribothymidine.

Brønsted catalysis law an expression of the relation between the rate constant, k, of a catalysed reaction, and the pK of the catalyst: $\log k = c + b\mathrm{p}K$, where b and c are constants and the value of b provides a measure of the balance between nucleophilic and general base character of the catalysis. [After Johannes Nicolaus Brønsted (1879–1947), Danish physical chemist.]

Brønsted–Lowry acid any chemical species capable of donating a hydron to a base. Examples are H_2O, H_3O^+, CH_3COOH, $H_2PO_4^-$, NH_4^+. [After J. N. Brønsted (*see* **Brønsted catalysis law**) and Thomas Martin Lowry (1874–1936), British chemist.]

Brønsted–Lowry base any chemical species capable of accepting a hydron from an acid. Examples are OH^-, H_2O, CH_3COO^-, HPO_4^{2-}, NH_3.

Brookhaven database *see* **PDB**.

broth *(in bacteriology)* any liquid culture medium, especially nutrient broth.

Brown, Michael Stuart (1941–), US physician and molecular geneticist notable for his discovery (in collaboration with J. L. Goldstein) of cellular receptors for low-density lipoproteins and their role in the removal of cholesterol from the bloodstream; Nobel Laureate in Physiology or Medicine (1985) jointly with J. L. Goldstein 'for their discoveries concerning the regulation of cholesterol metabolism'.

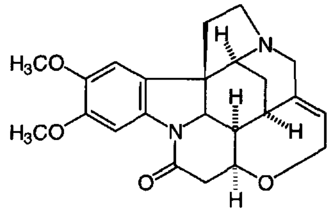
brown adipose tissue a highly specialized tissue with a high content of lipid and cytochromes found in some animals, particularly hibernating animals and the newborn of some species. It is highly vascular and consists of small polygonal cells, each containing many separate lipid droplets and many mitochondria. Its function is thermogenesis during the arousal period after hibernation or, in the young, to provide heat before shivering has developed. It is active also in normal but not in obese humans. The colour is due to the high cytochrome content. Heat is generated by lipid oxidation through electron transport not coupled to oxidative phosphorylation. The uncoupling is mediated by **brown fat uncoupling protein**. *Compare* **white adipose tissue**.

brown fat uncoupling protein *abbr.*: UCP; *other names*: mitochondrial UCP; thermogenin; a dimeric integral mitochondrial inner membrane protein that forms an H^+-channel, is unique to brown adipose tissue mitochondria, and is important for thermogenesis of this tissue. By causing the membrane to leak protons, it abolishes the proton gradient that drives oxidative phosphorylation, so that electron transport results solely in heat production. Example from mouse: database code UCP_MOUSE, 306 amino acids (33.08 kDa).

Brownian movement the peculiar random movements shown by microscopic particles in a disperse phase, i.e. when suspended in a liquid or gas. It is caused by the continuous irregular bombardment by the molecules of the surrounding medium. [After Robert Brown (1773–1858), British botanist who first observed it in 1827.]

browning reaction any of a group of complex reactions, both enzymic and nonenzymic, that occur in some foods during processsing and/or storage and cause a brownish discoloration. Members of this group include the oxidation of phenolic compounds (which is believed to be enzymic) together with caramelization, ascorbate decomposition, and the **Maillard reaction** (all of which are believed to be nonenzymic).

brucine 2,3-dimethoxystrychnidin-10-one; a highly toxic bitter-tasting alkaloid obtained from *Strychnos* seeds. It is used as a resolving agent for some racemates. *Compare* **strychnine**.

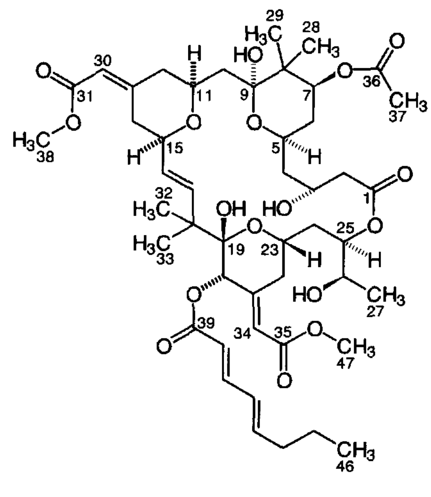


BrUra *symbol for* bromouracil.

BrUrd *symbol for* bromouridine (alternative to B). *Compare* **BrdUrd**.

brush border the dense covering of **microvilli** on the apical surface of epithelial cells in the intestine and kidney. The microvilli aid absorption by increasing the surface area of the cell.

bryostatin any of a number of highly potent activators of protein kinase C found in bryozoans, especially *Bugula neritina*. They were isolated as the active anti-leukemic agent of bryozoan extracts. They block phorbol ester-induced differentiation of human promyelocytic leukemia cells and other actions of phorbol esters.



bryostatin 1

BSA *abbr. for* bovine serum albumin. *See* **serum albumin**.

BSE *abbr. for* bovine spongiform encephalopathy.

B side the side of the nicotinamide ring of NADH or NADPH at which projects the *pro-S* hydrogen atom (known as H_B) at the 4 position. *Compare* **A side**.

BTEE *abbr. for* *N*-benzoyl-L-tyrosine ethyl ester (a substrate for the assay of chymotrypsin).

B-transferase EC 2.4.1.37; *recommended name*: fucosylglycoprotein 3-α-galactosyltransferase; *other name*: [histo-blood group] B transferase; an enzyme that catalyses the reaction:

UDPgalactose + glycoprotein α-L-fucosyl-(1,2)-D-galactose = UDP + glycoprotein α-D-galactosyl-(1,3)-[α-L-fucosyl-(1,2)]-L-galactose,

thus adding galactose to the H antigen of the **ABH antigen** system, leading to formation of the B antigen. The protein is a product of one allele of the *ABO* gene. It has virtual identity of sequence with that catalysing **A-transferase** activity, differing only at residues 176 (A has Arg, B has Gly), 235 (A Gly, B Ser), 266 (A Leu, B Met), and 268 (A Gly, B Ala). Database code BGAT_HUMAN, 354 amino acids (40.89 kDa).

Btu *abbr. for* British thermal unit.

Bu *symbol for* the butyl group.

BU *(sometimes) abbr. for* bromouracil (BrUra is recommended).

bubble column a column-shaped bioreactor in which the reaction medium is kept mixed and aerated by the introduction of air at the bottom.

Büchner, Eduard (1860–1917), German chemist and biochemist renowned for his seminal discovery that alcoholic fermentation could be initiated with a cell-free press juice from brewers' yeast, the active principle of which he considered to be a protein denoted zymase; Nobel Laureate in Chemistry (1907) 'for his biochemical researches and his discovery of cell-free fermentation'.

Büchner funnel a cylindrical funnel for filtration, usually of porcelain or plastic, that includes a perforated plate on which a filter paper is placed. It is generally used with a vacuum. *See also* **Hartley filter funnel**.

bud **1** a small lateral or terminal protuberance on the stem of a plant that contains undeveloped foliage or floral leaves. **2** a budlike protuberance on the surface of a yeast cell or other simple organism. **3** to form a bud. **4** to reproduce asexually, as in yeasts, by the process of **budding**.

Buddha suite of computer programs for structure refinement and energy minimization.

budding **1** the production of a bud or buds. **2** a form of asexual reproduction, occurring in certain bacteria and fungi (e.g. yeasts) and some primitive animals in which an individual arises from a daughter cell formed by pinching off a part of the parent cell. The budlike outgrowths so formed may sometimes remain attached to the parent cell.

BUdR *or* **BUDR** *(sometimes) abbr. for* bromodeoxyuridine (the symbol BrdUrd is preferred).

bufadienolide any of various naturally occurring doubly unsaturated lactones of certain steroids with important pharmacological effects on heart muscle (*see* **cardiac glycoside**). They are so named because they were originally found in the venomous secretion of the skin glands of some toads (*Bufo*), and are hence also known as **toad poisons**. They also occur in certain plants, e.g. *Digitalis*. *See also* **bufogenin B**.

buffer **1** any substance or mixture of substances that, when dissolved (usually in water), will maintain its solution at approximately constant pH despite small additions of acid or base. The commonest examples are moderately strong solutions containing both a weak acid and its conjugate base (or a weak base and its conjugate acid). A substance is useful as a buffer over a range of about one pH unit either side of its pK, but is most effective at or near the pK. Buffer substances used for biochemical or biological purposes include: acetate, bicarbonate, bis-tris propane, borate, citrate, dimethylmalonate, glycinamide, glycylglycine, imidazole, phosphate, succinate, and **Tris** together with any **Good buffer substances**. By extension, the term may be applied to agents controlling the activities of various other specified entities, e.g. redox buffer, carbon-dioxide buffer, metal-ion buffer. Also used attributively: e.g. buffer action; buffer salt; buffer solution. **2** a solution of a **buffer** (def. 1). **3** a short-term storage facility (e.g. as part of the memory of a computer), especially one whose patterns or rates of input and output can differ. **4** to treat with or to incorporate a buffer (def. 1); to act as a buffer. *See also* **buffering capacity**, **buffer value**. —**buffered** *adj.*; **buffering** *n.*

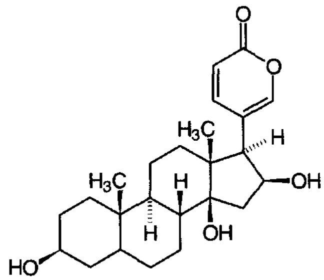
buffering capacity *or* **buffering power** **1** the number of gram-equivalents of either hydrogen ions or hydroxide ions required to change the pH of 1 litre of 1 M buffer solution by one unit. Buffering capacity = $(1/m)$ (dn/dpH), where m is the number of moles of buffer, and dpH is the pH change produced by addition of dn equivalents of hydrogen ions or hydroxide ions. **2** *an alternative term for* **buffer value**.

buffer solution *see* **buffer** (def. 2).

buffer value *or* **Van Slyke buffer value** *or* **buffering capacity** *symbol*: β; the amount of acid or base, in gram-equivalents, needed to change the pH of 1 litre of a buffer solution by one unit at any pH; i.e. $\beta = db/dpH$, where b is the molar concentration of base in the solution. [After Donald Dexter Van Slyke (1883–1971), US biochemist, who described it in 1922.]

buffy coat the layer of white cells that forms between the layer of red cells and the plasma when unclotted blood is centrifuged or allowed to stand.

bufogenin B 3β,14,16β-trihydroxy-5β-bufa-20,22-dienolide; a **bufadienolide** found in the Chinese drug Ch'an Su, prepared from Chinese toads (*Bufo asiaticus*).



bulge loop a structure in a polynucleotide duplex in which one strand contains a nonterminal extra sequence that is not able to base-pair with the second strand, thereby forming a bulge on one side of the duplex.

BUN *abbr. for* blood urea nitrogen, an index of the blood-urea concentration.

bundle sheath a parenchymal sheath surrounding a vascular bundle in plants.

bungarotoxin any of various neurotoxins derived from the venom of the elapid snake, *Bungarus multicinctus*. The chief components are: (1) α bungarotoxin, a single polypeptide chain of 74 amino-acid residues; it is an irreversible antagonist of nicotinic cholinergic receptors and causes paralysis; (2) β bungarotoxin, a multicomponent protein composed of two polypeptide chains: a long chain (120 amino-acid residues) and a short chain (60 residues); it prevents the release of acetylcholine from cholinergic neurons.

Bunsen coefficient *symbol*: α; the absorption coefficient of a gas in a solution. It is defined as the volume of gas in litres, reduced to 273.15 K and 1 atm (101 325 Pa) pressure, that dissolves in 1 litre of liquid when the partial pressure of the gas in the gas phase is 1 atm. [After Robert Wilhelm Bunsen (1811–99), German chemist.]

bunyavirus any of a group of RNA animal viruses consisting of enveloped particles, 90–100 nm in diameter with helical nucleocapsids, and containing segmented RNA (minus strand) of 3–4 MDa. The group has been recognized as different from **togaviruses** and includes arthropod-borne viruses that can cause encephalitis.

buoyant density the density of a solute molecule as determined by density-gradient ultracentrifugation. It is the density of the solution, ρ, at the point in the gradient where: $\rho = 1/\gamma$, where γ is the **partial specific volume** of the solute in question.

buoyant force *symbol*: F_B; the force acting on an object suspended in a liquid due to the liquid displaced by the object. It is given by: $F_B = V_P \rho_M g$, where V_P is the volume of the object, ρ_M is the density of the displaced liquid M, and g is the acceleration due to gravity.

burette *or (US)* **buret** a graduated (often glass) tube with a pinch-clamp, stopcock, or valve at one end used for measuring or dispensing known volumes of fluids, especially liquids. A graduated syringe used for the same purpose may also be termed a burette.

Burkitt's lymphoma a malignant tumour of the lymphatic system, most commonly affecting children in tropical Africa within 15° north and south of the Equator. **Epstein–Barr virus** can be demonstrated in a proportion of cultured tumour cells though it is unlikely that the virus is the sole cause of the malignancy. [After Denis Parsons Burkitt (1911–93), British surgeon.]

Burnet, (Sir) Frank Macfarlane (1899–1985), Australian physician, virologist, and immunologist notable in particular for formulation of the clonal selection theory of acquired immunity; Nobel Laureate in Physiology or Medicine (1960) jointly with P. B. Medawar 'for discovery of acquired immunological tolerance'.

Burnet's theory of immunity *see* **clonal selection theory**.

bursacyte *an alternative name (archaic) for* **B lymphocyte**.

bursa of Fabricius *or* **cloacal bursa** a saclike lymphoepithelial structure opening into the dorsal part of the cloaca in young birds. It usually degenerates as the birds reach maturity. It is associated with humoral immunity and the lymphocytes processed by it are termed **B lymphocytes**. In mammals it is likely that hemopoietic tissue itself fulfils the equivalent role, providing the appropriate microenvironment for the maturation of B lymphocytes from precursor stem cells.

bursectomy removal of the **bursa of Fabricius**, either by surgery *in ovo* or shortly after hatching, or by destruction *in ovo* by application of, e.g., testosterone. Bursectomy inhibits the formation of **B lymphocytes** and hence circulating antibody.

bursin the tripeptide Lys-His-Gly-NH_2; a selective B-cell differentiating hormone from the bursa of Fabricius of chickens.

burst 1 the initial pre-steady-state liberation of the first product, B, in an enzymic reaction of the type: $E + AB = EAB \rightarrow EA + B$ and $EA \rightarrow E + A$, where E is the enzyme and A is the second product. **2** the sudden release of phage particles accompanying lysis of a phage-infected bacterial cell. *See also* **burst size, respiratory burst**.

burst size the mean number of bacteriophage particles set free per infected bacterium upon lysis of phage-infected cells.

butanediol fermentation a type of fermentation effected by some members of the Enterobacteriaceae, e.g. *Enterobacter*, *Erwinia*, *Klebsiella*, and *Serratia*, in which glucose is fermented with the production of 2,3-butanediol and other substances.

butanoyl *another name for* **butyryl**.

Butenandt, Adolf Friedrich Johann (1903–95), German organic chemist and biochemist renowned for his pioneering studies of steroid hormones and pheromones, especially for his isolation and determination of the structures of the first sex hormones (androsterone, estrone, and progesterone) and the first insect hormone (ecdysone), and for the isolation of the first pheromone (bombykol); Nobel Laureate in Chemistry (1939) 'for his work on sex hormones' [prize shared with L. Ružička].

butyl *symbol*: Bu; the alkyl group, $CH_3–[CH_2]_2–CH_2–$, derived from butane.

butyl-PBD *see* **biphenylylphenyloxadiazole**.

butyrate 1 *trivial and preferred name for* **butanoate**; the anion $CH_3–[CH_2]_2–COO^-$, derived from butyric acid (butanoic acid), a saturated, unbranched, aliphatic acid. **2** any mixture of free butyric acid and its anion. **3** any salt or ester of butyric acid.

butyric acid fermentation a type of fermentation effected by some saccharolytic species of *Clostridium*, e.g. *C. butyricum*, in which glucose is fermented, with the production of acetic acid, butyric acid, carbon dioxide, and dihydrogen.

butyrophenone any of a group of antipsychotic drugs, e.g. **haloperidol**, that are more selective than the **phenothiazine** group. Butyrophenones interact rather more specifically with dopamine D_2 receptors and reduce the firing of dopaminergic neurons.

butyrophilin a type I **membrane protein** of mammary gland, probably involved in stimulating the secretion of droplets of fat in milk. Example (bovine, precursor): database code BUTY_BOVIN, 526 amino acids (59.20 kDa).

butyryl *symbol*: Br; *trivial and preferred name for* **butanoyl**; the acyl group, $CH_3–[CH_2]_2–CO–$, derived from butyric acid (butanoic acid).

BX-C *abbr. for* the bithorax complex, one of two (*see* **ANT-C**) major families of **homeotic genes** in *Drosophila*, mutations in which affect thoracic and abdominal segments.

byte a single unit of information handled by a computer; the binary unit required for storage of a character, usually comprising eight **bits**.

Bz *symbol for* the benzoyl group.

bZIP *abbr. for* basic zipper; a fingerprint of a family of proteins, named after their two main features: a basic region (b) and a **leucine zipper** domain (ZIP). The basic region harbours a DNA-contact surface, and the leucine zipper domain is essential for homodimer formation or formation of heterodimers with other leucine zipper proteins. The family includes the **AP**, **ATF**, and **CREB protein** transcription factors.

Bzl *symbol for* the benzyl group.

BZLF1 one of a group of transcription factors related to the *jun* and *fos* gene products. It is involved in the switch from latency to virus-particle production following infection by **Epstein–Barr virus**. Database code BZLF_EBV, 245 amino acids (26.83 kDa).

Cc

c 1 *symbol for* **a** centi+ (SI prefix denoting 10^{-2}). **b** **cyclic** (def. 3) or **cyclo+** (denoting cyclic compounds, as in cAMP for cyclic AMP or Hx_c for cyclohexyl. **c** *(obsolete)* **curie** (use Ci). **2** *abbr. for* **a** complementary (as in **cDNA**). **b** ***cis*** (not recommended). **c** *(obsolete)* cubic (i.e. to the third power).

c- *(in genetics) prefix* denoting cellular, as in c-*myc*, the cellular as opposed to viral (v-) version of a (proto)oncogene; *see*, e.g., ***MYC***.

c *symbol for* **1** amount-of-substance concentration; that for a particular substance may be denoted by adding a subscript, e.g. c_B for amount concentration of a substance B. **2** speed (alternative to *v*, *u*, or *w*). **3** (bold italic) velocity (alternative to ***v***, ***u***, or ***w***).

c- *abbr. for* ***cis-***.

c_0 *symbol for* speed of light *in vacuo*. The subscript may be omitted when there is no risk of ambiguity.

C **1** *symbol for* **a** carbon; *see* **carbon-11**, **carbon-12**, **carbon-13**, **carbon-14**. **b** designating a specific carbon atom, e.g. C-1, C-2. **c** a residue of the α-amino acid L-cysteine (or L-half-cysteine) (alternative to Cys). **d** a residue of the base cytosine in a nucleic acid sequence. **e** a residue of the ribonucleoside cytidine (alternative to Cyd). **f** **coulomb**. **g** *or (formerly)* C′ complement (the main components of complement are designated C1–C9). **2** a high-level computer language based on function evaluation and much used in the development of computer software for molecular biology.

°C *symbol for* degree Celsius (*formerly* degree centigrade). *See* **Celsius temperature**.

$\mathbf{C_H}$ *symbol for* the constant region of an immunoglobulin **heavy chain**. The particular constant domain in question may be indicated as C_H1, C_H2, C_H3.

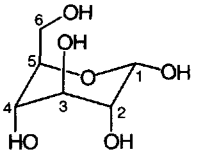
$\mathbf{C_L}$ *symbol for* the constant region of an immunoglobulin **light chain**.

C *symbol for* **1** capacitance. **2** heat capacity (C_p at constant pressure, C_V at constant volume, C_m molar heat capacity). **3** number concentration (C_B for a substance B).

C- *(in chemical nomenclature)* a locant indicating substitution on carbon; a particular carbon atom may be specified by a right superscript.

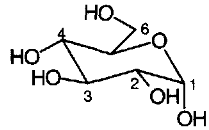
-*C* a conformational descriptor designating the **chair conformation** of a six-membered ring form of a monosaccharide or monosaccharide derivative. Locants of ring atoms that lie on the side of the structure's reference plane from which the numbering appears clockwise are indicated by left superscripts and those that lie on the other side of the reference plane by right subscripts; e.g. $-{}^1C_4$, $-{}^4C_1$. *See also* **conformation** (def. 2).

$-{}^1C_4$ *or (formerly)* **-1*C*** conformational descriptor for an aldopyranose in the chair conformation, with C2, C3, C5, and O5 in the structure's reference plane, and with C1 above and C4 below the plane; e.g. *α*-D-glucopyranose-${}^1C_4$.



${}^1C_4$

$-{}^4C_1$ *or (formerly)* **-*C*1** a conformational descriptor for an aldopyranose in the chair conformation, with C2, C3, C5, and O5 in the structure's reference plane, and with C1 below and C4 above the plane; e.g. *α*-D-glucopyranose-${}^4C_1$.



${}^4C_1$

Ca *symbol for* calcium.

CAA a codon in mRNA for L-glutamine.

CAAT box *or* **CAT box** a conserved sequence in DNA found within the **promoter** region of the protein-encoding genes of many eukaryotes. It has the consensus sequence GGCCAATCT and is believed to determine the efficiency of transcription from the **promoter**.

Cab-O-Sil *proprietary name for* **fumed silica** supplied by Cabot Carbon Ltd.

CaBP3 *see* **calreticulin**.

CAC a codon in mRNA for L-histidine.

cachectic factor a 24 kDa proteoglycan isolated from the murine adenocarcinoma MAC16 that has been implicated in the production of **cachexia**. The peptide has the N-terminal sequence YDPEAASAPGSGDPSHEA and has *N*- and *O*-glycans. Intravenous injection of the proteoglycan produces rapid weight loss.

cachectin *see* **cachexia, tumour necrosis factor-**α.

cachexia a condition caused by chronic disease, such as cancer, and characterized by wasting, emaciation, feebleness, and inanition. It led to the name 'cachectin' for the protein now known as **tumour necrosis factor** but a 24 kDa proteoglycan has more probably been implicated in this. *See* **cachectic factor**.

cacodylate **1** dimethyl arsinate, the anion $(CH_3)_2{-}AsOO^-$ derived from cacodylic acid. It is useful as a buffer substance ($pK_a = 6.2$). **2** any salt of cacodylic acid.

cadaverine 1,5-pentanediamine; a substance formed by microorganisms in decaying meat and fish by the decarboxylation of lysine. It also occurs as an intermediate in the biosynthesis, via lysine, of some quinolizidine alkaloids (e.g. lupinine) in plants.

cadherin any member of a family of calcium-dependent cell adhesion proteins that preferentially interact in a homophilic manner in cell–cell interactions. Cadherins may thus contribute to the sorting of heterogeneous cell types; they are type I **membrane proteins**. The names of different classes indicate the tissues in which they were first found: E-cadherin is present on many types of epithelial cells; N-cadherin is present on nerve, muscle, and lens cells; and P-cadherin is present on placental and epidermal cells. Typically, cadherins have five similar extracellular domains, the outermost three of which have Ca^{2+}-binding sites, and an intracellular C-terminal domain that interacts with the actin cytoskeleton. Examples (all human) E-cadherin precursor: database code CADE_HUMAN, 882 amino acids (97.35 kDa); N-cadherin precursor: database code CADN_HUMAN, 906 amino acids (99.75 kDa); P-cadherin precursor: database code CADP_HUMAN, 829 amino acids (91.32 kDa); three motifs.

cadmium mycophosphatin a cadmium-binding phosphoglycoprotein, M_r 12 000; it lacks sulfur-containing amino acids, but is rich in aspartic and glutamic acids and phosphoserine.

CAD protein a multifunctional protein found in many eukaryotes and containing domains for three enzymes of pyrimidine biosynthesis: **carbamoyl phosphate synthase** (glutamine-hydrolysing); **aspartate transcarbamylase**; and **dihydroorotase**. Example from *Drosophila melanogaster*: database code PYR1_DROME, 2236 amino acids (248.95 kDa).

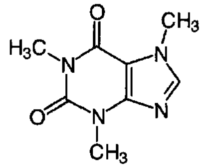
Caenorhabditis elegans a nematode (roundworm), with a genome of about 97 million base pairs. Its cellular anatomy has been fully described, and the nucleotide sequence of the genome determined, revealing over 19 000 genes, 40% of the predicted protein products having significant matches in other organisms.

caerulein *the Brit. spelling of* **cerulein.**

caeruloplasmin *the Brit. spelling of* **ceruloplasmin.**

caesium *alternative spelling (esp. Brit.) for* cesium.

caffeine 1,3,7-trimethylxanthine, 3,7-dihydro-1,3,7-trimethyl-1*H*-purine-2,6-dione; a substance found in certain plants, notably in tea and coffee and the beverages derived from them. It is an adenosine (A_1 and A_2) receptor antagonist and phosphodiesterase inhibitor, and a stimulant of the central nervous system, affecting the cardiovascular system and causing diuresis. Its actions are similar to those of **theophylline** but less potent.



caffetannic acid *an alternative name for* **chlorogenic acid.**

CAG a codon in mRNA for L-glutamine.

cage carrier *see* **ionophore.**

cage compound **1** *an alternative name for* **clathrate** (def. 1). **2** any compound possessing a nonplanar bicyclic or polycyclic molecular structure that encloses a cavity.

caged ATP a type of protected ATP analogue, e.g. adenosine-5′-triphospho-1-(2-nitrophenyl)ethanol (or the methanol-containing equivalent), that releases ATP in good yield when photolysed by a short pulse of light of 360 nm wavelength. Similarly, guanosine-5′-triphospho-1-(2-nitrophenyl)ethanol is used as **caged GTP**. These compounds can be introduced into cells prior to photolysis. Caged ATP has been used, e.g., to study muscle contraction on the millisecond timescale.

caged GTP *see* **caged ATP.**

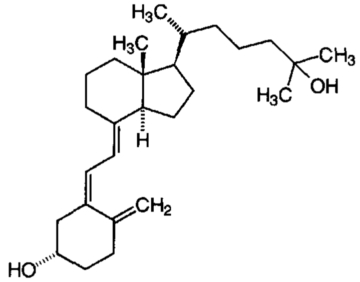
Cahn–Ingold–Prelog rule *or* **Cahn–Ingold–Prelog system** *an alternative name for* **sequence rule.**

cal *symbol for* calorie.

calbindin *other names*: cholecalcin; calbindin D9K; CABP; vitamin D-dependent calcium-binding protein; an **EF-hand** Ca^{2+}-binding protein from intestine. It is present in a variety of cells, including absorptive cells of the duodenum (in which its synthesis is vitamin D_3-dependent), in hippocampal cells, and in kidney. Example from *Rattus norvegicus*: database code CABI_RAT, 78 amino acids (8.90 kDa). Example from human: database code CABI_HUMAN, 78 amino acids (8.88 kDa).

calcemic factor *or (esp. Brit.)* **calcaemic factor** the prohormone of **parathyrin.**

calcidiol *the recommended trivial name for* calcifediol, 25-hydroxycholecalciferol. It is formed in liver from cholecalciferol, and is the major store of the vitamin in the body, being present largely in plasma. It is the precursor of the hormonal form of the vitamin, calcitriol. *See also* **vitamin D.**



calciductin *a name proposed for a* 23 kDa sarcolemmal protein that, when phosphorylated by a cAMP-dependent protein kinase, causes an approximately two-fold stimulation of the slow Ca^{2+} inward current.

calcifediol 25-hydroxyvitamin D_3; *another name for* **calcidiol.** *See also* **vitamin D.**

calciferol *an alternative (older) name for* **ergocalciferol**, also known as vitamin D_2. *See also* **vitamin D.**

calcification the deposition of calcium salts in the tissues. It is part of the normal process of bone formation in which **hydroxyapatite** crystals are deposited, in the neighbourhood of osteoblast cells, in a pre-existing collagen-containing matrix.

calcified describing a tissue that has been solidified by deposition of calcium; having undergone **calcification.**

calcimedin *see* **annexin.**

calcimycin *see* **A23187.**

calcineurin *or* **protein phosphatase 2B** (*abbr.*: PP-2B) *or* **CaM-BP$_{80}$** a major heat-labile calmodulin-binding protein isolated from bovine brain but later found in all cells from yeast to mammals. It has protein serine/threonine phosphatase activity, and is classified as a protein phosphatase 2B (EC 3.1.3.16). It plays an important role in terminating synaptic transmission, and in T-cell activation. It is only weakly inhibited by **okadaic acid**, but potently inhibited by certain pyrethrin compounds. It consists of two subunits, A and B, the A subunit having the catalytic activity and the B subunit conferring calcium sensitivity. Examples, from human, A subunit: database code P2B1_HUMAN, 514 amino acids (58.01 kDa); B subunit: database code CALB_HUMAN, 169 amino acids (19.17 kDa). The B subunit has a myristoylation site at the N terminus.

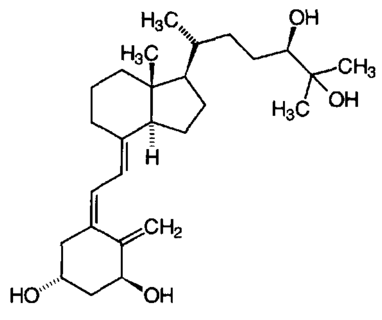
calciol *an alternative trivial name for* **cholecalciferol.**

calciosome a discrete cytoplasmic organelle in non-muscle cells, and a candidate for the inositol 1,4,5-trisphosphate-sensitive Ca^{2+} store. It has a high content of a **calsequestrin**-like protein.

calciotropic describing any hormone or other agonist acting on calcium metabolism.

calciphorin a calcium ionophore polypeptide, M_r 3000, isolated from the inner membrane of calf mitochondria.

calcitetrol *the recommended trivial name for* 1α,24*R*,25-trihydroxycholecalciferol. It is the inactivation product of calcitriol, from which it is formed by calcitriol 24-hydroxylase. *See also* **vitamin D.**



calcitonin *or* **thyrocalcitonin** a 3.4 kDa polypeptide hormone, that regulates the levels of calcium and phosphate in blood. In all species studied it consists of a single polypeptide chain of 32 amino-acid residues with a C-terminal prolinamide residue and an N-terminal seven-membered ring formed by a disulfide bridge between hemicystine residues at positions 1 and 7. The single letter code structure for the polypeptide is

CGNLSTCMLGTYTQDFNKFH-
TFPQTAIGVGAP–NH_2.

Calcitonin is secreted by the **C cells**, which in mammals occur primarily in the thyroid gland; in more primitive vertebrates they are found mainly in the ultimobranchial bodies and the lung. The hormone is also found to be concentrated in the hypothalamus of humans and some other vertebrates, and in the primitive brain of the chordate organism *Ciona intestinalis*. Calcitonin causes a rapid but short-lived drop in the level of calcium and phosphate in blood by promoting the incorporation of these ions in the bones. Overall its action is antihypercalcemic, opposing that of **parathyrin**. The response of plasma calcitonin to stimulation by administered **pentagastrin** or calcium may be used to screen for medullary cell carcinoma of the thyroid; the level rises to abnormally high values after stimulation, though normal in its absence. The precursor (human example below) and **calcitonin gene-related peptide** precursor are formed by alternative splicing of the same gene: database code CAL0_HUMAN, 141 amino acids (15.45 kDa); calcitonins have three motifs.

calcitonin gene-related peptide *abbr.*: CGRP; a pleiotropic 3.8 kDa polypeptide of 37 amino-acid residues, found in parts of the nervous system and endocrine system, and in some other organs. It is a potent vasodilator and hypotensive agent, and is regarded as a neuromodulatory peptide. The mRNA is formed by **alternative splicing** of the transcript of the **calcitonin** gene, which has exons coding for a region common to both CGRP and calcitonin precursors together with specific calcitonin and CGRP exons. CGRP was the first peptide to be identified by a molecular approach in the absence of biological information; its existence was predicted, in 1983, on the evidence of a second, structurally distinct transcript resulting from calcitonin gene transcription. Initially a 12.5 kDa pro-CGRP is formed, from which a 55-residue 'amino terminal flanking peptide' is generated. The mature polypeptide has the structure (rat)

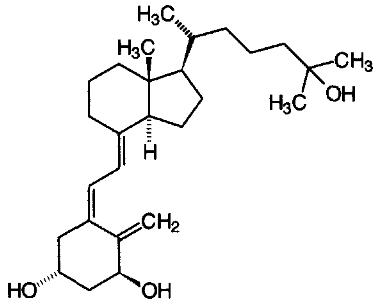
SCNTATCVTHRLAGLLSRSGGVV
KDNFVPTNVGSEAF-NH_2,

a form known as α-CGRP. Another form, known as β-CGRP, is identical except for Lys instead of Glu at position 35, and is encoded by a separate gene related to the calcitonin/CGRP gene but lacking any exon encoding calcitonin. α-CGRP is found in the central nervous system in a distribution distinct from that of any other known neuropeptide; it is present in trigeminal and spinal sensory ganglion cells, in olfactory and gustatory systems, and at neuromuscular junctions. CGRP is also found in the endocrine system – in a subset of adrenal medullary cells and in small amounts in thyroid C cells – in bronchiolar cells, and in intestinal cells. Administration of synthetic rat CGRP causes a drop in blood pressure (CGRP is the most potent known hypotensive gene peptide), and increase in heart rate. *In vitro* it can relax arteries from many vascular beds taken from a variety of species, and is one of the most potent vasodilators known. It also inhibits gastrointestinal motility and causes catecholamine release and gastric hypoacidity. *See also* **amylin**.

calcitonin gene-related peptide receptor any of several membrane proteins that bind calcitonin gene-related peptide (CGRP) and mediate its effects. $CGRP_1$ and $CGRP_2$ subtypes have been proposed; the antagonist $CGRP_{8-37}$ is selective for $CGRP_1$ receptors, and $[Cys(Acm)^{2,7}]$CGRP for $CGRP_2$ receptor types. CGRP receptors are widely distributed in the nervous system, following the pattern of distribution of CGRP, except that receptor numbers are high in cerebellum. They are also widespread in the cardiovascular system, as well as in adrenal, pituitary, exocrine pancreas, kidney, and bone. CGRP receptor stimulation leads to cyclic AMP accumulation in a number of tissues (heart, liver, muscle, pancreatic acinar cells) indicating coupling to one or more G-proteins.

calcitonin receptor any of the membrane proteins that bind calcitonin and mediate its effects. They are seven-transmembrane-helix receptors, normally coupled to G-proteins. In many systems they can activate both adenylate cyclase and the phosphatidylinositol cycle. In some systems, which second messenger system is activated depends entirely on conditions. For example in synchronized cells of a kidney cell line calcitonin activates adenylate cyclase during the G_2 phase, but activates the phosphoinositol cycle during S phase. Activation of the phosphatidylinositol cycle also causes activation of protein kinase C. In other systems, e.g. in brain, adenylate kinase is inhibited, but not through a G-protein. Example (precursor) from human ovarian small cell carcinoma: database code CALT_HUMAN, 490 amino acids (57.3 kDa).

calcitriol *the recommended trivial name for* 1α,25-dihydroxycholecalciferol. *See* **cholecalciferol**, **vitamin D**.



calcium *symbol*: Ca; an alkaline-earth metal of group 2 of the IUPAC **periodic table**; relative atomic mass 40.08, atomic number 20; it occurs naturally only in an ionized (Ca^{2+}) or combined state, and is a mixture of stable nuclides of relative mass 40 (96.97 atom percent) and 44 (2.06 atom percent) with small proportions of nuclides of relative mass 43, 46, and 48. Calcium is the fifth most abundant element of the Earth's crust and is an essential component of all living material. It occurs in bone, shell, and teeth and low concentrations of ionic calcium play many important roles in the regulation of diverse cellular processes. The most abundant mineral in the human body, most of it is in the skeleton. It has especially important functions in bone, in the control of nervous, muscle, and other excitable tissue, and as a **second messenger** and regulator of enzyme activity. Calcium homeostasis depends on the action of: (1) **parathyroid hormone**, which increases tubular reabsorption of calcium, and releases calcium from bone, thereby having overall a hypercalcemic action; (2) **vitamin D** (calcitriol), which causes absorption of calcium from the gut and its release from bone, and also therefore has a hypercalcemic action; and (3) **calcitonin**, which reduces calcium resorption from bone. The range of plasma calcium in normal human adults is 2.2–2.6 mmol L^{-1}. *See also* **hypercalcemia**, **hypocalcemia**.

calcium-45 *symbol*: ^{45}Ca; an artificial radioactive nuclide of calcium, with a half-life of 165 days. It emits β^--particles (i.e. electrons) of 0.252 MeV max. and no γ-radiation.

calcium-47 *symbol*: ^{47}Ca; an artificial radioactive nuclide of calcium, with a half-life of 4.54 days. It emits β^--particles (i.e. electrons) of two energy ranges (1.98 and 0.67 MeV max.) and γ-radiation of three energies (1.31, 0.815, and 0.49 MeV).

calcium channel any of several proteins that permit the controlled (gated) passage of calcium ions through membranes. There are several types. L-type channels are involved in excitation–contraction coupling in muscle; they bind 1,4-dihydropyridine (DHP). The DHP receptor is a protein located in transverse (T-) tubules that specifically binds DHP compounds such as **nifedipine**. This modulates the slow L-type channel in skeletal muscle, neurons, and cardiac cells. N-type channels, identified only in neurons, participate in neurotransmitter release and are blocked by **ω-conotoxin**. T-type channels influence pacemaker activity in the heart and repetitive spike activity in neurons; they are blocked by octanol. P-type channels, identified in some CNS neurons and prominent in Purkinje cells, are blocked by ω-agatoxin, much less so by ω-conotoxin. The L-type channel from skeletal muscle is hetero-oligomeric, consisting of five subunits: α1, database code CIC1_RABIT, 1873 amino acids (211.78 kDa); α2, database code CIC2_RABIT, 1106 amino acids (124.90 kDa); β, database code CICB_RABIT, 524 amino acids (57.76 kDa); γ, database code CICG_RABIT, 222 amino acids (25.03 kDa); and δ (M_r ≈27 000). The largest subunit (α1) is responsible for many of the functional features, including DHP binding and the voltage-gated pore. It is structurally similar to the Na^+ channel α subunit, with four internal repeats of 200–300 amino acids exhibiting sequence homology. Each of these repeats contains six putative transmembrane segments, S1 to S6; segment S4 in all repeats has positively charged groups in -(K/R)XX- repeats (four to six K or R residues per segment), that have been implicated in sensing voltage changes; potential *N*-glycosylation sites (N residues 79 and 257, rabbit skeletal muscle) and seven potential cAMP-dependent phosphorylation sites (S residues 687, 1502, 1575, 1757, 1772, 1854, T residue 1552) have been identified on the α_1 protein. *See also* **verapamil**.

calcium-dependent regulator protein *or* **calcium-dependent modulator protein** *abbr.*: CDR; *former name for* **calmodulin**.

calcium ion (Ca^{2+})-release channel *another name for* the **ryanodine receptor**.

calcium pump any system that transports calcium ions across a biological membrane from a lower to a higher concentration with the consumption of energy. The term is used especially of the pump in the sarcoplasmic reticulum of muscle of which a transmembrane **calcium-transporting ATPase** is an integral part.

calcium-regulated actin bundling protein a protein that is probably involved in the formation of filopodia. Example from *Dictyostelium discoideum*: database code ACTB_DICDI, 295 amino acids (33.31 kDa).

calcium-transporting ATPase EC 3.6.1.38; an integral membrane protein that forms an essential component of the **calcium pump** in sarcoplasmic reticulum. It catalyses the hydrolysis of ATP to ADP and orthophosphate, simultaneously transporting two calcium ions from the cytosol into the sarcoplasmic reticulum for each ATP hydrolysed. Probably comprising 10 transmembrane domains, it is related to other metal-ion-transporting ATPases. A number of types exist. Example, SR/ER type from *Drosophila melanogaster*: database code ATCB_DROME, 1002 amino acids (109.47 kDa).

calcium-trigger protein any of a group of small calcium-binding **trigger** proteins, including **calmodulin**, troponin C (*see* **troponin**), and **parvalbumin**.

calculus (*pl.* **calculi**) a concretion of material that forms within the body. It often resembles a small pebble, hence is called a 'stone'. Calculi are most common in the gall bladder or kidney, and are composed variously of organic or inorganic salts, frequently of calcium; cholesterol calculi are gallstones of pure cholesterol.

calcyclin *other names*: prolactin receptor associated protein, growth factor-inducible protein 2A9, S100 calcium-binding protein A6; a small protein that copurifies with **prolactin receptor**. It is induced in fibroblasts by growth factors and is overexpressed in acute myeloid leukemia. It binds specifically **annexin** XI. Example from human fibroblasts: database code S106_HUMAN, 90 amino acids (10.18 kDa).

caldecrin a pancreatic protein that lowers serum calcium. It also has chymotrypsin-like activity, which is not needed for the calcium-lowering effect.

caldesmon an F-actin crosslinking protein in thin filaments in smooth muscle and in stress fibres in (nonmuscle) fibroblasts. It also binds myosin and tropomyosin. It appears to be a homodimer comprising two polypeptide chains each with a single actin-binding site and a single self-association site. It controls actin–myosin interactions and inhibits the actin-activated ATPase of myosin. Its action is attenuated by calcium-calmodulin and potentiated by tropomyosin. Caldesmon will interact with Ca^{2+}-calmodulin (but not with calmodulin alone), an interaction that causes caldesmon to dissociate from F-actin. Example from *Gallus gallus*: database code CALD_CHICK, 756 amino acids (86.96 kDa). *See also* **caldesmon kinase**, **caldesmon phosphatase**.

caldesmon kinase EC 2.7.1.120; the name given to a reaction now known to result from the autophosphorylation of **caldesmon** (by ATP) in the presence of calcium. This autophosphorylation abolishes the ability of caldesmon to bind actin.

caldesmon-phosphatase EC 3.1.3.55; an enzyme that hydrolyses the phosphate from caldesmon phosphate.

caldolysin an extracellular protease isolated from *Thermus aquaticus* strain T-351, an extreme thermophile. It contains 13% carbohydrate by weight, and one atom of zinc; M_r 21 000.

calelectrin *see* **annexin IV**, **annexin V**, **annexin VI**.

calgizzarin an **EF-hand** Ca^{2+}-binding protein isolated from chicken gizzard smooth muscle. It is also abundant in rabbit lung and is found in other tissues. It has homology with **annexin** II light chain and is a homodimer (disulfide-linked). It is overexpressed in human colorectal tumours. Example from pig: database code ICAL_PIG, 713 amino acids (77.04 kDa).

callose a linear (1,3)-*β*-D-glucan insoluble in water but which dissolves in dilute alkali. The glucan chains form stable triple helices. Callose is a ubiquitous higher plant polysaccharide occurring as a component of specialized, and often transient, cell walls, especially in reproductive tissues and is a component of the cell plate. Callose is rapidly deposited on plasma membranes as drops, plugs, or plates on wounding or physiological stress. Callose deposits on cell walls stain specifically with Aniline Blue or its fluorochrome. Specific hydrolases, e.g. glucan endo-1,3-*β*-D-glucosidase (EC 3.2.1.39) for callose and membrane-bound callose synthetase (EC 2.4.1.34) are found in plants. Functionally callose may act as a temporary wall matrix, as a special permeability barrier or as a wall strengthening agent. Other (1,3)-*β*-D-glucans include yeast glucan, pachyman, lamarin, scleroglucan, curdlan, leucosin, mycolamarin, paramylon, chrysolamarin, and lentinan.

callus **1** a mass of relatively unspecialized tissue that develops at wound sites in plants, forming a protective covering. Callus cells are used in tissue culture as the starting material for the propagation of plant clones. **2** tissue formed during the healing of broken bone.

calmitine a mitochondrial Ca^{2+}-binding protein of fast-twitch muscle fibres with a characteristic 28-residue signal. The mature protein is identical to **calsequestrin** of the sarcoplasmic reticulum.

calmodulin *abbr.*: CAM; a small, heat-stable, acidic, calcium-dependent modulator protein that binds four calcium ions per molecule and is then able to stimulate a variety of eukaryotic enzymes or enzyme systems. These include brain adenylate cyclase, cytosolic cyclic nucleotide phosphodiesterase, myosin light-chain kinase, erythrocyte Ca^{2+},Mg^{2+}-ATPase, plant NAD^+ kinase, and a number of calcium-dependent protein kinases including phosphorylase kinase (to which it contributes the δ subunits); various other cellular processes such as membrane phosphorylation and microtubule disassembly also are stimulated. Calmodulin consists of a single-chain polypeptide of known sequence, aspartic or glutamic accounting together for about 30% of the total residues; it contains one residue of N^6-trimethyllysine per mole but no cysteine, hydroxyproline, or tryptophan residues. Calmodulin appears to be ubiquitous within the animal and plant kingdoms, to be without species or tissue specificity, and to be almost invariant in structure, being conserved nearly perfectly throughout evolution; it has **EF-hand** domains and is similar to, but not identical with, troponin C from muscle. It was known previously by a variety of provisional names including modulator protein, phosphodiesterase activator protein, troponin C-like (modulator) protein, and calcium- (or Ca^{2+})-dependent regulator (or modulator) protein (CDR protein). Examples (1) from black rat: database code NRL_3CLN, 143 amino acids (16.13 kDa); 3-D structure known; (2) from human, rabbit, other rat species, *Xenopus laevis*, chicken, the sea urchin *Arbacia punctulata*, and *Bos taurus*: database code CALM_HUMAN, 148 amino acids (16.13 kDa).

calmodulin-dependent protein kinase II *recommended name*: Ca^{2+}/calmodulin-dependent protein kinase, EC 2.7.1.123; *other names*: Ca^{2+}-CAM kinase II; microtubule-associated protein 2 kinase; multifunctional Ca^{2+}-CAM kinase. An enzyme with broad specificity and activated by Ca^{2+} that catalyses the phosphorylation of Ser and Thr residues in proteins by ATP. It is involved in cell regulation. An oligomeric polypeptide, it is composed of various combinations of α (M_r 60 000) and β (M_r 55 000) subunits in a dodecahedral structure. On activation by Ca^{2+}-CAM, autophosphorylation occurs as a result of which the enzyme converts to a Ca^{2+}-CAM-independent form. Substrates include synapsin I, tryptophan hydroxylase, skeletal muscle glycogen synthetase, and microtubule-associated proteins tau and MAP-2. A synthetic peptide substrate is PLSRTLSVSS. Examples from yeast: type I, database code KCC1_YEAST, 445 amino acids (50.08 kDa); type II (potentially autophosphorylation enzyme), database code KCC2_YEAST, 447 amino acids (50.36 kDa).

calmodulin-like domain protein kinase *abbr.*: CDPK; EC 2.7.1.–; *other name*: Ca^{2+}-binding Ser/Thr protein kinase. An enzyme that is subject to autophosphorylation and resembles calmodulin-dependent protein kinases. Example from soybean: database code CDPK_SOYBN, 508 amino acids (57.10 kDa).

calnexin a calcium-binding protein of the endoplasmic reticulum. It appears to play a role in the processing of endoplasmic reticulum proteins, in monitoring assembly, and in retaining unassembled or incorrectly folded proteins. It has a single transmembrane helical region. Example from human (precursor): database code CALX_HUMAN, 592 amino acids (67.57 kDa); residues 1–20 are the signal, 21–592 calnexin.

calomel electrode the mercury–mercurous chloride (mercury–calomel) ($Hg;Hg_2Cl_2$, Cl^-) half-cell that is reversible to chloride ions. Because of its stability and simplicity, it is frequently used in conjunction with a saturated-KCl bridge as a reference electrode in pH meters, etc.

calorie *or (sometimes)* **gram calorie** *or* **small calorie** *symbol*: cal; any of several units of heat or energy in the CGS system. A calorie originally represented the quantity of heat required to raise one gram of water through one degree centigrade (i.e. Celsius). However, the energy represented by a calorie varies slightly according to the initial temperature of the water. Hence, the **International Table calorie** (*symbol*: cal_{IT}) is now taken to equal 4.1868 J exactly, while the **15 °C calorie** (*symbol*: cal_{15}) is equal approximately to 4.1855 J. Thermochemical calculations have often used a slightly different value; one **thermochemical calorie** (*symbol*: cal_{th}) is now taken to equal 4.184 J exactly. In nutrition and physiology, the term 'calorie' is often used to mean the kilocalorie or Calorie (initial capital), equal to 10^3 cal_{IT}. Use of all these units is now deprecated, the joule now being preferred.

Calorie *or (sometimes)* **kilogram calorie** *or* **large calorie** the name used, especially in nutrition, for the kilocalorie, i.e. 10^3 cal_{IT}. *Compare* **calorie**.

calorific of, relating to, or generating heat.

calorific value the amount of heat produced by the complete combustion of a given mass of a substance, such as a fuel or foodstuff. It is usually expressed in J kg^{-1}.

calorigenesis the production or increased production of heat in an organism. —**calorigenic** *adj.*

calorimeter an instrument for determining quantities of heat evolved, transferred, or absorbed.

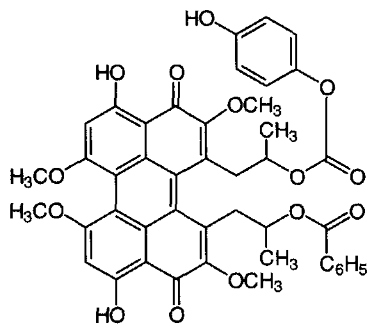
calorimetry the measurement of the amount of heat evolved, transferred, or absorbed by a system. —**calorimetric** *adj.*

calpactin *see* **annexin**.

calpain EC 3.4.22.17; a Ca^{2+}-activated, neutral, thiol proteinase enzyme that preferentially cleaves the bonds: Tyr-|-Xaa, Met-|-Xaa, or Arg-|-Xaa with Leu or Val as the P2 residue. Calpains are cytoplasmic mammalian enzymes, of three main types; these have an identical small regulatory **EF-hand** subunit and large catalytic subunit but differ in their sensitivity to Ca^{2+}. Example (human) calpain II large subunit (responds to millimolar Ca^{2+}): database code CAP2_HUMAN, 700 amino acids (79.92 kDa).

calpastatin a protein, found in liver and erythrocytes of several mammals, that is a specific calpain inhibitor. There are four characteristic domains involved in the inhibitory action. Example (precursor) from pig: database code ICAL_PIG, 713 amino acids (77.04 kDa).

calphostin C one of a group of compounds isolated from *Cladosporium cladosporioides* having a unique structural feature in possessing a perylene quinone. It was identified in a search for inhibition of protein kinases. It inhibits protein kinase C by action on the regulatory domain and has a relatively high degree of specificity, but acts equally on all isoforms having the common regulatory domain. It is a highly potent and specific inhibitor (IC_{50} = 50 nM) of protein kinase C. At higher concentrations it inhibits other kinases, including myosin light chain kinase, and cyclic GMP- and cyclic AMP-dependent kinases.



calphotin a Ca^{2+}-binding protein found in the cytoplasm of photoreceptor cells of *Drosophila melanogaster*. The C-terminal region contains a leucine zipper uninterrupted by prolines. The protein contains >50% proline, alanine, and valine. Database code CPN_DROME, 865 amino acids (84.78 kDa).

calponin a thin filament-associated protein implicated in regulation and modulation of smooth muscle contraction. It is found as an actin-, calmodulin-, and tropomyosin-binding protein present in many vertebrate smooth muscles, and is related to troponin T in immunological and biochemical characteristics. Examples from rat; calponin H1: database code CLP1_RAT, 297 amino acids (33.31 kDa); acidic isoform: database code CLPA_RAT, 330 amino acids (36.39 kDa).

calpromotin a cytoplasmic protein of red cells that activates Ca^{2+}-dependent potassium transport.

calregulin *see* **calreticulin**.

calreticulin *other names*: calregulin; Crp55; CaBP3; HACBP; erp60; a Ca^{2+}-binding protein of the endoplasmic reticulum lumen. Example (precursor) from human: database code CRTC_HUMAN, 417 amino acids (48.09 kDa); amino acids 1–17 constitute the signal; the C terminus contains the **KDEL** sequence.

calsequestrin a 44 kDa calcium-binding protein of low isoelectric point, found on the interior surface of the membrane of the **sarcoplasmic reticulum** of muscle. It acts as an internal calcium store in muscle and releases Ca^{2+} at **calcium channels**. It has only a moderate affinity for calcium ions, but a high capacity (>40 Ca^{2+} ions per molecule); selective binding of Ca^{2+} in preference to other cations is promoted by ATP; it also exhibits protein kinase activity. Calsequestrin makes up some 19% of the protein of isolated sarcoplasmic vesicles, and is thought to assist in depleting the sarcoplasm of free calcium ions during relaxation. Example, human muscle isoform: database code CAQS_HUMAN, 390 amino acids (44.53 kDa), very acidic (47 Asp, 56 Glu; has a sequence of eight aspartic-acid residues at the C terminus).

calspectin *an alternative name for* **fodrin**.

calstorin a calcium-binding protein of the microsomal lumen of rat brain, similar to **calsequestrin**.

caltractin *or* **centrin** an **EF-hand** Ca^{2+}-binding protein of the centrosome of interphase and mitotic cells that plays a fundamental role in microtubule-organizing centre structure. It is present in the centrosome/basal body apparatus of the green alga *Chlamydomonas reinhardtii*. It contains four EF-hands, and belongs to the calmodulin/parvalbumin/troponin C superfamily. Example from the salt bush *Atriplex nummularia*: database code CATR_ATRNU, 167 amino acids (19.22 kDa).

caltrin *abbr. for* calcium transport inhibitor; *other name*: seminalplasmin; a small basic protein of male seminal vesicle fluids. It binds to calmodulin and inhibits Ca^{2+} uptake by epididymal spermatozoa. It has antimicrobial properties. Example (precursor, bovine): database code CALT_BOVIN, 80 amino acids (8.97 kDa).

Calvin, Melvin (1911–97), US chemist and biochemist distinguished for pioneering the use of radioactive isotopes, especially carbon-14, as tracers in metabolism; Nobel Laureate in Chemistry (1961) 'for his research on the carbon dioxide assimilation in plants'.

Calvin cycle *see* **reductive pentose phosphate cycle**.

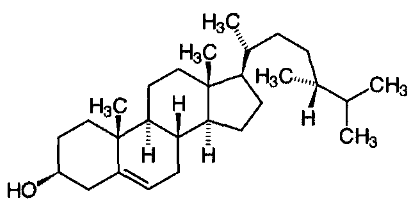
Cam *symbol for* the **carbamoylmethyl** group.

CAM *abbr. for* **1 calmodulin. 2** crassulacean acid metabolism.

CaM-BP$_{80}$ *an alternative name for* **calcineurin**.

cAMP *abbr. for* cyclic AMP; *see* **adenosine 3′,5′-phosphate**.

campesterol (24*R*)-ergost-5-en-3β-ol; a sterol found in small amounts in rapeseed and other vegetable oils.

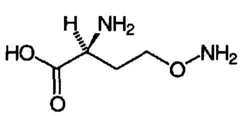


camphor 1,7,7-trimethylbicyclo[2.2.1]heptan-2-one; a compound obtained from the camphor tree, *Cinnamomum camphora*, which is indigenous to Taiwan. It is widely used in ointments and liniments. It is also used as a starting material in organic chemistry.

cAMP receptor protein *see* **catabolite (gene) activator protein**.

cAMP-regulatory protein *see* **catabolite (gene) activator protein**.

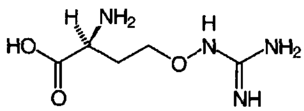
canaline O^4-aminohomoserine; 2-amino-4-(aminooxy)-butanoic acid; $H_2N–O–[CH_2]_2–CH(NH_2)–COOH$; a basic α-amino acid found as the L-enantiomer in the jack bean, *Canavalia ensiformis*, and in some other legumes containing L-canavanine. It is formed from **canavanine** by deamidination or transamidination. It inhibits pyridoxal-dependent enzymes.



canavalin a globular protein isolated from ripe seeds of the jack bean, *Canavalia ensiformis*. It is a storage protein, with sequence similarity to **phaseolin**, **vicilin**, and related proteins. *See also* **concanavalin A**. Example from *Canavalia gladiata*: database code CANA_CANGL, 445 amino acids (50.29 kDa).

canavanase *see* **arginase**.

canavanine O^4-guanidinohomoserine; 2-amino-4-(guanidinooxy) butanoic acid; a basic α-amino acid, occurring as the L-enantiomer in certain legumes esp. the jack bean, *Canavalia ensiformis*; an arginine antagonist. In seeds of the jack bean it constitutes up to 8% of their dry mass and forms their main storage compound for nitrogen. On germination of the seeds the nitrogen is released for synthetic purposes by hydrolysis of L-canavanine to L-canaline and urea, from each of which ammonium is then formed by further hydrolysis. *See also* **canaline**.



cancellate *or* **cancellated** *or* **cancellous** having a porous or spongy structure; meshlike or lattice-like.

cancer any malignant **neoplasm**. Cancers are usually divided into **carcinomas**, derived from epithelial tissue, and **sarcomas**, derived from connective tissue.

cancer-associated retinopathy protein *see* **recoverin**.

candela *symbol*: cd; the SI base unit of **luminous intensity**, defined as the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of 1/683 watt per steradian.

Candida a genus of (in many cases) dimorphic yeasts containing species of medical (*C. albicans*) and industrial importance: *C. cylindrica* is a source of **lipases** and is a rare example of a eukaryote with a non-standard genetic code for nuclear/cytoplasmic gene expression.

cane sugar *an old and common name for* **sucrose**.

cannabinoid any of about 30 derivatives of 2-(2-isopropyl-5-methylphenyl)-5-pentylresorcinol found in the Indian hemp, *Cannabis sativa*, among which are those responsible for the narcotic actions of the plant and its extracts. The most important cannabinoids are cannabidiol, cannabinol, *trans*-Δ^9-tetrahydrocannabinol, *trans*-Δ^8-tetrahydrocannabinol, and Δ^9-tetrahydrocannabinolic acid. *See also* **cannabis**.

cannabinoid receptor any of several membrane proteins that bind cannabinol and structurally similar compounds and mediate their intracellular action. Two types have been characterized, both of which are seven-transmembrane-domain G-protein-coupled receptors. CB1 receptors are found in brain and testis. CB2 receptors are found in spleen and not in brain. For both types arachidonoylethanolamide (anandamide) is a putative endogenous ligand and both types are negatively coupled to adenylate cyclase decreasing intracellular cyclic AMP levels. Examples from *Mus musculus*: CB1, database code CB1R_MOUSE, 473 amino acids (52.94 kDa); CB2, database code CB2R_MOUSE, 347 amino acids (38.21 kDa).

cannabinol *see* **cannabinoid**.

cannabis the dried flowering or fruiting tops of Indian hemp, *Cannabis sativa*; numerous synonyms exist, including **marijuana**, **bhang**, and **maconha**. A resin, known variously as cannabin, hashish, or charas, is also obtained from the plant. Both preparations contain active principles, **cannabinoids**, and are used as recreational drugs, producing relaxation, euphoria, and enhanced awareness. Occasionally, anxiety or mental disturbance may result, and, rarely, loss of consciousness or even death. The major active ingredients are tetrahydrocannabinols. Cannabis is a controlled substance in the US Code of Federal Regulations.

Cannizzaro reaction the base-catalysed dismutation of aldehydes with no α hydrogen atoms into the corresponding acids and alcohols:

$$2R–CHO + OH^- \rightarrow R–COO^- + R–CH_2OH.$$

[After Stanislo Canizzaro (1826–1910).]

canonical form *or* **canonical structure** *an older name for* **contributing structure**.

canonical sequence a nucleotide or amino-acid sequence regarded as the archetype with which variants are compared.

cap **1** *or* **cap structure** a structural feature present at the 5′ end of most eukaryotic (cellular or viral) mRNA molecules and also of some virion mRNA molecules but not of bacterial mRNA molecules. It consists of a residue of 7-methylguanosine and a triphosphate bridge linking it 5′–5′ to the end of the polynucleotide chain; in addition, the first one or two nucleotide residues in the chain may possess a 2′-*O*-methyl group and the first, if adenylyl, a 6-*N*-methyl group. The cap structure is thought to protect the 5′ end of the mRNA from degradation by phosphatases or nucleases and to facilitate initiation of translation of mRNA by the eukaryotic (but not by the bacterial) ribosome. **2** a cluster or patch of aggregated proteins at one site on the surface of a cell, e.g. a lymphocyte. Surface components of the cell membrane are aggregated, e.g., by the action of polyvalent ligands, and may be swept to the posterior area of the cell when the cell moves, forming a cap at that site. **3** to form or place a cap on a molecule (or structure).

CAP *abbr. for* **1** catabolite (gene) activator protein. **2** chloramphenicol (use deprecated, full name preferred). **3** *Clostridium* (*histolyticum*) aminopeptidase, EC 3.4.11.13. **4** cyclic AMP receptor protein; *see* **CRP**.

capacitance *or* **electric capacitance** *symbol*: *C*; the ability of a system to store electric charge or a measure of this capacity for a particular system. $C = Q/U$ where Q is the quantity of charge stored to raise the system's potential by U volts. The SI derived unit of capacitance is the **farad**.

capacitance minimization method a simple method for the determination of asymmetric surface potentials in lipid bilayers, based on the dependence of bilayer capacitance on transmembrane voltage. The capacitance is measured by rectifying the 90° component of an applied a.c. current signal. A superimposed slow triangular wave results in a hysteresis-like time course of capacitance. The centre of the hysteresis figure is shifted along the voltage axis by an amount equal to the **capacitance minimization potential**, the difference between the dipole plus surface-charge potentials on the two sides of the membrane.

CAPB *see* **calbindin**.

cap-binding protein any protein (e.g. a 24 kDa protein obtainable from rabbit reticulocytes) that specifically binds to the **cap** (def. 1) of eukaryotic capped mRNA. Such proteins may facilitate the translation of capped mRNA molecules.

CAP factor *see* **catabolite (gene) activator protein**.

capillarity the phenomenon, resulting from surface tension, in which liquids rise up capillary tubes and which also causes them to form a concave or convex meniscus at their surface where it contacts a solid.

capillary **1** of, or relating to, a hair or hairlike structure. **2** of, or relating to, a tube with a very fine bore. **3** any of the very fine blood vessels that form a network between the arterioles and the venules throughout the body. **4** a capillary tube.

capillary column a chromatography column that has a very fine bore (0.1–0.25 mm), the inner surface of which may be coated with **stationary phase** or other support. Such columns are used in gas chromatography, supercritical fluid chromatography, or capillary electrophoresis. The column may be up to a few metres in length, coiled, and made of glass or fused silica. In high-performance liquid chromatography, columns of somewhat wider bore and shorter length are referred to as capillary columns.

capillary electrophoresis *abbr.*: CE; a very-high-resolution method of electrophoresis, also known as **capillary zone electrophoresis** (*abbr.*: CZE) *or* **high-performance electrophoresis**. It is based on the use of long (up to 100 cm), narrow (<100 μm) silica columns for separation of peptides, etc. by electroosmotic flow. By using sensitive detectors (e.g. laser-induced fluoresence), sensitivity in the attomole range can be achieved.

capillary filtration a technique for **membrane filtration** in which the membrane is supported inside porous capillary tubes (capillary membranes). Bundles of such tubes make up a larger filtration unit. *Compare* **hollow-fibre technique**.

capillary viscometer an instrument for measuring the viscosity of a liquid by determining the rate of flow of the liquid through a capillary tube of known dimensions. An **Ostwald viscometer** is a simple capillary viscometer; an **Ubbelholde viscometer** is a viscometer with several different capillary tubes, so that measurements may be made at different shear values.

capped mRNA an mRNA molecule that has a **cap** (def. 1).

capping the act or process of forming or placing a **cap** on a molecule or structure.

capping enzyme any guanylyltransferase enzyme concerned in the formation of a **cap** (def. 1) on an RNA molecule.

caprate *or* **caprinate** *a trivial name for* **decanoate**, or for any salt or ester of decanoic acid.

capric acid *or* **caprinic acid** *a trivial name for* decanoic acid; *see* **decanoate**.

caprin *a trivial name for* any of the glyceryl esters of decanoic acid (*see* **decanoate**); these often occur in milk fat, especially the triester tricaprin, $(CH_3–[CH_2]_8–COO)_3C_3H_5$.

caprinate *see* **caprate**.

caprinyl (*formerly*) *an alternative name for* **caproyl** (def. 1).

caproate *a trivial name for* **hexanoate**, or any salt or ester of hexanoic acid.

caproic acid *a trivial name for* hexanoic acid (*see* **hexanoate**).

caproin *a trivial name for* any of the glyceryl esters of hexanoic acid (*see* **hexanoate**); these often occur in milk fat, especially the triester tricaproin, $(CH_3–[CH_2]_4–COO)_3C_3H_5$.

caproyl **1** *or (formerly)* **caprinyl**; *symbol*: Dco; *trivial name for* decanoyl, $CH_3–[CH_2]_8–CO–$, the univalent acyl radical derived from decanoic acid (*formerly* capric acid *or* caprinic acid). Use of the term caproyl is disfavoured because of confusion with **caproyl** (def. 2) and **capryloyl**. **2** *an old trivial name for* hexanoyl, the univalent group derived from hexanoic acid (*formerly* caproic acid), a saturated unbranched, acyclic, aliphatic acid (i.e. fatty acid). It occurs naturally in acylglycerols in, e.g., coconut and palm nut oils, and milk fat. Use abandoned (*see* **caproyl** (def. 1)).

capryl **1** *an old trivial name for* **octyl**. **2** *an old trivial name for* *sec*-octyl. **3** *symbol*: Dec; *a trivial name sometimes used for* **decyl**.

caprylate *former name for* **octanoate** or for any salt or ester of octanoic acid.

caprylic acid *a former name for* octanoic acid; *see* **capryloyl**.

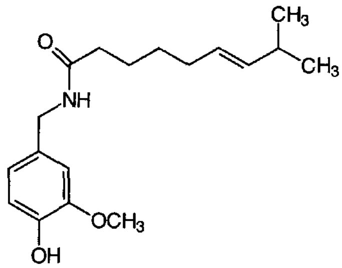
caprylic alcohol **1** *or* ***n*-caprylic alcohol** *a former name for* (*n*-) octyl alcohol (octan-1-ol). **2** *or* ***sec*-caprylic acid** *a former name for sec*-octyl alcohol (octan-2-ol). *See also* **caprylic acid**.

caprylin *a former name for* any of the glyceryl esters of **octanoic acid**; these occur in milk fat, especially the triester tricaprylin, $(CH_3–[CH_2]_6–COO)_3C_3H_5$.

capryloyl *or* **caprylyl** *an old trivial name for* **octanoyl**.

Caps *or* **CAPS** *abbr. for* 3-(cyclohexylamino)propanesulfonic acid; a compound with properties similar to a **Good buffer substance**, pK_a (20 °C) = 10.4.

capsaicin (*E*)-*N*-(4-hydroxy-3-methyoxybenzyl)-8-methylnon-6-enamide; the active principle in chilli pepper (*Capsicum*), paprika, and cayenne that causes a burning sensation by stimulation of nociceptive (pain sensory) neurons followed by desensitization of nociceptive responses and neuronal depletion of **substance P**.



capsaicin

capsanthin (3*R*,3′*S*,5′*R*)-3,3′-dihydroxy-β,κ-caroten-6′-one; the main carotenoid pigment of red pepper (*Capsicum annuum*).

capsid the protein coat that surrounds the infective nucleic acid in some virus particles. It comprises numerous regularly arranged subunits, or **capsomeres**.

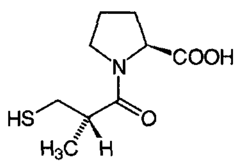
capsomere *or* **capsomer** any of the (protein) subunits that comprise the closed shell or coat (capsid) of certain viruses.

cap structure *see* **cap** (def. 1).

capsular antigen *or* **capsular polysaccharide** *or* **capsular substance** any of the antigens, usually polysaccharide in nature, that are carried on the surface of bacterial capsules. The term **K antigen** is used for those that mask somatic (O) antigens.

capsule 1 an outer layer, rich in polysaccharides, that is present in certain types of bacteria. Presence of a capsule is often linked with **virulence** and the capsular material is often immunogenic. 2 a membrane or sheath of connective tissue that encloses an organ or tissue.

captopril D-3-mercaptomethylpropanoyl-L-proline, (*S*)-1-(3-mercapto-2-methyl-1-oxopropyl)-L-proline; an antihypertensive drug that acts as an inhibitor of **angiotensin converting enzyme**. It was designed to fit the known active site of carboxypeptidase A; the mercapto group binds to the enzyme zinc ion, the amide carbonyl to a hydrogen-bonding site, and the proline carboxylate to an electrophilic centre. A related compound, **enalapril**, is more potent when administered orally and has fewer side effects.



captopril

capture cross-section the probability of a particle that impinges on an atomic nucleus being captured by that nucleus multiplied by the cross-sectional area of the nucleus (measured in **barns**).

carba+ *comb. form (in chemical nomenclature)* indicating the replacement of a heteroatom by a carbon atom.

carbamide *former name for* **urea**.

carbamido *an alternative term for* **ureido**.

carbamino the chemical group –NH–COO⁻, which is stable only in the salt form. Carbamino compounds are formed by spontaneous reversible reaction of CO_2 with unprotonated primary aliphatic amino groups; e.g. with α-amino acids to form carbamino acids, or with the N termini of proteins, as in the formation of carbamino hemoglobin from hemoglobin. *Compare* **carbamoyl**.

carbamoyl *or (formerly)* **carbamyl** *or* **carboxamido** *symbol*: Cbm; the acyl group, $-CO-NH_2$, corresponding to carbamic acid (an acid unknown in the free state). *See also* **carboxamide**.

carbamoylmethyl *symbol*: Cam; the chemical group 2-amino-2-oxoethyl, $-CH_2-CO-NH_2$.

carbamoyl-phosphate synthase 1 *or* **carbamoyl-phosphate synthase (glutamine-hydrolysing)** EC 6.3.5.5; *other names*: carbamoyl-phosphate synthetase (glutamine-hydrolysing); GD-CPSase. A glutamine-dependent enzyme that is the first step in the pathway for pyrimidine biosynthesis. It catalyses a reaction between 2 ATP, glutamine, CO_2, and H_2O to form 2 ADP, orthophosphate, glutamate, and carbamoyl phosphate; this enzyme is part of the **CAD protein** in many eukaryotes. 2 *or* carbamoyl-phosphate synthase (ammonia) EC 6.3.4.16; *other names*: carbamoyl-phosphate synthetase (ammonia); carbamoyl-phosphate synthetase I, carbon-dioxide–ammonia ligase. An enzyme that incorporates free ammonium ion into carbamoyl phosphate, which then enters the **ornithine–urea cycle**. It catalyses a reaction between 2 ATP, NH_3, CO_2, and H_2O to form 2 ADP, orthophosphate, and carbamoyl phosphate. Example of latter type from human mitochondria: database code CPSM_HUMAN, 1500 amino acids (164.64 kDa).

carbamyl *former name for* **carbamoyl**.

carbanion any anion containing an even number of electrons in which a significant part of the excess negative charge is located on one or more carbon atoms.

carbapenem antibiotic any of various broad-spectrum β-lactam antibiotics, such as thienamycin derived from *Streptomyces cattleya*, that inhibit peptidoglycan synthesis. Semisynthetic derivatives include imipenem.

carbene 1 any chemical species of the type R_2C: containing an electrically neutral, bivalent carbon atom with two nonbonding electrons. The spins of such electrons may be either antiparallel (in the singlet state) or parallel (in the triplet state). Carbenes are highly reactive short-lived groups formed as intermediates in the course of certain organic reactions. 2 *or (formerly)* **methylene** the chemical group H_2C:, the simplest of the carbenes.

carbenium ion any real or hypothetical **carbocation** with one important contributing structure containing a tervalent carbon with a vacant *p*-orbital. These structures have often been described as **carbonium ions**.

carbenoxolone glycyrrhetic acid hydrogen succinate, a 3-(3-carboxy-1-oxopropoxy)-11-oxoolean-12-en-29-oic acid, steroid agent with anti-ulcerative effects, the mechanism of which is unresolved.

carbobenzyloxy *or* **carbobenoxy** *former name for* **benzyloxycarbonyl**.

carbocation a cation with an even number of electrons with a significant portion of the excess positive charge located on one or more carbons. This general term includes **carbenium ions**, **carbonium ions**, etc.

carbocycle any compound or molecular structure that is **carbocyclic** (def. 1); a **carbocyclic** (def. 2).

carbocyclic 1 describing any cyclic molecular structure containing only carbon atoms in the ring or rings; describing any compound having such a stucture. *See also* **homocyclic** (def. 1). *Compare* **heterocyclic** (def. 1). 2 *or* **carbocycle** any such compound.

carbodiimide any organic compound of general structure R–N=C=N–R, widely used for forming amide (peptide) or phosphodiester linkages in the laboratory. Water-soluble carbodiimides, e.g. 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide, are used to couple carboxylic acid residues to ω-aminoalkyl agarose, while for reactions in nonaqueous media *N*,*N*′-dicyclohexylcarbodiimide is much used.

carbohydrase *former name for* **glycosidase**.

carbohydrate any of a group of organic compounds based on the general formula $C_x(H_2O)_y$. The group comprises the **monosaccharides**, **oligosaccharides**, and **polysaccharides**, and is usually extended to include their derivatives and, sometimes, the **cyclitols**. Some of the simplest members of the group may, notionally, be considered as hydrates of carbon. However, note that the very important carbohydrate, 2-deoxy-D-ribose ($C_5H_{10}O_4$), does not fit the empirical formulation.

carbon *symbol*: C; a nonmetallic element of group 14 of the IUPAC **periodic table**; atomic number 6; relative atomic mass 12.011. It exists in two main allotropic forms, diamond and graphite, and as a third allotrope, fullerene; there are also several amorphous forms, such as carbon black and charcoal. Carbon also forms a vast array of compounds, the organic compounds, based on chains and rings of covalently bonded carbon atoms. There are two stable nuclides, the predominant one being carbon-12. This has a relative abundance in natural carbon of 98.89 atom percent. Its mass is used as the reference standard for all relative atomic masses of nuclides. The other stable nuclide is **carbon-13**. There are also four radioactive nuclides: carbon-10, **carbon-11**, **carbon-14**, and carbon-15.

carbon-11 a radioactive nuclide of carbon, $^{11}_{6}C$. It emits a positron (β^+-particle) (0.97 Mev), no γ-ray, and has a half-life of 20.3 min.

carbon-13 *or* **heavy carbon** the stable nuclide $^{13}_{6}C$. Relative abundance in natural carbon 1.108 atom percent. It is used as a tracer in studies of reaction mechanisms and of carbon metabolism; now often replaced by **carbon-14** unless double labelling is required. Carbon-13 was/is used when a non-radioactive isotope is essential. *See also* **carbon-13 nuclear magnetic resonance spectroscopy**.

carbon-14 a radioactive nuclide of carbon, $^{14}_{6}C$. It emits an electron (β^--particle) (0.156 Mev), no γ-radiation, and has a half-life of 5730 years, decaying to $^{14}_{7}N$. A minute proportion occurs in atmospheric CO_2 due to the action of cosmic rays, and it forms the basis of **carbon dating**. It is used very widely as a radioactive tracer, for which it is produced artificially.

carbonate dehydratase EC 4.2.1.1; *systematic name*: carbonate hydro-lyase; *other names*: carbonic dehydratase; carbonic anhydrase; an enzyme that catalyses the reversible hydration of carbon dioxide to carbonic acid (or to bicarbonate ion at certain pH values) in a wide range of species. Zinc is a cofactor. It is an intracellular enzyme, present in erythrocytes, and is needed to accelerate the reaction, which, although it occurs spontaneously, would not be sufficiently rapid for the needs of higher animals. CO_2 entering the blood from tissues is taken up by the erythrocytes and released as bicarbonate. At least seven enzymatic forms are known, designated carbonic anhydrases I to VII (*abbr.*: CA-I to CA-VII). Example of CA-I from *Chlamydomonas reinhardtii* (precursor): database code CAH1_CHLRE, 377 amino acids (41.63 kDa). Example from human (type I): database code CAH1_HUMAN, 260 amino acids (28.71 kDa). *See also* **dehydratase**.

carbon cycle the sum total of chemical and biochemical processes whereby carbon is cycled between carbon dioxide in the atmosphere (and oceans) and organic compounds in organisms and their remains. Atmospheric carbon dioxide is fixed during photosynthesis by green plants and other photoautotrophic organisms, using solar energy to split water molecules and liberate O_2. The carbon dioxide is thus converted into glucose and other organic compounds, which are used as metabolic fuel and building blocks by the majority of organisms. Aerobic organisms, in particular, are able fully to oxidize organic compounds during respiration to release carbon dioxide to the atmosphere. The carbon stored in the remains of dead organisms is liberated by the respiration of decomposers or, especially that in fossil fuels, by combustion. Hence, photosynthetic organisms and aerobic heterotrophic organisms exist in a state of syntrophy. However, green plants and other aerobic autotrophs also liberate carbon dioxide during respiration. *Compare* **oxygen cycle**.

carbon dating *or* **radiocarbon dating** a technique for determining the age of specimens of biological origin, e.g. wood. It is based on determining the amount of **carbon-14** remaining in the specimen, the known half-life of carbon-14, and the assumption that the abundance of this radionuclide in the atmosphere has remained constant since its incorporation into the material from atmospheric CO_2.

carbon-dioxide buffer any aqueous solution of a water-soluble salt, pK_a 9–10, that, when equilibrated with CO_2, is capable of maintaining a constant partial pressure of CO_2 (pCO_2) in the gas phase in a closed vessel. Carbon-dioxide buffers are used especially in maintaining a constant pCO_2 in manometric experiments on reactions involving the steady production of that gas.

carbon-dioxide electrode an electrode system for determining CO_2 tension in a gas or a liquid. It consists of a combination pH electrode bathed in a thin layer of a dilute bicarbonate solution, the whole being separated from the sample by a rubber or silicone-rubber membrane permeable to CO_2 but not to water or ionizable solutes. The pH of the layer of bicarbonate solution, and hence the electrode's response, varies logarithmically with the CO_2 tension in the sample.

carbon-dioxide fixation the process whereby the carbon atoms of carbon dioxide are incorporated into hexoses by the **reductive pentose phosphate cycle**.

carbon-dioxide snow solid carbon dioxide in powder form. *See also* **dry ice**.

carbonic anhydrase *common name for* **carbonate dehydratase**.

carbonium ion any real or hypothetical cation, containing an even number of electrons and at least one penta-coordinate carbon atom, in which a significant part of the excess positive charge is located on one or more carbon atoms. *See also* **carbenium ion, carbocation**.

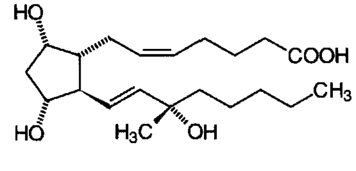
carbon monoxide CO; a colourless, odourless, toxic gas that combines with hemoglobin to form **carboxyhemoglobin**; it also combines with and inhibits cytochrome oxidase. In very small amounts it functions as a neurotransmitter.

carbon-13 nuclear magnetic resonance spectroscopy *or* **^{13}C-NMR spectroscopy** a variant of **nuclear magnetic resonance spectroscopy** in which resonances in **carbon-13** nuclei in natural carbon of organic molecules are examined instead of proton responses. Carbon-12 has no NMR spectrum, and with a natural abundance of carbon-13 of 1.11%, a carbon-13 NMR spectrum has a several thousandfold disadvantage in signal-to-noise ratio compared with a proton NMR spectrum. However, the carbon-13 spectrum is much simpler and therefore has a particular advantage in large molecules. Since ^{13}C–^{13}C coupling can be observed, a double labelling tracer technique is possible, using e.g. $^{13}CH_3$–$^{13}COOH$ as a precursor in studies of **polyketide** biosynthesis.

carbon skeleton the arrangement of the carbon atoms in an organic molecule.

carbonyl the bivalent group =C=O (often written =CO); it is a characteristic group in aldehydes, ketones, and carboxylic acids.

carboprost (15*S*)-15-methylprostaglandin $F_{2\alpha}$; an oxytocic agent used to terminate pregnancy. *See also* **prostaglandins**.



Carbowax *proprietary name for* a group of polyethylene glycols of general formula $H[OCH_2–CH_2]_n–OH$ and of average M_r 400 to 20 000.

carboxamide an amide derived from a carboxylic acid. The carboxamide group is –CO–NH_2. *See also* **carbamoyl**.

carboxamido *former name for* **carbamoyl**.

carboxy+ *prefix (in chemical nomenclature)* denoting the presence of a carboxyl group.

carboxyethyl *symbol*: Cet; the –CH_2–CH_2–COOH group.

4-carboxyglutamic acid *or* **γ-carboxyglutamic acid** *symbol*: Gla; 1-aminopropane-1,3,3-tricarboxylic acid; a **noncoded amino acid**, one or more residues of which are present in a number of calcium-binding proteins, e.g. **osteocalcin** and **blood coagulation** factors II, VII, IX, and X. It is formed subsequent to biosynthesis of the proteins by vitamin K-dependent carboxylation of glutamic acid residues.

carboxyhemoglobin a form of hemoglobin in which carbon monoxide has displaced oxygen from the heme group of hemoglobin, to which it binds with much greater affinity and thus less reversibly than oxygen. The formation of carboxyhemoglobin thus reduces the oxygen-carrying capacity of the blood and deprives tissues of oxygen. It is the basis of carbon monoxide poisoning.

carboxyl the chemical group –COOH in an organic molecule; it is acidic in character.

carboxylate 1 the anionic group –COO^-, formed by dissociation of a **carboxyl** group. **2** any salt, anion, or ester formed from a **carboxylic acid**. **3** to perform or undergo **carboxylation**.

carboxylation the introduction of a molecule of CO_2 into an organic compound.

carboxylic acid any organic acid containing one or more carboxyl (–COOH) groups.

carboxyl protease *see* **aspartic proteinase**.

carboxyl proteinase *or* **acid proteinase** *or* **aspartic proteinase** any enzyme of the sub-subclass EC 3.4.23 (aspartic endopeptidases), a common feature of which is a low pH optimum and an aspartic acid residue in the active centre. It includes **pepsins**, **rennins**, and **cathepsin D**.

carboxyl terminus *or* **carboxyl terminal** *an alternative name for* the **C terminus** of a polypeptide chain (also called, less correctly, carboxy terminus or carboxy terminal). **—carboxy-terminal** *adj.*

carboxy-lyase any enzyme of sub-subclass EC 4.1.1 that catalyses a decarboxylation reaction.

carboxymethyl *symbol*: Cm; the –CH_2–COOH group.

carboxymethylation the process of introducing **carboxymethyl** groups into a substance; e.g. cellulose may be carboxymethylated to produce a cation-exchanger.

carboxypeptidase any enzyme of the sub-subclasses serine carboxypeptidases (EC 3.4.16) or metallo-carboxypeptidases (EC 3.4.17). Both types hydrolyse C-terminal amino-acid residues from oligopeptides or polypeptides. *See* **carboxypeptidase A**, **carboxypeptidase B**, **carboxypeptidase H**.

carboxypeptidase A EC 3.4.17.1; *other name*: carboxypolypeptidase. A **carboxypeptidase** enzyme that catalyses the hydrolysis of peptidyl-L-amino acid to peptide and L-amino acid; zinc is a cofactor.

carboxypeptidase B EC 3.4.17.2; *other name*: protaminase. A **carboxypeptidase** enzyme that catalyses the hydrolysis of peptidyl-L-lysine (or L-arginine) to peptide and L-lysine (or L-arginine); zinc is a cofactor.

carboxypeptidase E *see* **carboxypeptidase H**.

carboxypeptidase H EC 3.4.17.10; *other names*: carboxypeptidase E; enkephalin convertase; *(wrongly)* enkephalinase. A **carboxypeptidase** enzyme involved in prohormone processing. It catalyses the hydrolysis of peptidyl-L-lysine (or L-arginine) to peptide and L-lysine (or L-arginine); zinc is a cofactor. Example from human: database code CBPH_HUMAN, 476 amino acids (53.09 kDa).

carboxypeptidase-tubulin *see* **tubulinyl-Tyr carboxypeptidase**.

carboxy-terminal *see* **carboxyl terminus**.

carboxy terminus *a less correct alternative term for the* **C terminus** of a polypeptide chain. *See also* **carboxyl terminus**.

carcinoembryonic antigen *abbr.*: CEA; any member of a subfamily of immunoglobulin-like proteins on the outer cell surface and grouped in the cluster CD66 (*see* **CD markers**). **CD66e**, *other name*: **meconium antigen 100**, mass 180–200 kDa, was first found in extracts of tumour tissue and of normal fetal colon. It is a glycoprotein, attached to the membrane by a GPI-anchor. It is present in elevated concentrations in many patients with colorectal cancer, but its presence is too variable in this and other diseases for screening for the condition. However, it can be used for monitoring therapy. Example, human (precursor): database code CCEM_HUMAN, 702 amino acids (76.8 kDa). **CD66d**, *other names*: **CGM1, CEA gene member 1**, is present on leukocytes and granulocytes, but not lymphocytes, monocytes, or tumours. Several forms are produced by alternative splicing. Example, human (precursor): database code CGM1_HUMAN, 252 amino acids (27.08 kDa). **CD66b** (*formerly* CD67), *other names*: **CGM6, CEA gene member 6**, is expressed in leukocytes of chronic myeloid leukemia patients and bone marrow. It is attached to the membrane by a GPI-anchor. Example, human (precursor): database code CGM6_HUMAN, 349 amino acids (38.13 kDa).

carcinogen any agent that directly or indirectly induces the transformation of a normal cell into a neoplastic cell. Such agents include various substances of small M_r (often termed **chemical carcinogens**), **oncogenic** viruses, ionizing radiations, and ultraviolet light. *See also* **precarcinogen**, **proximate carcinogen**. **—carcinogenic** *adj.*

carcinogenesis the process(es) involved in the production of **cancers**, including the action of **carcinogens** on living cells. *See also* **oncogenesis**.

carcinogenicity the measure or extent of the potency of a chemical **carcinogen**.

carcinoid (tumour) *or* **argentaffin carcinoma** *or* **argentaffinoma** any tumour arising from **argentaffin cells**. Carcinoids arise from any site of such cells, notably in the ileocecal region, and produce large amounts of 5-hydroxytryptophan and 5-hydroxytryptamine, the former mainly by bronchial tumours and the latter by the decarboxylase-containing intestinal tumours. Histamine, kinins, and substance P may also be produced.

carcinoma any cancer that arises from epithelial tissue. Carcinomas form malignant tumours that tend to invade lymph spaces in surrounding connective tissue and to spread by lymphatic permeation and embolism; they also enter the bloodstream to form **metastases**. **—carcinomatous** *adj.*

cardenolide any of a group of C_{23} cardiotonic steroids characterized by a 14β-hydroxyl group and an unsaturated γ-lactone ring on C-17. They occur free or as the aglycons of **cardiac glycosides** in plants and insects.

cardiac 1 of, relating to, or affecting the heart. **2** of, or relating to, the upper part of the stomach.

cardiac glycoside any of a group of toxic steroid glycosides, derived from plants, that in pharmacological doses increase the power of the contractions of the mammalian heart; some are inhibitors of Na^+,K^+-ATPase. The steroid aglycon carries a 17β unsaturated pentenyl lactone ring and a 3β-hydroxyl group to which the carbohydrate moiety is attached. Examples are **digitoxin** from *Digitalis purpurea*, **ouabain** from *Strophanthus gratus*, and **strophanthin** from *Strophanthus kombe*.

cardiac muscle the specialized muscle of which the walls of the vertebrate heart are made. It is composed of a network (syncytium) of branching elongated cells (fibres) with cross striations, and undergoes spontaneous rhythmical contractions.

cardiac puncture a convenient and quick method for obtaining reasonable volumes of blood from an animal by inserting a syringe needle directly into the heart.

Cardice *proprietary name for* solid carbon dioxide. *See* **dry ice**.

cardio+ *or (before a vowel)* **cardi+** *comb. form* denoting heart.

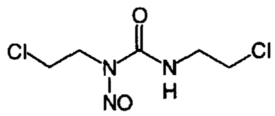
cardioid *(in mathematics)* a type of expression that yields a heart-shaped curve.

cardiolipin an immunogenic 1,3-bis(3-phosphatidyl)glycerol, extracted from beef heart, that is used as an active antigen in the **Wassermann reaction** and other serological tests for syphilis. It may carry antigenic determinants similar to *Treponema pallidum* or be similar to products of tissue necrosis formed when the spirochaete attacks human tissues.

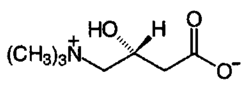
cardiotonic having the effect of improving the tone of heart muscle. Cardiotonic steroids (**cardiac glycosides**) act directly on heart muscle to restore the power of the heart's contractions.

cardiovascular of, or relating to, the heart and the system of blood vessels.

carmustine 1,3-bis(2-chloroethyl)-1-nitrosourea; an antineoplastic derivative of chloroethylnitrosourea with immunosuppressive and antifungal activity.



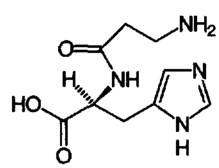
carnitine γ-amino-β-hydroxybutyric acid trimethylbetaine, 3-carboxy-2-hydroxy-*N*,*N*,*N*-trimethyl-1-propanaminium hydroxide, inner salt; $(CH_3)_3N^+{-}CH_2{-}CH(OH){-}CH_2{-}COO^-$; a compound that is found in striated muscle, liver, and other tissues and participates in the transfer of acyl groups across the inner mitochondrial membrane. **Carnitine acyltransferases** catalyse the transfer of acyl groups to and from acyl-CoA molecules to form *O*-acylcarnitine, which can exchange across the inner mitochondrial membrane with unacylated carnitine. This cyclic movement of carnitine is sometimes referred to as the **carnitine shuttle**.



L-carnitine

carnitine acyltransferase *or (sometimes)* **carnitine acylase**. any of various enzymes that catalyse a reaction between acyl-CoA and carnitine to form CoA and *O*-acylcarnitine. **Carnitine *O*-acetyl transferase**, EC 2.3.1.7, is a soluble enzyme found (in eukaryotes) in the mitochondrial matrix, peroxisomes, and the lumen of the endoplasmic reticulum. It is specific for short-chain fatty acyl-CoA (C_2–C_{14}), having no activity with palmitoyl-CoA. In transferring acyl groups to carnitine, it maintains the concentration of free coenzyme A, and also provides a mechanism for the transport of acetyl groups, as acetylcarnitine, through the mitochondrial membrane. Example from yeast: database code CACP_YEAST, 670 amino acids (77.19 kDa). **Mitochondrial carnitine palmitoyltransferase**, EC 2.3.1.21, exists as two isoforms. One isoform, CPT_1 *or* CPT-I, is a mitochondrial outer membrane protein involved in regulating fatty acid β-oxidation by controlling the rate of formation of carnitine acyl esters; database code CPT1_RAT, 773 amino acids (88.04 kDa); another isoform, CPT_2 *or* CPT-II, is located in the mitochondrial inner membrane and re-forms acyl-CoA esters in the mitochondrial matrix after the carnitine esters have been transported through the inner membrane by acylcarnitine:carnitine antiporter.

carnosine *N*-β-alanyl-L-histidine; a substance that occurs in the skeletal muscle of some animals, including humans (up to 30 mmol kg^{-1}). *Compare* **anserine**.



carnosine

carotene *see* **carotenoid**.

carotenoid any of a group of naturally occurring tetraterpenes that are widely distributed in plants and animals but only synthesized by higher plants, algae, fungi, and bacteria. They may be represented formally as consisting of eight isoprenoid (ip) residues and formed by the joining tail to tail of two units each comprising four ip residues joined head to tail: ip.ip.ip.ip.pi.pi.pi.pi. There are two groups of carotenoids: **carotenes**, which are hydrocarbons, and **xanthophylls**, which contain oxygen in various forms. They occur in all photosynthetic tissues, where they function as protectors against photosensitization and as antenna pigments in photosynthesis.

carrageenan *or* **carrageen** *or* **carrageenin** a mixture of sulfated galactans, also known as **Irish moss**, extracted from red seaweeds (class Rhodophyceae), particularly *Chondrus crispus* and *Gigartina stellata*. It shows blood anticoagulant activity and is used as a gelling and stabilizing agent in foods and pharmaceuticals. Three major types are known: ι-carrageenan is the most highly sulfated and contains alternate units of 3-linked β-D-galactopyranose 4-sulfate and 4-linked 3,6-anhydro-α-D-galactopyranose 2-sulfate; κ-carrageenan is precipitated by potassium chloride and contains only one sulfate per disaccharide unit; λ-carrageenan is largely devoid of the 3,6-anhydro component. ι-Carrageenan forms a kinked double-helical structure. [After Carragheen, near Waterford, Republic of Ireland.]

κ-carrageenase EC 3.2.1.83; an enzyme that catalyses the hydrolysis of 1,4-β-D-linkages between D-galactose 4-sulfate and 3,6-anhydro-D-galactose in various carrageenans.

carrier 1 any known or putative component of a biological membrane that effects the transfer of a specific substance or group of related substances from one side of the membrane to the other. The mechanism is supposedly a cyclic process, involving combination of the carrier with the substance(s) on one side of the membrane, diffusion of the combined form across the membrane, release of the substance(s) on the far side, and diffusion of the carrier back across the membrane in uncombined form. *See also* **facilitated diffusion**, **permease**, **translocation**, **transport**. 2 any component of a biological fluid (e.g. a blood protein) that can combine with a specific substance (e.g. a hormone or metabolite) and transport it from one part of the organism to another. 3 *(in enzymology)* any substance undergoing reversible combination with a metabolite or reversible oxidation–reduction and thereby carrying the metabolite, electron(s), or hydrogen atom(s) between enzymes or cellular organelles, or into cells. Some **coenzymes** may act as carriers. *See* **electron carrier**, **hydrogen carrier**. 4 a substance that is mixed with a reagent or other substance to facilitate the chemical or physical manipulation of the latter. The carrier is commonly of similar constitution and/or properties to the reagent, and present at relatively high concentration. The term is used especially for the nonradioactive chemical counterpart of a radionuclide or of a compound containing a radionuclide, whether deliberately added or present from the time of initial preparation of the radioactive substance. *Compare* **carrier-free**. 5 *(in chromatography)* the **mobile phase** or a component thereof. *See* **carrier ampholyte**, **carrier displacement chromatography**, **carrier gas**. 6 *or* **matrix** the inert support material to which a small molecule, biopolymer, cell fragment, or intact cell is artificially linked to form an immobilized or insolubilized derivative. *See also* **microcarrier**. **7** any vesiculated structure within which a biologically active substance (e.g. a drug or enzyme) is artificially incorporated, commonly to protect the substance from degradation and/or to direct it to a target organ upon injection into the bloodstream. Such a carrier may be either artificial (e.g. a **liposome**) or of natural origin (e.g. an erythrocyte **ghost**). **8** *(in immunochemistry)* any substance of high M_r (e.g. a protein) to which a non- or poorly immunogenic substance (e.g. a **hapten**) is naturally or artificially linked to enhance its immunogenicity. **9** *(in genetics)* any individual who is heterozygous for a specified recessive allele and who thus does not normally display any characteristic (e.g. a disease) associated with that allele but who may, with appropriate mating, produce homozygous offspring that exhibit the recessive characteristic. **10** *(in pathology)* any individual who harbours a specified infectious microorganism without showing signs of disease, and is capable of transmitting the microorganism to others. *Compare* **vector** (def. 2). *See also* **vehicle**.

carrier ampholyte a mixture of amphoteric electrolytes used in **isoelectric focusing** to generate a linear and stable pH gradient, and in **isotachophoresis** and **ampholyte displacement chromatography**. The component electrolytes are commonly of similar constitution, low molecular mass, and closely spaced isoelectric points. *See also* **Ampholine**.

carrier displacement chromatography a variant of **displacement chromatography** in which substance(s), either related or unrelated, are added to the mixture being chromatographed to assist in the manipulation of the components separated.

carrier-free describing a preparation of a radionuclide, or of a radionuclide-labelled compound, to which no carrier has been added, and for which precautions have been taken to minimize contamination with other isotopic nuclides. It does not necessarily imply, however, 100% isotopic abundance. Material of high specific activity is often wrongly referred to as 'carrier-free'; more correctly it would be of high isotope abundance.

carrier gas the inert gas that forms the mobile phase in **gas–liquid chromatography** or gas–solid chromatography and that carries the eluted substances in gaseous form into the detector.

carrier lipid *see* **undecaprenol**.

carrier protein any protein that functions as a **carrier** (def. 8), e.g. **acyl carrier protein**. The term has been applied more specifically to describe **cytochome b_5**, ADP,ATP carrier protein, and **GDC** (Graves' disease carrier protein).

Carr-Price reaction the reaction of an anhydrous solution of antimony trichloride in chloroform with retinol (vitamin A) to produce a transient blue colour, maximal at about 10 s. It can be used to estimate retinol.

CARS *abbr. for* **coherent anti-Stokes Raman spectroscopy**.

Cartesian of, or relating to, **Descartes** or his works.

Cartesian coordinates a system of coordinates that defines the location of a point in a plane (two dimensions) or in a space (three dimensions) in terms of its perpendicular distance from each of a set (two or three) of mutually perpendicular axes.

Cartesian diver *or* **Cartesian devil** a device, devised as a toy, adapted as an ultramicrorespirometer, consisting of a small piece of capillary glass tubing open at one end and closed and slightly expanded at the other. A sample of tissue, in approximately 2 μL buffer, is placed in the vessel which is then filled with gas and the neck sealed with a drop of oil. The diver is placed in a constant-temperature vessel filled with a solution in which the diver just floats. If the pressure in the vessel is reduced the diver will rise and if the pressure is increased the diver will fall. The buoyancy of the diver is determined by the gas volume within it, and changes in the latter may be measured by adjusting the pressure in the vessel so that the diver floats in the same position.

cartilage a tough, elastic type of **connective tissue**, found in most vertebrates. It is a major component of the embryonic skeleton, but in higher vertebrates it is mostly converted to bone during development. In adult humans, it is largely confined to the ears, nose, trachea, the anterior ends of the ribs, and the articular surfaces of bones. Cartilage consists of a firm resilient matrix, formed by chondroblasts (which become enclosed as chondrocytes), and composed of glycosaminoglycans including **chondroitin sulfate** A and C, hyaluronic acid, and keratosulfate II, together with varying amounts of **collagen**. The entire skeleton of some lower vertebrates, notably the cartilaginous fishes (class Chondrichthyes), consists of cartilage. —**cartilaginous** *adj.*

cartilage link protein a protein that stabilizes the aggregates of proteoglycan monomers with hyaluronic acid in the extracellular cartilage matrix. Example (precursor) from human: database code PLK_HUMAN, 354 amino acids (40.12 kDa).

cartilage oligomeric matrix protein *abbr.*: COMP; a pentameric disulfide-linked Ca^{2+}-binding **EF-hand** protein of the cartilage matrix. Example (precursor) from *Rattus norvegicus*: database code COMP_RAT, 755 amino acids (82.57 kDa).

cartilage-specific proteoglycan core protein a major proteoglycan of the extracellular matrix of cartilaginous tissues. The N-terminal region binds hyaluronic acid. Chondroitin sulfate is the main glycan component, but keratan sulfate and O-linked and N-linked oligosaccharides are also present. It also has an epidermal growth factor-like domain and a **Sushi domain**. Example (precursor) from human: database code PGCA_HUMAN, 2415 amino acids (250.19 kDa).

caryon *a variant spelling of* **karyon**.

caryopsis *see* **grain**.

CAS *abbr. for* carbonic anhydrase.

cascade sequence *or* **waterfall sequence** a sequence of successive activation reactions involving either enzymes (**enzyme cascade**) or hormones (**hormone cascade**), or both. There are various examples: (1) the series of transformations of proteolytic enzymes in **blood coagulation** in which each enzyme activates the next until the final substrate, fibrinogen, is reached; (2) the series of reactions involved in the formation of, and mediation of the action of second messengers, e.g. in the activation of **glycogen phosphorylase**; (3) the process of activation of **steroidogenesis** in the adrenal gland by hypothalamic and pituitary hormones. The term has also been applied to other sequential phenomena, e.g. DNA → RNA → protein. Cascade sequences are characterized by a series of amplifications of a weak initial stimulus, and may be activated at more than one point along the sequence.

cascara the dried bark of the small tree *Rhamnus purshiana*. It is a potent purgative due to its constituent hydroxyanthracene derivatives (anthroglycosides), including cascarosides and related glycosides. These yield active anthracene derivatives after metabolism by colonic bacteria.

casein **1** *or (esp. Brit.)* **caseinogen** the principal protein fraction of milk; it is present as a colloidal suspension and may be precipitated by acidification. It is a mixture of several related proteins – α-, β-, γ-, and κ-caseins – which differ in amino-acid composition and which may be separated electrophoretically. The α- and β-caseins are rich in phosphate, present mainly as *O*-phosphoserine residues, which bind calcium ions. Casein is easily prepared and is valuable as a dietary supplement. **2** *or (esp. US)* **paracasein** the product formed by coagulation of milk (or **casein** (def. 1)) by treatment with chymosin (rennin). This enzyme removes a 64-residue carbohydrate-rich hydrophilic C-terminal moiety of κ-casein, thereby destroying the latter's protective action on the casein micelles and thus allowing them to be precipitated by the calcium ions present. Examples from *Bos taurus*, β-casein (precursor): database code CASB_BOVIN, 224 amino acids (25.11 kDa); κ-casein (precursor): database code CASK_BOVIN, 190 amino acids (21.27 kDa).

caseinogen *an alternative name (esp. Brit.) for* **casein** (def. 1).

casomorphin *or* **exomorphin** any of a number of opioid peptides isolated from an enzymatic digest of bovine casein; activity corresponds to residues 60–66 of bovine β-casein. *See also* **morphiceptin**.

caspase *see* **apoptosis**.

castor oil an oil derived from seeds of the plant *Ricinus communis*. It contains triglycerides of ricinoleic, oleic, linoleic, palmitic, and stearic acids.

CAT *abbr. for* **1** computer of average transients. **2** chloramphenicol *O*-acetyltransferase. **3** computer-assisted tomography.

catabolic dead-end a point in a **catabolic pathway** where, in a particular organism, a metabolite is neither catabolized further nor excreted but accumulates intracellularly. Consequently the pathway is unable to function as an energy-yielding process. The enzyme that can form the metabolite thus lacks an obvious role in that organism.

catabolic pathway any degradative, usually exergonic, sequence of reactions. *Compare* **anabolic pathway**. *See also* **amphibolic**.

catabolism *or* **katabolism** any metabolic process involving the breakdown of complex substances into smaller products, including the breakdown of carbon compounds with the liberation of energy for use by the cell or organism. *Compare* **anabolism**. —**catabolize** *or* **catabolise** *vb*.; **catabolic** *adj*.

catabolite any product of a **catabolic pathway**.

catabolite (gene) activator protein *abbr*.: CAP; *other names* **cAMP receptor protein, cAMP-regulatory protein** (*abbr*.: CRP); a protein that binds to and is activated by cAMP. It then acts as a transcription-initiation factor, binding near the operator of bacterial operons that are subject to **catabolite repression** and being necessary for their efficient transcription. It belongs to the **CRP** family of transcription factors. Example from *Haemophilus influenzae*: database code CRP_HAEIN, 224 amino acids (25.15 kDa).

catabolite inhibition a general regulatory mechanism, found in *Escherichia coli*, that controls the activity of an early reaction in carbohydrate metabolism. For instance when *E. coli* is grown in synthetic medium with radioactive galactose or lactose as the carbon source, the addition of glucose rapidly inhibits utilization of the radioactive substrate. Subsequent removal of the glucose rapidly reverses the inhibition.

catabolite repression a mechanism of genetic regulation in bacteria in which the accumulation of catabolites in the cell represses the formation of enzymes that contribute to the formation of the catabolites.

catalase EC 1.11.1.6; a widely distributed hemoprotein enzyme that catalyses the formation of O_2 + 2 H_2O from two molecules of H_2O_2; heme and manganese are cofactors. It plays an important role in removing H_2O_2 formed in peroxisomes in animals and plants and in **glyoxysomes** in plants. The molecule has seven motifs. Examples: (1) homotetramer from tomato: database code CAT1_LYCES, 492 amino acids (56.44 kDa); (2) bovine enzyme: database code NRL_7CATA, 498 amino acids (56.61 kDa); 3-D structure is known; α helical and β barrel regions are present.

catalatic of, or pertaining to, **catalase**.

catalyse *or (esp. US)* **catalyze** to influence a reaction through **catalysis**.

catalysis an increase in the rate of a chemical reaction brought about by a **catalyst**. —**catalytic** *adj*.; **catalytically** *adv*.

catalysome **1** an enzyme-containing cap found on lipid droplets in the fat body of some insects. **2** a mitochondrion-like organelle specialized for lipid metabolism, found in adipose tissue.

catalyst any substance that increases the rate of a chemical reaction but is itself unchanged at the end of the reaction. Catalysts are usually present in very low concentrations relative to those of the substances whose reaction they are catalysing.

catalytic activity an index of the actual or potential activity of a catalyst. The catalytic activity of an enzyme or an enzyme-containing preparation is defined as the property measured by the increase in the rate of conversion of a specified† chemical reaction that the enzyme produces in a specified† assay system. [†This word, in one or other position, has been misprinted as 'specific' in certain published versions of this definition.] Catalytic activity is an extensive quantity (*see* **extensive property**) and is a property of the enzyme, not of the reaction mixture; it is thus conceptually different from rate of conversion although measured by and equidimensional with it. The unit for catalytic activity is the **katal**; it may also be expressed in mol s^{-1}. Dimensions: N T^{-1}. Former terms such as catalytic ability, catalytic amount, and enzymic activity are no longer recommended. Derived quantities are **molar catalytic activity**, **specific catalytic activity**, and **catalytic activity concentration**.

catalytic activity concentration *or* **catalytic concentration** the **catalytic activity** of an enzyme per unit volume, where volume refers to that of the original enzyme-containing preparation, not that of the assay system. It may be expressed in katals per litre.

catalytic amount the amount of a substance used in a chemical reaction as a catalyst, primer, or sparker. It is generally much smaller than the stoichiometric amounts of either reactants or products. For an enzyme the term is now obsolete and replaced by **catalytic activity**.

catalytic assay a technique for measuring low concentrations (≈1 μM) of substrates in certain enzyme-catalysed reactions (e.g. $NAD(P)^+$ or coenzyme A), where simple kinetic methods are not feasible because too-rapid consumption makes reaction rates impossible to measure reliably. Conditions are arranged so that the concentration of the substrate in question is kept constant by a regeneration reaction, the substance being made to behave as an intermediate catalyst. The regeneration reaction may act also as the indicator reaction. For example, NAD^+ may be determined by coupling the reaction catalysed by alcohol dehydrogenase (EC 1.1.1.1) to the combined regeneration/indicator reaction catalysed by cytochrome *c* reductase (EC 1.6.99.3), according to the equations:

$$NAD^+ + C_2H_5OH \rightarrow NADH + CH_3CHO + H^+$$

and

$$NADH + H^+ + 2\ Cyt\ c^{3+} \rightarrow NAD^+ + 2\ Cyt\ c^{2+} + 2\ H^+.$$

The rate of increase in absorbance of Cyt c^{2+} at 550 nm is measured. If the coupled regeneration reaction does not act as an indicator reaction, the technique is known as **enzymic cycling**. *Compare* **coupled assay**.

catalytic concentration *an alternative term for* **catalytic activity concentration**.

catalytic constant *symbol*: k_{cat} *or* k_0; a measure of the catalytic potential of an enzyme, customary unit s^{-1}. In general it is the value of the velocity of the reaction, *v*, divided by the stoichiometric concentration of active sites, $[E]_0$, obtained by extrapolating all substrate concentrations to infinity; alternatively it can be expressed as $V_{max}/[E]_0$, where V_{max} is the **limiting rate**. k_{cat} is also known as the **turnover number**, because it represents the number of reactions catalysed per unit time by each active site.

catalytic domain *see* **domain** (def. 3).

catalytic exchange method a method for randomly labelling a compound with tritium. A solution of the compound in a tritiated hydroxylic solvent is allowed to stand in the presence of metal catalysts and either acid or base.

catalytic facilitation a process whereby the product formed by one enzyme in a complex is passed directly to the catalytic site of the next enzyme in a reaction sequence more efficiently than when the second substrate is supplied from the environment.

catalytic factor a term sometimes used of a specified enzyme to define the quantitative ratio between the rates of a chemical reaction in the presence and the absence of the enzyme but otherwise under the same conditions.

catalytic intermediate a chemical species produced during the course of a catalytic reaction, intermediate between reactants and products.

catalytic power a measure of the ability of an enzyme to accelerate a chemical reaction.

catalytic residue any of the amino-acid residues in an enzyme that are directly involved in making or breaking covalent bonds while the enzyme is acting on a substrate.

catalytic site the site on a catalyst that is concerned with catalysis, especially as part of the **active site** of an enzyme.

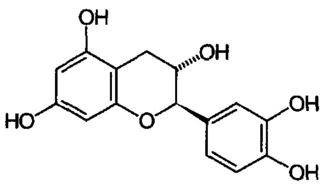
catalyze *the US spelling of* **catalyse**.

CAT assay assay of **chloramphenicol *O*-acetyltransferase**, frequently employed to detect expression of its gene in cDNA technology.

catastrophe theory *see* **error catastrophe hypothesis**.

CAT box *see* **CAAT box**.

catechin the type compound for one of the 12 classes of flavanoids. It is characterized by oxidation state (OH at C-3) of the central pyran ring. Some 70 catechins are known; e.g. (+)-catechin is (2*R*,2*S*)-3,3′,4′,5,7-flavanpentol. Catechins are among the polyphenols widely distributed in vegetable foods, arising biosynthetically from the **shikimate** pathway. (+)-Catechin produces artificial cross-links in collagen *in vitro* and has anti-tumour activity. Catechin and **curcumin** have been found to be non-toxic, non-mutagenic, non-teratogenic and excellent anti-mutagens and anti-carcinogens which are being clinically tested by the National Cancer Institute.



(+)-catechin

catechol 1 *former name for* **pyrocatechol**. 2 any aromatic *o*-diol; any compound containing a pyrocatechol nucleus or substituent. *Compare* **hydroquinone** (def. 2), **quinone** (def. 2).

catecholamine any of a group of physiologically important **biogenic amines** that possess a **catechol** (def. 2) (i.e. 3,4-dihydroxyphenyl) nucleus and are derivatives of 3,4-dihydroxyphenylethylamine. They include **adrenaline**, (epinephrine) **noradrenaline** (norepinephrine), and **dopamine**. Catecholamines have various physiological roles, particularly as neurotransmitters, and produce effects in the brain, the cardiovascular system, and other organs; they also help to regulate carbohydrate and fat metabolism.

catechol melanin any of various deep-brown or black plant pigments, so called because they yield catechol on alkali fusion. *See also* **melanin**.

catechol-*O*-methyltransferase EC 2.1.1.6; *systematic name*: *S*-adenosyl-L-methionine:catechol *O*-methyltransferase. An enzyme found primarily in non-neuronal tissues and catecholamine-releasing neurons, especially in the periphery; it effects *O*-methylation of catecholamines and their deaminated metabolites, but is less active against catechols. It catalyses the reaction between *S*-adenosyl-L-methionine and catechol to form *S*-adenosyl-L-homocysteine and guaiacol; in catecholamines it methylates the phenolic group at position 4. This inactivates catecholamine neurotransmitters and catechol hormones. Example from human: database code COMT_HUMAN, 271 amino acids (30.00 kDa), is the membrane-bound form; a cytosolic form (residues 52–271) lacks the transmembrane signal anchor.

catechol oxidase *see* **polyphenol oxidase**. *See also* **monophenol monooxygenase**.

catenane any type of compound containing two or more rings that are interlocked but not covalently joined; i.e. resembling the links of a chain. An example is an interlocked circular duplex DNA molecule found in mitochondria. *Compare* **concatemer**.

catenate 1 to interlock, without covalent bonding, two or more circular structures, e.g. cyclic molecules. 2 linked as the interlocked rings of a chain. *Compare* **concatenate**. **—catenated** *adj.*; **catenation** *n.*

cathepsin any of a group of intracellular peptide hydrolases.

cathepsin B EC 3.4.22.1; *other name*: lysosomal cysteine endopeptidase. An enzyme that catalyses the hydrolysis of proteins with broad specificity for peptide bonds; preferentially it cleaves -Arg-Arg-|-Xaa- bonds in small substrates. It is a glycoprotein dimer comprising a heavy chain and a light chain cross-linked by a disulfide bond. Example (human) precursor: database code CATB_HUMAN, 339 amino acids (37.76 kDa).

cathepsin C EC 3.4.14.1; *recommended name*: dipeptidyl peptidase I. An enzyme that catalyses the hydrolysis of dipeptides from the N terminus of a polypeptide, except where the N-terminal residue is Arg or Lys, or where the adjacent amino acid is Pro. It also catalyses the transfer of dipeptide residues.

cathepsin D EC 3.4.23.5; a lysosomal aspartic endopeptidase that is active in intracellular protein breakdown. It has a specificity similar to, but narrower than, that of **pepsin A**. The molecule comprises a light chain and a heavy chain. Example (human) precursor: database code CATD_HUMAN, 412 amino acids (44.50 kDa).

cathepsin E EC 3.4.23.34; an aspartic endopeptidase with similar activity to **cathepsin D**, but found in lymphoid-associated tissue. It may have a role in immune function. Example (human) precursor: database code CATE_HUMAN, 396 amino acids (42.74 kDa).

cathepsin G EC 3.4.21.20; a glycoprotein serine endopeptidase with a specificity similar to that of **chymotrypsin**. It is found in the lysosomes of polymorphonuclear leukocytes. Example (human) precursor: database code CATG_HUMAN, 255 amino acids (28.80 kDa).

cathepsin H EC 3.4.22.16; a lysosomal cysteine endopeptidase that also acts as an aminopeptidase on peptide substrates with a free N terminus. It is composed of a minichain and a large chain; the large chain may be split into heavy and light chains; all chains are held together by disulfide bonds. Example (human) precursor: database code CATH_HUMAN, 335 amino acids (37.36 kDa).

cathepsin L EC 3.4.22.15; *other name*: major excreted protease (*abbr.*: MEP). A lysosomal cysteine endopeptidase that hydrolyses proteins but not acylamino acid esters, and has no peptidyl-dipeptidase activity – the activity is similar to that of **papain**. It is found in rat-liver and other mammalian lysosomes, and is inhibited by **leupeptin**. The molecule is a dimer of a heavy and a light chain linked by disulfide bonds. Example (human) precursor: database code CATL_HUMAN, 333 amino acids (37.52 kDa).

cathepsin S EC 3.4.22.27; a lysosomal single-subunit cysteine endopeptidase with similar activity to **cathepsin L** but more activity for -Val-Val-Arg-|-Xaa. Example (human) precursor: database code CATS_HUMAN, 331 amino acids (37.44 kDa).

cathode *or (formerly)* **kathode** 1 the electrode of an electrolysis cell or electrophoresis apparatus towards which cations of the electrolyte migrate under the influence of an electric field applied between the electrodes. 2 negative electrode: the electrode, pole, or terminal of a discharge tube, solid-state rectifier, thermionic valve, or other apparatus by which conventional electric current returns to an external source; i.e. the electrode through which electrons flow into such a device. 3 the positive pole or terminal of an electric (primary or secondary) cell or battery (of cells); i.e. the pole from which electrons emerge from such a device. *Compare* **anode**. **—cathodal, cathodic,** *or* **cathodical** *adj.*; **cathodally** *or* **cathodically** *adv.*

cathode-ray tube a vacuum tube in which a beam of electrons is emitted and focused by an electron gun onto a fluorescent screen. The beam can be deflected by horizontal and vertical electrostatic or magnetic fields to produce a visual display of electric signals, as in a television tube or cathode-ray oscilloscope.

catholyte in isoelectric focusing, the electrolyte that is in immediate contact with the cathode.

cation *or (formerly)* **kation** any **ion** with a net positive charge. In a solution being electrolysed, a cation migrates towards the **cathode**. *Compare* **anion**. **—cationic** *adj*.

cation exchange the process of **ion exchange** in which one cation is exchanged for another cation.

cation exchanger any **ion exchanger** that is able to exchange cations with the surrounding solution. *See also* **ion-exchange resin**.

cationic acid any acid that carries a positive charge, i.e. is a **cation**.

cationic detergent any **detergent** in which the polar part of the molecule carries one or more positive charges.

cationized ferritin a polycationic derivative of **ferritin** prepared by coupling horse-spleen ferritin with *N,N*-dimethyl-1,3-propanediamine, most of the carboxyl groups on the ferritin molecule thereby being converted into positively charged tertiary amino groups. It is useful for labelling negatively charged groups on cell surfaces, where the ferritin particles may easily be identified by electron microscopy because of their electron-dense iron cores.

cationmotive describing any enzyme, enzyme system, or other process that produces the vectorial movement of cations, especially across a biomembrane. The particular cation(s) may be specified, e.g. protonmotive, Na^+/K^+-motive.

CAU a codon in mRNA for L-histidine.

caudal a homeodomain transcription factor which forms a concentration gradient in the opposite direction to that of **bicoid protein**, possibly as a result of the translation control exerted on the caudal mRNA by bicoid.

cauliflower mosaic virus a double-stranded DNA virus that is a potential recombinant vector for plants.

caulimovirus any double-stranded DNA plant virus belonging to the *Caulimovirus* group, the type member of which is the **cauliflower mosaic virus**.

caustic soda *see* **soda**.

caveola (*pl.* **caveolae**) a closed plasmalemmal vesicle that is thought to remain attached to the plasma membrane while extruding into the cytoplasm materials taken up from the outside of the cell. It is rich in glycosylphosphatidylinositol (GPI)-anchored proteins. [From Latin *cavea*, a small cavity.]

caveolin a protein that lines the cytoplasmic surface of a **caveola**. Example from dog: database code CAVE_CANFA, 178 amino acids (20.6 kDa).

CBF *abbr. for* centromere binding factor (*see* **centromere binding proteins**).

CBG *abbr. for* cortisol binding globulin. *See* **transcortin**.

CBL a protooncogene related to v-*cbl*, the nuclear oncogene of Cas NS-1 retrovirus, named from Casitas B-lineage lymphoma. The function of the normal protein is unknown, but it may be a transcriptional regulator. v-*cbl* encodes a truncated form of Cbl, which has lost a C-terminal leucine zipper and a 208-residue proline-rich region. Example, human protooncogene product c-Cbl: database code CBL_HUMAN, 906 amino acids (99.65 kDa).

Cbm *symbol for* carbamoyl.

CBN *abbr. for* (IUPAC-IUB) Commission on Biochemical Nomenclature; now succeeded by **JCBN**.

Cbz *symbol for* the benzyloxycarbonyl group.

cc *symbol for* cubic centimetre (cm^3 is preferred).

CC *abbr. for* cholecalciferol (**vitamin D_3** is preferred).

CCA a codon in mRNA for L-proline.

C_{55} carrier lipid *or* **C_{55} lipid carrier** *an alternative name for* **undecaprenol**.

CCC 1 a codon in mRNA for L-proline. 2 *abbr. for* β-chloroethyltrimethylammonium chloride; an inhibitor of gibberellin synthesis in plants.

cccDNA *see* **DNA**.

CCCP *abbr. for* carbonyl cyanide *m*-chlorophenylhydrazone, a powerful uncoupler of oxidative phosphorylation.

C cell the distinctive type of apud cell that secretes the polypeptide hormone **calcitonin**. In mammals C cells are found predominantly in the thyroid gland (when they are also known as **parafollicular cells**), and in some mammals and birds they also occur in the parathyroid gland and the thymus. In other vertebrates the cell type is present mainly in the discrete **ultimobranchial body** and the lung. [The name C cell was chosen to signify calcitonin-secreting cell.]

CCG a codon in mRNA for L-proline.

CCK *abbr. for* cholecystokinin.

CCK cell *see* **I cell**.

CCK-PZ *abbr. for* cholecystokinin–pancreozymin. *See* **cholecystokinin**.

cCMP *abbr. for* cyclic CMP.

C_1 compound *or* **one-carbon compound** *or* **single-carbon compound** any (real or hypothetical) organic compound that consists of a single carbon atom with attached hydrogen atom(s), and sometimes also an oxygen atom or imino group, and that functions as a unit in metabolism. Examples include the methyl group of *S*-adenosylmethionine or the iminomethyl-formimidoyl- (or formimino-), formyl-, hydroxymethyl-, methyl-, methylene-, and methenyl- groups of tetrahydrofolate coenzymes.

C3/C5 convertase *see* **complement**.

C5 convertase *see* **complement**.

CCP *abbr. for* Collaborative Computational Programs; an initiative, started in the UK, for the development and distribution of software for scientific applications. Programs include: CCP4 (crystallographic software), CCP5 (molecular dynamics), and CCP11 (sequence analysis).

CCU a codon in mRNA for L-proline.

C_3 cycle *an alternative name for* **reductive pentose phosphate cycle**.

C_4 cycle *an alternative name for* the **Hatch–Slack pathway**.

cd *symbol for* candela.

Cd *symbol for* cadmium.

CD *abbr. for* 1 circular dichroism. 2 cluster of differentiation *or* cluster determinant; *see* **CD marker**. 3 compact disc.

c-di-GMP *abbr. for* cyclic diguanylic acid.

CD4 a **CD marker** that occurs on T-helper cells and is involved in MHC class II restricted interactions. It is down-regulated by HIV infection. Example, T-cell CD4 antigen (precursor) from human: database code CD4_HUMAN, 458 amino acids (51.05 kDa); 3-D structure known; a type I membrane glycoprotein of the immunoglobulin superfamily.

CD8 a **CD marker** that occurs on human cytotoxic T-cells and suppressor T-cells. It is associated with MHC class I restricted interactions. It is a heterodimer of an α and β chain and isoforms of the β chain result from alternative splicing. The α chain binds to the MHC α_3 domain. Examples, α chain (precursor): database code CD8A_HUMAN, 235 amino acids (25.73 kDa); β_1 chain (precursor): database code C8B1_HUMAN 210 amino acids (23.72 kDa).

CD44 *other names*: phagocyte glycoprotein I; lymph node homing receptor; HERMES antigen; extracellular matrix receptor III. A cell **adhesion molecule** that is constitutively expressed on the surface of B and T cells, monocytes, neutrophils, epithelial cells, fibroblasts, glial cells, and myocytes. It has a molecular mass of 85–90 kDa and plays a general role in cell adhesion, including lymphocyte homing; it binds to collagen and hyaluronate. Example from horse: database code CD44_HORSE, 359 amino acids (38.99 kDa).

CD59 a glycoprotein that is a potent inhibitor of complement membrane attack complex (MAC); it acts after C5b8 assembly (*see* **complement**). It is also involved in signal transduction for T-cell activation, being complexed to a protein tyrosine kinase. It is attached to the membrane by a **GPI-anchor**. Example (human) precursor: database code CD59_HUMAN, 128 amino acids (14.16 kDa).

CD66 *see* **carcinoembryonic antigen.**
CD66d *see* **carcinoembryonic antigen.**
CD66e *see* **carcinoembryonic antigen.**
CD79 *see* **B cell antigen receptor complex associated protein.**
***cdc* gene** *abbr. for* cell-division-cycle gene. Many genes are now known that encode proteins essential for cell division. These were initially identified in yeast, using temperature-sensitive mutants that divide at low temperature but which at high temperature arrest at a point in the cell-division cycle at which expression of the mutated gene would normally be required. The yeast genes were named in a series *cdc1*, *cdc2*, etc. Homologues of these genes have since been found in many other species, but the *cdc* nomenclature is used specifically for the fission yeast, *Schizosaccharomyces pombe*. The convention Cdc followed by a numeral is used for the protein products of these genes. Many products identified by this method were found not to be directly involved in the control of cell division, being proteins such as DNA polymerases. Some of those that are involved in cell-division control are (all database examples are from yeast):

Cdc2, a major cell regulatory protein, with counterparts in all eukaryotic cells. *cdc2* is homologous to *CDC28* in budding yeast. Cdc2 is a Ser/Thr protein kinase, alternatively denoted $p34^{cdc2}$, and regulated by cyclin B. $p34^{cdc2}$ and cyclin B comprise **maturation promoting factor** (MPF). Cdc2 is the patriarch of a family of cyclin-dependent kinases; Cdc2 is Cdk1. Example from *S. pombe*: database code CC2_SCHPO, 297 amino acids (34.36 kDa).

cdc25 encodes a protein phosphatase that positively regulates $p34^{cdc2}$ by dephosphorylation of Tyr^{15}. Entry to mitosis is negatively regulated by ***wee 1***, a gene in *S. pombe* that encodes a protein tyrosine kinase. Its action on mitosis results from phosphorylation of Tyr^{15} of $p34^{cdc2}$. Note: *CDC25* from budding yeast encodes a guanine nucleotide exchange protein.

Cdc28, a Ser/Thr protein kinase required for completion of both the START transition of the cycle and also the G_2/M transition: database code CC28_YEAST, 298 amino acids (34.02 kDa);

Cdc31, an **EF-hand** protein involved in microtubular organization: database code CC31_YEAST, 298 amino acids (18.73 kDa);

Cdc46, a protein mobilized from the cytoplasm to the nucleus after mitosis, and a potential ATP-binding protein: database code CC46_YEAST, 775 amino acids (86.31 kDa);

Cdc48, a putative ATP-binding protein involved in spindle formation: database code CC48_YEAST, 834 amino acids (91.77 kDa).

CDE *abbr. for* centromere DNA element; CDE-I is a group of DNA sequences, RTCACRTG, found in centromeres and some promoters. *See also* **centromere-binding proteins.**
CD59 glycoprotein precursor *see* **membrane attack complex inhibition factor.**
CDH *abbr. for* ceramide dihexoside ($Cer(Hex)_2$ is preferred).
CDI *abbr. for* *N,N′*-carbonyldiimidazole; 1,1′-carbonylbis-1*H*-imidazole, a reagent used in the synthesis of nucleotide triphosphates and peptides, and for cross-linking proteins.
Cdk *abbr. for* cell division protein kinase.
CD marker *(in immunology)* any of a series of antigenically distinct molecules occurring on the surface of leukocytes and other cell types, and used in the characterization of such cells. More than 80 individual CD marker molecules are recognized, as identified by **monoclonal antibodies**. The most frequently encountered CD molecules are those used to distinguish T cells. CD2 is present on all T cells and is involved in antigen-nonspecific activation. CD3 is also present on all T cells and is an invariant part of the T-cell antigen receptor involved in antigen-specific T-cell activation. **CD4** occurs on T-helper cells and is involved in MHC class II restricted interactions, **CD8** occurs on cytotoxic T cells and is involved in MHC class I restricted interactions. *See also* **CD44**, **CD59**. Other CD markers listed here under other names include CD16 (*see* **Fc receptor**), CD29 (*see* **VLA**), CD31 (*see* **PECAM**), CD32 (*see* **Fc receptor**), CD49a and CD51 (*see* **VLA**), CD50 and CD54 (*see* **ICAM**), CD62 (*see* **selectin**), CD64 (*see* **Fc receptor**), CD102 (*see* **ICAM**), and CD106 (*see* **VCAM**). [From 'cluster of differentiation'.]
cDNA *abbr. for* complementary DNA.
C-DNA *abbr. for* C form of DNA.
cDNA clone any bacterial cell transformed by a plasmid containing a **complementary DNA** copy of an RNA molecule.
cDNA library a collection of cloned recombinant cDNA, usually in bacteria or yeast. The clones represent cDNAs prepared from all the mRNAs in any particular species or organ.
CDP *abbr. for* **1** cytidine 5′-diphosphate *or* the group cytidine(5′)diphospho. **2** cardiodilatin related peptide; *see* **natriuretic peptide.**
CDPcholine *see* **cytidine(5′)diphosphocholine.**
CDPdiacylglycerol *see* **cytidine(5′)diphosphodiacylglycerol.**
CDPdiacylglycerol–inositol 3-phosphatidyltransferase EC 2.7.8.11; *other names*: phosphatidylinositol synthase; CDPdiacylglyceride–inositol phosphatidyltransferase; an enzyme that catalyses the formation of phosphatidyl-1-D-*myo*-inositol from CDPdiacylglycerol and *myo*-inositol with release of CMP. It is a component of the pathway for the biosynthesis of phosphatidylinositol and also for its resynthesis after activation of the **phosphatidylinositol cycle**. Example from yeast: database code PIS_YEAST, 220 amino acids (24.80 kDa).
CDPdiacylglycerol–serine *O*-phosphatidyltransferase EC 2.7.8.8; *systematic name*: CDPdiacylglycerol:L-serine 3-*O*-phosphatidyltransferase; *other names*: phosphatidylserine synthase; CDPdiglyceride–serine *O*-phosphatidyltransferase. It is an enzyme of the pathway that effects *de novo* synthesis of phosphatidylserine. It catalyses the formation of *O-sn*-phosphatidyl-L-serine from CDPdiacylglycerol and L-serine with the release of CMP. Example from *Escherichia coli*: database code PSS_ECOLI, 452 amino acids (52.76 kDa). Example from *Saccharomyces cerevisiae*: database code PSS_YEAST, 275 amino acids (30.67 kDa).
CDPethanolamine *see* **cytidine(5′)diphosphoethanolamine.**
CDP-glycerol pyrophosphorylase *see* **glycerol-3-phosphate cytidylyltransferase.**
CDPK *abbr. for* calmodulin-like domain protein kinase.
CDR *abbr. for* calcium-dependent regulator protein (i.e. **calmodulin**).
CDTA *abbr. for* *trans*-cyclohexylene-1,2-diamine-*N,N,N′,N′*-tetraacetate, the anion of *trans*-1,2-diaminocyclohexane-*N,N,N′,N′*-tetraacetic acid. A **chelating agent** with a high affinity for Mg^{2+}; the abbreviation is also used for its partially or fully protonated forms.
Ce *symbol for* cerium.
C_xE_y a series of non-ionic detergents useful for the solubilization of membrane-bound proteins in their native state. They include $C_{10}E_6$, hexaethyleneglycol mono-*n*-decyl ether; $C_{10}E_8$, octaethyleneglycol mono-*n*-decyl ether; $C_{12}E_6$, hexaethyleneglycol mono-*n*-dodecyl ether; $C_{12}E_8$ octaethyleneglycol mono-*n*-dodecyl ether; and $C_{12}E_9$, nonaethyleneglycol mono-*n*-dodecyl ether; the latter two are constituents of the proprietary detergent Lubrol-PX a detergent useful for the solubilization of membrane-bound proteins in their native state.
CEA *abbr. for* carcinoembryonic antigen.
CEA gene member 1 *see* **carcinoembryonic antigen.**
Cech, Thomas Robert (1947–) US chemist, biochemist, and molecular biologist; Nobel Laureate in Chemistry (1989) jointly with S. Altman 'for their discovery of catalytic properties of RNA'.
cecropin any of a group of small, basic polypeptides with potent antibacterial activity. Cecropins are active against *Escherichia coli* and several other Gram-negative species, and are isolated from the cecropia moth, *Hyalophora cecropia*, and several other insects.
CEF10 *or* **CEF-10** a secreted protein that is implicated in

growth regulation. It is induced by v-*src* and belongs to a family of growth regulators that includes **connective tissue growth factor**. Example (precursor) from chicken: database code CE10_CHICK, 375 amino acids (40.60 kDa).

Cek a chick gene encoding receptor-type protein tyrosine kinases, and named from chicken embryo kinase. The protein product Cek-1 is the chicken **fibroblast growth factor** receptor; Cek-2 and Cek-3 are related to ***Bek***; Cek-4 and Cek-5 are homologous to ***Eph***.

celiac *or (esp. Brit.)* **coeliac** of, or belonging to, the abdominal cavity.

celiac disease *or (Brit.)* **coeliac disease** *or* **gluten enteropathy** *or* **nontropical sprue** a condition caused by sensitivity of the mucosa of the upper small intestine to the α-gliadin component of **gluten** in wheat and rye. Characterized by diarrhoea and malabsorption, it can be relieved by elimination of gluten from the diet.

Celite *proprietary name for* preparations of amorphous diatomaceous silica, used as a filter-aid and in partition chromatography.

cell 1 the basic structural unit of all living organisms; it typically comprises a small, usually microscopic, discrete mass of organelle-containing cytoplasm, bounded externally by a membrane. Each cell is capable of interacting with other cells and performing all the fundamental functions of life. In plants the cell usually also includes the cell wall. Originally the term referred to the bounding wall of a plant cell that had lost its living content. *See also* **+cyte**, **cyto+**. 2 a container for liquid during measurement or separation procedures, such as dialysis, electric conductivity measurement, electrolysis, microdiffusion, moving boundary electrophoresis, analytical ultracentrifugation, ultrafiltration, and (spectro)photometry; *see also* **cuvette**. 3 or **electric cell** a device for producing electric current by converting chemical energy into electric energy; it consists of two electrodes in contact with an electrolyte. 4 *(in crystallography) see* **unit cell**. —**cellular** *adj.*

cell-adhesion molecule *see* **adhesion molecules**.

cell body *an alternative term for* **perikaryon**.

cell-bound antibody *or* **cell-fixed antibody** any antibody molecule that is bound to the surface of a cell, e.g. an **incomplete antibody** (def. 1) or a **cytophilic antibody**.

cell coat *another name for* the **glycocalyx**.

cell cortex the part of the microtrabecular lattice lying just under the outer cell membrane of certain protozoans, e.g. paramecia.

cell counter any device for determining the number of cells in a known volume of a liquid suspension. *See* **Coulter counter**, **flow cytometer**, **hemocytometer**.

cell culture *see* **tissue culture**.

cell cycle the period between the formation of a cell by division of its mother cell and the time when the cell itself divides to form two daughter cells. *See also* **cell-division cycle**, **DNA cycle**.

cell-cycle engine all the biochemical components, and the reactions between them, that cause the periodic activation and inactivation of **M-phase-promoting factor** during cell maturation.

cell differentiation *see* **differentiation**.

cell-division cycle the pattern of DNA synthesis in dividing eukaryotic cells. It is divided into four periods: G_1 is the first 'gap' in DNA synthesis between **telophase** and the period of DNA synthesis, designated S. This S phase is followed by G_2, the second gap in DNA synthesis. This runs on into the period of **mitosis**, M, which lasts from early **prophase** to late **telophase**.

cell division protein kinase *abbr.*: CDK; any of various protein kinase enzymes involved in the regulation of **cyclin** activity, such as CDC20. *See also* ***cdc***.

cell-fixed antibody *an alternative name for* **cell-bound antibody**.

cell fractionation the fractionation of disrupted cells into their subcellular components.

cell-free extract a fluid, made by breaking open cells, that contains most of the constituents of the cells in question but no intact cells.

cell-free system any experimental system composed of subcellular fractions and/or cell-free extracts. Cell-free systems for amino-acid incorporation (protein synthesis) generally contain ribosomes, natural or synthetic mRNA, tRNAs, enzymes, amino acids, an ATP generating system, GTP, buffer, certain inorganic salts, and some organic compounds.

cell fusion the formation of a single hybrid cell containing the nuclei and cytoplasms from different cells. It may be induced by treatment of a mixed cell population with certain **fusogens** (e.g. killed Sendai virus or polyethylene glycol). It is an important step in the technique of forming **hybridoma cells** for the production of **monoclonal antibodies**.

cell junction a specialized region of connection between two cells or between a cell and the extracellular matrix. *See also* **desmosome**, **gap junction**, **hemidesmosome**, **tight junction**.

cell line a population of cells cultured *in vitro* that are descended through one or more generations (and possibly subcultures) from a single primary culture. The cells of such a population share common characteristics.

cell locomotion *see* **cell migration**.

cell matrix *see* **extracellular matrix**.

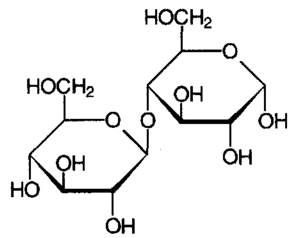
cell-mediated immunity specific immunity that depends on the presence of T lymphocytes. It is responsible for, e.g., allograft rejection, delayed hypersensitivity, and tuberculin test reactions, and is important in the organism's defence against viral and some bacterial infections.

cell membrane any **membrane** found in a cell. The term is sometimes used loosely to indicate the **plasma membrane**. *See also* **cytoplasmic membrane**.

cell migration the active movement of a cell from one location to another, particularly the migration of a cell over a surface.

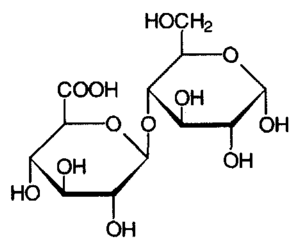
β-cellobiase *another name for* β-glucosidase; *see* **glucosidase**.

cellobiose 4-*O*-β-D-glucopyranosyl-D-glucose; a disaccharide that represents the basic repeating unit of **cellulose**; it is liberated by **cellulase**.



α-anomer

cellobiuronic acid 4-*O*-β-D-glucopyranuronosyl-D-glucose; *condensed form*: GlcA(β1-4)Glc. It is a component of, e.g., type III pneumococcal bacterial capsule polysaccharides.



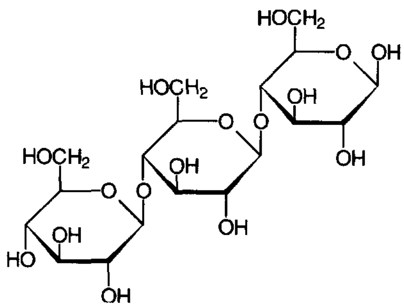
α-anomer

cellophane a form of transparent, flexible sheeting made from regenerated **cellulose**; originally a proprietary name.

Cellosolve *proprietary name for* a range of ethanediol monoalkyl ethers, especially 2-ethoxyethanol. They are useful as solvents, e.g. in liquid scintillation mixtures, to facilitate the dispersion of aqueous samples in organic solvents.

cellotetraose a tetrasaccharide composed of D-glucose units in β1–4 linkage.

cellotriose the trisaccharide *O*-β-D-glucopyranosyl-(1–4)-*O*-β-D-glucopyranosyl-(1-4)-D-glucose; *condensed form*: Glc(β1-4)Glc(β1-4)Glc.



β-D-pyranose form

cell plate the nascent cell membrane and cell wall structure that forms between the two daughter nuclei near the centre of a dividing plant cell. It develops at the equatorial region of the **phragmoplast**. It grows outwards to join with the lateral walls and form two daughter cells.

cell sorter any **flow sorter** for sorting cells in liquid suspension, on the basis of some characteristic that is detectable by the sorter.

cell-surface protein *abbr.*: CSP; *an alternative name for* **fibronectin**.

cellubrevin a **synaptobrevin** homologue that is involved in membrane traffic as part of a constitutively recycling pathway. With VAMP proteins it is a component of the machinery controlling docking and fusion of secretory vesicles with their target membrane. It is a ubiquitous substrate for **tetanus toxin** light chain. Example from *Rattus norvegicus*: database code S35077, 103 amino acids (11.47 kDa).

cellular affinity the tendency of cells to adhere specifically to cells of the same type, but not to cells of a different type. This property is lost in cancer cells.

cellular immunity 1 the increased ability of phagocytic cells to destroy parasitic organisms, i.e. macrophage immunity. 2 *an alternative name for* **cell-mediated immunity**.

cellulase 1 EC 3.2.1.4; *other names*: endoglucanase; endo-1,4-β-glucanase; carboxymethyl cellulase. An enzyme that catalyses the endohydrolysis of 1,4-β-D-glucosidic linkages in **cellulose**, thereby hydrolysing cellulose to cellobiose. It causes abscission in plants, and its synthesis is stimulated by ethylene. Microbial cellulases play an important part in the digestion of cellulose by ruminants. 2 any lytic enzyme whose substrate is cellulose. *See* **cellulose 1,4-β-cellobiosidase**, **glucosidase**. Example from *Cellulomonas uda*: database code GUN_CELUD, 359 amino acids (40.64 kDa).

cellule 1 (*archaic*) a minute cavity or very small **cell**. 2 *a former name for* a **liposome** bounded by a single lipid bilayer.

cellulose a linear β1–4 **glucan** of molecular mass 50–400 kDa with the pyranose units in the **-$^4C_1$** conformation. It is predominantly a plant polysaccharide though it also occurs in some tunicates. Cellulose microfibrils are major components of plant cell walls. Cellulose is insoluble in water but will dissolve in ammoniacal solutions of cupric salts. It is often said to be the most abundant organic compound in the biosphere, but this distinction has also been accorded to the **hopanoids**.

cellulose 1,4-β-cellobiosidase EC 3.2.1.91; *other names*: exoglucanase; exocellobiohydrolase; 1,4-β-cellobiohydrolase. An enzyme that catalyses the hydrolysis of 1,4-β-D-glucosidic linkages in cellulose and cellotetraose, releasing cellobiose from the nonreducing ends of the chains. Example (precursor) from *Trichoderma reesei*: database code GUX1_TRIRE, 513 amino acids (54.01 kDa); 3-D structure of C-terminal domain: database code NRL_1CBH, 36 amino-acid fragment (3.74 kDa).

cellulosic 1 of, or pertaining to, cellulose; of the nature of cellulose; composed (primarily) of cellulose; derived from or made by chemical modification of cellulose. 2 a substance made from cellulose or a derivative of cellulose.

cellulolysis the process of hydrolysing cellulose. —**cellulolytic** *adj.*

cell wall the rigid or semi-rigid envelope lying outside the **cell membrane** of plant, fungal, and most prokaryotic cells maintaining their shape and protecting them from osmotic lysis. In prokaryotes it lies inside any capsule and slime layer, usually consists mainly of **peptidoglycan**, and can be removed by various techniques with retention of its three-dimensional form. In fungi the cell wall is composed largely of polysaccharides, while in plants it is made up of cellulose and, often, lignin.

Celsius temperature *symbol*: θ *or* t; a temperature defined as the excess of the thermodynamic temperature, T, over 273.15 K; i.e. $\theta = T - 273.15$, where θ is measured in °C (i.e. **degrees Celsius**, formerly called degrees centigrade) and T is measured in **kelvins**. It is identical with centigrade temperature. [After Anders Celsius (1701–44), Swedish astronomer.]

centi+ *symbol*: c; SI prefix denoting 10^{-2}.

centigrade temperature *former name for* **Celsius temperature**.

centimetre–gram–second system *see* **cgs system**.

centimorgan *or* centiMorgan *abbr.*: cM; a unit of **map distance** used in genetic mapping, equal to 1% crossover found in recombination studies; 1 cM equals 1 map unit. Hence, if two genes are separated by 1 cM, there is a 1% probability of recombination in a single meiotic event.

central dogma a fundamental principle of molecular biology, first articulated by Francis **Crick** in 1958. He wrote that the central dogma states that once ‘information’ has passed into protein *it cannot get out again*. In more detail, the transfer of information from nucleic acid to nucleic acid, or from nucleic acid to protein may be possible, but transfer from protein to protein, or from protein to nucleic acid is impossible. Information means here the *precise* determination of sequence, either of bases in the nucleic acid or of amino-acid residues in the protein. The dogma has been taken to relate primarily to the following transfers of information that can occur in all cells: DNA → DNA, DNA → RNA, and RNA → protein. However, the transfer of information from RNA to RNA occurs in the replication of RNA viruses. The transfer of information from RNA to DNA occurs following infection of a cell with a retrovirus; the DNA is then incorporated into the host genome. When challenged that the reverse transfer of information from RNA to DNA by retroviruses contradicted the dogma, Crick refuted the challenge by drawing attention to the fact that as originally formulated the dogma permitted the transfer of information from (any) nucleic acid to (any other) nucleic acid (although the action of retroviruses was not known at that time) and that the essential point was that *sequence* information could not be passed from protein to nucleic acid, or from protein to protein.

central nervous system *abbr.*: CNS; the part of the nervous system of an animal that contains a high concentration of cell bodies and synapses and is the main site of integration of nervous activity. In higher animals it is isolated from the circulation by the **blood–brain barrier**. It consists essentially of a brain or cerebral ganglia, and a nerve cord, which may be dorsal or ventral, single or double. *Compare* **peripheral nervous system**.

centrifugal 1 moving, acting, or tending to act away from a centre or axis. 2 of, pertaining to, or operated by a **centrifugal force**. 3 *(in botany)* developing outwards from a centre, as the flowers of certain types of inflorescence. 4 *(in physiology)* sending nerve impulses from a central location to parts supplied by that nerve; **efferent**. *Compare* **centripetal**.

centrifugal analyser *or* **centrifugal fast analyser** an apparatus widely used in clinical chemical laboratories for the simultaneous and very rapid chemical, biochemical, or immunochemical determination of one particular constituent in each of a large number of samples. Basically it consists of an appropriately designed transfer disk into which reagent(s) and samples are discretely and automatically pipetted. When loaded, the transfer disk is placed in the centre of a cuvette rotor in the centrifuge, and the rotor is accelerated. Reagent(s) and each sample are mixed together as they are radially propelled along specially shaped channels into the cuvettes. Absorbance, fluorescence, or other measurements are then made automatically on the cuvettes as they are rotating. The whole process, including data handling, is controlled by a computer.

centrifugal elutriation a technique in which cells or other particles are separated by **elutriation** in a specially constructed centrifuge rotor; the increased gravitational field resulting from centrifugation speeds up separation of the particles.

centrifugal field any space in which a **centrifugal force** is operating, as in the spinning rotor of a centrifuge.

centrifugal force a force acting radially outwards on any body moving in a curved path, equal and opposite to the **centripetal force**. The centrifugal force on a body of mass m, moving in a circular path of radius r, with velocity v, is: mv_2/r.

centrifuge 1 an apparatus in which fluids may be rotated rapidly so that substances (solutes or dispersed particles) of different densities may be separated by centrifugal force. 2 to rotate rapidly in a centrifuge. —**centrifugation** *n.*

centrin *an alternative name for* **caltractin**.

centriole a cellular organelle, found close to the nucleus in many eukaryotic cells, consisting of a small cylinder with microtubular walls, 300–500 nm long and 120–150 nm in diameter. It contains nine short, parallel, peripheral microtubular fibrils, each fibril consisting of one complete **microtubule** fused to two incomplete microtubules. Cells usually have two centrioles, lying at right angles to each other. At division, each pair of centrioles generates another pair and the twin pairs form the poles of the mitotic spindle. *Compare* **basal body**.

centripetal 1 moving, acting, or tending to act towards a centre or axis. 2 of, pertaining to, or operated by a **centripetal force**. 3 *(in botany)* developing inwards from the outside, as the flowers of certain inflorescences. 4 *(in physiology)* sending nerve impulses towards a central location from peripheral parts supplied by that nerve; **afferent**. *Compare* **centrifugal**.

centripetal force a force acting radially inwards on any body moving in a curved path; it constrains the body to a curved path. It is equal and opposite to the **centrifugal force**.

centromere the region of a eukaryotic chromosome that is attached to the spindle during nuclear division. It is defined genetically as the region of the chromosome that always segregates at the first division of meiosis; i.e. the region of the chromosome in which no **crossing over** occurs. At the start of M phase, each chromosome consists of two sister chromatids with a constriction at a point which forms the centromere. During late prophase two **kinetochores** assemble on each centromere, one kinetochore on each sister chromatid.

centromere-binding proteins proteins from, e.g., the yeast **kinetochore** that are able to bind centromere DNA; examples from yeast: centromere-binding protein 1: database code CBF1_YEAST, 351 amino acids (39.40 kDa); a protein that binds to a sequence in centromere DNA known as CDE-I; it is a helix-turn-helix protein; (2) centromere-binding protein 2: sequence of the 110 kDa subunit: database code CBF2_YEAST, 956 amino acids (111.79 kDa); a complex involving a larger (240 kDa) subunit of this protein also binds to CDE-I and is involved in spindle formation.

centrosome a structure occurring close to the nucleus during **interphase** in many eukaryotic cells. It comprises a pair of **centrioles**, satellite bodies, and a cytoplasmic zone from which the mitotic microtubule spindle is organized. The structure of the centrosome in animal cells changes continually during the **cell-division cycle**.

centrosphere a more or less well delineated part of the cytoplasm at the poles of the spindle (*see* **mitotic spindle**).

cephaeline an alkaloid derived from *Uragoga ipecacuanha*. It is a gastric irritant, inducing emesis. See also **emetine**, **ipecachuanha**.

cephalin *a former name for* 1 **phosphatidylethanolamine**. 2 **phosphatidylserine**.

cephalosporin any of a heterogeneous group of natural and semisynthetic, often β-lactam, antibiotics. Cephalosporins C, N, and P are obtained from cultures of moulds such as *Cephalosporium acremonium*. Cephalosporins are active against a range of Gram-positive and Gram-negative bacteria.

cephalosporinase *an alternative name for* **β-lactamase** II.

Cer *symbol for* a ceramide residue. *See also* **Cer(Hex)**.

ceramidase EC 3.5.1.23; *other name*: acylsphingosine deacylase; an enzyme involved in the degradation of sphingolipids. It catalyses the hydrolysis of *N*-acylsphingosine to a fatty acid and sphingosine.

ceramide *symbol*: Cer; any *N*-acylated **sphingoid**.

ceramide cholinephosphotransferase EC 2.7.8.3; an enzyme involved in the biosynthesis of sphingomyelin by the reaction of CDPcholine with *N*-acylsphingosine with the release of CMP.

ceramide glucosyltransferase EC 2.4.1.80; an enzyme that catalyses the formation of glucosylceramides. The reaction involves the formation of D-glucosyl-*N*-acylsphingosine from UDPglucose and *N*-acylsphingosine with the release of UDP.

cercidosome an atypical mitochondrion or oxidizing body, found, e.g., in trypanosomes, whose function may primarily be to regulate the $NAD^+/NADH$ ratio of the cytosol. [From Greek *kerkis, kerkido*, weaver's shuttle, + **+some**.]

cerebellin *see* **precerebellin**.

cerebral 1 of, or relating to, the cerebrum. 2 *(loosely)* of, or relating to the brain.

cerebrodiene a lipid, resembling **sphingosine**, that is obtained from the brain of sleep-deprived cats.

cerebroglycan an integral membrane heparan sulfate proteoglycan unique to the developing nervous system and expressed specifically during neuronal differentiation. Example from neonate rat brain: database code RATCRBGLVC, 579 amino acids (63.28 kDa).

cerebronate 2-hydroxytetracosanoate; $CH_3(CH_2)_{21}CH(OH)$-COO^-; it is the 2-hydroxy derivative of lignocerate. It is enriched in brain cerebrosides, the fraction containing this fatty acid being known as **phrenosin**.

cerebrose *see* **galactose**.

cerebroside any **glycosphingolipid** that contains a monosaccharide, normally glucose or galactose, in 1-*O*-β-glycosidic linkage with the primary alcohol of an *N*-acyl **sphingoid** (ceramide). In plants the monosaccharide is normally glucose; in animals it is normally galactose, though this may vary with the tissue, glucose being present in blood cerebroside. In plants, the sphingoid is usually phytosphingosine, in animals sphingosine or dihydrosphingosine (sphinganine). Cerebrosides frequently contain large amounts of longer-chain fatty acids such as behenic (22:0), lignoceric (24:0), and nervonic (24:1) acids, and 2-hydroxy fatty acids such as cerebronic (α-OH24:0) and 2-hydroxynervonic acids. Cerebroside synthesis may proceed either by acylation of psychosine or transfer of glucose or galactose to ceramide.

cerebroside-sulfatase EC 3.1.6.8; *other name*: arylsulfatase A; an enzyme involved in the catabolism of sulfatides. It catalyses the hydrolysis of a cerebroside 3-sulfate to a cerebroside and sulfate. Example (precursor) from human: database code ARSA_HUMAN, 507 amino acids (53.53 kDa).

cerebrospinal fluid *abbr.*:CSF; a clear fluid, containing little protein and few cells, that fills the subarachnoid space and ventricles of the brain, and the central canal of the spinal cord.

Cerenkov counter *a variant spelling of* **Cherenkov counter**.

Cerenkov radiation *a variant spelling of* **Cherenkov radiation**.

Cer(Hex) *symbol for* ceramide monohexoside. The di- and trihexosides are symbolized by $Cer(Hex)_2$ and $Cer(Hex)_3$ respectively. *See also* **Cer**.

cerinic acid *see* **cerotoyl**.

ceroid an autofluorescent, insoluble lipopigment that accumulates in the livers of rats with nutritional cirrhosis.

cerotoyl *symbol*: Crt; *the trivial name for* hexacosanoyl, $CH_3–[CH_2]_{24}CO–$, the univalent acyl group derived from hexacosanoic acid (*also called* **certoic acid** *or (formerly)* **cerinic acid**), a saturated, unbranched, acyclic, aliphatic acid, which occurs naturally in wool fat.

certoic acid *see* **cerotoyl**.

cerulein *or (Brit.)* **caerulein** a decapeptide, pGlu-Gln-Asp-Tyr(SO_3H)-Thr-Gly-Trp-Met-Asp-PheNH_2, isolated from the skin of various frogs, e.g. *Hyla caerulea*. It is often accompanied by the nonapeptide **phyllocerulein** (*or* **phyllocaerulein**). Both have the same C-terminal pentapeptide amide sequence as porcine **cholecystokinin**. Cerulein-like peptides have been isolated from the antrum and small intestine of amphibians, in which they stimulate strongly the secretion of gastric acid.

cerulenin (2*R*,3*S*)-epoxy-4-oxo-7*E*,10*E*-dodecadienamide; an antifungal antibiotic isolated from *Cephalosporium caerulens* and other sources. It inhibits fatty-acid biosynthesis by binding with 3-oxoacyl-[acyl carrier protein] synthase (*see* **fatty acid synthase complex**) and also interferes with sterol biosynthesis by inhibiting **hydroxymethylglutaryl-CoA synthase** activity.

cerulin *a variant spelling of* **cerulein**.

ceruloplasmin *or (Brit.)* **caeruloplasmin** a bright-blue copper-containing protein found in blood plasma. It has ferroxidase activity, EC 1.16.3.1, and catalyses the reaction:

$$4\ \text{iron(II)} + 4\ H^+ + O_2 = 4\ \text{iron(III)} + 2\ H_2O.$$

It binds six or seven Cu^{2+} ions per molecule. It is important in iron metabolism and is abnormally low in hepatolenticular degeneration (Wilson's disease). It is one of the acute phase proteins, its level rising late in the acute phase. Example from human: database code CERU_HUMAN, 1065 amino acids (122.2 kDa).

ceryl *the common name for* hexacosyl, $CH_3–[CH_2]_{24}–CH_2–$, a saturated, unbranched, acyclic, alkyl group.

cesium chloride *or (esp. Brit.)* **caesium chloride** a salt that forms dense solutions in water and so is used in **isopycnic centrifugation** to separate DNA molecules of different densities.

Cet *symbol for* the carboxyethyl group, $–CH_2–CH_2–COOH$.

cetaceum *an alternative term for* **spermaceti**.

Cetavlon *a proprietary name for* **cetrimide**.

cetoleic acid *see* **docosenoic acid**.

cetrimide *or* **cetrimidum** a detergent disinfectant consisting of a mixture of alkylammonium bromides, principally **cetyltrimethylammonium bromide**. *See* **quaternary ammonium compounds**.

cetyl *or* **palmityl** *the common name for* hexadecyl, $CH_3–[CH_2]_{14}–CH_2–$, a saturated, unbranched, acyclic, alkyl group (hexadecyl is preferred).

cetyltrimethylammonium bromide *abbr.*: CTAB; hexadecyltrimethylammonium bromide; *N*,*N*,*N*-trimethyl-1-hexadecanaminium bromide; a cationic detergent useful in the solubilization of membrane proteins; aggregation number 170, CMC 1 mM. It is the principal component of **cetrimide**.

Cf *symbol for* californium.

C form of DNA *abbr.*: C-DNA *or* DNA-C; the molecular conformation adopted by fibres of the lithium salt of duplex DNA at low relative humidity (below approximately 44%). It consists of a right-handed double helix containing about 9.3 nucleotide residues per turn; otherwise it is similar to the **B form**. *See also* **A form** (def. 1), **Z form**.

CFTR *abbr. for* cystic fibrosis transmembrane conductance regulator.

CG *abbr. for* chorionic gonadotropin (*see* **choriogonadotropin**).

cg clone any crown gall **clone** (def. 1); *see* **crown gall disease**.

CGM1 *see* **carcinoembryonic antigen**.

cGMP *abbr. for* cyclic GMP; *see* **guanosine 3′,5′-phosphate**.

CGRP *abbr. for* calcitonin gene-related peptide.

cgs system *or* **CGS system** *abbr. for* centimetre–gram–second system; a metric system of units based on the centimetre (length), gram (mass), and second (time). It has now been superseded by the **SI** system.

Chaikoff homogenizer a hydraulically operated tissue or cell homogenizer in which the sample is forced through an annulus by a piston. The piston diameter relative to that of the annulus is chosen to suit the size of the component that it is desired to isolate with minimal damage.

Chain, (Sir) Ernst Boris (1906–79), German-born British chemical pathologist and biochemist remembered particularly for his part in the development of penicillin (with H. W. Florey) and semisynthetic variants of penicillin as clinically useful antibacterial agents; Nobel Laureate in Physiology or Medicine (1945) jointly with A. Fleming and H. W. Florey ' for the discovery of penicillin and its curative effect in various infectious diseases'.

chain reaction a chemical reaction in which one or more reactive intermediates are continuously regenerated, often through a repetitive cycle of elementary 'propagation' steps.

chain-termination codon *an alternative name for* **termination codon**.

chain-termination method *or* **dideoxy method** *or* **Sanger method** a rapid technique for determining nucleotide sequences in DNA in which the 2′,3′-dideoxy (or arabinonucleoside) analogues of the normal deoxynucleoside triphosphates are used as specific chain-terminating inhibitors of DNA polymerase. Single-stranded fragments obtained by nuclease action on the DNA are replicated by DNA polymerase I in the presence of a primer and a mixture of one dideoxynucleoside triphosphate and the four normal nucleoside triphosphates, one of which is labelled with ^{32}P. Separate incubations are carried out using analogous terminators for the other nucleotides. The effect is to yield radioactively-labelled stretches of DNA corresponding in length to the number of bases between the 5′ end and the position of the base corresponding to the terminator, which will be incorporated once into each stretch of DNA in such a way that it yields an appropriate stretch of DNA for every position occupied by the base in question. The resulting products from each incubation are separated in parallel by gel electrophoresis; from the pattern of radioactive bands obtained the sequence of the fragment can be read off. The method is reasonably accurate for sequences of 15 to about 200 nucleotides.

chair configuration (of a cyclobutadipyrimidine) *see* **pyrimidine dimer**.

chair conformation any **conformation** of a nonplanar, six-membered, saturated ring compound in which the ring atoms in relative positions 1, 2, 4, and 5 lie in one plane, and those in relative positions 3 and 6 lie on opposite sides of that plane. For a monosaccharide or monosaccharide derivative the conformational descriptor *-C* may be added to its name, and the reference plane is so chosen that the lowest-numbered carbon atom in the ring is exoplanar. See also $^1C_4$, $^4C_1$ *Compare* **boat conformation**.

chalcone 1 phenyl styryl ketone;1,3-diphenyl-2-propen-1-one. 2 any of a group of hydroxylated derivatives of chalcone (def. 1); they are important intermediates in plants in the biosynthesis of flavanones, flavones, flavonols, and anthocyanidins. The chalcone derivative **isoliquivitigenin** has anti-platelet activity, inhibiting 12-lipoxygenase. It stimulates soluble guanylate cyclase. It has anti-tumour activity and potently prevents tumour promotion by phorbol esters.

challenge 1 *(in immunology)* the act or process of injecting antigenic material into an immunized animal to test for immunity or to provoke a further immune response. **2** *(in endocrinology)* the act or process of administering an agent to evaluate hormonal responses. It is used to test hormonal control and function.

chalone any inhibitor affecting the cell cycle before the onset of mitosis that is tissue-specific and present in the tissue of action; it need not be species-specific. [From Greek *khalon*, to make slack.]

channel 1 a passage, often tubular or trough-shaped, along or through which movement of substances, especially fluids, can occur; e.g. a pore through a membrane. **2** the span between selected settings of two discriminators in a pulse counter, e.g. a scintillation counter, that defines the range of pulse intensities that will be recorded. **3** to guide into or cause to move through a particular channel or along a particular route; e.g. to channel a metabolite along a particular metabolic pathway. **4** descriptive of a membrane protein that transports ions across the membrane in the direction of the concentration gradient. *See also* **calcium channel**, **potassium channel**, **sodium channel**. — **channelled** *or* (*US*) **channeled** *adj.*

channel carrier *see* **ionophore**.

channel-forming integral membrane protein 28 *abbr.*: CHIP 28; *other name*: aquaporin 1; an integral membrane protein that forms a water-specific channel that provides the plasma membranes of red cells and kidney proximal tubules with high permeability to water. It is a homotetramer. Example from human: database code AQP1_HUMAN, 269 amino acids (28.49 kDa).

channelling *or* (*US*) **channeling** (*in column chromatography*) the uneven flow resulting from inhomogeneous packing of the column.

channel protein an **integral (cell membrane) protein** that permits the controlled (gated) movement of ions through a membrane. *See* **calcium channel**, **potassium channel**, **sodium channel**.

channels-ratio method a method of quench correcting in liquid scintillation counting in which two channels are used to measure the average energies of beta particles both before and after quenching.

chaoptin an extracellular membrane glycoprotein required for *Drosophila melanogaster* photoreceptor cell morphogenesis and containing a **leucine-rich repeat** domain. It is a cell-type specific adhesion molecule, exclusively localized to photoreceptor cells, linked to the outer surface of the membrane by a **GPI-anchor**. Example (precursor) from *D. melanogaster*: database code CHAO_DROME, 1134 amino acids (130.57 kDa).

Chaotropase *the proprietary name for* a mixture of endopeptidases isolated from *Streptomyces griseus*, so named for their stability and activity in the presence of the **chaotrope** 6 M guanidinium chloride.

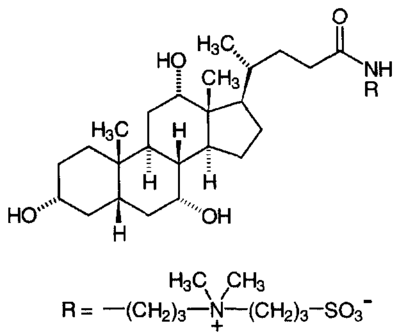
chaotrope *or* **chaotropic agent** any substance that increases the transfer of apolar groups to water because of its ability to decrease the 'ordered' structure of water and to increase its lipophilicity. Chaotropes are usually ions, e.g. SCN^-, CNS^-, ClO_4^-, I^-, Br^-, CH_3–COO^-. They cause the dissolution of biological membranes, the solubilization of particulate proteins, changes in the secondary, tertiary, and quaternary structure of proteins, and denaturation of nucleic acids.

chaotropic dissociation assay a technique used to measure the heterogeneity of antibody affinities in, e.g., serum. Immune complexes are dissociated in buffers of varying strengths and pH or in chaotropic agents (*see* **chaotrope**). Low-affinity antibodies dissociate more readily.

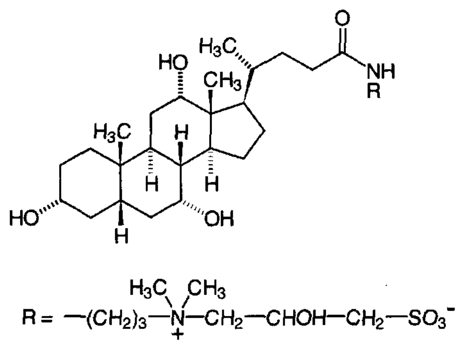
chaperone any of a functional class of unrelated families of proteins that assist the correct non-covalent assembly of other polypeptide-containing structures *in vivo*, but are not components of these assembled structures when they are performing their normal biological functions. *See also* **chaperonin**, **heat-shock protein**.

chaperonin any of a ubiquitous subclass of molecular **chaperones** implicated in the folding of other proteins. They include two kinds of proteins: (1) chaperonin 60 kDa (*also called* chaperonin-60 *or* cpn 60; the counterpart in other structures of the product of the ***groEL*** gene of *Escherichia coli*); five motifs identify such chaperonins; example from *Brassica napus*, plastid chaperonin-60: database code BNA60CPNA, 546 amino acids (57.63 kDa); and (2) chaperonin 10 kDa (*also called* chaperonin-10 *or* cpn 10; the counterpart in, e.g., *Mycobacterium* of the product of the ***groES*** gene of *E. coli*). *See also* **heat-shock protein**.

CHAPS 3-[(3-cholamidopropyl)dimethylammonio]-1-propanesulfonate; a zwitterionic detergent combining the properties of sulfobetaine and bile-salt types of detergent. It is used for membrane solubilization. Aggregation number 4–14, CMC 6–10 mM.



CHAPSO 3-[(3-cholamidopropyl)dimethylammonio]-2-hydroxy-1-propanesulfonate; zwitterionic detergent combining the properties of sulfobetaine and **bile-salt** types of detergent; it is more soluble than **CHAPS**. Aggregation number 11, CMC 8 mM.



characteristic group *(in chemistry)* any atom or group that is incorporated into a parent compound otherwise than by a direct carbon–carbon linkage, but including the groups –C≡N and =C=X where X is O, S, Se, Te, NH, or substituted NH.

characteristic time a measure of the time required for a system to change from one state of motion to another, usually taken as the time required for the system to approach within one half of the new state of motion.

characteristic X-ray *see* **X-rays**.

Charcot–Marie–Tooth disease *see* **myelin proteins**.

Chargaff's rules a set of quantitative rules describing the base composition of duplex DNA: (1) [A] = [T] and (2) [G] = [C], where the square brackets denote the concentrations of the bases in moles per cent; minor bases, if present, are included with the appropriate major base. Three corollaries follow: (1) [A]/[T] = [G]/[C]; (2) total purines = total pyrimidines, i.e. [A] + [G] = [C] + [T]; and (3) total 6-aminobases = total 6-keto-bases, [A] + [C] = [G] + [T]. A further consequence of these relationships is that the [A + T]/[G + C] ratio is a characteristic property of individual DNAs. [After Erwin Chargaff (1905–), US biochemist.]

charge 1 to load with the proper or appropriate quantity, e.g. to charge a tRNA molecule with an aminoacyl group, or to charge (an accumulator, capacitor, etc.) with electricity. 2 *see* **electric charge**. 3 the quantity of electricity held in, e.g., an accumulator or a capacitor. 4 a load. *Compare* **discharge**.

charge density *symbol* ρ; SI unit C m^{-3} (coulombs m^{-3}).

charged tRNA any **aminoacyl-tRNA** molecule.

charge-relay system a catalytic triad found in the serine proteases comprising, in chymotrypsin, the residues His^{57}, Ser^{195}, and Asp^{102}, the histidine being hydrogen bonded to the carboxylate of the buried aspartate. Spatially, these residues occur in positions such that the electronegativity of the aspartate carboxyl group can be relayed through the histidine nitrogen atoms to the serine hydroxyl group, enhancing its reactivity.

chargerin II *other names*: A61; protein 8 of H^+-transporting ATP synthase (EC 3.6.1.34); one of the chains of the nonenzymic component (CF_0 subunit) of the mitochondrial ATPase complex (membrane bound). Example from *Rattus norvegicus*: database code ATP8_RAT, 67 amino acids (7.63 kDa). *See also* **H^+-transporting ATP synthase**.

charge-shift electrophoresis a method of **electrophoresis** in a detergent solution in which the net charges of the complexes between amphiphilic proteins and the detergent depend on the charge of the latter, while the net charges on hydrophilic proteins are unaffected. This effect is exploited in, e.g., SDS–polyacrylamide gel electrophoresis.

charge-transfer chromatography a method of **liquid-gel chromatography** in which separation is due to electronic coupling between molecules being chromatographed and strong electron donor or acceptor groups attached to the supporting gel.

charge-transfer complex a complex that may be formed in an oxidation–reduction reaction, corresponding to electronic transition(s) to an excited state in which there is a partial transfer of electronic charge from the electron donor to the electron acceptor.

charging the covalent attachment of an aminoacyl group to a tRNA molecule to form an **aminoacyl-tRNA** molecule.

Charles's law *see* **gas laws**. [After Jacques Alexandre Cesar Charles (1746–1823), French physicist.]

charybdotoxin *abbr*.: ChTX; a 37-residue peptide isolated from the venom of the scorpion *Leiurus quinquestriatus hebraeus*. It is a potent inhibitor of some Ca^{2+}-activated K^+ channels and voltage-dependent K^+ channels, and probably acts at the outer face of the membrane, blocking the ion conductance channel. It has links between Cys residues 7–28, 13–33, and 17–35; amino-acid sequence:

XFTNVSCTTSKGCWSVCQRLHNTSR-
GKCMNKKCRCYS

(X = pyroglutamyl).

chase 1 the effective termination of incorporation of either a (radio)nuclide or a labelled compound into a substance, brought about by the addition of a large excess of an isotope of the (radio)nuclide or the unlabelled compound to the system, used especially after a pulse. 2 the quantity of isotope or unlabelled compound added to stop the incorporation of a (radio)nuclide or a labelled compound.

che the symbol for any of a class of genes that are implicated in chemotaxis in *Escherichia coli*. For example, *cheA* encodes a chemotaxis protein, protein phosphotransferase (EC 2.7.1.–), that is autophosphorylated and transfers its phosphate to CheB and CheC. Other two-component bacterial regulatory proteins are related. Example: database code CHEA_ECOLI, 654 amino acids (71.24 kDa). *cheB* encodes a chemotaxis protein, **protein-glutamate methylesterase** (EC 3.1.1.61), that catalyses the hydrolysis of protein L-glutamate *O*-methyl ester to protein L-glutamate and methanol. The protein is a **methyl-accepting chemotaxis protein** (MCP), the methylation state of which is crucial for sensory responses and adaptations. Example: database code CHEB_ECOLI, 349 amino acids (37.45 kDa). *cheR* encodes the chemotaxis protein methyltransferase (EC 2.1.1.80); it catalyses the reaction:

S-adenosyl-L-methionine + protein L-glutamate =
S-adenosyl-L-homocysteine + protein L-glutamate methyl ester.

The protein is an MCP, the methylation state of which is crucial for sensory responses and adaptations. Example: database code CHER_ECOLI, 286 amino acids (32.71 kDa). *cheW* encodes a purine-binding chemotaxis protein that is involved in the transmission of sensory signals from chemoreceptors to the flagellar motors. Example: database code CHEW_ECOLI, 167 amino acids (18.06 kDa). *cheY* encodes a chemotaxis protein involved in the transmission of sensory signals from chemoreceptors to flagellar motors and the regulation of clockwise rotation. Example: database code CHEY_ECOLI, 128 amino acids (13.95 kDa); the 3-D structure (NRL_3CHY) is known; the structure is largely α-helical, with some β-strands. *cheZ* encodes a chemotaxis protein thought to be involved in generating a regulating signal for bacterial flagellar rotation. Example: database code CHEZ_ECOLI, 214 amino acids (23.95 kDa).

checkpoint a point in the eukaryotic **cell division cycle** where the cycle can be halted until conditions are suitable for the cell to proceed to the next stage.

chel *abbr. for* chelate effect.

chelate 1 any chemical species in which there is **chelation**. 2 of, or denoting, a chelate (def. 1). 3 to form a chelate compound; to effect **chelation**. A chelate compound is a chemical compound in which a metal ion is combined with a substance that contains two or more electron donor groups so that the resulting structure contains one or more rings. —**chelated** *adj.*

chelate effect *abbr*.: chel; the stability difference between a chelate complex and the corresponding complex with simple ligands. For the displacement reaction:

$$MA_n + mL \rightarrow ML_m + nA,$$

$$\text{chel} = \log \beta_m (ML_m) - \log \beta_n (MA_n),$$

where ML_m and MA_n are complexes containing an equal number of the same donor atoms. Ligand L is multidentate, containing several donor atoms of the same element as the unidentate ligand A, and β_m and β_n are the overall stability constants.

chelating agent *or* **chelator** a substance that is able to form a **chelate compound** with a metal ion.

chelation the formation or presence of bonds from two or more atoms within the same ligand to a single central (metal) atom.

Chelex 100 *the proprietary name for* a synthetic ion-exchange resin consisting of a styrene divinylbenzene copolymer containing paired iminodiacetate groups that act as chelating groups in binding metal ions, especially transition-metal or other divalent ions.

chemical adsorption *an alternative term for* **chemisorption**.

chemical antagonism antagonism resulting from the chemical combination of an **antagonist** with the substance being antagonized.

chemical carcinogen any carcinogenic substance of low relative molecular mass. *See* **carcinogen**.

chemical cleavage method *or* **Maxam–Gilbert method** a rapid technique for determining nucleotide sequences in DNA (or in restriction fragments) that depends on labelling either end of a DNA strand with ^{32}P and the subsequent preferential, partial chemical cleavage of the labelled DNA specifically at positions occupied by one of the four possible bases. This results in a nested set of radioactive fragments extending from the ^{32}P label to each of the positions occupied by the particular base. The process is then repeated with cleavage, in turn, at each of the other three bases. The four sets of radioactive fragments are then separated, in adjacent lanes, by polyacrylamide gel electrophoresis, which arranges oligonucleotides in ascending order of the number of nucleotide residues contained. The base sequence of the original piece of DNA can be read directly from an autoradiograph of the resultant gel. The method can be used for sequences of 150 or more nucleotide residues. *Compare* **chain-termination method**. [After Allan M. Maxam and Walter Gilbert (1932–).]

chemical coupling hypothesis the hypothesis that the coupling of ATP synthesis to oxidation in oxidative phosphorylation is due to the formation of common 'high-energy' intermediates during electron transport that are subsequently used to phosphorylate ADP to ATP. It has been superseded by the **chemiosmotic coupling hypothesis**.

chemical equivalent the combining proportion of a substance, by mass, relative to a hydrogen standard. The chemical equivalent of an element is the number of grams of that element that will combine with or replace 1 g of hydrogen.

chemical exchange the changing magnetic environment of an atomic nucleus in a molecule that is rapidly undergoing reversible chemical or physical changes, which in turn produces changes in the **chemical shift** of that nucleus.

chemically induced dynamic nuclear polarization *abbr.*: CIDNP; a technique for the direct detection and identification of low concentrations of free radicals by **electron spin resonance**. It depends on strong polarization of certain nuclear spins by the unpaired electron during the molecule's existence as a free radical.

chemical messenger a term sometimes used loosely to mean variously a hormone, a neurocrine or paracrine transmitter, or a rheoseme. *See also* **second messenger**.

chemical oxygen demand *abbr.*: COD; a measure of the oxygen equivalent of that portion of the organic matter in a sample of water that is susceptible to oxidation by a strong chemical oxidant. Essentially the sample is boiled under a reflux condenser with potassium dichromate in strong sulfuric acid (with silver sulfate as catalyst); part of the dichromate is reduced by the contained organic matter and the remaining dichromate is titrated with ferrous sulfate. The result is expressed as milligrams oxygen absorbed from standard dichromate per litre of sample. *Compare* **biochemical oxygen demand**.

chemical pathology the branch of pathology dealing with the biochemistry of disease processes and the measurement of the amounts of substances present in body fluids and other samples as an aid to diagnosis.

chemical potential *symbol*: μ; the partial molar free energy of a chemical entity; the entity in question may be indicated by a subscript. For example, μ_B symbolizes the chemical potential of entity B in a mixture of entities B, C, ... etc. It is defined by:

$$\mu_B = (\partial G/\partial n_B)_{T,P,n_C},$$

where G is the Gibbs energy of the mixture, T is the thermodynamic temperature, P is the pressure, and n_B, n_C, ... etc. are the amounts of the entities B, C, ... etc. in the mixture.

chemical quenching *or* **impurity quenching** the **quenching** that occurs in liquid scintillation counting due to diffusion-controlled collisional interaction between the quencher molecules and the excited solvent and solute molecules. *Compare* **colour quenching**.

chemical reaction any single process or operation involving the interconversion of chemical species through changes in orbital electrons but not in the atomic nuclei.

chemical score a measure of the nutritional value of a protein. The limiting essential amino acid in the test protein is expressed as the percentage of the amount of the same amino acid present in egg albumin, the nutritionally perfect protein. Chemical score is numerically equal to **biological value**.

chemical shift the displacement of a resonance signal in NMR spectroscopy along the magnetic field coordinate resulting primarily from the magnetic effects of the chemical environment of the nuclei producing that signal. *See* **nuclear magnetic resonance**.

chemical species any set of chemically identical **molecular entities**, the members of which have the same composition and can explore the same set of molecular energy levels within the time scale of a particular experiment. Two conformational isomers may interconvert sufficiently slowly to be considered separate chemical species on an NMR time scale, but in a slow chemical reaction they may interconvert sufficiently quickly to be considered as a single chemical species.

chemical transmitter *(sometimes) an alternative term for* **neurotransmitter**.

chemiluminescence the production of visible light (luminescence) occurring as a result of a chemical reaction. It occurs, for example, during the **respiratory burst** in neutrophils and other phagocytic cells. The phenomenon is now exploited in sensitive enzyme assays and labelling methods in nucleic-acid hybridization, etc.

chemiosmotic coupling hypothesis a hypothesis proposed in 1961 by the British biochemist Peter Mitchell to explain ATP formation in mitochondria. Essentially it states that the energy-yielding reactions of the **respiratory chain** are coupled to the energy-requiring reactions of phosphorylation through the creation of a membrane electrochemical potential. An electrochemical gradient of H^+ ions, and a voltage gradient, is established across the mitochondrial inner membrane during electron flow through the respiratory chain. The electron carriers of the respiratory chain serve as an active-transport system or 'pump' to transport H^+ ions from the mitochondrial matrix across the inner membrane, which is an integral part of the coupling mechanism. The electrochemical gradient thus formed drives the synthesis of ATP from ADP and P_i. A similar mechanism is believed to drive oxidative phosphorylation in the chloroplast during the **light reactions of photosynthesis**.

chemism *(rare)* chemical action, operation, activity, or force.

chemisorption *or* **chemical adsorption** a type of **adsorption** in which the forces involved are valence forces of the same kind as those operating in the formation of chemical compounds. *Compare* **physisorption**.

chemoattractant *an alternative term for* **attractant**.

chemoautotroph *or* **chemolithotroph** any organism that is both an **autotroph** and a **chemotroph**. —**chemoautotrophic** *adj.*

chemoceptor *an alternative name for* **chemoreceptor** (esp. in immunology).

chemography an artefact in autoradiography due either to the creation of reduced crystals in the photographic emulsion by chemicals present in the specimen (**positive chemography**), or to the removal of latent images, also by chemicals in the specimen (**negative chemography**).

chemoheterotroph *or* **chemoorganotroph** any organism that is both a **chemotroph** and a **heterotroph**. —**chemoheterotrophic** *adj.*

chemokine *or* **intercrine** any of a subgroup of cytokines acting primarily on hemopoietic cells in acute and inflammatory processes and other immunoregulatory functions. They are chemotactic and related by primary structure, especially conservation of a motif of four cysteines, the first two of which are either adjacent or separated by one other residue.

chemolithotroph *an alternative term for* **chemoautotroph**.

chemoorganotroph *an alternative term for* **chemoheterotroph**.

chemoreceptor *or* **chemoceptor** any **receptor** on a cell that receives chemical stimuli and reacts to them or binds them.

chemostat a device for the continuous culture of bacterial (and other) cells. Growth occurs in an aerated fermenter vessel and its rate is controlled by the rate of addition of fresh nutrient from a reservoir (i.e. the dilution rate); this in turn controls the rate of removal of cells (and culture medium). In a chemostat, as opposed to an **auxostat**, the dilution rate is constant. At equilibrium, the rate of production of new cells by multiplication is equal to the rate of removal of grown cells.

chemosynthesis the synthesis of organic chemical compounds by an organism using energy obtained from the oxidation of inorganic chemical compounds rather than from light. *Compare* **photosynthesis**. —**chemosynthetic** *adj.*

chemotactic factor for macrophages *see* **lymphokine**.

chemotaxin *(sometimes) an alternative term for* **cytotaxin**.

chemotaxis **1** a movement of a motile cell or organism (*see* **taxis**) in response to a specific chemical concentration gradient. Movement may be towards a higher concentration (**positive chemotaxis**) or towards a lower concentration (**negative chemotaxis**). In *Escherichia coli* the ***che*** family of genes has been implicated in controlling certain aspects of chemotaxis, and **galactose/glucose binding protein** is also involved. *See also* **cytotaxis**. **2** *(in immunology)* the migration of polymorphonuclear leukocytes and macrophages towards higher concentrations of certain fragments of complement such as C3a and C5a. *See also* **lymphokine**. —**chemotactic** *adj.*

chemotaxonomy the classification of organisms on the basis of the nature, content, and/or the distribution of constituent chemical substances, e.g. DNA.

chemotherapy the treatment of disease, especially infections and neoplasms, with chemical agents (**chemotherapeutic agents**) that act specifically on infective organisms or tumours. The term also describes the treatment of psychiatric disease with chemical agents (as opposed to cognitive therapy). —**chemotherapeutic** *adj.*

chemotractant *see* **attractant**.

chemotroph any organism that derives its energy from exogenous chemical sources, independently of light. Chemotrophs may be subdivided into (1) chemoautotrophs (or chemolithotrophs) i.e. organisms that are both chemotrophs and **autotrophs** and (2) chemoheterotrophs (or chemoorganotrophs) i.e. organisms that are both chemotrophs and heterotrophs. *Compare* **phototroph**. —**chemotrophic** *adj.*; **chemotrophy** *n.*

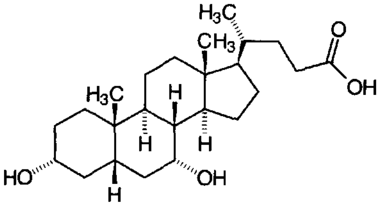
chemotropism the orientation of cells or organisms in response to chemical stimuli. The term is used especially of plants or plant organs. *See also* **tropism**. —**chemotropic** *adj.*

Cheng–Prussoff equation an equation relating the concentration, I_{50}, of a competitive ligand required to reduce the binding of a substrate to an enzyme by 50%, to the dissociation constant, K_i, of the enzyme–inhibitor complex; it is:

$$I_{50} = K_i (1 + S/K_m),$$

where S is the substrate concentration and K_m the Michaelis constant for the enzyme–substrate complex.

chenodeoxycholate *or* **chenodiol** the anion of (3α,5β,7α)-3,7-dihydroxycholan-24-oic acid; a major component of bile in some species (hens, geese) but a minor component in humans. It is used therapeutically to decrease the synthesis of cholesterol and to help dissolve cholesterol gallstones.

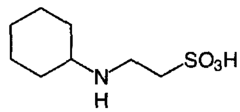


Cherenkov, **Pavel Alekseyevich** (1904–1990), Russian-born Soviet physicist noted for his study of the effects produced by high-energy charged particles; Nobel Laureate in Physics (1958) jointly with I. M. Frank and I. Y. Tamm 'for the discovery and interpretation of the Cherenkov effect'.

Cherenkov counter *or* **Cerenkov counter** a device for counting charged particles, or radionuclides emitting them, that depends on **Cherenkov radiation**.

Cherenkov radiation *or* **Cerenkov radiation** a radiation of bluish light, consisting of photons, emitted when charged particles, especially high-energy beta particles, pass through either a solid or liquid medium at velocities greater than that at which light passes through the same medium. The effect is analogous to the creation of a sonic boom that occurs when an object exceeds the speed of sound in a medium.

Ches *abbr. for* 2-(cyclohexylamino)ethanesulfonic acid; a compound with similar properties to a **Good buffer substance**; pK_a (25 °C) = 9.3.



chewing gum *see* **chicle**.

chi *symbol*: χ (lower case) *or* X (upper case); the twenty-second letter of the Greek alphabet. For uses *see* **Appendix A**.

chiasma (*pl.* **chiasmata**) **1** *(in genetics)* a connection formed between **chromatids**, visible during **meiosis**, thought to be the point of the interchange involved in crossing-over. **2** *or (esp. US)* **chiasm** (*pl.* **chiasms**) *(in anatomy)* a decussation or intersection, as of two nerves; e.g. optic chiasma.

chicle a mixture of *cis*- and *trans*-1,4-polyprenoids obtained from the evergreen tree, *Achras sapota*; the original chewing gum base.

chimera *or (esp. Brit.)* **chimaera** *(in genetics)* an organism comprising tissues of two or more **genotypes**. Chimeras can occur as a result of mutation, abnormal distribution of chromosomes, or grafting. *See* **mosaic**, **transgenic**, **chimeric molecule**. —**chimeric** *or* **chimaeric** *adj.*

chimeric molecule **1** any of the hybrid DNA molecules formed when DNAs from different sources are digested with the same **restriction endonuclease** and the digests are mixed. The DNA fragments in the mixture become associated by hydrogen bonds between complementary sequences to form new arrangements, which may be converted into covalently linked molecules by the action of DNA ligase. This process can produce **chimeric plasmids** and other chimeric structures. **2** a **chimeric protein** is a protein obtained by the insertion or substitution of a partial sequence from one protein into another, typically using cDNA technology. The resultant protein thus has elements of both the original proteins. **3** a recombinant antibody that has, e.g., structural characteristics of both human and mouse antibodies.

chimerin any of several related proteins that function as GTPase-activating proteins, specific for the p21 Ras-related Rac GTPase. They include n-chimerin, and α- and β-chimerins. n-Chimerin is a cerebellar protein and β-chimerin is expressed in testis. They are phorbol ester receptors, with an N-terminal domain homologous to the zinc-finger region of protein kinase C, and a C-terminal domain homologous to the product of ***BCR***. Through their action on Rac, they lead to changes in cytoskeletal organization. Example from human: database code A53764, 466 amino acids (53.81 kDa).

chimyl alcohol glycerol 1-hexadecyl ether, (+)-3-(hexadecyloxy)-1,2-propanediol; a hydrolysis product of **ether lipids**.

Chinese restaurant syndrome *see* **monosodium glutamate**.

CHIP 28 *abbr. for* channel-forming integral membrane protein 28.

chiral describing a chemical compound that displays **chirality**.

chiral drug a single pure enantiomer of a specified drug rather than the racemate; the enantiomers may have very different pharmacological effects, e.g. in the case of propoxyphene and levopropoxyphene.

chirality topological handedness; the property of nonidentity of an object with its mirror image. A chiral chemical compound, i.e. one that possesses chirality, is one that cannot be superimposed on its mirror image, either as a result of simple reflection or after rotation and reflection. If superposition can be achieved then the molecule is said to be **achiral**. Chirality is most commonly due to the presence of one or more **chiral centres** (formerly referred to as asymmetric carbon atoms) or the presence of a **chiral axis** (as in allene structures, some of which are found in natural products). Rarely, chirality may also result from the presence of a chiral plane. *See also* **prochiral**. [From Greek *kheir*, hand.]

chiral methyl a carbon atom carrying the three hydrogen isotopes, i.e. -$C^1H^2H^3H$.

chiral recognition the differentiation of the enantiomers of a compound. This can be achieved by living organisms, chiral molecules, enzymes, drug receptors, etc.

chiroid *(rare)* any molecule that has **chirality**.

chiroptical describing any of the phenomena that depend on the ability of chiral and other intrinsically asymmetric molecules to rotate the plane of polarized light, known collectively as **optical activity**. Such phenomena include **optical rotation**, **optical rotatory dispersion**, and **circular dichroism**. The term is also applied to any of the techniques used for investigating these phenomena.

***chi* sequence** *or* **chi site** a sequence of base pairs first discovered in a phage lambda mutant, called *chi*. These mutants were found to have single base pair changes creating sites that stimulate **recombination**. Characteristic of chi sites is a nonsymmetrical sequence of eight base pairs, consensus sequence:

5′ GCTGGTGG 3′.

A recombinational hot spot in *Escherichia coli* and a binding site for **Rec A** is a chi site.

chi site *see* **chi sequence**.

chi-square test a statistical test to determine whether an observed series of values differs from a series of values expected on a hypothesis, to a greater degree than would be expected by chance. If m is the expected value and $(m + x)$ is the observed value then $\chi^2 = \Sigma(x^2/m)$. The goodness of fit may be found from available tables of χ^2 and the number of degrees of freedom, n, in which the observed series may differ from the hypothetical.

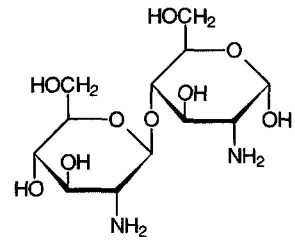
chi structure a structure formed when circular duplex DNA that is undergoing **recombination** is cut with a restriction enzyme. This yields two strands joined at a site intermediate between the ends, which by electron microscopy can yield a shape with the appearance of the Greek letter χ. The structure results from the fact that if two circular duplex DNAs undergo site-specific recombination, a single strand from each duplex will be opened and the two single strands ligated to each other to form a figure of eight. If each of the loops of this figure of eight is cut by the restriction enzyme the chi structure will result. *Compare* **Holliday junction**.

chitin a linear polysaccharide consisting of β-1,4-linked *N*-acetyl-D-glucosamine residues. It is found in annelid cuticle, arthropod exoskeleton, and in some plants and fungi. In fungi, chitin represents the microfibrillar component of the cell wall. *See also* **chitobiose**, **chitosan**.

chitinase EC 3.2.1.14; *other names*: chitodextrinase; 1,4-β-poly-*N*-acetylglucosaminidase; poly-β-glucosaminidase. An enzyme that catalyses the hydrolysis of the 1,4-β-linkages of *N*-acetyl-D-glucosamine polymers of chitin. Example, endo-chitinase from *Hordeum vulgare* (barley): database code NRL_1BAA, 243 amino acids (25.92 kDa); 3-D structure known, largely α helix.

chitin synthase EC 2.4.1.16; *other name*: chitin–UDP *N*-acetyl-glucosaminyltransferase; an enzyme that catalyses the reaction between UDP-*N*-acetyl-D-glucosamine and [1,4-(*N*-acetyl-β-D-glucosaminyl)]$_n$ to form UDP and[1,4-(*N*-acetyl-β-D-glucosaminyl)]$_{n+1}$. Example from *Neurospora crassa*: database code CHS3_NEUCR, 960 amino acids (106.78 kDa).

chitobiose 4-*O*-(2-amino-2-deoxy-β-D-glucopyranosyl)-2-amino-2-deoxy-D-glucose; *condensed form*: Glc*p*N(β1-4)GlcN; a disaccharide found in its acetylated form in chitin. This acetylated form, Glc*p*NAc(β1-4)GlcNAc, is the repeating unit of chitin, and should, logically, have the name chitobiose (by analogy with cellobiose, the repeating unit of cellulose). However, historically the nonacetylated form received the name chitobiose, a convention that has persisted.



α-anomer

chitosamine *see* **glucosamine**.

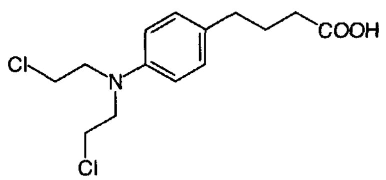
chitosan the cationic polymeric carbohydrate obtained by the deacetylation of **chitin**.

chitosome a body found in fungi and containing chitin synthase; it synthesizes chitin microfibrils.

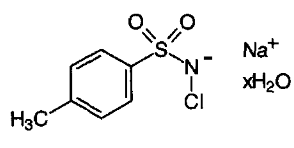
Chl *abbr. for* chlorophyll.

chlor+ *a variant of* **chloro+** (sometimes before a vowel).

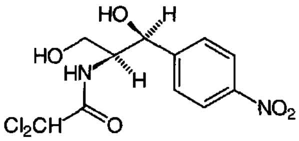
chlorambucil 4-[bis(2-chloroethyl)amino]benzenebutanoic acid; an alkylating agent that reacts with DNA causing the formation of covalent bonds that cross-link with the two strands of DNA, thereby preventing its replication.



chloramine T *N*-chloro-4-methylbenzenesulfonamide sodium salt; sodium 4-toluenesulfonchloramide hydrate; $CH_3–C_6H_4–SO_2NClNa{\cdot}xH_2O$. A white crystalline solid with a faint odour of chlorine. In aqueous solution it slowly decomposes to sodium hypochlorite and thus is a mild oxidant and antiseptic. It is widely used for the oxidation of radioiodide in the preparation at high specific radioactivity of radioiodine-labelled peptides and proteins for use in radioimmunoassay or other *in vitro* techniques. Microgram quantities of many proteins are labelled efficiently at or near pH 7; however, some proteins may suffer an unacceptable degree of oxidative damage.

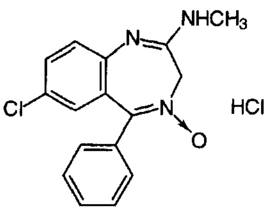


chloramphenicol *abbr.*: CAP; a broad-spectrum, chlorine-containing antibiotic produced by *Streptomyces venezuelae*; it is now produced by chemical synthesis. The configurations at C-1 and C-2 are (*S*), as shown in the structure. It inhibits prokaryotic protein synthesis by attaching to the 50S ribosomal subunit and inhibiting peptidyltransferase (EC 2.3.2.12), thereby preventing the formation of peptide bonds. Chloramphenicol also inhibits protein synthesis in mitochondria, and this accounts for its toxic effect in causing aplastic anemia. It is much used for the treatment of typhoid fever.



chloramphenicol O-acetyltransferase *abbr.*: CAT; EC 2.3.1.28; a bacterial enzyme, not found in eukaryotes, that catalyses the reaction between acetyl-CoA and chloramphenicol to form CoA and chloramphenicol 3-acetate. This reaction forms the basis for acquired chloramphenicol resistance in certain bacterial strains. Also the gene for CAT is used as a **reporter** because its expression can be detected by a sensitive radiochemical assay. In cDNA technology, for example, the expression of a eukaryotic promoter can be characterized by assay of the enzyme if CAT cDNA is included in the genes to be expressed. Example from *Escherichia coli*: database code CAT2_ECOLI, 213 amino acids (24.75 kDa).

chlordiazepoxide 7-chloro-*N*-methyl-5-phenyl-3*H*-1,4-benzodiazepin-2-amine 4-oxide; a CNS depressant noted primarily for its effective anti-anxiety effects at nonsedative doses. It is the prototypical **benzodiazepine**, used therapeutically as the hydrochloride. It is a controlled substance in the US Code of Federal Regulations. One proprietary name of the hydrochloride is Librium.



chloride 1 *symbol*: Cl^-; the anion of the element chlorine. 2 any salt of hydrochloric acid. 3 any compound containing a chlorine atom in organic linkage.

chloride channel any of a family of proteins that facilitate Cl^- transport through a membrane. These channels are of two types: (1) ligand-gated Cl^- channels associated with inhibitory neurotransmitter receptors such as the **GABA receptor** or the **glycine receptor**; and (2) voltage-gated Cl^- channels. These have 13 transmembrane helical regions, the last two being separated by a long (between 130 and 160 residue) cytoplasmic loop. In nerve membranes these channels ensure the high conductance of non-innervated membranes of the electrolyte necessary for efficient current generation caused by Na^+ influx through the acetylcholine receptor at the innervated membrane. Example from *Torpedo californica*: database code CICH_TORCA, 810 amino acids (89.45 kDa). In muscle cells voltage-gated Cl^- channels are involved in regulation of cell volume, membrane potential stabilization, and transepithelial transport. Examples: database code CIC1_HUMAN, 988 amino acids (108.74 kDa). A Cl^- channel in the kidney proximal tubule is involved in chloride reabsorption. Example from *Rattus norvegicus*: database code CICK_RAT, 686 amino acids (75.63 kDa). *See also* **cystic fibrosis transmembrane conductance regulator**.

chloride shift *or* **Hamberger shift** the phenomenon whereby, because of the **isohydric shift**, there is a higher concentration of hydrogencarbonate ions in the erythrocytes in blood leaving the tissues (venous blood) than in arterial blood, and hence the hydrogencarbonate/chloride ion ratio is higher in the erythrocytes than in plasma. This results in a movement of hydrogencarbonate ions out of the erythrocytes into the plasma and an equivalent movement of chloride ions from the plasma into the erythrocytes. In the lungs the reverse ion movements occur upon the oxygenation (arterialization) of the blood there.

chlorin *a trivial name for* 2,3-dihydroporphyrin.

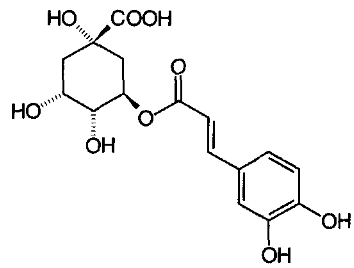
chlorinate 1 to treat or react (a substance) with dichlorine. 2 to introduce one or more chloro groups into an organic compound, whether by addition or substitution. 3 to disinfect (especially water) with dichlorine. —**chlorination** *n.*

chlorine 1 *symbol*: Cl; a halogen element of group 17 of the IUPAC **periodic table**; atomic number 17; relative atomic mass 35.453. The 15th most abundant element of the Earth's crust, occurring in the combined state, mostly as inorganic chlorides. 2 *symbol*: Cl_2; the common name for the yellowish-green gaseous substance, correctly known as **dichlorine**.

chloro+ *or (sometimes before a vowel)* **chlor+** *comb. form* 1 denoting a chlorine atom in organic linkage. 2 denoting the colour green.

chlorocruorin any green, approximately 3 MDa, hemochrome respiratory pigment found in the blood of certain polychaete annelids; it contains 96 heme groups.

chlorogenic acid 3-*O*-(3,4-dihydroxycinnamoyl)-D-quinic acid; a tannin found in tea and coffee.



chlorolabe a pigment sensitive to green light (535 nm) found in some retinal cones. It is lacking in subjects with green-blindness but present in those with red-blindness. *Compare* **cyanolabe, erythrolabe**.

***p*-chloromercuribenzenesulfonate** *abbr.*: PCMBS; 4-(chloromercuri)benzenesulfonate; $Cl-Hg-C_6H_4-SO_3^-$; a reagent that reacts with thiol groups; it is more soluble than *p*-chloromercuribenzoate.

***p*-chloromercuribenzoate** *abbr.*: PCMB; 4-(chloromercuri)-benzoate; $Cl-Hg-C_6H_4-COO^-$; a reagent that reacts with thiol groups. It is supplied either as *p*-chloromercuribenzoic acid or as *p*-hydroxymercuribenzoate sodium salt.

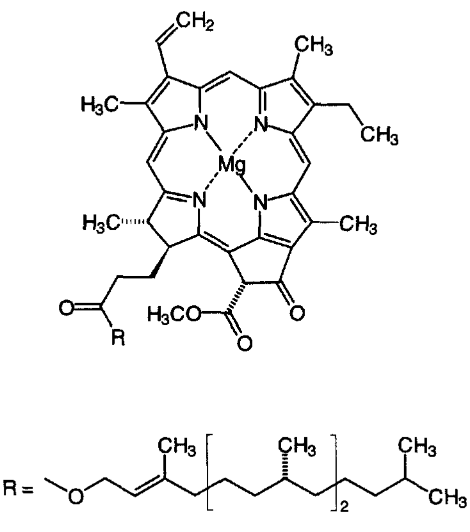
4-chloromercuriphenylsulfonate *see* ***p*-chloromercuribenzenesulfonate**.

chloroperoxidase EC 1.11.1.10; *other name*: chloride peroxidase. An enzyme that catalyses a reaction between H_2O_2, two molecules of Cl^- ion, and two molecules of an organic compound to form two molecules of an organic chloride and two molecules of H_2O. It is useful for the introduction of (radioisotopes of) chlorine, bromine, or iodine into tyrosine residues in proteins. It is involved in the synthesis of chlorine-containing compounds such as (1*S-trans*)-2,2-dichloro-1,3-cyclopentanedione (caldariomycin). Example (precursor) from

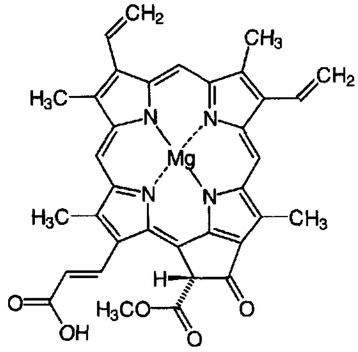
Caldariomyces fumago: database code PRXC_CALFU, 321 amino acids (35.20 kDa).

p′-chloro-p-phenylaniline *see* **4-amino-4′-chlorodiphenyl**.

chlorophyll any of the several green (or purple, *see* **bacteriochlorophyll**) pigments, found in plants and photosynthetic bacteria, that function in **photosynthesis** by absorbing light energy mainly in the red and violet-blue parts of the spectrum. Chlorophylls are magnesium complexes of various closely related porphyrins or chlorins. The main chlorophylls of land plants are chlorophylls *a* and *b*, and some algae contain chlorophylls *c*. Chlorophylls *a* and *b* are dihydroporphyrins, having no double bond at position 3 of ring B, while chlorophylls *c* are porphyrins. Photosynthetic bacteria contain various bacteriochlorophylls. The structure of chlorophyll *a* is shown. Chlorophyll *b* differs from chlorophyll *a* only in the presence of a formyl group in place of the methyl group at position 3 on ring B (Fischer system: *see* **bacteriochlorophyll**). Several chlorophylls *c* are known. They are accessory light-harvesting pigments found in eukaryotic algae that do not contain chlorophyll *b*. The structure of chlorophyll c_2 is shown, chlorophyll c_1 differing from this in having an ethyl group at position 4 of ring B, and chlorophyll c_3 differing from it in having a methylformyl group at position 3 of ring B. Most chlorophylls exist in **antenna complexes**, where their function is to absorb visible light and to transmit the energy so absorbed to a reaction centre in which other chlorophyll molecules are excited by the accumulated energy to transfer an electron to an electron transfer system known as a **photosystem**. Differences in structure have significant effects on the absorption spectrum of the chlorophyll molecule, i.e. on the wavelength of the light absorbed, and thus on the energy available as a result of the absorption (*see* **Planck constant** for the relationship). To restore its electronic state, the oxidized chlorophyll molecule can accept an electron from a water molecule (or, in photosynthetic sulfur bacteria, from hydrogen sulfide), the transfer being catalysed by a manganese-rich protein complex that is part of the photosystem. This reaction is responsible for the dioxygen (or sulfur) that is formed during photosynthesis, one molecule of dioxygen, four protons, and four electrons being produced from two water molecules. *See also* **P680**, **P690**, **P700**. [From the Greek *khloros*, yellowish or pale green, and *phyllon*, leaf.]

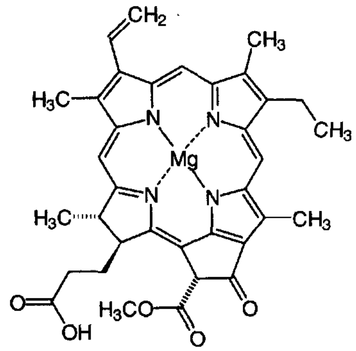


chlorophyll *a*



chlorophyll c_2

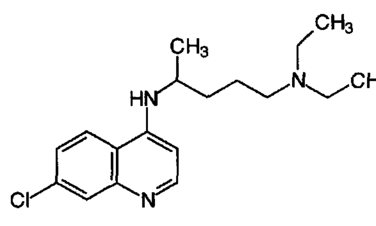
chlorophyllide a **chlorophyll** lacking the terpenoid side chain, usually phytol. The penultimate intermediate in chlorophyll biosynthesis. It is formed by the light-catalysed reduction of protochlorophyllide attached to a protein, protochlorophyllide holochrome.



chlorophyllide *a*

chloroplast *or (sometimes)* **chloroplastid** a chlorophyll-containing **plastid**, found in cells of algae and higher plants, that is the site of **photosynthesis**. Chloroplasts are characterized by a system of membranes embedded in a hydrophobic proteinaceous matrix, or stroma. The basic unit of the membrane system is a flattened single vesicle called the **thylakoid**; thylakoids stack into grana (*sing.* **granum**). All the thylakoids of a granum are connected with each other, and the grana are connected by intergranal lamellae.

chloroquine [N^4-(7-chloro-4-quinolinyl)-N^1,N^1-diethyl-1,4-pentanediamine; a widely used synthetic quinoline antimalarial drug, with action similar to **quinine**. It accumulates in the cells of the malaria parasites and prevents digestion of their host's hemoglobin; it may also possibly disrupt the parasite's ribonucleotide metabolism. *See also* **acridine**.

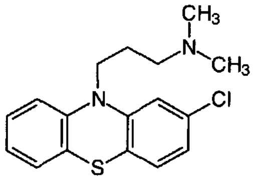


chlorosome a small compartment found in photosynthetic

Chlorobiaceae bacteria. It contains **bacteriochlorophyll** *c* and is attached to the cytoplasmic membrane.

chlorosulfolipid a class of **sulfolipid** found in the alga, *Ochramonas danica*. They are based on *n*-docosane-1,14-diol disulfate; chlorine atoms have been located at C-2, C-13, C-15, and C-16.

chlorpromazine 2-chloro-10-(3-dimethylaminopropyl)phenothiazine; a **neuroleptic** drug with sedative and antiemetic actions. It acts as an antagonist at dopaminergic (D_2), α-adrenergic, and many other receptors (*see* **adrenoceptor**). It was formerly classed as a **major tranquillizer**, a term now deprecated. Largactil is one proprietary name of the hydrochloride.

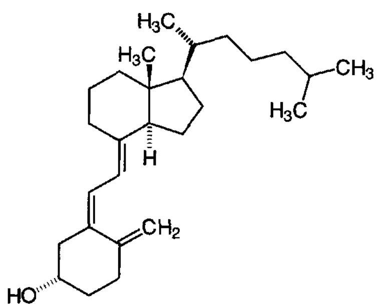


Cho *symbol for* a residue of choline.

CHO *abbr. for* Chinese hamster ovary (applied to cell cultures derived from this organ).

chole+ *or (before a vowel)* **chol+** *comb. form* denoting bile.

cholecalciferol *or* **calciol** *the recommended trivial name for* vitamin D_3; (5*Z*,7*E*)-(3*S*)-9,10-seco-5,7,10(19)-cholestatrien-3-ol. It is formed in skin by UV-irradiation of 7-dehydrocholesterol, and is metabolized to 1*α*,25-dihydroxycholecalciferol (calcitriol). The latter affects the absorption of calcium from the gastrointestinal tract and the metabolism of calcium in bone. *See* **vitamin D**.



cholecalcin an intracellular calcium-binding protein.

cholecystokinin *abbr.*: CCK; *formerly called* cholecystokinin–pancreozymin (*abbr.*: CCK-PZ); gall-bladder emptying peptide hormone from **I cells** of the duodenum and proximal jejunum of mammals, also present in most parts of the brain and in the nerves of the colon, ileum, and pancreas. It is identical with **pancreozymin**, the different names relating to the activities by which they were separately identified. The sequence of human cholecystokinin is

KAPSGRMSIVKNLQNLDPSHRISDRDY(-SO_3)-MGWMDF-NH_2.

Sulfation of the tyrosine is essential for activity. Highly heterogeneous, the various species are known by the number of amino-acid residues they contain, e.g. CCK-39, CCK-33 (e.g. human), CCK-12, CCK-8, CCK-4, etc.; all are truncated from the N terminus and thus contain the same C-terminal tetrapeptide amide sequence, which is identical to that of **gastrin** and in which the functional activity of gastrin largely resides. CCK is released when fat, peptones, amino acids, or $MgSO_4$ are placed on the intestinal mucosa; it causes contraction of the gall bladder and secretion of enzyme precursors from pancreatic acinar tissue, it appears to act as a neurotransmitter in a number of situations, and it elicits the behavioural sequence characteristic of satiety.

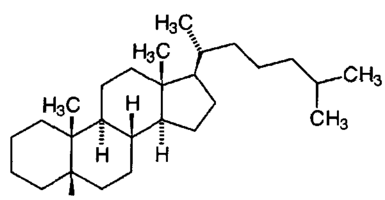
cholecystokinin receptor any of several membrane proteins that bind **cholecystokinin** forms and mediate their intracellular effects. Two types are recognized, designated CCK_A and CCK_B. Both types also bind gastrin, but type A binds cholecystokinin with 1000-fold higher affinity than gastrin. When CCK-8 and CCK-4 are used as agonists, CCK-8 is the most potent. Gastrin and CCK-4 are about equipotent, with much greater effect on CCK_B than CCK_A. CCK_B is regarded as the gastrin receptor. Both types of receptor are seven-transmembrane-segment receptors, coupled to G-proteins that stimulate the **phosphatidylinositol cycle**. Examples, human (precursors), type A: database code CCKR_HUMAN, 428 amino acids (47.84 kDa); type B: database code GASR_HUMAN, 447 amino acids (48.42 kDa).

choleragen *(sometimes) an alternative name for* **cholera toxin**.

choleragenoid a 57 kDa protein that occurs naturally in the cell-free culture fluid of *Vibrio cholerae* and is immunologically almost indistinguishable from **cholera toxin** but is nontoxic. It consists of the pentamer of the B subunit of the toxin without its A subunit.

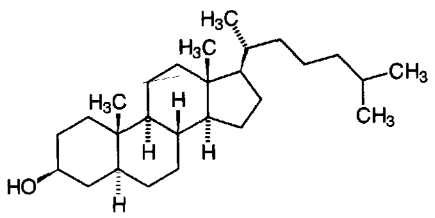
cholera toxin *or* **choleragen** the antigenic and potently diarrheagenic **enterotoxin** produced by the cholera bacterium, *Vibrio cholerae*. Pathogenic strains of the bacterium secrete the toxin into their growth medium. The toxin is an 84 kDa multimeric protein of exceptional stability and formed of two dissimilar subunits: the A subunit consists of two polypeptide chains (α, 22 kDa; γ, 5 kDa) linked by a single disulfide bond; the B subunit consists of four to six noncovalently associated β chains (11.5 kDa each). Binding of the toxin to its receptor ganglioside (G_{M1}) in the membranes of sensitive cells occurs via the B subunit; the A subunit then activates cellular adenylate cyclase; together with cellular **ADP-ribosylation factor** it constitutes an **ADP-ribosyltransferase**, which transfers ADP-ribose to the **G-protein** α_s subunit, inhibiting its GTPase and converting it to an irreversible activator of adenylate cyclase. Database code CHTA_VIBCH, 258 amino acids (29.30 kDa), is precursor for chains α (amino acids 19–211) and γ (amino acids 213–258); database code CHTB_VIBCH, 124 amino acids (13.94 kDa), is precursor for β chain (signal is amino acids 1–21).

cholestane the fully saturated carbon skeleton from which **cholesterol** is structurally derived; the configuration at C-5 can be *α* or *β*; 5*β*-cholestane is known as **coprostane**.



5*β*-cholestane

cholestanol 3*β*,5*α*-cholestan-3-ol; dihydrocholesterol; 3*β*-hydroxycholestane; a steroid that occurs in human feces, gallstones, and in eggs.

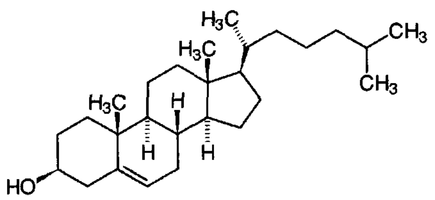


cholestanol

cholestasis the arrest (stasis) or impairment of bile flow. It may originate intrahepatically, due to hepatocellular defects, or extrahepatically, often due to obstruction of the bile duct by calculi.

cholesteric of, relating to, or resembling cholesterol or a derivative of cholesterol. *See also* **liquid crystal**.

cholesterol cholest-5-en-3β-ol; the principal sterol of vertebrates and the precursor of many steroids, including bile acids and steroid hormones. It is found in all animal tissues, especially brain and spinal cord, in the bile, and in gallstones, being a component of the plasma membrane lipid bilayer. It is also a component of the plasma lipoproteins and excessive levels in the plasma have been implicated in atherogenesis. It is synthesized from acetyl-CoA via hydroxymethylglutaryl-CoA and squalene.



cholesterol acyltransferase *see* **sterol *O*-acyltransferase**.

cholesterol desmolase *see* **cholesterol monooxygenase**.

cholesterol 7α-hydroxylase *see* **cholesterol 7α-monooxygenase**.

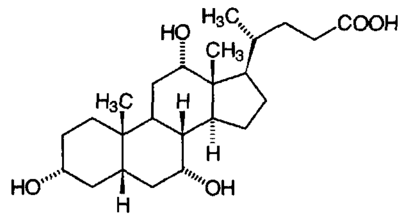
cholesterol 7α-monooxygenase EC 1.14.13.17; *other names*: cholesterol 7α-hydroxylase; cytochrome P450 VII; a cytochrome P450 enzyme that catalyses the hydroxylation of cholesterol to form 7α-cholesterol. NADPH and dioxygen are reactants, $NADP^+$ and H_2O being formed. This is the first step in the biosynthesis of bile acids from cholesterol and is rate-limiting. Example from human: database code CP70_HUMAN, 504 amino acids (57.66 kDa).

cholesterol monooxygenase (side-chain cleaving) EC 1.14.15.6; *systematic name*: cholesterol,reduced-adrenal-ferredoxin:oxygen oxidoreductase (side-chain-cleaving); *other names*: CYPXIA1 cholesterol desmolase; cholesterol side-chain cleavage enzyme; cytochrome $P450_{scc}$; a cytochrome P450 enzyme that catalyses the formation of pregnenolone from cholesterol, with release of 4-methylpentanal and H_2O. It is the first enzyme in the pathway for the biosynthesis of steroid hormones from cholesterol and is a rate-limiting step. It is a target for regulatory systems that stimulate steroid hormone biosynthesis. The reaction involves an oxidation by dioxygen, reduced adrenal ferredoxin being oxidized in the process. This is a heme-thiolate enzyme. Example from human: database code CPM1_HUMAN, 521 amino acids (60.11 kDa); a theoretical 3-D model for the bovine enzyme has been produced: database code NRL_1SCC, 482 amino acids (56.46 kDa).

cholesterol side-chain cleavage enzyme *see* **cholesterol monooxygenase (side-chain cleaving)**.

cholestyramine resin *see* **colestyramine resin**.

cholic acid 3α,7α,12α-trihydroxy-5β-cholan-24-oic acid; a metabolic derivative of cholesterol, produced in the liver and secreted as a **bile acid**. It is used as a detergent, as the sodium salt; aggregation number 2, CMC 9–15 mM.



choline *symbol*: Cho; *abbr.*: Ch; *N*,*N*,*N*-trimethylethanolammonium (ion); 2-hydroxyethyltrimethylammonium (ion); 2-hydroxy-*N*,*N*,*N*-trimethylethanaminium; $(CH_3)_3N^+$–CH_2–CH_2–OH. An amino alcohol that functions as a methyl group donor in metabolism; it also occurs in the neurotransmitter acetylcholine and in some phospholipids.

choline acetyltransferase EC 2.3.1.6; *recommended name*: choline *O*-acetyltransferase; *systematic name*: acetyl-CoA: choline *O*-acetyltransferase. An enzyme that catalyses the reaction between acetyl-CoA and choline to form *O*-acetylcholine and CoA. It is instrumental in the synthesis of the neurotransmitter acetylcholine. Example from rat: database code A48319, 644 amino acids (72.32 kDa). *See also* **cholinoceptor**.

choline kinase EC 2.7.1.32; an enzyme that catalyses the phosphorylation by ATP of choline to form *O*-phosphocholine and ADP. Phosphocholine (choline phosphate) is an important intermediate in phospholipid biosynthesis, reacting with CTP to form CDPcholine. Example from human: database code S23104, 456 amino acids (52.01 kDa). *See also* **choline-phosphate cytidylyltransferase**.

choline-phosphate cytidylyltransferase EC 2.7.7.15; *systematic name*: CTP:choline-phosphate cytidylyltransferase; *other name*: phosphorylcholine transferase. An enzyme that catalyses the formation of **cytidine(5′)diphosphocholine** (CDP-choline) from CTP and choline phosphate with release of pyrophosphate. Example from yeast: database code CTPT_YEAST, 424 amino acids (49.33 kDa).

cholinergic describing nerve fibres that, when activated, release acetylcholine or an acetylcholine-like substance from their endings. *Compare* **adrenergic**, **peptidergic**, **purinergic**. *See also* **cholinoceptor**.

cholinesterase 1 *recommended name for* EC 3.1.1.8; *other names*: choline esterase II, pseudocholinesterase. An enzyme that hydrolyses a variety of choline esters to choline and a carboxylic acid. In mammals, the enzyme is found in the liver, pancreas, heart, serum, and the white matter of brain. Its function is unknown. In humans, individuals with abnormally low activity of cholinesterase fail to hydrolyse succinylcholine, often given as a muscle relaxant in anaesthesia. The gene (E_1) encoding the enzyme can exist in at least four allelic forms, the normal ($E_1{}^u$), and three variants, $E_1{}^a$ (atypical), $E_1{}^f$ (fluoride resistant), and $E_1{}^s$ (silent). Homozygotes for the atypical gene have a dibucaine-resistant enzyme which is weakly active against most enzyme substrates, and homozygotes for the silent gene exhibit minimal enzyme activity. Individuals having low enzyme activity from whatever cause may suffer prolonged respiratory paralysis after succinylcholine has been administered. Example from *Canis familiaris*: database code CHLE_CANFA, 141 amino acids (15.09 kDa). 2 alternative name for **acetylcholinesterase**.

cholinoceptor any receptor for acetylcholine on cholinergic or other neurons; such receptors occur on the postjunctional membrane of synaptic clefts or on terminal axons and mediate the neurotransmitter action of acetylcholine. Cholinoceptors may be divided into two main types: (1) **muscarinic cholinoceptors** of smooth muscle, cardiac muscle, endocrine glands, and the central nervous system at which muscarine, but not nicotine, can readily mimic the action of acetylcholine; and (2) **nicotinic cholinoceptors** of skeletal muscle, ganglia, and the central nervous system at which nicotine, but not muscarine, can readily mimic the action of acetylcholine.

Muscarinic acetylcholine receptors for which structures are known are all G-protein-coupled and of the single polypeptide chain, seven-transmembrane-helix class. They fall into five types: four are designated M_1–M_4; a fifth, which has been cloned, acts through phosphoinositide-specific phospholipase C when expressed in CHO cells, but although mRNA has been isolated from brain, a translation product has not been found *in vivo*; its database code is ACM5_HUMAN, 532 amino acids (60.00 kDa). Of the remainder, M_1 (database code, human, ACM1_HUMAN, 460 amino acids (51.36 kDa)) and M_3 (database code, human, ACM3_HUMAN, 590 amino acids (66.05 kDa)) activate phosphoinositide-specific phospholipase C; M_2 (database code, human, ACM2_HUMAN, 466 amino acids (51.65 kDa)) and M_4 (database code, human, ACM4_HUMAN, 479 amino acids (53.00 kDa)) are negatively coupled to adenylate cyclase; M_2 also increases potassium channel conductance through the action of G-proteins.

Nicotinic acetylcholine receptors are glycosylated multisubunit pentamers with an intrinsic cation-permeable channel with little selectivity among cations; subunit variants include α, β, γ, δ, and ε. After binding acetylcholine, this type of receptor responds by an extensive change in conformation that affects all subunits and leads to opening of an ion-conducting channel across the plasma membrane. *Torpedo* electric organ and embryonic muscle pentamers have $\alpha_2\beta\gamma\delta$, adult mammalian muscle $\alpha_2\beta\delta\varepsilon$, and mammalian neurons $\alpha_2\beta_3$ stoichiometry; subunits are of the four-transmembrane-helix class (the segments being termed M_1 to M_4); bungarotoxin is a selective antagonist. Examples (precursors) from mouse: α, database code ACHA_MOUSE, 457 amino acids (51.88 kDa); β, database code ACHB_MOUSE, 501 amino acids (56.87 kDa); γ, database code ACHG_MOUSE, 519 amino acids (58.68 kDa); δ, database code ACHD_MOUSE, 520 amino acids (59.08 kDa). The neuronal receptor has two subunits (α and non-α); example, precursors α-3 and β-2 (non-alpha) respectively from human: database code ACH3_HUMAN, 502 amino acids (57.05 kDa); and database code ACHN_HUMAN, 502 amino acids (56.95 kDa).

cholinomimetic describing a drug with an acetylcholine-like action.

cholyl the acyl group derived from **cholic acid**.

chondrioma *or* **chondriome** 1 the entire mitochondrial content of a cell. 2 all the hereditary determinants that are localized in mitochondria.

chondriosome *a former name for* **mitochondrion**.

chondroblast a cell that secretes the matrix of **cartilage**.

chondrocalcin the C-terminal peptide formed from amino acids 1173–1418 of human procollagen α_1. Database code (of procollagen α_1) CA12_HUMAN, 1418 amino acids (134.34 kDa).

chondroclast a large, multinucleate cell that digests the cartilage matrix.

chondrocyte a cartilage cell, formed from a **chondroblast**.

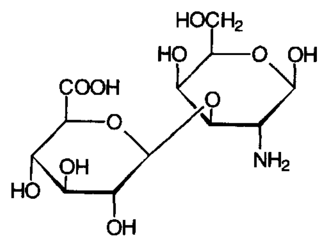
chondroitinase *see* ***N*-acetylgalactosamine-4-sulfatase**, ***N*-acetylgalactosamine-6-sulfatase**.

chondroitinsulfatase *see* ***N*-acetylgalactosamine-4-sulfatase**, ***N*-acetylgalactosamine-6-sulfatase**, **iduronate-2-sulfatase**.

chondroitin sulfate any member of a group of 10–60 kDa glycosaminoglycans, widely distributed in cartilage and other mammalian connective tissues, the repeat units of which consist of β(1–4)-linked D-glucuronylβ(1–3)*N*-acetyl-D-galactosamine sulfate. They usually occur linked to a protein to form proteoglycans. Two subgroups exist, one in which the sulfate is on the 4-position (chondroitin sulfate A) and the second in which it is in the 6-position (chondroitin sulfate C). They often are polydisperse and often differ in the degree of sulfation from tissue to tissue. The chains of repeating disaccharide are covalently linked to the side chains of serine residues in the polypeptide backbone of a protein by a glycosidic attachment through the trisaccharide unit galactosyl–galactosyl–xylosyl. Chondroitin sulfate B is more usually known as **dermatan sulfate**.

chondrosamine *former name for* the D-enantiomer of **galactosamine**, reflecting its presence in **chondrosine**.

chondrosine the disaccharide 2-amino-2-deoxy-3-*O*-β-D-glucopyranuronosyl-D-galactopyranose; *condensed form*: Glc*p*A(β1-3)GalN. In sulfated form it is the core unit of **chondroitin sulfate**.



chopper 1 a device for chopping tissue e.g. the McIlwain chopper. 2 *(in optics)* a device for interrupting a light beam in a regular manner. For example, in a spectrophotometer it consists of a rotating wheel with alternate silvered and cut-out sectors that is placed in the light beam to allow the light to pass alternately through the sample and the reference cells, the photocell thereby generating an alternating current.

choriocarcinoma a malignant tumour, typically occurring in the uterus, originating as a proliferation of chorionic villi. It may follow pregnancy or develop from a hydatidiform mole. **Choriogonadotropin** is an excellent marker for the presence of this tumour.

choriogonadotropin *or* **chorionic gonadotropin** *or* **choriogonadotrophin** *abbr.*: CG; *the recommended name for* a placental **glycoprotein hormone** that has the gonadotropic activities of both luteinizing hormone (LH) and follicle-stimulating hormone (FSH). It is a member of the family of glycoprotein hormones that includes LH and FSH, the β subunit being specific to choriogonadotropin. It is produced by the corpus luteum after fertilization of the ovum; the blood level rises to a detectable concentration 7–9 days after conception and CG may be detected in urine 1–2 days later, which yields a highly sensitive and specific test for pregnancy. Its administration to males causes increased secretion of testosterone; the lack of such a response indicates testicular failure. It is also a good marker for **choriocarcinoma**. It is a heterodimer of an α chain (identical with that of LH and FSH) and a β chain that confers specificity. Example from *Equus asinus* β chain (precursor): database code LHSB_EQUAS, 169 amino acids (17.94 kDa).

choriomammotropin *or* **chorionic mammotropin** *or* **choriomammotrophin** *abbr.*: CS; *the recommended name for* a placental hormone similar in action to **somatotropin** and **prolactin** and of similar amino-acid sequence to pituitary gonadotropin. It is also known as **chorionic somatomammotropin** or **placental lactogen** (*abbr.*: PL). Example, human lactogen (precursor): database code PLC_HUMAN, 217 amino acids (25.02 kDa).

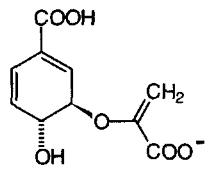
chorion 1 the highly vascular outermost embryonic membrane of higher vertebrates. In placental animals it has a villous part that enters the placenta. **2** membrane of an animal egg, especially the hardened external membrane of an insect's egg.

chorionic gonadotropin *or* **chorionic gonadotrophin** *an alternative name for* **choriogonadotropin**.

chorionic somatomammotropin *or* **chorionic somatomammotrophin** *an alternative name for* **choriomammotropin**.

chorion villus sampling *abbr.*: CVS; a technique for a biopsy of the chorion frondosum during pregnancy to obtain fetal tissue for prenatal diagnosis.

chorismate the anion of (3*R-trans*)-3-[(1-carboxyethenyl)oxy]-4-hydroxy-1,5-cyclohexadiene-1-carboxylic acid; the unsymmetrical ether derived from phospho*enol*pyruvate and 5-phosphoshikimic acid (*see* **shikimate pathway**); it is formed as an intermediate in the biosynthesis of aromatic amino acids and many other compounds. The common pathway to aromatic amino acids bifurcates at chorismate: one branch, the prephenate pathway, leads to phenylalanine and tyrosine; the other branch, the anthranilate pathway, leads to tryptophan. There are many other branchpoints in the shikimate pathway. (Note: the configurations can be stated as 3*R*,4*R*.)



chorismate mutase EC 5.4.99.5; an enzyme that catalyses the conversion of chorismate to prephenate. It is the first enzyme in the branch of the **shikimate pathway** that leads to synthesis of phenylalanine. Isoenzymes are found in cytoplasm and in chloroplasts. The chloroplast isoenzyme is activated by tryptophan and inhibited by phenylalanine and tyrosine. Example from yeast: database code CHMU_YEAST, 256 amino acids (29.71 kDa). For an example see **prephenate dehydratase**.

chorismate synthase EC 4.6.1.4; *other name*: 5-*enol*pyruvylshikimate-3-phosphate phospholyase; an enzyme that catalyses the formation of chorismate from 5-*O*-(1-carboxyvinyl)-3-phosphoshikimate with release of orthophosphate. Example from *Escherichia coli*: database code AROC_ECOLI, 360 amino acids (38.96 kDa). *See also* **shikimate pathway**.

Christmas disease *or* **hemophilia B** a sex-linked inherited disorder of **blood coagulation** due to a deficiency of coagulation factor IX (Christmas factor) activity in the blood. It occurs in humans, dogs, cats, and other species, but is less common than classical **hemophilia** (hemophilia A) from which it can only be differentiated by laboratory tests.[After S Christmas, the first described sufferer of the disease.]

Christmas factor *an alternative name for* **factor IX**. *See also* **blood coagulation**, **Christmas disease**.

chrom+ *see* **chromo+**.

chromaffin *or* **chromaffine** describing a tissue, cell, or subcellular granule that stains deeply with chromium salts.

chromaffin tissue a tissue made up of modified neural cells (chromaffin cells) that synthesize, store, and secrete catecholamines. The catecholamines are stored in special subcellular organelles, **chromaffin granules**, ready for release. Chromaffin tissue is found in all vertebrates, at various bodily locations but especially in the medulla of the adrenal gland.

chromat+ *see* **chromato+**.

chromatic of, or related to, colour.

chromatid one of the two daughter strands of a duplicated chromosome that become apparent between early prophase and metaphase in mitosis and between diplotene and second metaphase in meiosis.

chromatin the network of fibres of DNA and protein that make up the **chromosomes** of the eukaryotic nucleus during interphase. It stains strongly with basic dyes. Some parts of the chromosomes are highly dispersed (called **euchromatin**) and other parts are highly condensed (called **heterochromatin**). Chromatin is made up of repeating units, each unit consisting of some 200 DNA base pairs and two each of **histones** H2A, H2B, H3, and H4; most of the DNA is wound around the outside of a core of histones to form **nucleosomes**. The remainder of the DNA, linker DNA, joins adjacent nucleosomes, thereby contributing flexibility to the **chromatin fibre**, a flexibly jointed chain of nucleosomes.

chromatin body a body, containing ribosomal DNA, that occurs during early oogenesis in some water beetles. It disappears at about the time of vitellogenesis and its contents become dispersed in the nuclear interior.

chromatin fingerprinting a two-dimensional electrophoretic technique for simultaneously mapping the total distributions of DNA sizes, protein compositions, and nuclease-mediated precursor–product relationships present in heterogeneous populations of mono- and polynucleosomes. Chromatin, partially digested with micrococcal nuclease, is electrophoretically separated in the first dimension in the presence of inactive DNase I. Subsequently the enzyme is activated by incubating the gel in a magnesium ion-containing buffer and the histones are then separated by electrophoresis in the second dimension.

chromato+ *or (before a vowel)* **chromat+** *comb. form* **1** denoting colour or coloured. **2** denoting **chromatin**. **3** indicating **chromatogram**, **chromatograph**, or **chromatography**.

chromatofocusing a column-chromatographic technique of high capacity and high selectivity for separating proteins according to their isoelectric points. A pH gradient is formed in a column containing an anion exchanger, by use of a special eluting buffer containing a large number of differently charged species. Proteins emerge from the column as a series of sharply focused zones in descending order of their p*I* values, with peak widths in the range 0.04–0.05 pH unit.

chromatogram the result of a chromatographic separation. It may be directly visible, e.g. in the form of a column or strip of material, or, after processing of data, take the form of a graph.

chromatograph **1** to effect a separation by **chromatography**. **2** an apparatus with which to carry out a particular form of chromatography. **—chromatographic** *adj.*; **chromatographically** *adv.*

chromatography any technique, analytical or preparative, for separating the components of a mixture by differential adsorption of compounds to adsorbents, partition between stationary and mobile immiscible phases, ion exchange, or a combination of these. *See* individual entries for the many different types of chromatography; these include: adsorption, affinity, affinity-elution, ampholyte-displacement, argentation, ascending, biospecific-elution, charge-transfer, chromatography, circular, countercurrent, covalent, descending, dye-ligand, electro, exclusion, frontal, gas, gas–liquid, gel-filtration, gel-permeation, high-performance liquid-affinity, high-performance liquid, high-pressure liquid, hydrophobic, ion-exchange, ion-exclusion, ionic-interaction, ion-moderated partition, ligand-mediated, liquid, liquid–liquid, metal-chelate affinity, molecular-exclusion, molecular-sieve, negative, paper, partition, permeation, positive, pseudo-affinity, reverse(d)-phase, salting-out, sievorptive, steric-exclusion, subunit-exchange, thermal-elution, thin-layer, triazine-dye affinity.

chromatophore **1** a pigment-containing cell, especially one in the integument of an animal, that is capable of changing the apparent skin colour by contracting or expanding. **2** any of the particles isolated from photosynthetic organisms that contain photosynthetic pigments, e.g. a **chromoplast**, or membrane fragments from photosynthetic bacteria.

chromatopile a stack of filter papers, of the same diameter, that is compressed in a chromatographic column in large-scale separations.

chromatoplate the plate that is covered with the support during **thin-layer chromatography**.

chromatosome a **nucleosome** lacking the linker DNA.

+chrome *comb. form* denoting colour, coloured, or pigment.

chromium *symbol*: Cr; the element of proton number 24. It is a paramagnetic, metallic transition element of group 6 of the IUPAC **periodic table**, with oxidation states of II, III, and IV. Natural chromium has a relative atomic mass of 51.966 and is a mixture of four stable nuclides with nucleon numbers of 50, 52, 53, and 54, the predominant one being chromium-52 (relative abundance 83.789 atom percent). A number of artificial radioactive nuclides are known, of which **chromium-51** is the longest lived. Chromium has an abundance of about 0.02% of the igneous rocks of the earth and it occurs in nature exclusively in the combined form, the mixed oxide chromite $Fe_2Cr_2O_4$ being the chief ore. It is an essential dietary trace metal, but functions in the body primarily as a component of **glucose-tolerance factor**, the chief symptom of experimental chromium deficiency in animals being impaired glucose tolerance.

chromium-51 the artificial radioactive nuclide of chromium, $^{51}_{24}Cr$. It decays by electron capture, 9.8% of nuclei emitting γ-radiation (0.32 MeV); it also emits X-rays (0.005 MeV, 22%) but no β-particles; it has a half-life of 27.7 days. It is used as chromate for labelling blood cells, and as the 1:1 complex of Cr^{2+} with EDTA for measuring glomerular filtration rate.

chromo+ *or (before a vowel)* **chrom+** *comb. form* **1** denoting colour, coloured, or pigment. **2** indicating **chromium**.

chromobindin *see* **annexin I**, **annexin II**, **annexin IV**, **annexin VI**.

chromogen **1** the colourless precursor of a dye or pigment, that is converted into the dye or pigment, e.g. by oxidation. **2** a compound containing a **chromophore** that is not itself coloured but becomes so when an auxochromic group (*see* **auxochrome**) is introduced into the molecule. **3** any microorganism that produces pigment.

chromogenic **1** producing colour. **2** of, or pertaining to, a **chromogen**.

chromogenic substrate a substrate that, when acted on by an enzyme, gives rise to a coloured product.

chromogranin *abbr.*: Cg; either of two soluble proteins, originally discovered in bovine chromaffin granules (*see* **chromaffin tissue**) but now known to be widely distributed in endocrine (including neuroendocrine) tissues. They are designated chromogranin A (CgA) and chromogranin B (CgB; *also called* **secretogranin I**). Both proteins are acidic (pI 4.5–5.0), but homology is limited except in the N-terminal and C-terminal regions, and there is a lack of immunological reactivity. Both have been detected immunologically in various mammals, amphibians, birds, and fishes; CgA also occurs in *Drosophila*. CgA is a 430 amino acid Ca^{2+}-binding glycoprotein; it is hydrophilic with no hydrophobic stretches, and is co-secreted with many hormones from, e.g., pancreas and thyroid C cells. Both proteins have **dibasic cleavage sites** for endopeptidases, the action of which leads to the formation of several active peptides: CgA yields **parastatin** (porcine $CgA_{347-419}$), an inhibitor of low calcium-stimulated parathyroid cell secretion; **pancreastatin** (porcine $CgA_{240-288}$), an inhibitor of glucose-induced insulin release from the pancreas; **chromostatin** (bovine $CgA_{124-143}$), an inhibitor of catecholamine release by chromaffin cells; **vasostatin** (bovine CgA_{1-76}); **beta-granin** (bovine CgA_{1-114}), an inhibitor of parahormone secretion; and a peptide WE-14. CgB is co-secreted from adrenal medulla with catecholamines, from sympathetic nerve with norepinephrine, and from some cell lines; proteolytic cleavage releases several peptides of unknown function. Example (incomplete but contains entire CgA sequence) from pig: database code CMGA_PIG, 446 amino acids (48.83 kDa).

chromomere any of the beadlike concentrations of **chromatin** that are arranged linearly along a **chromosome**.

chromonema (*pl.* **chromonemata**) *cytological term for* **1** all of the threads which make up the **nuclear reticulum**. **2** any of the smallest strands of DNA in a chromosome or chromatid. **3** a twisted chromatid thread within the chromosome.

chromophore any functional grouping within a molecular entity that gives rise to the colour (or characteristic absorption spectrum) of the substance.

chromoplast any coloured **plastid** containing carotenoid pigments but not chlorophyll. Chromoplasts give the yellow to red colours of many flowers (e.g. daffodil), fruits (e.g. tomato), and some roots (e.g. carrot).

chromoprotein any of a class of proteins that contain a colour-producing group or **chromophore**. Both chlorophylls and carotenoids exist in chloroplasts as chromoproteins.

chromosome a structure composed of a very long molecule of **DNA** and associated proteins (e.g. **histones**) that carries hereditary information. Chromosomes are especially evident in plant or animal cells undergoing mitosis or meiosis, where each chromosome becomes condensed into a compact, readily visible thread. In nondividing cells chromosomes typically assume a more dispersed form called **chromatin**. The number of chromosomes is characteristic for the species concerned. In a bacterium only one chromosome is evident as the cell is about to divide. After DNA replication, the two new chromosomes attach to a specialized site on the bacterial plasma membrane for segregation to the two daughter cells. —**chromosomal** *adj.*

chromosome mutation a type of **mutation** (def. 1) involving alteration(s) in chromosome structure.

chromosome walking a method used to identify which clone in a gene bank contains a desired gene or sequence that cannot easily be selected for. The gene bank must contain the entire DNA sequence of the chromosome as a series of overlapping fragments. A series of **colony hybridizations** is carried out, starting with a cloned gene that has already been identified and that is known to be on the same chromosome as the desired gene. This identified gene is used as a probe to pick out clones containing adjacent sequences. These are then used in turn as probes to identify clones carrying sequences adjacent to them, and so on. At each round of hybridization one 'walks' further along the chromosome from the identified gene.

chromostatin a 20-amino-acid peptide that can be released *in vitro* by the proteolytic cleavage of **chromogranin** A. It potently inhibits catecholamine release by **chromaffin** cells induced by carbamoylcholine or by depolarizing K^+ concentration, and it inhibits secretagogue-induced calcium influx. Its physiological status is uncertain – it is not flanked by **dibasic cleavage sites** in the precursor.

chronic **1** continuing over a long time; repeatedly recurring. **2** *(of a disease)* developing slowly or lasting for a long time.

chrono+ *or (before a vowel)* **chron+** *comb. form* denoting time.

chronobiochemistry the branch of biochemistry concerned with the biochemical aspects of **chronobiology**, the study of the inherent rhythmicity or periodicity of living organisms and their activities.

chrysolamarin *see* **callose**.

chrysotherapy the use of gold-containing substances in the treatment of disease.

CHS *(in clinical biochemistry) abbr. for* cholinesterase (def. 1).

ChTX *abbr. for* charybdotoxin.

chyle the white lymph found in the lymph vessels of the intestine (lacteals) during the digestion of fats. It contains globules with a high fat content. *See also* **chylomicron**.

chylomicron a large lipoprotein particle, of molecular mass up to 10^7 kDa, measuring up to 100 nm in diameter, and of density < 1.006. The approximate composition (% by weight) of chylomicrons is 2% cholesterol, unesterified; 5% cholesterol, esterified; 7% phospholipid; 84% triacylglycerol; 2% protein. Their apolipoprotein composition (% by weight total apolipoprotein) is 7.4% A-I; 4.2% A-II; 22.5% B-48; 66% C-I + C-II + C-III. Chylomicrons are formed in the mucosa of the small intestine, and contain mostly triacylglycerols re-esterified in the mucosal cells from dietary long-chain fatty acids associated with apolipoproteins. They then enter the lymphatic capillaries (lacteals) of the intestinal villi, and thence the bloodstream. They are the vehicle by which fat is absorbed into the body.

chymase EC 3.4.21.39; *other names*: mast cell protease I; skeletal muscle (SK) protease. An enzyme that catalyses preferential cleavage of the bonds: Phe-|-Xaa > Tyr-|-Xaa > Trp-|-Xaa > Leu-|-Xaa. It is a trypsin-like serine proteinase, and a homologue of chymotrypsin; it is found in mast cell granules (heart, lung, etc.). Example, human (precursor): database code MCP1_HUMAN, 247 amino acids (27.29 kDa).

chyme the semifluid mass of partially digested food that passes through the pylorus of the stomach into the small intestine.

chymopapain EC 3.4.22.6; *other name*: papaya proteinase II; the major endopeptidase enzyme of papaya (*Carica papaya*) latex; its specificity is similar to that of **papain**. The protein from that species is: database code PAP2_CARPA, 218 amino acids (23.66 kDa).

chymosin *or* **rennin** EC 3.4.23.4; an aspartic proteinase enzyme that cleaves a single bond in κ **casein** (Ser-Phe^{105}-|-Met-Ala) and is responsible for clotting of milk. It is produced from a precursor in the mucosa of the fourth stomach of calves. It is now produced by cDNA technology. Example (bovine): database code NRL_1CMS, 323 amino acids (35.61 kDa); 3-D structure is known. (*Note*: use of the synonym rennin is now deprecated because of possible confusion with **renin**.)

chymotrypsin a serine endopeptidase, EC 3.4.21.1, formed from the proenzyme **chymotrypsinogen** by the action of trypsin. In vertebrates, chymotrypsins are contained in **pancreatic juice**, and hydrolyse protein in the small intestine. Chymotrypsinogen is synthesized in the acinar cells of the exocrine pancreas and stored in **zymogen granules** until released. Several forms of chymotrypsin are found: **Chymotrypsin A** is formed from bovine and porcine chymotrypsinogen A. It will hydrolyse peptides, amides, and esters, preferentially at the carbonyl end of Tyr, Trp, Phe, and Leu residues. **Chymotrypsin B**, formed from chymotrypsinogen B, is homologous with chymotrypsin A. **Chymotrypsin C**, EC 3.4.21.2, is formed from porcine chymotrypsinogen C and from bovine subunit II of procarboxypeptidase A. It preferentially cleaves at the carbonyl end of Leu, Tyr, Phe, Met, Trp, Gln, and Asn residues. Enzymes homologous to and with similar specificities to chymotrypsin have been isolated from many species. Chymotrypsinogens are synthesized as a single 245-residue polypeptide chain cross-linked by five disulfide bonds. This is fully activated when the bond between Arg^{15} and Ile^{16} is cleaved by trypsin to yield π-chymotrypsin; subsequent autocatalytic cleavage of more peptide bonds (to remove 14–15, 147–148) converts it into α-chymotrypsin. Amino acids 1–13 form chymotrypsin A-chain, 16–146 form chymotrypsin B-chain, and 149–245 form chymotrypsin C-chain, the three chains being covalently linked by disulfide bonds, 1–122, 42–58, 136–201, 168–182, and 191–220. The mechanism involves a **charge-relay system** of His^{57}, Asp^{102}, and Ser^{195}. Chymotrypsin is more active than trypsin against certain esters, e.g. *N*-acetyl-L-tyrosine ethyl ester. Chymotrypsin binds α_1-antitrypsin. Example (bovine): database code CTRA_BOVIN, 245 amino acids (25.64 kDa). 3-D structure known.

chymotrypsinogen *see* **chymotrypsin**.

chymotryptic of, or relating to, **chymotrypsin**; **chymotryptic digestion** is the partial hydrolysis of a protein by chymotrypsin.

Ci *symbol for* curie.

cI *or* *c*1 *symbol for* lambda repressor.

CI *abbr. for* Colour Index (especially as a prefix to the identifying number for a particular dye or stain).

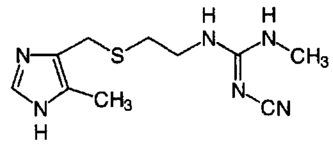
CIDNP *abbr. for* chemically induced dynamic nuclear polarization.

CIE *abbr. for* counterimmunoelectrophoresis.

ciliatine *see* **phosphono+**.

cilium (*pl.* **cilia**) a specialized eukaryotic locomotor organelle that consists of a filiform extrusion of the cell surface. Each cilium is bounded by an extrusion of the cytoplasmic membrane, and contains a regular longitudinal array of **microtubules**, anchored basally in a **centriole**. *Compare* **flagellum**. — **ciliary** *adj.*

cimetidine *N*-cyano-*N'*-methyl-*N''*-[2-[[(5-methyl-1*H*-imidazol-4-yl)methyl]thio]ethyl]guanidine; a competitive antagonist of histamine H_2 receptors used clinically as an anti-ulcer drug; it inhibits gastric acid secretion and reduces the output of pepsin. One proprietary name is Tagamet.



cIMP *abbr. for* cyclic IMP; *see* **inosine 3',5'-phosphate**.

cin4 the gene for the enzyme ***trans*-cinnamate 4-monooxygenase** found in many plants.

cinchonine an alkaloid obtained from cinchona bark and possessing antimalarial properties. *See also* **quinine**.

cinnamate the anion of 3-phenyl-2-propenoic acid; *trans*-cinnamate (3-phenyl-2*E*-propenoate) is a compound formed from phenylalanine by phenylalanine ammonia-lyase, EC 4.3.1.5; it is the precursor of ferulic and sinapic acids in the pathway for **lignin** synthesis.

***trans*-cinnamate 4-monooxygenase** *abbr.*: CA4H; EC 1.14.13.11; *systematic name*: *trans*-cinnamate,NADPH:oxygen oxidoreductase (4-hydroxylating); *other names*: cinnamic acid 4-hydroxylase; cinnamate 4-hydroxylase. An enzyme that catalyses a reaction between *trans*-cinnamate, NADPH, and dioxygen to form 4-hydroxycinnamate, $NADP^+$, and H_2O. It is an endoplasmic reticulum enzyme of high specificity – it does not act on amino acids, phenylacetate, or benzoate – and is the first enzyme in the pathway from **cinnamate** to **lignin**. Example from *Catharanthus roseus*: database code CRCIN4HX, 505 amino acids (58.21 kDa).

cinoxacin azolinic acid; 1-ethyl-1,4-dihydro-4-oxo[1,3]dioxolo[4,5-*g*]cinnoline-3-carboxylic acid; a **4-quinolone antibiotic** with activity against Gram-negative bacteria through the inhibition of DNA gyrase.

ciprofloxacin a fluorinated **4-quinolone antibiotic** active against both Gram-positive and Gram-negative bacteria although more so against the latter and also active against *Chlamydia* and mycobacteria.

circadian describing biological activity (e.g. behavioural, physiological, metabolic) that exhibits an endogenous periodicity of approximately 24 hours independently of any daily variation in the environment. *Compare* **diurnal**, **infradian**, **ultradian**. [From Latin *circa*, about, + *dies*, day.]

Circe effect the phenomenon underlying any physicochemical description of the action of an enzyme upon a substrate. It refers to the strong attractive forces that bind the substrate into the active site of the enzyme, where it undergoes a radical transformation of form and structure. It is named after the sorceress, Circe, who according to Greek mythology transformed the men of Odysseus into beasts.

circular birefringence the **birefringence** produced by left and right circularly polarized light.

circular chromatography a technique of paper chromatography in which the substances to be separated are allowed to travel radially from the centre of a filter-paper disk. The material is applied to the centre of the disk from which a small sector is cut out and folded down to allow it to dip into the eluting solvent.

circular chromosome a circular DNA duplex, the form taken by the chromosomes of some prokaryotic organisms, e.g. *Escherichia coli*, *Bacillus subtilis*, and *Pseudomonas aeruginosa*.

circular dichroism *abbr.*: CD; the difference in absorption of left and right circularly polarized light. The shape and magnitude of the CD curve as a function of wavelength of solutions of proteins (and other macromolecules) are sensitive to changes in conformation of these solutes. *See also* **Cotton effect, ellipticity, molecular ellipticity, optical rotatory dispersion.**

circular DNA any single- or double-stranded DNA molecule with a circular (but not necessarily geometrically circular) structure. If single-stranded, the circle is closed covalently, but if double-stranded, one or both strands may be open. *See also* **covalent (closed) circle of DNA.**

circularly polarized light *see* **polarized light.**

cis *abbr.*: c; *conformational descriptor* denoting synperiplanar (use not recommended); *see* **conformation.**

cis- *abbr.*: *c*-; *prefix (in chemical nomenclature)* denoting that two specified substituents lie on the same side of a reference plane in the molecule. (*cis* means 'on this side'.) In systematic names it is denoted by the symbol *Z*. *See* ***cis–trans*** **isomerism, *E/Z* convention.** *Compare* ***trans-***.

***cis*-acting** *(in genetics)* describing a regulatory genetic element whose effects are sensitive to its position relative to the gene being regulated, i.e. on the same molecule of DNA. Examples include bacterial **operators.**

cis* configuration** *(in genetics)* describing the configuration of two linked heterozygous genes in which both wild-type alleles occur on one of the paired homologous chromosomes while the mutant alleles occur on the other homologue; i.e. (++/*ab*). *Compare* ***trans **configuration.**

cis* isomer** *see* ***cis–trans **isomer.**

CISP *abbr. for* corticotropin-induced secreted protein; now called thrombospondin-2 (*see* **thrombospondin**).

cisplatin *cis*-diammineplatinum(II) dichloride; *cis*-diaminedichloroplatinum; a substance that interacts with, and forms cross-links between, DNA and proteins. It is used as a neoplasm inhibitor to treat solid tumours, primarily of the testis and ovary.

cisterna (*pl.* **cisternae**) **1** *(in cytology)* the flattened, saclike space enclosed between paired membranes of the **endoplasmic reticulum** and the **Golgi apparatus. 2** *(in anatomy)* an enlarged space, such as the cisterna magna in the rear of the brain or the cisterna chyli between the crura of the diaphragm. —**cisternal** *adj.*

***cis–trans* complementation test** a genetic test to determine whether two gene mutations (*a* and *b*) occur in the same functional gene (or **cistron**). It involves comparison of heterozygotes in which the relevant mutations are in the same chromosome, i.e. in the *cis* configuration (++/*ab*), with heterozygotes in which the mutations are in separate chromosomes, i.e. the *trans* configuration (*a*+/+*b*).

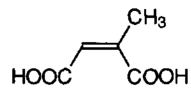
cis–trans* isomer** *or (sometimes)* **geometric isomer** *or* **geometrical isomer** either of a pair of **diastereoisomers** that differ only in the positions of atoms relative to a specified reference plane in the molecules, in cases where these atoms are, or are considered as if they were, parts of a rigid structure. The two isomers are designated the *cis* isomer and the *trans* isomer when the two particular atomic groupings lie respectively on the same or on opposite sides of the reference plane. *See also* ***cis–trans **isomerism.** —***cis–trans*-isomeric** *adj.*

cis–trans* isomerism** *or (sometimes)* **geometrical isomerism** the phenomenon of the existence of *cis–trans* isomers in general, or the occurrence of paired *cis* and *trans* isomers in a particular molecule. Atoms or groups are termed *cis* or *trans* to one another when they lie respectively on the same or on opposite sides of a reference plane identifiable as common among stereoisomers. For compounds containing only doubly bonded atoms the reference plane contains the double-bonded atoms and is perpendicular to the plane containing these atoms and those directly attached to them. For cyclic compounds the reference plane is that in which the ring skeleton lies or to which it approximates. *See also* ***E/Z **convention.**

cistron a unit of DNA sequence that codes for a single polypeptide or protein. It may be a smaller unit than the **gene**, which contains the unit of DNA determining a single character. Whether or not a cistron is the same as a gene can be determined by the ***cis–trans*** **complementation test**, hence the name cistron (*cis* means 'on this side').

Cit *symbol for* a residue of the α-amino acid L-citrulline.

citraconate 1 *the trivial name for* methylmaleate, *cis*-1,2-propylenedicarboxylate, (*Z*)-2-methyl-2-butenedioate, the dianion of citraconic acid, (*Z*)-2-methyl-2-butenedioic acid. **2** any mixture of citraconic acid and its mono- and dianions. **3** any salt or ester of citraconic acid.



citraconic acid

citraconoyl the bivalent *cis*-acyl group, $-CO-CH=C(CH_3)-CO-$, derived from citraconic acid. *Compare* **citraconyl.**

citraconyl either of the isomeric univalent *cis*-acyl groups, $HOOC-C(CH_3)=CH-CO-$ or $HOOC-CH=C(CH_3)-CO-$, derived from citraconic acid. *See also* **citraconylation.**

citraconylation introduction of one or more **citraconyl** groups into a substance by acylation. Similarly to **maleylation** (and succinylation), citraconylation of a protein with citraconic anhydride is used to acylate its free lysine (and other) amino groups in order to change their charge at neutral pH from positive to negative and to render the adjacent peptide bond resistant to hydrolysis by trypsin. Reaction occurs at pH 8 and the citraconyl groups can be removed at pH 3.5 (both operations at 20°C). Introduction of citraconyl groups is slower than is the case with maleyl groups, but their removal is faster and more complete. —**citraconylate** *vb.*; **citraconylated** *adj.*

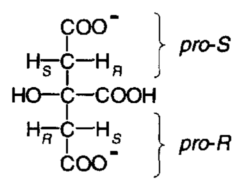
citrase *see* **citrate lyase.**

citratase *see* **citrate lyase.**

citrate 1 *the trivial name for* β-hydroxytricarballylate, 2-hydroxy-1,2,3-propanetricarboxylate,

$$HO-C(CH_2-COO^-)_2-COO^-,$$

the trianion of citric acid, 2-hydroxy-1,2,3-propanetricarboxylic acid. **2** any mixture of free citric acid and its anions; the trianion is the predominant form at physiological pH values. It is widely distributed in nature and is an important intermediate in the **tricarboxylic-acid cycle** and the **glyoxylate cycle**. Citrate (alone or in mixtures) is useful as a buffer; pK_{a1} (25 °C) = 3.13, pK_{a2} (25 °C) = 4.76, pK_{a3} (25 °C) = 6.40. It is useful also as a component of anticoagulant solutions and has many commercial applications in the chemical, food, pharmaceutical, and other industries. Industrial production of citrate is usually by fermentation of inexpensive sources of glucose or sucrose with selected strains of *Aspergillus niger*; world production in 1992 was 5×10^8 tons. **3** any (partial or full) salt or ester of citric acid.



citrate aldolase *see* **citrate lyase.**

citrate cleavage enzyme EC 4.1.3.8; *recommended name*:

ATP–citrate (*pro-S*)-lyase. An enzyme that catalyses a reaction between ATP, citrate, and CoA to form acetyl-CoA, oxaloacetate, ADP and orthophosphate. In mammals it participates in the release of acetyl-CoA in the cytosol, particularly as a precursor for fatty acid synthesis, there being no system for transporting excess acetyl-CoA out of the mitochondrion. The acetyl carbons are thus transported across the mitochondrial membrane as citrate and then released by this enzyme. Example from rat, homotetramer: database code ACLY_RAT, 1100 amino acids (120.50 kDa).

citrate cycle *another name for the* **tricarboxylic-acid cycle**.

citrate (*pro-3S*)-lyase EC 4.1.3.6; *other names*: citrase; citratase; citritase; citridesmolase; citrate aldolase; a bacterial enzyme that catalyses the formation of acetate and oxaloacetate from citrate. Example from *Klebsiella pneumoniae*: database code CILB_KLEPN 289 amino acids (31.32 kDa). ATP–citrate lyase is another name for **citrate cleavage enzyme**.

citrate (*Re*)-synthase EC 4.1.3.28; an enzyme from some bacteria that forms citrate with the incoming $-CH_2-COOH$ group in the *pro-R* position of citrate. The addition is to the *Re*-face of the carbonyl group of oxaloacetate.

citrate (*Si*)-synthase EC 4.1.3.7; *other names*: citrogenase; oxaloacetate transacetase; condensing enzyme; citrate condensing enzyme. An enzyme that catalyses a reaction between acetyl-CoA, H_2O, and oxaloacetate to form citrate and CoA. It synthesizes citrate from oxaloacetate and acetyl-CoA, placing the incoming $-CH_2-COOH$ in the *pro-S* position in citrate (*see* **oxaloacetate**). The addition is to the *Si*-face of the carbonyl group of oxaloacetate. This is the usual mammalian citrate synthase; mitochondrial enzymes are closely related homodimers (six motifs). Example from chicken: database code CISY_CHICK, 433 amino acids (46.88 kDa); 3-D structure known.

citric acid *see* **citrate**.

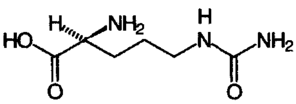
citric-acid cycle *an alternative name for* **tricarboxylic-acid cycle**.

citridesmolase *see* **citrate lyase**.

citritase *see* **citrate lyase**.

citrovorum factor *(formerly) an alternative name for* **folinic acid**.

citrulline N^5-carbamoyl-L-ornithine; 2-amino-5-ureiodovaleric acid; an α-amino acid, not found in proteins. L-Citrulline is an intermediate in the **urea cycle** in animals and in a modified urea cycle in plants. It was first isolated from the juice of watermelon, *Citrullus vulgaris*.



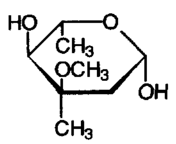
citrullinemia *or* **citrullinuria** an autosomal recessive genetic disease involving a deficiency of the enzyme **argininosuccinate synthase**. It is characterized by a high plasma citrulline level and excessive secretion of the amino acid in the urine. It results in neurological damage and mental retardation.

CJD *abbr. for* Creutzfeldt–Jakob disease.

C-kinase *see* **protein kinase C**.

Cl *symbol for* chlorine.

cladinose 2,6-dideoxy-3-*C*-methyl-3-*O*-methyl-L-*ribo*-hexose; 3-*O*-methyl mycarose; a component of **erythromycins** A and B.



β-L anomer

cladistics a method of classification that attempts to reconstruct a **phylogeny** using characters that are unique to each taxonomic group. The term **cladistic** is currently used to describe an evolutionary or phylogenetic tree based on gene sequence similarities and gene association.

cladogenesis a process of adaptive evolution leading to the development of a greater variety of organisms; branching evolution. —**cladogenetic** *adj.*

cladogram a diagram depicting **cladogenetic evolution**, i.e. the evolution of some character through the descendants of a common ancestor.

clarify to make or become clear, especially to remove finely divided suspended material from a solution, e.g. by centrifugation or filtration. —**clarified** *adj.*; **clarification** *n.*

Clark and Lubs buffers one of the early series of buffer mixtures of predetermined pH, covering the range pH 2.2–10.0. [After William Mansfield Clark (1884–1964), US biochemist, and Herbert A. Lubs.]

Clark electrode a type of **oxygen electrode**. [After Leland Charles Clark (1918–).]

class a taxonomic category ranking below the **phylum** (or division) and containing several **orders** (or sometimes only a single order).

classical pathway of complement activation *see* **complement**.

class switch a change (switch) in the expression of B lymphocytes from one class of antibody to another.

clastic dividing into parts, especially of a chemical reaction in which one molecular species is divided into parts; e.g. phosphoroclastic, thioclastic.

clastogen an agent that is able to bring about breakage in chromosomes. —**clastogenic** *adj.*

clathrate 1 *or* **cage compound** *(in chemistry)* a solid complex or an addition compound in which the crystal lattice or structure of one component (the **host molecule**) completely encloses spaces in which a second component (the **guest molecule**) is located. There is no chemical bonding between the two components. *Compare* **inclusion complex**. 2 designating or relating to a clathrate. 3 *(in biology)* lattice-shaped.

clathrin a 180 kDa protein that is the main component of the coat of **coated vesicles** and **coated pits**, which are concerned in the transfer of molecules between the membranous components of eukaryotic cells. Clathrin also occurs in **synaptic vesicles**. The clathrin coat is a polyhedral lattice of subunits; each subunit, or clathrin **triskelion**, comprises three heavy chains and three light chains. Examples from rat; light chain A: database code CLCA_RAT, 248 amino acids (26.95 kDa); light chain B: database code CLCB_RAT, 229 amino acids (25.09 kDa); heavy chain: database code CLH_RAT, 1675 amino acids (191.38 kDa).

clathrin uncoating protein *or* **uncoating ATPase** a protein from brain or erythrocyte cytosol that disassembles **clathrin** coats to **triskelions** in an ATP-dependent manner. Present in substantial excess of clathrin triskelions, it binds to a C-terminal domain of the heavy chain of clathrin triskelions.

Claude, Albert (1899–1983), Belgian-born US cell biologist distinguished for his development of methods of separating cellular components by differential centrifugation and for identifying mitochondria as the primary cellular site for biological oxidation; Nobel Laureate in Physiology or Medicine (1974) jointly with C. R. M. J. de Duve and G. E. Palade ‘for their discoveries concerning the structural and functional organization of the cell’.

clavulanic acid 3-(2-hydroxyethylidene)-7-oxo-4-oxa-1-azabicyclo[3.2.0]heptane-2-carboxylic acid; a weak antibacterial agent but potent *β*-lactamase inhibitor produced by *Streptomyces clavulingerus*. It protects lactamase-sensitive but otherwise potent antibiotics from deactivation, and is used in combination with *β*-lactam antibiotics to increase their efficacy. It has no antibacterial activity but by inactivating penicillinases it makes a combination active against penicillin-resistant bacteria. These include *Staphylococcus aureus*, 50% of many *Bacteroides*, and *Klebsiella* spp. A commonly used proprietary combination is Augmentin.

clearance *(in physiology) symbol*: *C*; a measure of the efficiency of the kidney in removing a substance from the blood. $C = UV/P$, where U and P are the concentrations of the substance in the urine and plasma, respectively, and V is the rate of urine flow in ml min^{-1}. Clearance is therefore a rate, with the dimensions of ml plasma per min.

clearing 1 *(in histology)* the process of preparing tissue samples for microscopical examination by removing the dehydrating agent and rendering the sample transparent. 2 *(in physiology)* the removal of a substance from the body or from the blood; *see* **clearance**.

clearing factor lipase *an alternative name for* **lipoprotein lipase**; an enzyme found in the capillary endothelium of various tissues, notably heart, lung, and adipose tissue, so called because it clears the opalescence from lipemic plasma.

cleavage 1 the tendency of a crystal to split in certain preferred directions with the formation of smooth surfaces, or the act of so splitting a crystal; hence **cleavage plane**, **cleavage face**. 2 the rupture of a (usually specified) chemical bond in a molecule with the formation of smaller molecules. 3 *or* **segmentation** *(in embryology)* the series of mitotic divisions by which a fertilized animal ovum changes, without any overall change in size, into a ball of smaller cells constituting the primitive embryo.

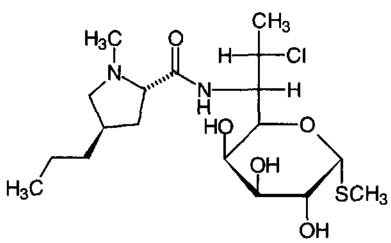
cleavage division the mitotic division of a fertilized egg up to the point at which the regions of the egg shift relative to each other.

cleave to split or cause to divide, especially along the line of a natural weakness; to split a chemical bond; to effect or undergo **cleavage**. —**cleavable** *adj.*

Cleland's reagent *see* **dithiothreitol**. [After William Wallace Cleland (1937–).]

climacteric 1 *(in medicine)* the female menopause, or the corresponding time of life in men. 2 *(in botany)* the marked increase in respiration in fruit just prior to and during ripening.

clindamycin (7*S*)-7-chloro-7-deoxylincomycin; a semisynthetic antibiotic that is useful against anaerobic bacteria; for structure and action *see* **lincomycin**, **lincosamide**.



clinical of, or pertaining to, the course of disease in an individual, or the examination and treatment of an individual directly, as opposed to the laboratory investigation of disease.

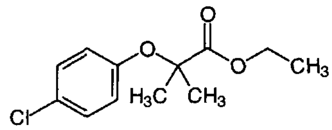
clinical chemistry a branch of applied biochemistry concerned with the nature and determination of chemical substances of interest in the investigation of diseases.

CLIP *abbr. for* 1 corticotropin-like intermediate peptide. 2 Class-II-associated invariant-chain peptide (*see* Class II MHC antigens in **major histocompatibility complex**).

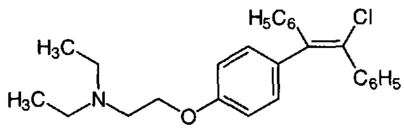
CLIP-170 *see* **restin**.

clofazimine *N*,5-bis(4-chlorophenyl)-3,5-dihydro-3-[(1-methylethyl)imino]-2-phenazinamine; an iminophenazine dye with anti-inflammatory, tuberculostatic, and leprostatic activity. It is used to treat multibacillary or sulfone-resistant leprosy.

clofibrate ethyl 2-(*p*-chlorophenoxy)-2-methylpropionate; 2-(4-chlorophenoxy)-2-methylpropanoic acid ethyl ester; a drug that reduces high plasma levels of triacylglycerols, very-low-density lipoproteins (VLDL), and (less so) cholesterol. Its actions are probably mediated by the stimulation of **lipoprotein lipase**, which reduces the triacylglycerol composition of VLDL and stimulates clearance of low-density lipoproteins by liver. It is used primarily in the treatment of type III (but also in type IV, type V, and type IIb) hyperlipoproteinemia. *Compare* **gemfibrozil**.



clomifene *or* **clomiphene** 2-[4-(2-chloro-1,2-diphenylethenyl)phenoxy]-*N*,*N*-diethylethanamine; an antiestrogenic aminoether derivative of stilbene; it blocks estradiol receptors in the hypothalamus and may stimulate the secretion of gonadotropin-releasing hormone. It is active as an antitumour agent against mammary carcinomas, and can be used to induce ovulation in infertile women. Both its structure and actions are similar to those of **Tamoxifen**.



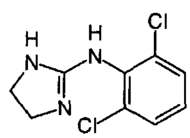
clonal culture any culture of cells produced in such a way that an individual **clone** (def. 1) can be selected.

clonal selection theory *or* **Burnet's theory of immunity** a theory to account for the phenomenon of **acquired immunity** and the ability of the immune system to produce large quantities of highly specific antibody for any particular immunogen. According to the theory, the ability to produce a given antibody is a pre-existing and genetically determined characteristic of a particular clone of B lymphocytes. An encounter with the antigen triggers proliferation of the corresponding clone, producing a greatly expanded population of cells that secrete specific antibody. Some members of the expanded clone assume the role of memory cells, which respond to any subsequent encounter with the same immunogen which accounts for the secondary response being greater than the primary response. *See also* **Burnet**.

clone 1 the descendants produced vegetatively or by apomixis from a single plant, or asexually or parthenogenetically from a single animal; a group of cells produced by division from a single cell. The members of a clone are of the same genetic constitution, unless **mutation** has occurred amongst them. 2 a homogeneous population of DNA molecules; *see* **molecular cloning**. 3 to propagate a selected organism or cell, or to replicate a particular DNA molecule so as to produce a clone. —**clonal** *adj.*; **cloning** *n.*

cloned library *an alternative name for* **gene library**.

clonidine 2,6-dichloro-*N*-2-imidazolidinylidenebenzenamine; an antagonist at α_2 (presynaptic) adrenergic receptors, P_1 purinergic receptors, and H_2 histamine receptors. It is used as a central antihypertensive drug; it also abolishes most symptoms of opiate withdrawal.



cloning vector *or* **cloning vehicle** the DNA of any transmissible agent (e.g. plasmid or virus) into which a segment of foreign DNA can be spliced in order to introduce the foreign DNA into cells of the agent's normal host and promote its replication and transcription therein. (1) **Plasmid vectors** contain (i) an origin of replication, so that the plasmid can be replicated; (ii) an antibiotic resistance gene to allow selection of transformed host cells; (iii) often, a 'polylinker' containing several different restriction enzyme recognition sites and a second marker gene (e.g. *lacZ*; *see* **X-gal**) which will be inactivated when foreign DNA is inserted therein, thus allowing identification of transformants bearing plasmids containing inserts, rather than 'empty' vectors. Examples include **pBR322** and the **pUC vectors**. *See also* **expression vectors**. (2) **Bacteriophage vectors**, usually based on **lambda phage**, can accommodate longer fragments of foreign DNA. In insertion vectors (e.g. λgt10) the foreign DNA is inserted into the phage DNA while in replacement vectors (e.g. λEMBL4) the foreign DNA is ligated between two phage fragments, or arms. (3) Hybrid vectors, **cosmids**, **phagemids** and **phasmids** (def. 1), with features of both plasmids and phage, have also been constructed. *See also* **molecular cloning**.

closed circle (of DNA) *see* **covalent (closed) circle (of DNA)**.

closed double-stranded DNA *see* **covalent (closed) circle (of DNA)**.

closed duplex DNA *or* DNA I a closed double-stranded DNA molecule in which the DNA helical structure is preserved.

closed-loop system a type of system, e.g. a metabolic pathway or an electrical circuit, in which control is exerted by **feedback**. *Compare* **open-loop system**.

closed system any (thermodynamic) system that cannot gain or lose matter, heat, or work. *Compare* **open system**.

close packing the structure of a solid in which nonbonded atoms (or molecules) are surrounded by other nonbonded atoms (or molecules) in such a way that the distance between them is close or equal to the sum of their van der Waals contact radii (*see* **van der Waals radius**).

clostridiopeptidase A the **collagenase** obtained from *Clostridium histolyticum*.

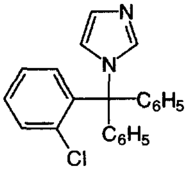
Clostridium a genus of spore-forming rod-shaped bacteria of the family Bacillaceae. Its members are chemoorganotrophic, obligately anaerobic, and typically Gram-positive. They are widespread in soil, mud, and in the intestinal tract of animals. Some species are pathogenic if they gain access to the tissues, and produce a range of toxins.

***Clostridium histolyticum* collagenase** *see* **microbial collagenase**.

clostripain *or* **clostridiopeptidase B** EC 3.4.22.8; a cysteine proteinase enzyme from *Clostridium histolyticum*. It preferentially cleaves Arg-|-Xaa and Arg-|-Pro bonds and is inactivated by substituted arginine and lysine chloromethyl ketones. It is a heterodimer of α and β chains. Example from *C. histolyticum* (α chain): database code CLOS_CLOH1, 526 amino acids (59.73 kDa).

clot retraction the contraction of a blood clot after its formation, accompanied by the expression of serum.

clotrimazole 1-[(2-chlorophenyl)diphenylmethyl]-1*H*-imidazole; an **imidazole** used topically as an antifungal agent. It inhibits ergosterol synthesis and disrupts the transport of amino acids into the fungus.

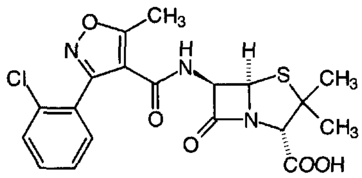


clotting factor *an alternative term for* coagulation factor; *see* **blood coagulation**.

clotting time *an alternative term for* **coagulation time**.

clover-leaf structure one of the models for the secondary structure of **transfer RNA** (tRNA). The nucleotide sequences of different tRNAs can be arranged in similar two-dimensional representations, resembling a clover leaf in shape, in which there are four bihelical regions, containing the maximal number of base pairs, connected by three loops. Some tRNAs have a small extra loop. The clover leaf structure has been largely supported by the tertiary structure of those tRNAs for which structure has been determined by X-ray crystallography.

cloxacillin a semisynthetic penicillin that is resistant to penicillinase and is useful against certain staphylococci.



Clp protease a proteinase/ATPase from *Escherichia coli* with little protein specificity (EC 3.4.21.-). It is a **heat-shock protein** but supposedly a housekeeping proteinase; counterparts occur in many organisms (five motifs). The enzyme is a heterodimer with subunits named from the *clp* genes. The proteolytic subunit is ClpP, database code CLPP_ECOLI, 207 amino acids (23.16 kDa); the ATP-binding subunit is one of ClpA (database code CLPA_ECOLI, 758 amino acids (84.07 kDa)) or ClpX (database code CLPX_ECOLI, 423 amino acids (46.17 kDa)).

CLUSTALV a computer program for the sequence alignment of proteins, etc.

cluster **1** a group of iron and sulfur atoms, e.g. Fe_2S_2 or Fe_4S_4, in an iron–sulfur protein. The term is sometimes extended to include other metal atoms, as in the $MoFe_{6-8}S_4$ cluster in nitrogenase. **2** *or* **cluster of enzymes** *or* **enzyme cluster** the phenomenon of the emergence of new enzymes, or the upsurge of enzymes already present in low concentration, at certain (usually defined) periods along the age axis of the organism.

cluster analysis a method of statistical analysis used, e.g., in **phylogeny** deduced from phenetic characters.

cluster-crystal any of the globular aggregates of calcium oxalate crystals found in some plants.

clustering the occurrence of a number of objects, like or unlike, in clusters; the term is used especially of components of cell membranes, such as receptors.

cm *symbol for* centimetre, i.e. 10^{-2} m.

Cm *symbol for* **1** the carboxymethyl group, $-CH_2-COOH$. **2** curium.

CM *abbr. for* carboxymethyl; e.g. CM-cellulose.

CMAR a gene, cloned from a colon cancer cell line, encoding an 82-amino-acid protein that enhances the attachment of cells to collagen and laminin; the gene is named from cell matrix adhesion regulator. The protein has an N-terminal myristoylation motif (underlined in the partial sequence) and a tyrosine phosphorylation site (bold in the partial sequence), this tyrosine being required for activity. Sequence:

MLRGSDMKGP..............TAL**RIVEILY**.

CMC *abbr. for* **1** critical micellar concentration. **2** carboxymethylcellulose.

CMH *abbr. for* ceramide monohexoside (Cer(Hex) preferred).

CMP *abbr. for* cytidine monophosphate, i.e. cytidine phosphate.

CMR (spectroscopy) *(sometimes) abbr. for* ^{13}C-nuclear magnetic resonance (spectroscopy).

CMU *abbr. for* 3-(*p*-chlorophenyl)-1,1′-dimethylurea; monuron;

a herbicide that acts by blocking noncyclic photophosphorylation.

^{13}C-NMR (spectroscopy) *abbr. for* carbon-13 nuclear magnetic resonance (spectroscopy).

CNP *abbr. for* C-type natriuretic peptide; it is similar to atrial **natriuretic peptide**.

CNS *abbr. for* central nervous system.

co+ *prefix* denoting **1** together, joint or jointly; in partnership, an equality; to the same degree. **2** *(in mathematics)* of the complement of an angle.

Co **1** *symbol for* cobalt. **2** *abbr. for* coenzyme.

Co I *(obsolete) abbr. for* coenzyme I; *see* **nicotinamide adenine dinucleotide**.

Co II *(obsolete) abbr. for* coenzyme II; *see* **nicotinamide adenine dinucleotide phosphate**.

CoA *symbol for* (free or combined) coenzyme A.

coacervate any viscous phase obtained by **coacervation** of solutions of two or more polymers.

coacervate droplet any of the droplets that, it has been suggested, may have formed in the **primordial soup** by **coacervation** and, perhaps, entrapped primitive catalyst(s); they may have given rise to the first cells. *Compare* **protobiont**.

coacervation any process in which there is a spontaneous separation of a continuous one-phase aqueous solution of a highly hydrated polymer into two aqueous phases, one with a relatively high polymer concentration, the other with a relatively low polymer concentration.

coactone an **ecomone** active in the process of relationship between an active and directing organism and a passive and receiving organism.

coagulant any substance that produces, or aids the production of, coagulation.

coagulase any of a number of serologically distinguishable bacterial enzymes that catalyse the coagulation of citrated or oxalated blood plasma. The extracellular bacterial protein specifically forms a complex with human **prothrombin**; this complex can clot fibrinogen without any proteolytic cleavage of prothrombin. Example from *Staphylococcus aureus*: database code STC1_STAAU, 658 amino acids (74.42 kDa).

coagulase/fibrinolysin precursor *see* **plasminogen activator**.

coagulate **1** to cause a fluid or part of a fluid to change into a solid or semisolid mass by the action of, e.g., heat or chemical substances; to clot. **2** the solid or semisolid mass produced by coagulation. —**coagulable** *adj.*; **coagulation** *n.*

coagulation factors *see* **blood coagulation**.

coagulation time *or* **clotting time** **1** the time taken for a freshly drawn specimen of blood or blood plasma to coagulate (clot) when measured under standard, controlled conditions; it is an indication of the state of the coagulation processes in the specimen. **2** the time taken for a specimen of milk, or of a casein fraction, to coagulate in standard, controlled conditions.

coagulogen a ≈16 kDa fibrinogen-like protein that is found in the hemocyte (amoebocyte) of horseshoe crab (*Limulus polyphemus*) hemolymph and participates in hemostasis. Coagulogen consists of a single, basic polypeptide chain and constitutes about 50% of the protein in horsehoe-crab hemocyte lysates.

coagulum (*pl.* **coagula**) any semisolid coagulated mass or clot, especially that formed by clotting of blood.

coarctation the state of being pressed together or constricted, especially of the aorta or other blood vessel.

coarse descriptive of **1** things that are rough in texture or structure. **2** particles that are large or powders composed of such particles. **3** threads, filaments, or fibres of relatively large diameter. **4** means of adjustment (as of instruments or processes) that provide only inaccurate or gross control, or the adjustments so made.

CoASAc *symbol for* acetyl-coenzyme A.

CoASH *symbol for* (free) coenzyme A.

coated pit an invagination of the cell membrane of many eukaryotic cells, concerned in receptor-mediated selective transport of many proteins and other macromolecules across the cell membrane. During endocytosis it is converted into a **coated vesicle**. The coat is of **clathrin**.

coated-tube method a solid-phase **radioimmunoassay** method in which the specific antibody is immobilized on the inner surface of a small test tube.

coated vesicle a vesicle, formed from a **coated pit**. Observed in the cytoplasm of many eukaryotic cells, coated vesicles measure 50–250 nm in diameter and are characterized by a 'bristle' coat on the outer surface of their lipid membrane. This coat is made up of a polyhedral lattice of **clathrin** subunits together with smaller amounts of other proteins, notably ones of about 100 kDa (**adaptins**), 50 kDa and 20 kDa. Coated vesicles are concerned with the rapid and continuous transport of molecules between specific membranous organelles of the cell and to and from the cell membrane.

coatomer a large protein complex that forms part of the coat of specific Golgi intercisternal transport vesicles that are involved in constitutive vesicular transport between endoplasmic reticulum, Golgi apparatus, and plasma membrane. It has seven subunits known as COPs (from coat protein). COPI-coated vesicles are formed by combination of **ARF** with coatomer. They participate in retrograde traffic from the Golgi apparatus to the endoplasmic reticulum. COPII-coated vesicles are formed from other proteins that include the gene products Sec 13p and Sec 23p. Example: the yeast coatomer is a complex (700–800 kDa); one component is the transport protein product of *SEC21*, database code SC21_YEAST, 935 amino acids (104.71 kDa).

coat protein **1** one of the outer structural proteins of a coated **virus**. **2** any of those subunits of the **coatomer** known as COPs.

cobalamin *the trivial name for* many members of the **vitamin B_{12}** group and their derivatives. A cobalamin is a **cobamide** in which 5,6-dimethylbenzimidazole is the aglycon that forms a link between the cobalt atom and the α1-position of the ribose residue. The sixth position of the cobalt atom may be (artefactually) filled by a variety of groups, e.g. a cyano group giving cyanocobalamin. The oxidation state of the cobalt may be indicated by using a Roman numeral, e.g. cob(I)alamin for Co^{I}.

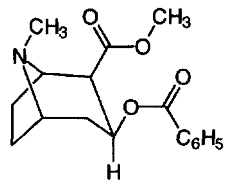
cob(I)alamin adenosyltransferase EC 2.5.1.17; *other name*: aquacob(I)alamin adenosyltransferase; an enzyme that catalyses the formation of adenosylcobalamin from ATP, cob(I)alamin, and H_2O, with release of orthophosphate and pyrophosphate. The enzyme requires manganese. Example from *Escherichia coli*: database code BTUR_ECOLI, 196 amino acids (21.97 kDa).

cob(II)alamin reductase EC 1.6.99.9; *other name*: vitamin B_{12r} reductase; a flavoprotein enzyme that catalyses a reaction between NADH and two molecules of cob(II)alamin to form two molecules of cob(I)alamin and NAD^+.

cobamide any member of a group of compounds (including the **cobalamins**) that have molecules consisting of a highly substituted tetrapyrrole ring system (corrin) in which a cobalt atom is linked to the four pyrrole nitrogen atoms and with an α-D-ribofuranose 3-phosphate residue attached to one of the side chains.

cobamide coenzyme any of the biochemically active forms of **vitamin B_{12}**. They are **cobalamins** in which either a methyl or 5′-deoxy-5′-adenosyl residue is attached to the sixth position on the cobalt atom.

cocaine 2β-carbomethoxy-3β-benzoxytropane; an alkaloid obtained from dried leaves of the South American shrub *Erythroxylon coca* or by chemical synthesis. It is a cerebral stimulant and narcotic, and has been used as a local anaesthetic. Cocaine acts by blocking neuronal reuptake of, especially, **norepinephrine**, hence its potent vasoconstrictor effects, which can lead to tissue necrosis. Its use is restricted in many countries.



cocaine

cocarboxylase *a former name for* thiamine diphosphate (*see* **vitamin B$_1$**).

cocarcinogen any noncarcinogenic agent that enhances the effect of a **carcinogen**.

cochromatography a chromatographic technique in which an unknown substance is applied to a chromatographic support together with one or more known compounds, in the expectation that the relative behaviour of the unknown and known substances will assist in the identification of the unknown one.

cocktail a mixture of substances prepared according to a recipe, especially reagent mixtures such as those for use in electrofocusing (**electrofocusing cocktails**), in liquid scintillation counting (**scintillation cocktails**), and in cell-free translation systems (**translation cocktails**). Many such cocktails are commercially available.

COD *abbr. for* chemical oxygen demand.

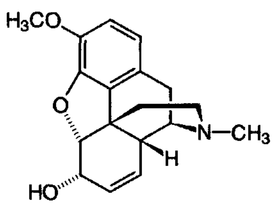
code 1 any system of symbols, together with the rules for their association, that can be used to represent or transfer information; e.g. the **genetic code**. 2 to contain, to be arranged in, or to be expressed through a code; used especially of DNA and mRNA. A gene is said to code for (or encode) its protein product. *Compare* **decoding**.

coded amino acid any of the 20 α-amino acids whose occurrence in proteins is under direct genetic control, i.e. for which a **codon** exists.

codehydrase I *or* **codehydrogenase I** *an obsolete name for* **nicotinamide adenine dinucleotide**.

codehydrase II *or* **codehydrogenase II** *an obsolete name for* **nicotinamide adenine dinucleotide phosphate**.

codeine methylmorphine; (5α,6α)-7,8-didehydro-4,5-epoxy-3-methoxy-17-methylmorphinan-6-ol; an opioid found in opium and manufactured by the methylation of **morphine**. It acts as an agonist at μ, δ, and **κ-opioid receptors** but is generally less potent than morphine, being metabolized in the liver to morphine and norcodeine. It is a narcotic analgesic with **antitussive** activity and is a controlled substance in the US Code of Federal Regulations.



coding ambiguity an adjustment to the prediction of Francis **Crick** that a single nucleotide triplet (codon) would not code for more than one particular amino acid, which was shown by subsequent work to require modification. Thus two codons, AUG and GUG, code not only for the amino acids, methionine and valine respectively, but also are important in chain initiation. Furthermore, there are differences between codon usage in the cytoplasm and mitochondria. In mitochondria UGA codes for Trp rather than chain termination, AUA for Met rather than Ile, and AGA for termination rather than Arg. There is also the problem of proteins containing selenocysteine which is coded by a unique tRNA^{Sec}, the codon being UGA which usually codes for termination.

'coding' problem the description given by Francis **Crick** and co-workers in 1957 to the crucial puzzle of the **genetic code**, namely how a sequence employing combinations of only four different nucleotides (i.e. genes) can determine sequences containing combinations of up to 20 different amino acids (i.e. proteins). Crick predicted that groups of three nucleotides would code for each amino acid providing the possibility of 64 codons, more than sufficient for the 20 amino acids. (A binary code of two nucleotides would only provide 16 codons.) The prediction was confirmed with the elucidation of the genetic code in the early 1960s.

coding region *or* **coding sequence** 1 (of a DNA molecule) *an alternative name for* **exon**. 2 (of a messenger RNA molecule) the portion of its complete nucleotide sequence that is translatable into polypeptide.

coding strand the strand in genomic duplex DNA that contains the same sequence of bases as messenger RNA transcribed from the DNA (excepting uracil in RNA in place of thymine in DNA). Its complement is the **noncoding strand**. In some bacteria and viruses, alternate segments of both strands of duplex DNA may be coding, in some instances with overlap. *Other names*: antitemplate strand; codogenic strand; nontranscribing strand; plus strand; sense DNA; sense strand.

coding triplet *or* **coding unit** *an alternative name for* **1 codon**. **2 anticodon**.

cod-liver oil a fish oil obtained from the liver of cod, especially the Atlantic cod (*Gadus morhua*). It is a good source of long-chain *n*-3 **fatty acids**. The typical composition (% total) of major acids is: 16:0, 14%; 16:1 *n*-7, 12%; 18:1 *n*-9, 22%; 20:1 *n*-9, 12%; 22:1 *n*-11, 11%; 20:5 *n*-3, 7%; 22:6 *n*-3, 7%. *See also* **fish oil**.

codogenic strand *an alternative name for* **coding strand**.

codon any group of three consecutive nucleotide bases (base triplet) in a given messenger RNA molecule that, by its composition and sequence, specifies either a particular amino-acid residue in the polypeptide chain synthesized by translation of that messenger RNA, or signals the beginning or the end of the message (i.e. **start codon**, **stop codon**). The term is also used loosely for base triplets in the genomic nucleic acid – in the case of duplex DNA, either on the **coding strand** or **noncoding strand**. For the full set of codons *see* **genetic code**.

coefficient of danger (of a drug or other agonist) the percentage lethal responses to be expected at a dose level that is effective in 95% of trials.

coefficient of variation *or* **coefficient of variability** a parameter used in comparing dispersion in distributions. It is the ratio of the **standard deviation** to the **mean**, i.e. $sx/\bar{x}$. The coefficient of variation is an abstract number independent of the units of measurements; it may also be expressed as a percentage, i.e. $100sx/\bar{x}$.

coefficient of viscosity *see* **viscosity**.

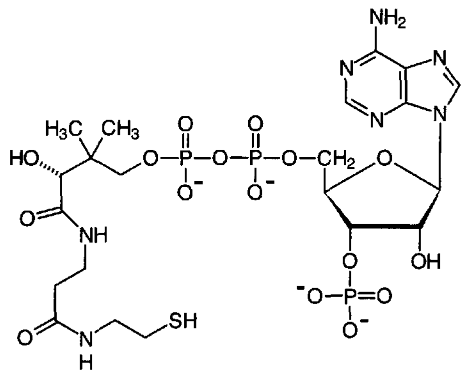
coeliac *a variant spelling (esp. Brit.) of* **celiac**.

coenzyme any of various nonprotein organic **cofactors** that are required, in addition to an enzyme and a substrate, for an enzymic reaction to proceed. Compared with a **prosthetic group** a coenzyme is more easily removed from the **apoenzyme**. The term 'coenzyme' is frequently used imprecisely, although it is now possible to give more informative chemical names to such substances and to define their roles in specific enzymic reactions.

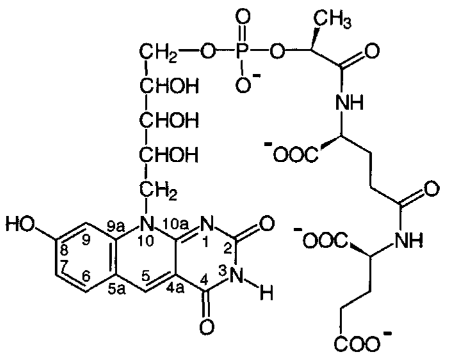
coenzyme I *an obsolete name for* **nicotinamide adenine dinucleotide**.

coenzyme II *an obsolete name for* **nicotinamide adenine dinucleotide phosphate**.

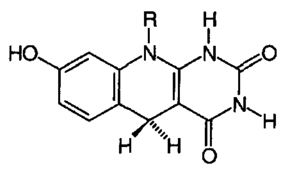
coenzyme A *symbol*: (combined form) CoA *or* (free form) CoASH; 3′-phosphoadenosine-(5′)diphospho(4′)pantetheine; a heat-stable compound that functions as an acyl carrier in a number of biochemical acylation and acyl-transfer reactions, in which the intermediate is a thiol ester. It was originally known as the **acetylation coenzyme**, hence its present name.



coenzyme C a name sometimes given to a polyglutamate derivative of **tetrahydrofolate**.
coenzyme F *abbr.*: CoF; a name sometimes given to **tetrahydrofolate** or some of its derivatives.
coenzyme F_{420} the *N*-(*N*-L-lactyl-γ-glutamyl)-L-glutamic acid phosphodiester of 7,8-didemethyl-8-hydroxy-5-deazariboflavin that functions as a coenzyme for several oxidoreductases e.g. methylenetetrahydromethanopterin dehydrogenase. It was first found in **methanogenic** Archaea, but later also in other Archaea, and some Bacteria and Eukarya. The oxidized form acts also as a photosensitizer in DNA photolyases. Reduction results in addition of hydrogen at C-1 and C-5, in all cases so far reported being stereospecific for the *Si*-face of the coenzyme.



oxidized form



reduced form (R = substituent at N_{10})

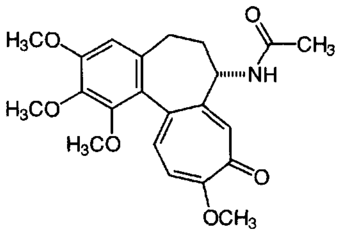
coenzyme F_{430} *see* **factor F_{430}**.
coenzyme factor *an alternative name for* **diaphorase**.
coenzyme M *abbr.*: CoM; 2-thioethanesulfonate, $HSCH_2–CH_2SO_3^-$; it is involved as a coenzyme in the pathway of methanogenesis in **methanogenic** bacteria. In the terminal step of the pathway, a methyl group is transferred from methyltetrahydromethanopterin (*see* **tetrahydromethanopterin**) to CoM to form CH_3–S–CoM, from which methane is formed in a reductive step involving **factor F_{430}**.
coenzyme Q *abbr.*: CoQ; *an obsolete name for* **ubiquinone** (the abbreviation Q, is, however, frequently used in diagrams).
coenzyme R *(sometimes) an alternative name for* **biotin**.
CoF *abbr. for* coenzyme F; *see* **tetrahydrofolate**.
cofactor **1** any accessory, nonprotein substance, commonly of low molecular mass, that is necessary for the activity of an enzyme. Metal-ion cofactors have been called **activators**; organic cofactors that are easily removable from the enzyme (e.g. by dialysis) may be called **coenzymes**, but if they are tightly bound they are often termed **prosthetic groups**. *See also* **apoenzyme**, **holoenzyme**. **2** any amino acid that must be present in the medium to allow certain phages to be adsorbed by host bacteria.
cofilin a pH-sensitive nuclear actin depolymerization factor, similar to **destrin** (five motifs in common). Example from pig: database code COFI_PIG, 166 amino acids (18.50 kDa).
cognitive neuroscience the area of neurobiology that aims to explain how the brain enables an animal to interact successfully with its environment. The term cognition has come to refer to that level of description that links molecular science to behaviour.
Cohen, Stanley (1922–) US biochemist; Nobel Laureate in Physiology or Medicine (1986) jointly with R. Levi-Montalcini 'for their discoveries of growth factors'.
coherent *(in physics)* describing two or more waves having the same wavelength and the same phase, e.g. coherent light. The laser is a source of coherent light.
coherent anti-Stokes Raman spectroscopy *abbr.*: CARS; a nonlinear optical technique for obtaining **Raman spectra** with high efficiency. The signal levels, signal-to-noise ratios, resolution, and fluorescence rejection are far better than with normal Raman spectroscopy. CARS can be used for substances at about 1 mM concentration in aqueous solution.
cohesive end *or* **cohesive terminus** *or* **sticky end** the protruding (3′ or 5′) single-stranded terminus on a double-stranded DNA fragment that can associate with a complementary, similarly protruding sequence at the opposite end of the same DNA fragment, or on another fragment, to form a circular, or larger, DNA molecule.
Cohn fractionation a method for the fractionation of human blood plasma proteins to produce minimally altered proteins suitable for clinical use. It involves fractional precipitation with ethanol at temperatures near or below 0°C, with careful control of pH and ionic strength and additions of certain bivalent metal ions, especially Zn^{2+}. The numbered **Cohn fractions** contain the following main constituents: I, fibrinogen; II, γ-globulins; III, β-globulins; IV, α-globulins; V, albumin; VI, glycoproteins. The method has also been applied to the fractionation of plasma proteins of other animals. [After Edwin Joseph Cohn (1892–1953), US biochemist.]
coiled coil *an alternative name for* **supercoil**.
coincidence **1** *(in physics)* the occurrence of radioactive nuclide disintegrations within a time span too short for them to be resolved by a radiation counter. **2** *(in genetics)* the ratio of the number of observed double crossovers to the number expected from the random combination of single crossovers among three or four linked genes.
coincidence correction a correction applied in counting radioactive events for **coincidence losses**, i.e. those events occurring at high count rates when two or more pulses arrive within the resolving time of the counter.
coincidence counter a counter for radioactive events that consists of two opposed detectors and appropriate electronic circuitry so that only those pulses observed by both detectors

are registered. This arrangement, common in liquid scintillation spectrometers, considerably reduces the number of background counts attributable to random thermionic noise arising in the detectors.

co-ion any ion of low molecular mass present in a colloidal solution or substance that has a charge of the same sign as the (net) charge of a (specified) colloidal ion under particular conditions. *Compare* **counterion**.

colcemide *or* **demecolcine** a derivative of **colchicine** in which the amino group (attached to the non-tropolone seven-membered ring) is methylated instead of acetylated. It binds tubulin and inhibits formation of the mitotic spindle.

colchicine a major alkaloid from the autumn crocus, *Colchicum autumnale* L. It is used experimentally to inhibit the polymerization of **tubulin**, to which it binds specifically and tightly, and to encourage depolymerization of **microtubules**. This action inhibits spindle formation, arrests mitosis at metaphase, and promotes polyploidy in plants. Colchicine also disrupts neutrophil migration and phagocytosis, and inhibits leukotriene B_4 synthesis; it is useful in the treatment of gout. The molecule is noted for a tropolone ring, which is rather rare in natural products. *See also* **colcemide**.



cold 1 of low, or lower than normal, temperature. 2 *(in radiochemistry)* containing no, or an insignificant amount of, a radioactive nuclide; the nonradioactive counterpart of a radionuclide-labelled substance. *Compare* **hot**.

cold agglutinin *see* **cold autoantibody**.

cold autoantibody *or* **cold agglutinin** an antibody that attaches to red cells at low temperatures, reacting better at 4 °C than at 37 °C, and dissociates as the red cells are warmed. Cold autoantibodies are mainly directed against antigens of the **Ii system** of blood groups.

cold-insoluble globulin a **cryoglobulin** present in blood plasma or serum. It is normally present at only very low concentrations in human serum but may be elevated in certain rare diseases such as systemic lupus erythematosis. *See also* **fibronectin**.

cold-labile enzyme *or* **cold-sensitive enzyme** any enzyme that, unlike most enzymes, is less stable at 0 °C than at room temperature. In some, but not all, instances activity slowly returns on warming. The phenomenon can be due, e.g., to exothermic dissociation of a stabilizing cofactor from the enzyme or to dissociation into inactive subunits.

cold-sensitive mutation any mutation producing a gene that is functional at a high temperature (the permissive temperature) but inactive at a low temperature (the restrictive temperature). *See* **temperature-sensitive mutation**.

cold-shock protein any of various proteins, synthesized by microorganisms in response to a sudden drop in temperature. These are typically DNA-binding transcription activators, many of which have a characteristic **cold-shock domain** (*abbr.*: CSD); the protein has three motifs. Example from *Escherichia coli*: database code CSPA_ECOLI, 69 amino acids (7.26 kDa). *Compare* **heat-shock proteins**.

coleoptile the protective sheath covering the first leaves of seedling grasses.

colestipol a basic anion-exchange resin that when given orally as the Cl^- form, exchanges anions of chloride in the intestine with those of bile salts, which are then excreted and thus removed from the enterohepatic circulation as an insoluble complex. The resulting reduction in cholesterol absorption culminates in reduced circulating levels of low-density lipoproteins but can increase plasma triacylglycerols. It has therapeutic application in relieving hyperlipoproteinemia.

colestyramine resin *proprietary name*: Dowex 1-X2-Cl; a basic anion-exchange resin that consists predominantly of polystyrene trimethylbenzylammonium chloride. If administered orally, it exchanges anions of chloride with those of intestinal bile salts, which are then excreted as an insoluble complex. The resulting reduction in cholesterol absorption culminates in reduced circulating levels of low-density lipoproteins (LDLs), hence its therapeutic application in relieving hyperlipidemia and LDL-induced hypercholesterolemia.

Col factor *(formerly) an alternative name for* **Col plasmid**.

colforsin *an alternative name for* **forskolin**.

colicin *or (formerly)* **colicine** any member of a range of 40–80 kDa proteins secreted by those strains of the Enterobacteriaceae that kill, but do not lyse, other susceptible strains within this family; a subcategory of **bacteriocin**. Different colicins are designated B, E_1, E_2, I, K, or V, and may be distinguished by their diffusibility and immunogenic and host specificities; their mechanisms of action may vary. They may be coded for by a **Col plasmid**.

colicin factor *or* **colicinogenic factor** *(formerly) an alternative name for* **Col plasmid**.

colicinogenic describing bacteria that are able to produce **colicins**.

colicinogenic factor *see* **Col plasmid**.

coliform 1 describing any bacterium belonging to the genera *Escherichia* or *Klebsiella*. 2 describing any Gram-negative, lactose-fermenting, enteric bacillus or, more loosely, any Gram-negative, enteric bacillus or related organism. 3 any coliform microorganism.

colipase a cofactor of **pancreatic lipase** that inhibits the surface denaturation of lipase and anchors the protein to the lipid–water interface of lipid micelles to prevent inhibition and displacement by bile salts (four motifs). Example (precursor) from human: database code COL_HUMAN, 112 amino acids (11.94 kDa): amino acids 1–17 form the signal; 18–22, enterostatin activation peptide; 23–112, colipase.

coliphage any **bacteriophage** that infects a strain of *Escherichia coli*.

colistin *or* **polymyxin E** a **polymyxin** antibiotic derived from *Bacillus colistinus* and comprising colistins A, B, and C. It acts as a cationic surface-active detergent, disrupting microbial cell wall structure.

colitose 3,6-dideoxy-L-*xylo*-hexose; 3,6-dideoxy-L-galactose; a sugar present in lipopolysaccharide of some strains of *Escherichia coli*. For the D enantiomer, *see* **abequose**.

collagen a group of fibrous proteins of very high tensile strength that form the main component of connective tissue in animals. Collagen is highly enriched in glycine (up to one-third of its amino-acid residues) and somewhat less so in proline plus 3- and 4-hydroxyproline (about 21%); prolines at the third position of the tripeptide repeating unit (G-X-P) are hydroxylated in some or all of the chains. Collagen of bones and skin is metabolically stable, but that of organs such as liver has a high metabolic turnover. Collagens are products of a superfamily of closely related genes found in multicellular animals. The products have been classified into types I to XIII in the order in which they were purified and characterized, and grouped into classes 1 to 4 according to their properties. All contain a typical triple-helical domain formed from three independent chains. Class 1 comprises types I to III, V, and XI. All of these types are fibre-forming, of approximately the same length, having a lengthy uninterrupted collagenous domain with three α chains in the form of a triple helix, and a large globular domain at the C-terminal end. Each of these types is synthesized as a procollagen molecule. Several types of α chain are found variously in the different types of class 1 collagen. All of these types aggregate into staggered arrays, first into microfibrils of five collagen molecules that then aggregate into fibres. Class 2 comprises types IX and XII. Type IX was first isolated from pig hyaline cartilage. It is composed of three different α chains, α-1, α-2, and α-3 and has three short triple-helical domains interrupted by non-collagenous domains.

At the N-terminal end there is a large globular domain. A glycosaminoglycan molecule, chondroitin sulfate, is attached to the α-2 chain. Example from human, α-1 chain (precursor): database code CA19_HUMAN, 931 amino acids (92.97 kDa); residues 1–23 are the signal, and 24–931 the α-1 chain. Type XII was discovered by screening a DNA library and has structural similarities to type IX. The function of these molecules is not known. Class 3 comprises types IV, VI, VII, and X, which serve unique functions. Type IV is the major component of basement membranes. It has long triple-helical domains with a complex globular domain at the C-terminal end. There are six α-x chain isoforms, numbered 1 to 6; example from human α-1 (IV) chain precursor: database code CA14_HUMAN, 1669 amino acids (160.61 kDa); residues 1–27 are the signal, 28–172 a propeptide, and 173–1669 the α-1 (IV) chain. Type VI is a short molecule having a triple-helical domain, 105 nm in length, separating two globular domains. It aggregates into tetramers stabilized by disulfide bonds. Carbohydrate chains consisting of glucosamine, mannose, galactose, fucose, and sialic acid are associated with type VI amounting to 10% of the total mass. Example from *Mus musculus*, α-1 (VI) (precursor): database code CA16_MOUSE, 1025 amino acids (108.49 kDa); residues 1–19 are the signal, 20–1025 the α-1 (VI) chain. Type VII is known as the **anchoring fibril network** and is the largest collagen yet described with a total molecular mass of the precursor of ≈1050 kDa. It has an N-terminal globular domain, a triple-helical domain 424 nm long, and a highly complex C-terminal globular domain. Anchoring fibrils are found within the sub-basal lamina, and appear to anchor the lamina densa to the underlying matrix. Example from human, α-1 (VII) chain (precursor): database code CA17_HUMAN, 2944 amino acids (295.21 kDa); residues 1–16 are the signal, 17–2944 the α-1 (VII) chain. Type X collagen has a restricted tissue distribution, being synthesized mainly by chondrocytes, and is 1438 nm in length. Class 4 comprises types VIII and XIII. Type VIII was first observed as a cell culture product of endothelial cells. It is a 130–140 nm rod found in specialized tissues including the sclera, periosta, perichondria, and cartilage growth plate. Type XIII was identified from a cDNA clone encoding short chains that form a triple-helical domain. Its function is unknown. *See also* **hydroxyproline**.

collagenase any of the proteinases that hydrolyse **collagen**, in particular *Clostridium histolyticum* collagenase (clostridiopeptidase A, EC 3.4.24.3), which preferentially cleaves at Xaa-|-Gly in the sequence

-Zaa-Pro-Xaa-Gly-Pro-Xaa.

It is widely used experimentally in the release of many types of cells from animal tissues, e.g. adipocytes and hepatocytes. *See also* **gelatinase**, **interstitial collagenase**, **microbial collagenase**.

collectin one of a family of plasma **lectins** with a trimeric structure similar to complement C1q.

colligation the formation of a covalent bond by means of two combining groups donating one electron each; the reverse of unimolecular **homolysis**. *Compare* **coordination** (def. 2). —**colligate** *vb.*

colligative property the property of any substance or system (e.g. an ideal solution) that depends only on the number of particles (atoms, molecules, and ions) and not on their nature. For example, gaseous pressure and osmotic pressure are colligative properties.

collimator any device, generally embodying a system of slits and (sometimes) lenses (optical, electric, or magnetic), that is used to produce a nondivergent beam of electromagnetic radiation or particles.

collinear 1 lying in or on the same (straight) line. 2 having a (straight) line in common. 3 describing the state of congruence between the order of amino acids along a polypeptide chain and the order of the corresponding codons along the polynucleotide chain encoding the polypeptide. —**collinearity** *n.*

Collip, James Bertram (1892–1965), Canadian biochemist and endocrinologist notable for his pioneering investigations in endocrinology but particularly for his role in developing (with J. J. R. Macleod) a method for the fractionation of pancreatic extracts to give insulin-containing preparations pure enough to permit the first clinical trials of the hormone to take place; this achievement was duly recognized by Macleod by dividing his share of the Nobel prize with Collip.

collision coupling hormone–receptor–enzyme model a model of enzyme activation summarized by the scheme:

$$\mathrm{H + R + E} \underset{}{\overset{K_{\mathrm{mH}}}{\rightleftharpoons}} \mathrm{HR + E} \overset{k_1}{\rightarrow} \mathrm{HRE} \overset{k_3}{\rightarrow} \mathrm{HR + E'}$$

where H is the hormone, R is the receptor, E is the inactive enzyme, E′ is the activated enzyme, K_{mH} is the hormone–receptor dissociation constant, k_1 is the bimolecular rate constant governing HRE formation, and k_3 is the rate constant governing enzyme activation. In this model the enzymically inactive intermediate, HRE, never accumulates and constitutes only a small fraction of the total receptor and enzyme concentrations.

collision frequency *or* **collision number** the number of collisions per second when there is only one molecule of reactant per cubic centimetre of gas or liquid e.g. in kinetics.

collodion a solution of 4 g pyroxylin (mainly nitrocellulose) in 100 ml of a mixture of 25 ml ethanol and 75 ml diethylether. When exposed to the atmosphere in thin layers it evaporates leaving a tough, colourless film that may be used as a **dialysis membrane**.

colloid *or* **colloidal system** 1 a state of subdivision of matter implying that the molecules or polymolecular particles (the dispersed phase) dispersed in a medium (the continuous phase) have, at least in one direction, a dimension roughly in the range 1 nm–1 μm or that in a system discontinuities are found at distances of this order. All three dimensions need not be in the colloidal range nor is it necessary for the units of a colloidal system to be discrete. Colloids may comprise many individual large molecules such as proteins or polysaccharides, or aggregations of smaller molecules such as colloidal gold. The colloids do not pass, or pass very slowly, through semipermeable membranes. 2 *(in histology)* the structureless substance seen in thyroid gland follicles. —**colloidal** *adj.*

colloidal dispersion any thermodynamically unstable colloidal system in which particles of colloidal size of any nature (e.g. solid, liquid, or gas) are dispersed in a continuous phase of different composition (or state), and which cannot easily be reconstituted after separation of the dispersed phase and the dispersion phase. *Compare* **colloidal solution**.

colloidal electrolyte any electrolyte that gives ions of which at least one is of colloidal size; it includes **polyelectrolytes**.

colloidal solution any thermodynamically stable solution consisting of colloidal macromolecules and solvent that can be easily reconstituted after separation of the macromolecules from the solvent. *Compare* **colloidal dispersion**.

colloid osmotic pressure that part of the osmotic pressure exerted by a solution that is due to dissolved colloids. *See* **osmotic pressure** (def. 1).

colon 1 *(in invertebrates)* the part of the large intestine that extends from the cecum to the rectum. 2 *(in insects)* the second part of the intestine.

colon bacillus *a former term for* ***Escherichia coli***.

colony *(in microbiology)* a circumscribed group of cells, normally derived from a single cell or small cluster of cells, growing on a solid or semisolid medium.

colony hybridization *or* **colony hybridisation** a blot-transfer method (*see* **blotting**) whereby a very large number of colonies of a bacterium carrying different hybrid plasmids can be rapidly screened to determine which hybrid plasmids contain a specified DNA sequence or gene(s). The bacterial colonies are grown on an agar plate, then replica plated onto nitrocellulose filters. A reference set of the colonies is retained. Then the colonies on the filter are lysed and their DNA is denatured and fixed to the filter *in situ*. The resulting DNA-prints of the colonies are then hybridized to a radioactive RNA probe for the sequence or gene of interest. DNA that binds the probe is then assayed by autoradiography or another detection method, enabling the selection of corresponding colonies from the reference plate.

colony stimulating factor *abbr*.: CSF; any of the **cytokines** that control the differentiation of hemopoietic stem cells, both in the **bone marrow** and in the periphery. The group consists of granulocyte, macrophage, and granulocyte/macrophage CSFs. These promote the development of their specific subsets of leukocytes. **Interleukin**-3 (IL-3), interleukin-5 (IL-5), and **erythropoietin** are also members of this group. IL-3 promotes the differentiation of all the leukocyte lineages, whereas IL-5 is necessary for eosinophil development. Erythropoietin promotes the expansion of erythroid colony forming units. **Granulocyte-colony stimulating factors** (G-CSFs) are cytokines that act in hemopoiesis by controlling the production, differentiation, and function of two related white cell populations: granulocytes and monocytes–macrophages. Example from human (precursor): database code CSF3_HUMAN, 207 amino acids (22.27 kDa). **Granulocyte/macrophage-colony stimulating factors** (GM-CSFs) are cytokine glycoprotein growth factors; they also act in hemopoiesis by controlling the production, differentiation, and function of the granulocytes and the monocytes–macrophages. Example (precursor) from human: database code CSF2_HUMAN, 144 amino acids (16.28 kDa); 3-D structure known. **Macrophage-colony stimulating factor** (M-CSF) is produced by fibroblasts, endothelium, and epithelium; it stimulates growth of macrophage colonies. The molecule is a homodimer; example (precursor) from human: database code CSF1_HUMAN, 554 amino acids (60.05 kDa).

colorimeter a device for measuring colour intensity or differences in colour intensity, either visually or photoelectrically. It is used for quantitative determination of coloured compounds in solutions.

colorimetry any method of quantitative chemical analysis in which the concentration or amount of a compound is determined by comparing the colour produced by the reaction of a reagent with both standard and test amounts of the compound, often using a **colorimeter**.

colostrum the milk secreted by the mammary gland just prior to and for a short period after parturition. It has high levels of protein and immunoglobulin, and its ingestion confers passive maternal immunity on the offspring of some species.

Colour Index *abbr*.: CI; a catalogue of commercially available dyes, identifying each by chemical class, colour, and generic name, and giving their chemical composition where disclosed. Each dye is assigned a five-digit number, prefixed by 'CI'. The preferred trivial name of each dye, its commoner synonyms, and the name(s) of its known manufacturer(s) also are listed.

colour quenching the **quenching** that occurs in liquid scintillation counting due to the absorption of the scintillation emission by the specimen, resulting in a decrease in the light-collecting factor. It can often be reduced by prior bleaching or dilution of the specimen before introduction into the scintillator. *Compare* **chemical quenching**.

Col plasmid *or (formerly)* **Col factor** *or* **colicin factor** *or* **colicinogenic factor** any **plasmid** that carries genetic information for the production of a **colicin**; a subcategory of **bacteriocinogenic factor**.

columbinic acid *or* **ranunculeic acid** (5*E*,9*Z*,12*Z*)-octadeca-5,9,12-trienoic acid; a fatty acid occurring in some seed oils. It can relieve many of the symptoms of essential fatty acid deficiency, although it does not act as a precursor for prostaglandin synthesis.

column an upright cylindrical shaft or tube, especially the tube used to contain the stationary phase in **column chromatography** or **column electrophoresis**, or to contain the packing material in fractional distillation. By extension, the term is also used for the tube (whether vertical, horizontal, or spiral) used in **gas chromatography**.

column chromatography any method of **chromatography** in which the stationary phase is a porous solid contained in a column through which the liquid or gas phase percolates. The principal examples are **adsorption chromatography**, **gel-filtration chromatography**, **gas–liquid chromatography**, and **ion-exchange chromatography**.

column electrophoresis any method of **electrophoresis** in which the support material is contained in a column. Such a technique is usually used for preparative scale separations by **electrofocusing** or **isotachophoresis**.

column volume *symbol*: *X*; the volume (empty) of that part of a chromatographic column that contains the packing, calculated from the inner diameter and the height or length of the column occupied by the stationary phase under the conditions of use. *Compare* **bed volume**.

CoM *abbr. for* coenzyme M.

combination electrode an electrode comprising a glass electrode and a reference electrode in the same envelope providing a smaller, more compact instrument and ensuring the two electrodes are in the same environment.

combinatorial describing any process that is governed by a specific combination of factors, rather than by any single factor, with different combinations having different effects.

combinatorial chemistry the use of a process to prepare large numbers of structurally diverse sets of organic compounds by combining sets of chemical building blocks (called monomers), often in every possible combination.

combined test of pituitary function a test for suspected hypopituitarism. Insulin, luteinizing hormone-releasing hormone (LHRH), and thyrotropin releasing hormone (TRH) are administered and the response of pituitary hormones is compared with their levels before insulin administration. The lowering of the blood glucose level should stimulate a rise in the blood levels of cortisol and growth hormone; LHRH should increase follicle stimulating hormone (FSH) and luteinizing hormone (LH); and TRH should increase thyrotropin and prolactin.

combining site the site on an **antibody** molecule that combines with the corresponding antigen.

combustion the act or process of burning, especially the quantitative conversion of organic compounds to carbon dioxide and water for elementary analysis, in the estimation of carbon-14 and tritium, or in the determination of a **heat of combustion**.

co-metabolism any oxidation of substances by a microorganism in which the energy yielded by the oxidation is not used to support microbial growth. The presence or absence of growth substrate therefore cannot be inferred from such an oxidation.

command language the language used for giving computers simple instructions. The commands can be assembled into files referred to as batch files (DOS), command files (VMS), or scripts (UNIX).

commensal **1** describing either of two different species of organism that live in close physical association with benefit to one partner but with little effect – either beneficial or harmful – on the other partner. **2** any organism in a commensal association. *Compare* **symbiosis**. **—commensalism** *n*.

committed step *or* **committing reaction** a reaction in a multienzyme reaction sequence (e.g. a metabolic pathway) that once it occurs, causes all the ensuing reactions of the sequence to take place. The sole metabolic role of the committed step is to provide precursors of the final product of the pathway; it is usually a physiologically irreversible step.

co-monomer the cross-linking agent added to the monomer in forming the gel used in **polyacrylamide gel electrophoresis**.

COMP *abbr. for* cartilage oligomeric matrix protein; now called thrombospondin-5 (*see* **thrombospondin**).

compactin *an alternative name for* **mevastatin**.

compartment **1** *(in tracer kinetics)* any anatomical, physiological, chemical, or physical subdivision of a system throughout which the concentration ratio of labelled to unlabelled substance is uniform at any given time. It is understood that the rate at which the tracer entering the compartment is mixed with tracer in the compartment is rapid compared with the rate at which the substance leaves the compartment. **2** any subdivision of a model used for the analysis of experimental observations.

compartmentation *or* **compartmentalization** the unequal distribution of a metabolite, enzyme, or metabolic pathway between different parts of a cell (e.g. in cell organelles) or of a whole organism. Consequently only a part of the total of any given chemical present is apparently available for a given reaction. It may refer to morphologically recognizable structures or to **metabolic pools**.

compatibility group *see* **incompatibility group**.

competence **1** the ability of a cell to take up a molecule of exogenously added DNA, and to be transformed by it (*see* **transformation**). **2** the ability of bacterial cells to incorporate phage DNA and to produce phage progeny (*see* **bacteriophage**). **3** the state of reactivity of a cell, tissue, or organism that allows it to respond to specific stimuli by forming new morphological structures or by synthesizing specific proteins (enzymes).

competence factor a secreted protein, or other substance added in culture, that is essential for **competence** in a bacterial or other cell. A 10 kDa basic protein of this type has been isolated from streptococci. It does not bind to DNA but interacts with the cell surface.

competition *(in chemistry)* rivalry between two or more different, but often similar, chemical species for a specific biochemical system, e.g. a receptor, enzyme, transport system, antibody molecule, or ion exchanger.

competition binding *see* **displacement binding**.

competitive antagonism the competition between an **agonist** and an **antagonist** for a receptor that occurs when the binding of agonist and antagonist is mutually exclusive. This may be because the agonist and antagonist compete for the same binding site, or combine with adjacent but overlapping sites. A third possibility is that different sites are involved but that they influence the receptor macromolecule in such a way that agonist and antagonist molecules cannot be bound at the same time. If the agonist and antagonist form only short-lived combinations with the receptor, so that equilibrium between agonist, antagonist, and receptor is reached during the presence of the agonist, the antagonism will be surmountable over a wide range of concentrations (*see* **reversible competitive antagonism**). In contrast, some antagonists, when in close enough proximity to their binding site, may form a stable covalent bond with it (*see* **irreversible competitive antagonism**); the antagonism becomes insurmountable when no spare receptors remain. More generally, the extent to which the action of a competitive antagonist can be overcome by increasing the concentration of agonist is determined by the relative concentrations of the two agents, by the association and dissociation rate constants for their binding, and by the duration of the exposure to each. The action of a competitive antagonist can therefore be surmountable under one set of experimental conditions and insurmountable under another.

competitive inhibition any inhibition of an enzyme activity by an inhibitor that binds to the same active site on the enzyme as does the substrate and when there is no interaction between the inhibitor and the enzyme–substrate complex, nor between the substrate(s) and the enzyme–inhibitor complex. Characteristically, analysis using a Lineweaver–Burk plot reveals that in the presence of a **competitive inhibitor** V_{max} is unchanged and K_m is increased.

competitive inhibitor any inhibitor that produces **competitive inhibition** of the activity of an enzyme. Inhibition produced by a given level of inhibitor can be overcome by an increase of substrate concentration.

competitive protein-binding assay *abbr.*: CPBA; a form of **saturation analysis** in which a naturally occurring proteinaceous binding substance is used.

complement *or (formerly)* **alexin(e)** a system of plasma proteins, found in the blood of vertebrates (and freshly prepared sera), that acts as an effector mechanism in immune defence against infection by microorganisms; the activation products of complement components cause lysis of antigenic cells, attract phagocytic cells to the site of activation, and assist the uptake and destruction of antigenic cells by phagocytes. Complement is also involved in immunological tissue injury. Activity in serum is lost on standing for several days at room temperature or for 30 min at 56 °C. The components of complement, mostly β- or γ-globulins, are generally designated by the symbol C and an identifying suffixed numeral; activated components are indicated by a superscribed bar.

C1, the first component of complement, is a serum proteinase of M_r 400 000 (Sepharose); its concentration in human serum is approximately 250 mg L^{-1}. C1 is a Ca^{2+}-dependent termolecular complex of C1q, C1r, and C1s in the molar ratio of 1:2:2. The C1q subcomponent is composed of nine subunits, six of which are disulfide-linked dimers of the A and B chains, and three of which are disulfide-linked dimers of the C chain. It is one of the few proteins that contain **hydroxyproline** C1r and C1s form a Ca^{2+}-dependent tetramer (M_r 350 000), which is composed of two molecules each of the zymogens C1r (M_r 83 000) and C1s (M_r 83 000). C1r has proteinase activity (EC 3.4.21.41), cleaving Lys(or Arg)–Ile bonds in C1s, which can then activate C2 and C4. Example C1QA: database code C1QA_HUMAN, 705 amino acids (80.08 kDa).

C2, the second component of complement, has an M_r of 110 000 and a concentration in human serum of approximately 15 mg L^{-1}. It is activated by C1s to form C2a and C2b. C2a combines with complement components C2b and C4b to form **C3/C5 convertase**, a serine proteinase (EC 3.1.21.43). C2 is a class II **MHC** protein sequence; example (human, precursor): database code CO2_HUMAN, 752 amino acids (83.14 kDa).

C3, the third component of complement, is a glycoprotein of M_r 180 000; its concentration in human serum is approximately 1200 mg L^{-1}. It activates the complement system, being processed by the removal of four Arg residues to form two chains, α and β, linked by a disulfide bond. C3/C5 convertase activates C3 by cleaving the α chain, releasing C3a anaphylatoxin and generating the α′ chain + β chain of C3b. C3b is split in two positions by complement factor I to form IC3b and C3f. IC3b is then slowly cleaved by factor I to form C3c and C3dg. C3 resembles C4, C5, and **α_2-macroglobulin**. Database code (human, precursor) CO3_HUMAN, 1663 amino acids (186.95 kDa).

C4, the fourth component of complement, has an M_r of 200 000 and a concentration in human serum of approximately 400 mg L^{-1}. The precursor is cleaved enzymatically to form three chains (α, β, and γ); the human form is polymorphic, the allotype being C4a. C4a alleles carry blood group Rogers and C4b alleles carry Chido. Total deficiency of C4a6 allotype is associated with hemolytic disease. C4a anaphylatoxin is a mediator of local inflammatory processes. It induces the contraction of smooth muscle, increases vascular permeability, and causes histamine release from mast cells and basophilic leukocytes. C4 is an MHC class III protein, similar to

C3, C5, and α_2-macroglobulin. Database code (human, precursor) CO4_HUMAN, 1741 amino acids (192.12 kDa).

C5, the fifth component of complement, is a glycoprotein of M_r 180 000; its concentration in human serum is approximately 80 mg L^{-1}. Activation of C5 initiates the spontaneous assembly of the late complement components, C5–C9, into the membrane attack complex (*see* **MAC** (def. 1)). C5b has a transient binding site for C6. The C5b–C6 complex is the foundation upon which the lytic complex is assembled. The C5 precursor is first processed by the removal of four basic residues, to form two chains, α and β, linked by a disulfide bond. C5 convertase activates C5 by cleaving the α chain, releasing C5a anaphylatoxin and generating the α′ chain + β chain of C5. C5 anaphylatoxin is a mediator of local inflammatory processes. C5a also stimulates the locomotion of polymorphonuclear leukocytes and directs their migration toward sites of inflammation. C5 is similar to C3, C4, and α_2-macroglobulin: database code (human, precursor) CO5_HUMAN, 1676 amino acids (118.12 kDa).

C6, **C7**, **C8**, and **C9**, the sixth, seventh, eighth, and ninth components of complement. These four components, together with C5b, constitute the membrane attack complex; all are related and similar to **perforin**. C6 is a 6S β_2-globulin, and contains nine Cys-rich structural units, believed to be linked by disulfide bonds: database code (human, precursor) CO6_HUMAN, 934 amino acids (104.73 kDa). C7 is a 5.2S β_2-globulin, and serves as a membrane anchor: database code (human, precursor) CO7_HUMAN, 843 amino acids (93.41 kDa). C8 is a γ_1-globulin of M_r 150 000; its concentration in human serum is 10–20 mg L^{-1}. The molecule is a heterotrimer comprising the following subunits: α (human, precursor), database code CO8A_HUMAN, 583 amino acids (64.84 kDa); β (human, precursor), database code CO8B_HUMAN, 591 amino acids (66.87 kDa); and γ (human, precursor), database code CO8B_HUMAN, 202 amino acids (22.19 kDa). C9, a 4.4S α_2-globulin, forms transmembrane channels. It is cleaved by thrombin to form C9a and C9b. Database code (human, precursor) CO9_HUMAN, 559 amino acids (63.09 kDa).

Factor B, a zymogen of M_r 100 000, is activated by factor D, which selectively cleaves an Arg-|-Lys bond when factor B is complexed with C3b to form factor Bb. Following cleavage the C3b–Bb complex acts as **alternative-complement-pathway C3/C5 convertase** (EC 3.4.21.47), which can convert C3 to C3a and C3b by cleaving an Arg-|-Ser bond, and C5 to C5a and C5b by cleavage of an Arg-|-Xaa bond. Database code (human, precursor) CFAB_HUMAN, 739 amino acids (82.91 kDa).

Factor $\overline{\mathbf{D}}$, is a serine proteinase (EC 3.4.21.46) of M_r 24 000; its concentration in human plasma is approximately 1.5 mg L^{-1}. Database code (human, precursor) CFAD_HUMAN, 246 amino acids (26.15 kDa); residues 19–246 are factor D.

Factor H (human, precursor: database code CFAH_HUMAN, 1231 amino acids, 138.97 kDa) is a cofactor for **factor I** (EC 3.4.21.45), a serine proteinase that cleaves the α chains of C4b and C3b in the presence of the cofactors for C4-binding; human, precursor: database code CFAI_HUMAN, 583 amino acids (65.65 kDa).

Properdin, a glycoprotein of M_r 220 000, is composed of four, probably identical, noncovalently linked polypeptide chains.

Two alternative reaction cascades for the activation of complement exist. In the so-called **classical pathway**, activation (which requires both Ca^{2+} and Mg^{2+}) is initiated by the binding of the C1q moiety of C1 to the C_H2 regions of antibody molecules in antibody–antigen aggregates or in antibody bound to cells, thus causing C1 to be converted to a proteinase, $C\overline{1}$. $C\overline{1}$ activates C4 and C2 leading to the formation of **C3 convertase**, a complex proteinase, $C\overline{42}$, consisting of the activated components $C\overline{4}$ and $C\overline{2}$. $C\overline{42}$ splits C3 to the chemotactic and anaphylatoxic fragment C3a and the fragment $C\overline{3}$, C3b, which associates with $C\overline{42}$ to give a $C\overline{423}$ complex, **C5 convertase**. The latter is a complex proteinase that in turn splits C5 to $C\overline{5}$ and the chemotactic and anaphylatoxic fragment C5a. In the so-called **alternative pathway**, activation (which requires Mg^{2+} but not Ca^{2+}) is initiated by polysaccharides such as those present on bacterial and yeast cell walls. The proteinase $\overline{D}$ is analogous to $C\overline{1}$ but is normally present in serum, where it causes a continuous slow activation, which is greatly enhanced when polysaccharides are present. $\overline{D}$ acts on B and $C\overline{3}$ to produce $\overline{C3B}$, an alternative C3 convertase. $\overline{C3B}$, which is stabilized by properdin, acts on C3 to produce $(C\overline{3})_n\overline{B}$, an alternative C5 convertase. In both pathways, $C\overline{5}$ production is the last proteolytic step, and there follows a spontaneous association of $C\overline{5}$ with C6, C7, C8, and C9 to form a lytic complex, $C\overline{56789}$. *See also* **lysine(arginine) carboxypeptidase**.

complementarity **1** the interrelationship between one or more units supplementing, being dependent on, or being in a polar position to another unit or units. **2** a reverse correspondence of part of one molecule to part of another molecule, especially of antigen to antibody, enzyme to substrate, or the partners in a **base pair**.

complementary base sequence a sequence in a polynucleotide chain in which all the bases are able to form **base pairs** with a sequence of bases in another polynucleotide chain.

complementary catalysis catalysis in the presence of a prototype molecule, leading to the formation of products with a structure that is in some way complementary to that of the prototype.

complementary DNA *abbr*.: cDNA; a single-stranded DNA molecule that has a **complementary base sequence** to a molecule of a messenger RNA, from which it is produced by the action of **reverse transcriptase** and is used for **molecular cloning**, or as a molecular **probe** (def. 3) in hybridization tests for a specific sequence in cellular DNA. It has the advantage that, unlike genomic DNA, it contains no introns. It must be distinguished from the complementary strand of duplex DNA (*see* **noncoding strand**).

complementary genes any genes that interact to produce a qualitative effect that is distinct from the effects of any one of them separately.

complementary RNA *abbr*.: cRNA; a single-stranded RNA molecule that has a **complementary base sequence** to a specific (chromosomal) DNA sequence (of either strand), from which it is produced by the action of DNA-directed **RNA polymerase**. Naturally occurring cRNA may represent **pre-messenger RNA**, **ribosomal RNA**, or **transfer RNA**. *Compare* **antisense RNA**.

complementary strand (of duplex DNA) *see* **noncoding strand**.

complementation **1** any instance in which the interaction of two incomplete, nonfunctional systems results in a fully functional system. **2** the interaction between viral gene products or gene functions in cells that are multiply or mixedly infected with virus(es), leading to an increased yield of infective virus of one or more parental types. **3** **genetic complementation**.

complementation test any experimental test for **genetic complementation**. *See* ***cis–trans*** **complementation test**.

complement-3 convertase *an alternative name for* C3 convertase; *see* **complement**.

complement-5 convertase *an alternative name for* C5 convertase; *see* **complement**.

complement decay-accelerating factor *abbr*.: DAF; a factor that interferes with catalysis of the conversion of component C2 and factor B of **complement**, and prevents the formation of the amplification convertases of the complement cascade. DAF is expressed on the plasma membranes of all cell types that are in intimate contact with plasma complement proteins, and is a glycosylphosohatidylinositol-anchored protein. Human proteins DAF1 and DAF2 are formed by alternative splicing of a transcript of the same gene. Example (precursor of DAF1) from human: database code DAF1_HUMAN, 440 amino acids (48.66 kDa).

complement fixation activation of the **complement** system, characteristically *in vitro* by an antigen–antibody interaction, in which complement components C1 to C9 are activated sequentially.

complement-fixation inhibition test a serological test in which the presence of a substance of known structure either may or may not inhibit the interaction of antibody with complex antigen, and hence also the fixation of **complement**. *Compare* **complement-fixation test**.

complement-fixation test a serological test for the presence of antibody that can fix **complement** on interaction with antigen. The presence or absence of residual complement is detected by an indicator system, e.g. added sheep erythrocytes sensitized with an antibody to sheep erythrocytes. If complement is still present, lysis of the erythrocytes occurs, but if no complement remains, no lysis takes place.

complement receptor any cell-surface receptor that binds activated **complement** components, e.g. the receptor for C3b found on macrophages, neutrophils, and B lymphocytes.

complete Freund's adjuvant *see* **Freund's adjuvant**.

complex 1 any distinct chemical species in which two or more identical or nonidentical chemical species (ionic or uncharged) are associated, usually in stoichiometric proportions, and bound together by weaker interactions than covalent bonds. The term is used for various entities, e.g. coordination complexes, antigen–antibody complexes, enzyme–substrate or enzyme–inhibitor complexes, multienzyme complexes, and receptor–agonist complexes; it is often employed where the precise nature of the interaction is uncertain. 2 to make a complex or incorporate into a complex, especially of a metal ion and a **chelating agent**; to chelate.

complex I, II, III, IV the protein complexes, mainly cytochromes, that with associated ligands including ubiquinone, Fe–S clusters, FMN, FAD, and nicotinamide nucleotides form the mitochondrial electron-transport chain or **respiratory chain**. Each complex can be isolated from the others, and, if suitable substrates and electron acceptors are provided, can function independently in transporting electrons. Complexes I, III, and IV are also involved in proton translocation, a function that requires the complex to be embedded in an orientated environment such as that provided by the intact lipid bilayer of the mitochondrial inner membrane. Purification has led to the identification of the components of each complex. Complex I contains altogether ≈25 different polypeptide subunits including **NADH dehydrogenase (ubiquinone)**, together with FMN and a number of different iron–sulfur clusters containing non-heme iron in the forms Fe–S, Fe_2–S_2, and Fe_4–S_4, the iron of which undergoes oxidation–reduction between oxidation states II and III. The reaction catalysed is the oxidation of NADH by ubiquinone. Complex II contains four subunits of **succinate dehydrogenase (ubiquinone)**, FAD, and iron–sulfur. It catalyses the oxidation of succinate by ubiquinone. Complex III contains ≈10 polypeptide subunits including the four redox centres: cytochrome b_{562}, cytochrome b_{566}, cytochrome c_1, and an Fe_2–S_2 cluster. It catalyses the oxidation of ubiquinol by oxidized cytochrome *c*. The transfer of electrons to cytochrome c_1 involves the **protonmotive Q cycle**. Complex IV contains the 13 polypeptide subunits of cytochrome *c* oxidase, including cytochromes *a* and a_3, and Cu^{2+}. It catalyses the oxidation of reduced cytochrome *c* by dioxygen.

complexation the act or process of forming a (chelate) **complex**, e.g. to render inorganic substances soluble in organic solvents.

complex closed DNA a DNA structure comprising catenated and concatenated circular polynucleotide oligomers. *See* **catenane, concatemer**.

complex formation *see* **complex** (def. 2), **complexation**.

complexity *(in molecular biology)* the number of different sequences in DNA as defined by hybridization kinetics.

complexometric titration any titration in which the titrant contains a complexing (chelating) agent that interacts with a metal ion or other chemical species in the test solution, the end point being determined either electrometrically or with an indicator.

complexometry any method for the determination of a metal ion (or other chemical species) involving the formation of a chemical **complex**.

complexone any complex-forming or chelating agent, especially any macrocyclic antibiotic that complexes metal ions.

Complexone *a proprietary name for* **EDTA**.

compound any substance containing two or more identical or nonidentical chemical atoms in fixed numerical proportions and held together by one or more kinds of chemical bond.

compound A, B, C (etc.) *see* **Kendall's compounds**.

compression *see* **band compression**.

Compton, Arthur Holly (1892–1962), US physicist noted for his studies of cosmic rays and X-rays; Nobel Laureate in Physics (1927) 'for his discovery of the effect named after him' [prize shared with C. T. R. Wilson].

Compton effect *or* **Compton scattering** the scattering of photons by free particles that results in an increased photon wavelength (i.e. a decrease in photon energy) and a corresponding increase in the particles' velocity (i.e. an increase in the particles' kinetic energy).

computer-assisted tomography *abbr*.: CAT; *see* **tomography**.

computer of average transients *abbr*.: CAT; an electronic device of many separate channels in which a set of signals (e.g. an NMR spectrum) is repeatedly scanned and the signals stored additively. The stored signals grow linearly with the number of scans but the noise, being random, grows only as the square root of the number of scans taken, thereby improving the signal-to-noise ratio.

COMT *abbr. for* catechol *O*-methyltransferase, EC 2.1.1.6.

comutagen any nonmutagenic agent that enhances the effect of a mutagen.

comutation the occurrence of a mutation near to, and at the same time as, another specified mutation.

con A *abbr. for* concanavalin A.

conalbumin *or* **ovotransferrin** a protein of ≈77 kDa that comprises about 12% of the total solids of egg white. It is distinguishable from **ovalbumin** by its lower thermal coagulation point. It binds Fe^{3+} ions. The amino-acid sequence of conalbumin is probably identical to that of chicken serum **transferrin**, but their carbohydrate moieties are different. Conalbumin precursor, example from *Gallus gallus* (chicken): database code TRFE_CHICK, 705 amino acids (77.76 kDa).

conantokin G *also called* **sleeper peptide**; a toxic peptide from the venom of *Conus geographicus* (fish-hunting cone snail). It is an antagonist of brain **NMDA receptors**. It has the sequence

GEXXLQXNQXLIRXKSN,

where X = γ-carboxyglutamate (Gla). The five Gla residues bind Ca^{2+}.

conarachin a minor globulin occurring, along with **arachin** (def. 1), in seeds of the peanut, *Arachis hypogaea*.

conc. *abbr. for* concentrated.

concanavalin A *abbr*.: con A; an agglutinating and mitogenic protein (*see* **lectin**) that constitutes the major component of the subsidiary globulin fraction (originally named **concanavalin** because it accompanied **canavalin** in purification) obtained from the ripe seeds of the **jack bean**. It is separable from **canavalin** and from the minor component **concanavalin B** by differences in solubility. It is a crystallizable tetramer with 27.5 kDa subunits, dissociating to a dimer below pH 5.8. Each subunit contains one binding site for Mn^{2+} (or other transition metal ion), one for Ca^{2+}, and one for carbohydrate (specifically α-D-mannopyranoses or α-D-glucopyranoses with unmodified hydroxyl groups at C-3, C-4, and C-6); carbohydrate does not bind until the metal binding sites are filled. It agglutinates erythrocytes, and precipitates polysaccharides containing a sufficient number of appropriate terminal residues. It is mitogenic, transforming lymphocytes to blast cells, and can agglutinate some animal tumour cells and inhibit their growth.

The precursor – database code CONA_CANEN, 290 amino acids (31.49 kDa) – is processed in an unusual way: the mature chain consists of residues 164–281 followed by 30–148; to form a mature chain the precursor undergoes further post-translational modification after removal of the signal sequence; cleavage after Asn at positions 148, 163, and 281 is followed by transposition and ligation (by formation of a new peptide bond) of residues 164–281 and 30–148. The 3-D structure is known (database code NRL_3CNA).

concanavalin B a minor crystallizable protein obtained from ripe seeds of the **jack bean** and separable from **canavalin** and **concanavalin A**.

concatemer *or* **concatenate** originally, an intermediate structure in the biosynthesis of certain DNA viruses, consisting of a number of genomes connected together end to end, and thought to be the immediate precursor to the mature viral genome. The term is now extended to include any linear or circular DNA structure composed of viral genomes joined end to end. The term is also used for circular DNAs which are physically inseparable, one being threaded through the other. *Compare* **catenane**. —**concatemeric** *adj.*

concatenate **1** to join or link together, end to end. **2** joined or linked together. **3** *an alternative term for* **concatemer**. *Compare* **catenate**. —**concatenated** *adj.*; **concatenation** *n.*

concatenated dimer *or* **circular dimer** any circular **concatemer** containing two DNA molecules.

concentrate **1** to reduce the volume of a solution or dispersion by removal of solvent or dispersing medium. **2** to make or become purer or more concentrated by removal of impurities from a substance or material. **3** a concentrated solution or material, especially of foodstuffs, e.g. a vitamin concentrate. —**concentrating** *n.*

concentrated **1** *abbr.*: conc.; *(of a solution, dispersion, or mixture)* containing a relatively high concentration (of a specified solute, kind of dispersed particle, or component); (of substance, particles, etc.) present at relatively high concentration (in a solution, dispersion, etc.). *Compare* **dilute** (def. 1). **2** *(of a solution, dispersion or mixture)* increased in concentration and reduced in volume by removal of solvent, dispersing medium, or admixed material. *Compare* **diluted**.

concentration *abbr.*: concn.; **1 (a) mass concentration** *or* **mass density** *symbol*: ρ or γ; an SI derived physical quantity; it equals the mass of a specified component per unit volume of the system in which it is contained. It is expressed in kg m^{-3}. **(b) amount-of-substance concentration** *or* **amount concentration** *or (in clinical chemistry)* **substance concentration** *symbol*: c; an SI derived physical quantity; it equals the number of moles of a specified component per unit volume of the system in which it is contained. Where there is no risk of confusion the term **concentration** may be used alone. It is expressed in mol m^{-3} or mol dm^{-3}. Amount concentration continues to be expressed often in terms of the litre (*symbol*: l *or* L), which is not an SI unit, although 1 L = 1 dm^3. The older, non-SI, term *molarity* is still commonly used for this quantity; a solution of, say, 1 mol dm^{-3} concentration may thus be referred to also as a 1 molar solution or denoted a 1 M solution, where M is treated as a symbol for mol dm^{-3} (or mol L^{-1}). – *see* **molar** (def. 2). **(c) surface concentration** *symbol*: Γ; an SI derived physical quantity; it equals the number of moles of a specified substance per unit area of the surface at which it is present. It is expressed in mol m^{-2}. **(d) number concentration** *or* **number density** *symbol*: C *or (sometimes)* n; an SI derived physical quantity; it equals the number of specified particles or molecular entities per unit volume of the system in which it is contained. It is expressed in m^{-3}. *Note*: The concentration of a solute in solution may sometimes be expressed in more convenient units esp. in a percentage; e.g. (for solids) as the mass of solute in grams per 100 millilitres of solution (*abbr.*: % w/v), or (for liquids) as the number of volumes of solute per 100 volumes of solution (*abbr.*: % v/v). *See also* **activity** (def. 3), **molality**, **osmolality**, **osmolarity**. **2** the act or process of concentrating. *See* **concentrate** (def. 1, 2).

concentration equilibrium constant any **equilibrium constant** expressed in terms of concentrations rather than activities.

concentration gradient the change of concentration with distance, e.g. the change in concentration of a solute across a membrane or along a gradient.

concentration of enzymic activity *(obsolete) see* **enzymic activity**.

concentration polarization *(in membrane ultrafiltration)* the accumulation of retained macrosolutes on the membrane, forming a secondary membrane, that results in restriction of transport through the filter and alteration of selectivity, which may lead to rejection of normally permeating species.

concentration quenching the **quenching** of fluorescence that becomes significant when the fluor concentration is increased beyond a certain point. It is of importance, e.g., in liquid scintillation counting and in experiments involving the introduction of fluorescent groups (probes) into macromolecules.

concentration ratio *or* **dose ratio** *symbol*: r; the ratio of the concentration of an agonist that produces a specified response (often but not necessarily 50% of the maximal response to that agonist in an assay) in the presence of an antagonist, to the agonist concentration that produces the same response in the absence of antagonist.

concentric cylinder viscometer *an alternative name for* **Couette viscometer**.

concerted reaction any chemical reaction in which two chemical processes occur within the same reaction step; i.e. one in which a new chemical bond is formed at the same time as, and as a direct result of, the breaking of another chemical bond.

concerted symmetry model a model of ligand binding to oligomeric proteins in which the characteristic assumption is that molecular symmetry of the **oligomer** is conserved so that only two states are accessible to it: either with all the **protomers** in the first conformational state or all the protomers in the second state. *See* **Monod–Wyman–Changeux model**.

concerted transition the simultaneous transition of all identical subunits (i.e. protomers) of an oligomeric molecule from the first conformational state to the second state and vice versa, as is required in any **concerted symmetry model** for allosteric proteins.

concn. *abbr. for* concentration.

concordance the condition in which both members of a twin pair demonstrate the same phenotype or trait.

condensate any product of a **condensation**, especially a liquid from a gas or vapour.

condensation **1** the act or process of condensing; the state of being condensed. **2** the product of condensing; a condensed mass. **3** any (usually stepwise) chemical reaction in which two or more identical or nonidentical molecular entities, or two parts of one entity, combine with the elimination of a molecule of water, ammonia, ethanol, hydrogen sulfide, or other simple substance. The combining entities or parts each (formally) contribute a moiety to the molecule eliminated.

condensation polymerization a polymerization by repeated condensation, i.e. with elimination of some simple substance, to form a **condensation polymer**.

condense **1** to make or to be made more compact or denser. **2** to change or cause to change a gas or vapour to a liquid or solid. **3** (of a compound) to undergo or cause to undergo a **condensation** (def. 3).

condensed **1** converted to a dense or denser form; e.g. **condensed DNA** in **nucleosomes** or in the early stages of maturation of T-even phages; or the **condensed form of mitochondria** in which the cristae are swollen, the matrix volume is much reduced, and the inner membrane space is increased. **2** reduced from the gaseous to the liquid state.

condenser **1** any device for converting a gas or vapour into a liquid (or solids) by cooling. **2** a lens or set of lenses that concentrates light into a small area. 3 *a former name for* capacitor; an electrical component that stores electric charge.

condensing enzyme 1 citrate condensing enzyme, EC 4.1.3.7; *see* **citrate (*si*)-synthase.** 2 β-oxoacyl-ACP synthase, EC 2.3.1.41; 3-oxoacyl-[acyl-carrier-protein]synthase (*see* **fatty acid synthase complex**).

conditional association constant *or* **conditional stability constant** an **association constant** determined in conditions similar to those in which the reaction under consideration is taking place, e.g. in the internal milieu of a cell.

conditional lethal mutation any mutation that produces a mutant (**conditional lethal mutant**) whose viability depends on the conditions of growth. It grows normally in permissive conditions but in restrictive conditions it does not grow, thereby expressing its lethal mutation. *See*, e.g., **temperature-sensitive mutation.**

conduct to allow the passage of electricity, heat, or electromagnetic radiation.

conductance *see* **electric conductance.**

conductance water *an alternative name for* **conductivity water.**

conductimetric *or* **conductiometric** *a variant spelling of* conductometric; *see* **conductometry.**

conductimetry *or* **conductiometry** *a variant spelling of* **conductometry.**

conduction 1 the transfer of heat or electricity through a material. 2 the transmission of an impulse, either electrical or chemical, along a nerve fibre.

conductivity 1 **electrical conductivity** *symbol*: κ; a measure of the ability of a material to conduct an electric current or impulse. It is defined as the reciprocal of the resistivity of the material, and is expressed in siemens m^{-1}. *See also* **electrolytic conductivity.** 2 **thermal conductivity.**

conductivity water *or* **conductance water** very pure water, of electrolytic conductivity $<1 \times 10^{-4}$ siemens m^{-1}.

conductometric titration any titration in which the end point is indicated by an inflection in the curve of conductivity against volume of titrant added. For instance when an HCl solution is titrated with an NaOH solution, the conductivity decreases as the faster moving H_3O^+ ions are replaced by slower moving Na^+ ions; after the equivalence point the conductivity increases as the concentration of the fast-moving OH^- ions increases.

conductometry *or* **conductimetry** *or* **conductiometry** any method of chemical analysis based on measurements of electrical conductivity. —**conductometric** *adj.*

conductor any substance or system that allows the passage of electricity, heat, or electromagnetic radiation.

cone cell *or* **cone** one of the two types of light-sensitive cell of the vertebrate retina (*compare* **rod cell**). Cone cells are conical or flask-shaped, and contain the photopigment **iodopsin** or (in freshwater and migratory fishes and some amphibians) **cyanopsin.** They are responsible for photopic (daylight) vision; and are densest in the fovea and absent from the margin of the retina. Human cone cells are thought to be of three types, each containing an iodopsin of different spectral sensitivity; hence they are responsible also for colour vision.

confidence limits *(in statistics)* the upper and lower limits of the range of values within which the true mean is expected to lie, on the basis of an estimate of the mean obtained from a sample of values. The confidence limits vary according to the (specified) probability that they will be correct.

configuration classically, the arrangement of the atoms of a molecule in space, without regard to arrangements that differ only as after rotation about one or more single bonds; the term is now sometimes limited so that no regard is paid to π-bonds or to bonds of a partial order between one and two, and sometimes further limited so that no regard is paid to rotation about bonds of any order, including double bonds. Interconversion of molecules of different configurations requires the formal making and breaking of covalent bonds, no account being taken of changes to hydrogen bonds. *Compare* **conformation.** *See also* **absolute configuration, pyrimidine dimer.** —**configurational** *adj.*

configurational base unit (of a polymer) a **constitutional base unit** of the polymer whose configuration is defined at the minimum at one site of steric isomerism in the main chain of the polymer.

configurational block a **block** in a polymer containing only one species of **configurational base unit.**

configurational inversion *see* **racemize, Walden inversion.**

configurational isomer any of two or more isomers that differ only in stereochemical **configuration.**

confluence *(in cell culture)* the state reached when one or more cells or groups of cells, grown in culture on (so-called) solid medium, have multiplied to cover the surface of the medium completely but without overlying each other. —**confluent** *adj.*

confocal microscope a microscope that produces a series of images, often called optical sections, that have much better resolution than images of the same specimens obtained using conventional epifluoresence light microscopy. Such a microscope depends on laser light focused through a pinhole onto a single point in the specimen. From a succession of such images a 3-D image can be created. This is primarily used in biological imaging for mapping the distribution of fluorescently labelled macromolecules, such as antibodies, in various tissues.

conformation the various arrangements of the atoms of a molecule (of defined **configuration**) in space that differ only as after rotation about single bonds; the term is now sometimes extended to include rotation about π-bonds or bonds of partial order between one and two, and sometimes further extended to include also rotation about bonds of any order, including double bonds. Interconversion of molecules of different conformations does not involve the breaking and making of chemical bonds (except hydrogen bonds) or changes in **chirality.** The following special terms and symbols distinguish different possible conformations of various classes or portions of molecules.

(1) **Cyclic compounds.** Nonplanar six-membered rings may exist in **boat conformation, chair conformation, half-boat conformation, half-chair conformation, skew conformation** (def. 1), or **twist conformation** (def. 1); and nonplanar five-membered rings may exist in **envelope conformation** or **twist conformation** (def. 2). For a six-membered or five-membered ring form of a monosaccharide or monosaccharide derivative these conformations may be designated by the following conformational descriptors suffixed to their chemical names: *-B*, boat; *-C*, chair; *-E*, envelope; *-H*, half-chair; *-S*, skew; *-T*, twist; the locants of the ring atoms that lie above and/or below the reference plane, when the numbering appears clockwise, may be added to these symbols as left superscripts and/or right subscripts, respectively. A bond to a tetrahedral atom in a six-membered ring is termed **axial** (*abbr.*: a) if it or its projection makes a large angle with the plane containing a majority of the ring atoms, and **equatorial** (*abbr.*: e) if this angle is small; the same terms are applied to atoms or groups attached to such bonds. The corresponding terms for a bond from an atom directly attached to a doubly bonded atom in a monounsaturated six-membered ring are **pseudoaxial** (*abbr.*: a′) and **pseudoequatorial** (*abbr.*: e′), respectively.

(2) **Nucleotides.** When atom C-2′ or C-3′ of the ribo- or deoxyribofuranose ring of a nucleotide lies out of the plane of the other four ring atoms, the ring conformation is designated by the descriptor *endo-* if either atom lies above the ring, i.e. towards the nitrogenous base, and by the descriptor *exo-* if it lies below the ring, i.e. away from the base. In addition the base and sugar rings may be in either of two relative positions: the conformation is said to be **anticlinal** (*abbr.*: *anti-*) if the groups at positions 2 and 3 of a pyrimidine residue or positions 1, 2, and 6 of a purine residue lie away from the sugar ring, and **synclinal** (*abbr.*: *syn-*) if these groups lie over the sugar ring.

(3) **Torsion angle**. If rotation about a given bond in a molecule is restricted for any reason, the conformation of atoms or groups about that bond may be described by the **torsion angle** (*symbol*: θ). When, in a **Newman projection**, two atoms or groups attached at opposite ends of a bond appear one more or less directly behind the other, i.e. $\theta \approx 0°$, these atoms or groups are termed **eclipsed**, and that portion of the molecule is described as being in the eclipsed conformation. If not eclipsed (in an ideal case, when all torsion angles equal 60°), the atoms or groups are termed **staggered**, and the molecule is described as being in the staggered conformation; such a conformation is termed **synperiplanar** (*abbr.*: *sp*) (or sometimes *cis* (*abbr.*: c)), **synclinal** (*abbr.*: *sc*) (or sometimes *gauche* (*abbr.*: g) or *skew*), **anticlinal** (*abbr.*: *ac*), or **antiperiplanar** (*abbr.*: *ap*) (or sometimes *trans* (*abbr.*: t)) according as the torsion angle is, respectively, within ±30°, ±60°, ±120°, or ±180° of 0°, the torsion angle being defined according to specified criteria. (*Note*: Because the terms *cis* and *trans* have other meanings, their use in this context is not recommended.) —**conformational** *adj.*

conformational analysis the application of a set of specific rules to determine the conformation of a molecule in which the strain will be minimal, i.e. the most stable and preferred conformation of the molecule.

conformational change 1 any change in conformation 2 any change in the three-dimensional structure of a macromolecule, membrane, or other entity.

conformational-coupling hypothesis a hypothesis (now superseded) that the energy yielded by the processes of electron transport, e.g. in mitochondria, is conserved in the form of a conformational change in an electron carrier protein or in the coupling factor (F_1 ATPase) molecule. The high-energy conformational state results from an energy-dependent shift in the number and location of weak bonds that maintain the three-dimensional conformation of the protein. The energy inherent in this 'energized' conformation is thought to be used to cause the formation of ATP from ADP and inorganic phosphate. This hypothesis was put forward after attempts to find a high-energy intermediate that coupled electron transport to oxidative phosphorylation, failed. It has been superseded by the **chemiosmotic coupling hypothesis**.

conformational formula a diagrammatic representation on a plane surface of the structural formula of a molecule that shows the relative dispositions in space of the constituent atoms or groups. Such representations are used especially to depict instances of conformational isomerism. For a monosaccharide or monosaccharide derivative, the conformational formula may be combined with a **Haworth representation**. *See also* **Newman projection**.

conformational isomer *or (sometimes)* **conformer** any of two or more isomers that differ only in their stereochemical **conformation**.

conformational map *an alternative name for* **Ramachandran plot**.

conformer 1 a molecule in a conformation into which its atoms return spontaneously after small displacements. 2 *an alternative term for* **conformational isomer**.

conformon a quantized energy package that, according to the **conformational-coupling hypothesis**, is associated with a localized conformational change in a protein in the mitochondrial membrane.

congener 1 any member of a (specified) group. 2 *(in taxonomy)* any member of a specified genus. 3 *(in chemistry)* **a** a chemical substance related to another, e.g. as a derivative or as an element of the same group of the periodic table. **b** a secondary product retained in an alcoholic beverage (or extract of a natural product) that contributes significantly to the final characteristics of the beverage (or the extract).

congenic of, or pertaining to, a **congener**; **homogenic**.

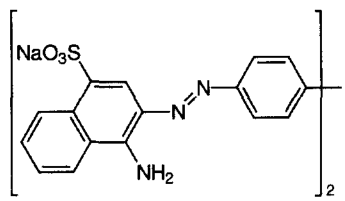
congenital describing a character or trait (phenotype) that exists at or dates from birth. It neither includes all inherited conditions nor excludes characters resulting from environmental causes.

congenital adrenal hyperplasia a syndrome embracing a group of inherited enzyme deficiencies of adrenal steroid hormone biosynthesis. A deficiency of 21-hydroxylase (steroid 21-monooxygenase, EC 1.14.99.10) accounts for the great majority of cases; it leads to failure to synthesize cortisol and results in overproduction of ACTH (due to release from negative feedback by cortisol) and consequently there is hyperplasia of the adrenal and overproduction of 17-hydroxyprogesterone and androstene 3,17-dione by overstimulated alternative pathways. Deficiency (usually partial) of 11β-hydroxylase (steroid 11β-monooxygenase, EC 1.14.15.4) accounts for most other cases; in this condition, 11-deoxycorticosterone accumulates. Very rarely, deficiency of 17-hydroxylase (steroid 17α-monooxygenase, EC 1.14.99.9) or 18-hydroxylase (corticosterone 18-monooxygenase, EC 1.14.15.5) may occur.

conglutination the agglutination of bacterial or red blood cells, that have been sensitized with specific antibody, in the presence of (nonhemolytic) **complement** and **conglutinin**.

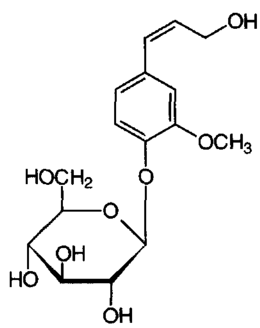
conglutinin a nonimmunoglobulin protein, present in the plasma of cattle and other bovids, that aggregates **complement**-coated immune complexes or cells.

Congo Red sodium diphenyldiazo-bis-α-naphthylaminesulfonate; CI 22120; Direct Red 28; an indicator dye that is blue-violet at pH 3.0 and red at pH 5.0. It is used for detecting and estimating free HCl in gastric contents, as an addition to culture media, and for staining **amyloid**.



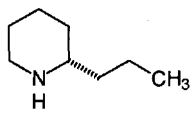
Congo Red fibrin a preparation obtained by heating fibrin to 80 °C in a slightly alkaline Congo Red solution and thoroughly washing. It was formerly used in the detection of proteolytic activity since the dye does not dissolve in aqueous solutions unless the fibrin is digested.

coniferin 4-(3-hydroxy-1-propenyl)-2-methoxyphenyl β-D-glucopyranoside; the major **glycoside** present in conifers and some other plants.



coniferyl alcohol the aglycon of **coniferin**; it occurs as a building unit of plant **lignins**.

coniine 2-propylpiperidine; the toxic alkaloid of poison hemlock, *Conium maculatum*. The natural enantiomer has *S* configuration.



conjugase γ-Glu-X carboxypeptidase, EC 3.4.19.9 (formerly EC 3.4.22.12); γ-glutamyl hydrolase; an enzyme that hydrolyses γ-glutamyl bonds in *N*-pteroyl-γ-oligoglutamate, pteroyl-γ-diglutamate being a prominent product.

conjugate 1 to effect or undergo **conjugation**. 2 describing either of a pair of interconvertible substances, e.g. a **conjugate acid–base pair**. 3 the product formed as a result of conjugation.

conjugate acid–base pair the relationship between two chemical species, BH^+ and B, that are related by the reaction: $BH^+ = B + H^+$. Then BH^+ is the **conjugate acid** of the base B, B is the **conjugate base** of the acid BH^+, and BH^+ and B are a conjugate acid–base pair.

conjugate base *see* **conjugate acid–base pair**.

conjugated 1 describing a compound that is formed by the linking of two compounds. 2 displaying or having undergone **conjugation**.

conjugated double bond any of two or more double bonds in a molecule where each double bond is separated from the next one by one single bond. *Compare* **conjugation** (def. 2).

conjugated protein any protein that contains a nonprotein component, often in stoichiometric proportion. The nonprotein component may be a metal ion, a lipid, a carbohydrate, or a nucleic acid, and may be either loosely or tightly bound to the polypeptide chain(s).

conjugation 1 the act of joining together; the state of being joined together. 2 an alternating sequence of multiple and single chemical bonds containing at least two multiple bonds with delocalization of pi-electrons and resultant additional chemical stability. 3 the covalent or noncovalent joining together of one (larger) molecule, e.g. a protein or bile acid, with a second (smaller) molecule. 4 a process of sexual reproduction occurring in various types of unicellular organisms. In bacteria, e.g. *Escherichia coli*, it involves the transfer of DNA from a donor cell to a recipient cell via a sex **pilus**; in protozoa, e.g. *Paramecium aurelia*, a true exchange of DNA occurs between the participating cells, which belong to different mating types.

conjugation labelling a procedure for introducing a **label** into a large molecule of interest by covalently coupling it, in a specific chemical reaction, to a small molecule containing the label (*see* **conjugation** (def. 3)). It is particularly useful for labelling with radioiodine any protein or peptide that is sensitive to oxidative procedures or to noxious components of commercial radioiodine, that lacks tyrosine residues reactive to iodine, or that requires to be labelled at a residue other than tyrosine. *See also* **Bolton and Hunter reagent**.

conjugative plasmid any **plasmid** that can bring about the transfer of DNA by **conjugation** (def. 3).

conjugon any genetic element that is required for bacterial **conjugation** (def. 4), e.g. **fertility factor**.

connective tissue any supporting tissue that lies between other tissues and consists of cells embedded in a relatively large amount of extracellular matrix.

connective tissue growth factor *abbr*.: CTGF; a growth factor of testis, spleen, kidney, lung, heart, and brain, that is similar to **CEF10**; one of a family of growth regulators that belong to a group of immediate-early genes expressed after induction by growth factors or certain oncogenes. The family includes **Cyr61**, βIG-M1, and βIG-M2. Example (precursor) from mouse: database code CTGF_MOUSE, 348 amino acids (37.75 kDa).

connectivity *(in chemistry)* the information in any molecular formula or model regarding the order in which the constituent atoms of the molecule are linked, irrespective of the nature of the linkages.

connexin the main protein component of a **connexon**. Each connexin contains four putative membrane-spanning α-helices, and six connexins make up each connexon. A number of different subtypes of connexin exist within each species, contributing different functional behaviour to different connexons. Example from human: database code CX26_HUMAN, 208 amino acids (24.27 kDa); four motifs.

connexon the structural subunit of a **gap junction**, the structure that forms a bridge between adjacent cells in certain tissues of vertebrates. Seen by electron microscopy or low-angle X-ray diffraction, an individual gap junction consists of a number ($10–10^4$) of connexons, often in hexagonal array, embedded in and protruding from either side of each of the opposed cell membranes in register and linked to one another. Each connexon is a cylindrical proteinaceous structure, about 6–8.5 nm in diameter and 7.5 nm long; it consists of six rod-shaped, essentially rigid subunits (**connexins**) of about 25 kDa arranged in an annulus. The diameter of the central opening can apparently be varied by radial displacement of the subunits at their cytoplasmic extremities. Each connexin spans one cell membrane, two with connecting ends being required to form a channel across both membranes. Such channels provide a regulable hydrophilic pathway permitting the passage of small molecules (up to about 800 Da) between adjacent cells; the specificity of each channel is determined by the type of connexin it utilizes.

conotoxin *abbr*.: ω-CT; any of several peptides of the family of ω-conotoxins isolated from the venoms of two marine snails, *Conus geographicus* and *C. magus*. All ω-conotoxins have a conserved pattern of cysteine residues linked by four disulfide bridges. They are neurotoxins that inhibit voltage-gated Ca^{2+} channels and neurotransmitter release. ω-Conotoxin GVIA has the structure:

[tricyclic (1→16, 8→19, 15→26)(H-Cys^1-Lys-Ser-Hyp-Gly-Ser-Ser-Cys^8-Ser-Hyp-Thr-Ser-Tyr-Asn-Cys^{15}-Cys^{16}-Arg-Ser-Cys^{19}-Asn-Hyp-Tyr-Thr-Lys-Arg-Cys^{26}-Tyr-NH_2)].

consensus sequence an idealized sequence of nucleotides, or their constituent bases, or amino acids, base, or amino acid that represents the nucleotide most likely to occur at each position in the sequence. Consensus sequences are used to identify RNA splicing sites, other sites, plasmids, and families of proteins.

conservation the retention of structure by a macromolecule, or by a specified segment of one, with variation of circumstance (environmental, genetic, etc.). When used of primary structure, it can be synonymous with **sequence homology**. The degree of retention of structure is usually specified. *See also* **conserved**.

conservation of energy (law of the) *see* **thermodynamics**.

conservative base change *or* **conservative base substitution** any mutational change, including substitution of a base, in a particular base triplet in a DNA molecule such that either the amino acid encoded by that base triplet is not altered, or there is no major change in the properties of the R group of the amino acids involved, e.g. Glu for Asp. *See* **genetic code**, **wobble hypothesis**.

conservative replacement *or* **conservative substitution** any replacement or substitution in a polypeptide chain of a particular amino acid residue by another one with similar properties, e.g. Arg for Lys, Phe for Tyr, Glu for Asp.

conserved describing a tendency to invariance in corresponding residues or sequences of residues of encoded macromolecules (e.g. proteins) obtained from specimens of genetically different sources. Macromolecules showing a high degree of invariance of their primary structure are said to be highly conserved; ones exhibiting considerable variation are described as poorly conserved. *See also* **conservation**, **homologous**.

constant region the region of a chain of an **immunoglobulin** molecule that is characterized by invariability of the amino-acid sequence from molecule to molecule, apart from certain residues at allotypic marker sites. The constant region of the light chain (corresponding to the C-terminal part) is designated C_L, while the constant region of the heavy chain is designated C_H. Within a constant region, individual domains may be further designated C_H1, C_H2, etc. In a more precise designation of a heavy chain the 'H' subscript is replaced by the Greek letter corresponding to the type of heavy chain that defines the particular class of immunoglobulins. Hence, the constant region of an **immunoglobulin G**, in which the heavy chain is designated γ, can be designated $C_\gamma1$, $C_\gamma2$, etc.

constitutional base unit the smallest possible regularly repeated unit (i.e. identical group of monomers) of a regular **polymer**.

constitutional block a **block** of a polymer that contains only one species of **constitutional base unit**.

constitutional isomer *or (formerly)* **structural isomer** any of two or more **isomers** that differ in their molecular constitution, i.e. in the nature and sequence of bonding of their atoms rather than in the arrangement of their atoms in space. For example, ethanol, $CH_3–CH_2–OH$, and dimethyl ether, $CH_3–O–CH_3$, are constitutional isomers. *Compare* **stereoisomer**.

constitutive enzyme any enzyme that is formed at a constant rate and in constant amount in a given cell, regardless of the metabolic state of the cell or organism. *Compare* **induced enzyme**.

constitutive genes genes that are expressed following interaction between a promoter and RNA polymerase without additional regulation. *Also known as* **housekeeping genes**, they are expressed in virtually all cells since they are essential for the cell's activity.

constitutive mutation any mutation that results in an increased constitutive synthesis (i.e. synthesis as if by a **constitutive gene**) by a prokaryote of several functionally related, inducible enzymes. In such mutations either an operator gene is modified so that the repressor cannot combine with it or the regulator gene is modified so that the repressor is not formed.

contact amino acid any of the amino-acid residues in the **active site** (def. 1) of an enzyme that are at some stage within one bond distance of some point on the substrate molecule in the enzyme–substrate complex. They may include both catalytic residues and specificity residues. *Compare* **auxiliary amino acid**.

contact hypersensitivity a form of immunological **hypersensitivity**, immediate or delayed, that is provoked in the skin or mucous surfaces by contact with specific agents acting as either antigens or haptens.

contact inhibition a phenomenon, seen in cultures of normal eukaryotic cells, in which cell division is inhibited by cell-to-cell contact. When one cell contacts another cell as a result of increased cell density, it first ceases to move, then metabolizes more slowly and ceases to divide. An outstanding feature of malignant cells is their loss of contact inhibition so that they multiply unchecked. *See also* **transformation** (def. 2).

contact residue *see* **contact amino acid**.

contact sensitivity the ability of an organism to respond to a biocide, e.g. an insecticide or a selective herbicide, applied to the integument.

containment the use of any structural or procedural safeguards to minimize the possible risks, to the experimenter or to the environment, in hazardous or potentially hazardous operations, especially genetic manipulations or the use of ionizing radiation.

contamination the presence of any contaminant; the act of introducing a contaminant; pollution; the presence of harmful or invasive microorganisms in a population of desirable ones. *Compare* **decontamination**.

context 1 any parts of a piece of writing or speech that precede or follow a specified word or passage and contribute to its full meaning. **2** *(in mycology)* the layers that develop between the hymenium and the true mycelium in certain fungi.

context effect any effect of nucleotide residues outside a codon in an mRNA molecule in influencing the efficiency of the reading of that codon by the appropriate tRNA molecule. *Compare* **context** (def. 1).

context mutation any mutation that brings about an alteration in the **context effect** of nucleotide residues outside a particular **codon** in an mRNA molecule.

contig one of a set of overlapping clones that represent a continuous region of DNA. Each contig is a genomic clone, usually in a **cosmid** or a **YAC**. They are used in **contig mapping**.

contig mapping a technique used in projects such as the Human Genome Project to enable the physical map of (part of) a chromosome to be determined. The method relies on the use of overlapping clones, referred to as **contigs**.

contiguous gene syndrome any of the complex disease phenotypes associated with a chromosomal microdeletion affecting multiple, unrelated genetic loci physically contiguous on a chromosome.

continuous culture any cell culture in which the cells are in the growth phase over a prolonged period of time. *See also* **chemostat**.

continuous distribution any nonquantized distribution; e.g. in a large population one measurement, x, may be considered to vary continuously, and the probability, y, of the occurrence of a particular value of x is given by $y = f(x)$.

continuous epitope an immunological determinant (i.e. **epitope**) on an antigen that is formed from a single linear stretch of amino acids. It is much less common than a **discontinuous epitope**.

continuous-flow centrifugation any preparative centrifugation technique, used for collecting materials from relatively large volumes of liquid, in which a liquid is continuously fed into a rotor, the sediment is collected, and the supernatant is withdrawn continuously as the rotor rotates. *See* **centrifugal elutriation**, **Sharples supercentrifuge**, **zonal centrifugation**.

continuous-flow electrophoresis a method of electrophoresis in which the protein (or other) solution is applied continuously to an inert support such as paper (then usually called **curtain electrophoresis**) or glass beads or is allowed to flow between two closely spaced parallel plates. The solution flows down the support by gravity and an electric potential applied at right angles causes migration of components away from the main stream. Fractions are collected at different positions across the bottom of the support.

continuous-flow technique a technique for following rapid chemical reactions in which two solutions, containing the reactants, are rapidly mixed and the mixture pumped at a predetermined speed through a tube of known dimensions that contains an observation chamber. By changing the rate of flow, the time between mixing and observation can be varied. Thus, given a method of analysis for reactants or products in the observation chamber, the extent of reaction as a function of time can be determined.

continuous process a method of cultivation in which nutrients are supplied and components of the culture medium are removed continuously at volumetrically equal rates, maintaining the cells in a condition of stable multiplication and constant growth rate.

continuous variation *or* **Job's method** a method for investigating metal–ligand (or other) equilibria in solution in which the mole ratio of, e.g., metal ion to ligand species is varied while the sum of their concentrations is kept constant and some property that varies in a fashion proportional to the concentration of one of the molecular entities present, e.g. absorbance or conductivity, is measured. The curve of the magnitude of the property observed versus the concentration ratio gives information about the stoichiometry of the metal–ligand (or other) complex formed and allows the stability (association) constant to be calculated.

contractile vacuole any of the small spherical vesicles, found in the cytoplasm of freshwater protozoans, that function to expel surplus water from the organism.

contrapsin a glycoprotein plasma serine protease inhibitor, similar to the **serpin**-type inhibitors. It specifically inactivates serine proteases of the trypsin class but not of the chymotrypsin class. It has 64% homology with human α_1-antichymotrypsin. Example (precursor) from *Mus musculus*: database code COTR_MOUSE, 418 amino acids (46.83 kDa).

contributing amino acid (in an enzyme) any amino-acid residue whose only function appears to be that of a foundation for the **contact amino acid** and **auxiliary amino acid** residues in the **active site**. They are required in the natural enzyme to maintain the framework to which the contact and auxiliary amino acids are attached.

contributing structure *or (formerly)* **canonical form** any one of two or more formal valence bond representations of a mesomeric molecular structure. *Compare* **mesomerism**.

control 1 any object or system in an experiment that is a standard of comparison. A control is prepared or carried out exactly as the other parts of an experiment but differs in respect of a single variable, allowing the significance of the latter to be assessed. *Compare* **experimental** (def. 3). 2 the regulation of a biochemical process such as a metabolic pathway or the expression of a gene.

controllability coefficient *former name for* **elasticity coefficient**.

controlled pore glass a type of glass in the form of rigid particles honeycombed with a great number of pores of selected, precisely defined size. It is useful as a supporting matrix in a large variety of chromatographic techniques, including affinity, gel-permeation, and ion-exchange methods.

control point the point – corresponding to a particular enzyme – in a metabolic pathway or other biochemical sequence where control is exercised.

control region *or* **control sequence** the part of a DNA molecule containing the **promoter** and the **operator** region of an **operon**.

control strength *former name for* **flux control coefficient**.

convergence *see* **convergent evolution**.

convergent evolution *or* **convergence** evolution such as to produce an increasing similarity in some characteristic(s) between initially different groups of, e.g., organisms or **gene products**. *Compare* **divergent evolution**.

convertin *an alternative name for* factor VIIa (*see* **blood coagulation**).

convertin enzyme (factor) *see* **serum converting enzyme**.

convolution 1 a twisting together, a coil. 2 a folding, such as those that produce the irregular ridges on the surface of the cerebrum of the brain.

Conway microdiffusion method a microchemical analytical method applicable to substances from which a gas, e.g. ammonia or carbon dioxide, can be liberated quantitatively by a specific reagent or enzyme; the method can also be used to assay an enzyme. The technique employs a special microdiffusion dish, variously termed a **Conway cell**, **Conway dish**, **Conway unit**, or **Conway vessel**. This consists of a shallow circular glass vessel, 40–70 mm in diameter, containing a central well and with a closely fitting lid sealed with grease; the lid closes the outer annular space but does not touch the wall of the central well. The sample is placed in the outer annulus, and a gas-absorbing reagent is placed in the centre well. The specific reagent is added to the sample, the lid is closed, and diffusion of the evolved gas allowed to proceed. Subsequently the remaining gas-absorbing reagent is titrated. [After Edward Joseph Conway (1894–1968), Irish biochemist.]

Coombs's test *an alternative name for* **antiglobulin test**. [After Robert Royston Amos Coombs (1921–), British immunologist.]

cooperative acting together or with others. *See also* **cooperativity**.

cooperative affinity the state of increased potency, receptor affinity, and resistance to destruction that is exhibited by agonist molecules covalently attached to a macromolecule in any situation where ligand and acceptor possess more than one mutual recognition or binding site, deployed in such a manner that simultaneous interaction is stereochemically feasible.

cooperative binding *see* **cooperative ligand binding**.

cooperative feedback inhibition the **feedback inhibition** of an enzyme by two or more end products of a reaction sequence such that the degree of inhibition caused by the mixture is greater than the sum of the effects of the individual end products.

cooperative hydrogen bonding the interaction between adjacent hydrogen bonds in a molecule such that the energy required to form them is less than the sum of the energies required to form the individual bonds, and the energy required to break these bonds is greater than the sum of the energies required to break the individual bonds. Such an interaction increases the stability of the molecular conformation in which it occurs.

cooperative ligand binding the binding of ligands to interacting sites on a macromolecule such that the binding of one ligand molecule to one site either increases (**positive cooperativity**) or decreases (**negative cooperativity**) the affinity of the other site(s) for ligand(s). Such effects may be either **homotropic**, i.e. one ligand species affects the binding of the same ligand species, or **heterotropic** (def. 1), i.e. one ligand species affects the binding of a different ligand species.

cooperativity *(in biochemistry)* any **cooperative** phenomenon, especially **cooperative ligand binding** or the situation in an enzyme that has more than one binding site for its substrate or allosteric effector where the binding of one molecule to the enzyme alters the affinity with which the others are bound.

cooperativity index *see* **saturation ratio**.

coordinate 1 to integrate or classify diverse elements or things. 2 *(in chemistry)* to be or become linked by **coordination** (def. 2). 3 *(in mathematics)* any of a set of values that defines the position of a point in space with reference to a system of axes.

coordinate bond *or* **coordinate covalence** *or* **coordinate link** *an alternative term for* **dipolar bond**.

coordinated enzymes enzymes whose rates of synthesis vary together; they are controlled by cistrons of the same **operon**.

coordinated enzyme synthesis the synthesis of enzymes that are controlled by the genes of one operon and are thus induced or repressed together.

coordinate induction/repression the simultaneous induction/repression of a number of enzymes that catalyse a sequence of either consecutive or related reactions and are controlled by scattered genes.

coordination 1 the balanced and effective interaction of processes. 2 the formation or existence of a covalent bond whose pair of electrons is regarded as originating from only one of the two parts of the molecular entity linked by it; linking by means of such a bond or bonds. *Compare* **colligation**. *See also* **dipolar bond**, **pi adduct**

coordination number *or* **ligancy** the number of other atoms directly linked to a specified atom in a chemical species.

cooxidation *an alternative term for* **co-metabolism**.

COP *abbr. for* coat protein; a protein subunit of a **coatomer**.

copeptin *see* **vasopressin–neurophysin 2–copeptin precursor**.

copigmentation the chemical association between a pigment molecule and another colourless molecule that results in altered absorption characteristics of the pigment and consequently a change in its shade or colour. Some flower petal colours are changed by hydrogen bonding between an anthocyanin pigment and a colourless flavonoid. For example, the anthocyanin in both purple and crimson roses is cyanidin 3,5-diglucoside, but in crimson flowers it is copigmented with large amounts of gallotannin. This results in a spectral shift of the absorption maximum by 5 nm to longer wavelengths.

copolymer the product of the polymerization of a mixture of more than one species of **monomer**.

copolymerization *or* **copolymerisation** the polymerization of a mixture of two or more species of **monomer**.

copper-chelate chromatography a type of **metal-chelate affinity chromatography** employing Cu^{2+} ions.

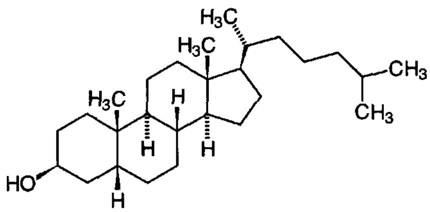
copro+ *or (before a vowel)* **copr+** *comb. form* indicating feces.

coprophagy the act of feeding on feces.

coproporphyrinogen III an intermediate in the biosynthesis of **metalloporphyrins** (hemes and chlorophylls).

coprostane 5β-cholestane; *see* **cholestane**.

coprostanol *or* **coprosterol** 5β-cholestan-3β-ol; a major **sterol** in feces.



copy-choice hypothesis a hypothesis to explain interchromosomal genetic recombination, according to which copying to form a new DNA strand alternates from one parent strand to the other. If the parent strands contain different alleles, a recombinant chromosome may be produced possessing some alleles from one parental strand and some from the other.

copy error a mutation arising from an error in DNA replication.

copy number the number of copies of any gene or plasmid in a given cell. *See* **plasmid**.

CoQ *symbol for* coenzyme Q (obsolete; *see* **ubiquinone**).

$CoQH_2$ *symbol for* reduced coenzyme Q (obsolete; the reduced form of **ubiquinone**).

cord blood blood taken from the umbilical cord.

cord factor 6,6′-dimycolyl trehalose; a highly toxic glycolipid extractable with nonpolar solvents from the cell walls of certain, especially virulent, strains of *Mycobacterium* that characteristically grow in long cords in liquid media.

cordycepin 3′-deoxyadenosine; a substance found in culture fluids of the microfungus *Cordyceps militaris*. It inhibits poly(A) synthesis and interferes with the processing of much eukaryotic mRNA and hence with protein synthesis. The α-^{32}P-labelled 5′-triphosphate is used to label the 3′ ends of DNA fragments in DNA sequence analysis.

cordycepose 3-deoxy-D-ribose; 3-deoxy-D-*erythro*-pentose; the carbohydrate component of **cordycepin**.

core the central, innermost, or least easily destroyed or removed part of a structure; e.g. the core of a partially digested biopolymer, the protoplast core of an endospore, or the core of a virus.

core debranching enzyme *see* **β-1,3-galactosyl-*O*-glycosyl-glycoprotein β-1,6-*N*-acetylglucosaminyltransferase**.

core enzyme the part of bacterial (*Escherichia coli*) **RNA polymerase** lacking the σ subunit. It will catalyse chain elongation but not initiation.

core glycosylation the first of two stages in the biosynthesis of the carbohydrate moiety of some glycoproteins; it takes place in the endoplasmic reticulum.

core particle 1 the RNA-rich particle obtained from a **ribosome** by the gentle removal of some of the ribosomal proteins. The particle retains peptidyltransferase activity. 2 any of the eukaryotic **chromatin** particles containing 140 base pairs of DNA and an octamer of the four conservative **histones** (two each of histones H2A, H2B, H3, and H4) but not histone H1. These particles are obtained by the treatment of chromatin with DNase. In the intact chromatin they are linked by stretches of variable length of so-called linker DNA.

core polymerase *see* **RNA polymerase**.

core protein any of the many types of protein in glycosaminoglycans to which oligosaccharide side-chains are covalently attached and which are themselves covalently attached to a very long oligosaccharide backbone. Each glycosaminoglycan will typically contain many core protein molecules. An example is **heparan sulfate proteoglycan core protein**.

corepressor any metabolite that by its specific combination with a **repressor** causes the activation of the latter.

Cori, Carl Ferdinand (1896–1984) and/or **Gerty Theresa Cori** (née Radnitz, 1896–1957), Czech-born US biochemists distinguished for their studies of glucose and glycogen metabolism, largely carried out in collaboration; joint Nobel Laureates in Physiology or Medicine (1947) 'for their discovery of the course of the catalytic conversion of glycogen' [prize shared with B. A. Houssay].

Cori cycle *or* **lactic acid cycle** an energy-requiring metabolic pathway in animals in which carbon atoms of glucose pass along the circular route: muscle glycogen → blood lactate → blood glucose → muscle glycogen. [After C. F. and G. T. **Cori** who together formulated it in 1929.]

Cori ester *an alternative name for* **α-D-glucopyranose 1-phosphate**. [After C. F. and G. T. **Cori** who reported its isolation in 1936.]

Cori's disease *an alternative name for* type III **glycogen disease**. [After G. T. **Cori** who first described it in 1953.]

cornea the transparent curved part of the front of the eyeball that refracts incident light onto the lens. It is composed of many layers, the thickest being the stroma, which is composed of uniformly fine collagen fibrils. These are kept in an orderly distribution by the expansive force between them of the contained glycosaminoglycan (mainly keratan sulfate I). The cornea has a high oxygen consumption and approximately 50% of the glucose utilized is oxidized via the phosphogluconate pathway. **—corneal** *adj.*

Cornforth (Sir) John Warcup (1917–) Australian-born British organic chemist and biochemist most noted for his elucidation of the stepwise biosynthesis of cholesterol from acetate; Nobel Laureate in Chemistry (1975) 'for his work on the stereochemistry of enzyme-catalyzed reactions' [prize shared with V. **Prelog**].

cornifin *or* **small proline-rich protein I** a cross-linked envelope protein of keratinocytes that first appears in the cell cytosol, but ultimately becomes cross-linked to membrane proteins by transglutaminase. Its presence correlates with squamous differentiation. It is down-regulated by retinoids. Structurally, it is rich in proline, glutamine, and cysteine and contains repeats of an octapeptide, consensus sequence, EPCQPKVP. Example from *Sus scrofa*: database code CORN_PIG, 97 amino acids (10.71 kDa).

corn-steep liquor *or* **corn-steep water** a concentrated fluid obtained by steeping corn (maize) grains in water (containing 0.2% SO_2) for 36–40 hours at 46–50 °C. It is used in the manufacture of **penicillin** to increase antibiotic yield.

corn sugar *a common name for* **D-glucose**.

corn syrup *or (sometimes)* **liquid glucose** a partial hydrolysate of corn (maize) starch containing **glucose** and glucose oligosaccharides.

coronand *an alternative name for* **crown compound**.

coronate *see* **crown compound**.

coronavirus any of a group of RNA animal viruses consisting of enveloped particles 80–120 nm long, with helical nucleocapsids. They include some of the viruses that cause common colds in humans.

corpuscle any living metazoan cell, especially the red corpuscles (erythrocytes) and the white corpuscles (leukocytes) of the blood.

corpuscular radiation *a former term for* **particulate radiation**.

corpus luteum (*pl.* **corpora lutea**) a yellow, progesterone-secreting, glandular body formed in the mammalian ovary from a **Graafian follicle** after extrusion of an ovum.

correlation *(in statistics)* the interdependence of two or more simultaneously variable quantities; or the nature or degree of this interdependence. If two variates both increase or both decrease together there is said to be **positive (direct) correlation** between them, whereas if they change in opposite directions to each other there is said to be **negative (inverse) correlation**.

correlation coefficient *(in statistics) symbol*: r; a measure of the correlation between two variates, x and y. If one variate has the values $x_1, x_2, \dots x_n$ and the corresponding values of the other variate are $y_1, y_2, \dots y_n$, with respective mean values of $\bar{x}$ and $\bar{y}$ then the correlation coefficient, r, is given by:

$$r = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) \Big/ \Big[\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2\Big]^{1/2}$$

If $r = 0$, x and y are completely independent. If $r = 1$ there is complete correlation and one variate may be calculated from the other variate.

correlation time *symbol*: τ_C; the average time between molecular collisions for molecules in some state of motion. The reciprocal of the correlation time is given by the equation:

$$1/\tau_C = 1/\tau_R + 1/\tau_M + 1/\tau_S$$

where $1/\tau_R$ is the rate at which the molecule is tumbling (τ_R being the **rotational correlation time**), $1/\tau_M$ is the rate at which two dipoles approach each other (τ_M being the **residence time**), and $1/\tau_S$ is the electron spin lattice relaxation rate (τ_S being the time for the upper electron spin state to dissipate its excess energy).

correndonuclease any correctional **endonuclease** that specifically acts on damaged DNA initiating a correctional pathway *in vivo*. The name is misleading because nuclease activity itself does not lead to repair.

corrin $C_{19}H_{22}N_4$, the fundamental heterocyclic skeleton of the **corrinoids**. It consists of four reduced pyrrole rings joined into a macrocyclic ring by links between their α positions; three of these links are formed by a one-carbon unit (methylidyne group) and the other by a direct $C\alpha$–$C\alpha$ bond. Corrin is so named because it is the core of the **vitamin B_{12}** molecule.

corrinoid any of a group of compounds containing the **corrin** skeleton. This group includes various B_{12} vitamins, factors, and derivatives. Some are more unsaturated than corrin itself; many have a regular pattern of substituents on the methylene carbon atoms of the reduced pyrrole rings and a cobalt atom in the centre of the macrocyclic ring.

corrole *a trivial name for* octadehydrocorrin; *see* **corrin**.

cortex (*pl.* **cortices**) **1** the outer layer of an animal organ, situated immediately beneath the capsule or outer membrane. It is usually morphologically distinct from the rest of the organ; e.g. adrenal cortex, cerebral cortex, renal cortex. **2** the unspecialized tissue in plant stems and roots lying between the vascular bundles and the epidermis; the outer layer of a stem. —**cortical** *adj.*

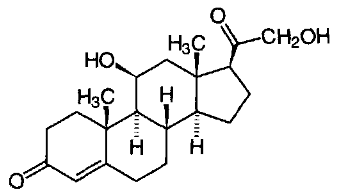
corticoid *an alternative name for* **corticosteroid**.

corticoliberin *an alternative name for* corticotropin-releasing hormone (*see* **CRH**).

corticosteroid *or* **corticoid** any of a large group of C_{21} steroids synthesized from cholesterol in the **adrenal cortex**, especially the adrenocortical hormones. Usage is often extended to include synthetic analogues with hormonal activity. Corticosteroid hormones are divided into **glucocorticoids**, i.e. cortisol and related compounds and **mineralocorticoids**, i.e. aldosterone and related compounds.

corticosteroid-binding globulin *or* **corticosteroid-binding protein** *an alternative name for* **transcortin**.

corticosterone 11β,21-dihydroxypregn-4-ene-3,20-dione; Kendall's compound B; Reichstein's substance H; a **glucocorticoid**, but on the biosynthetic route to aldosterone. Quantitatively it is the second most important corticosteroid of the normal human adrenal cortex. **Cortisol**, synthesized in the adrenal zona fasciculata, is the predominant corticosteroid in humans; corticosterone, synthesized in the adrenal zona fasciculata and glomerulosa, much less so, but it is the main corticosteroid in some rodents.



corticotrope a type of cell in the anterior pituitary that synthesizes **proopiomelanocortin**, a **polyprotein** from which **corticotropin** and a number of other peptide hormones are formed.

corticotrophin *(formerly) a variant spelling of* **corticotropin**.

corticotropin *or* **adrenocorticotropin** *or* **adrenocorticotropic hormone** (*abbr.*: ACTH) a 39-residue polypeptide hormone, secreted by the adenohypophysis of the pituitary, that stimulates growth of the adrenal cortex and the synthesis and secretion of various corticosteroids. It has the sequence (human):

[SYSMEHFRWGKPV]GKKRRPV-
KVYPNGAEDESAEAFPLEF.

The first 13 residues (in brackets) comprise, after proteolysis, α-**melanocyte-stimulating hormone**. Under normal conditions its own secretion is under feedback control from the plasma cortisol level, both directly and through hypothalamic regulatory factors. Biosynthesis occurs via the multihormone precursor peptide **proopiomelanocortin**. Hormonal activity is confined to the species-invariant N-terminal 24 amino acids (*see* **tetracosactide**). Corticotropin secretion is regulated by corticotropin-releasing hormone (**CRH**), is episodic (in bursts at intervals of a few hours), exhibits diurnal rhythm, and is increased by stress such as trauma, surgery, and hypoglycemia. The effects of fever may be mediated by cytokines 1, 2, and 6, which act indirectly by causing the release of CRH. *See also* **corticotropin-like intermediate lobe peptide, corticotropin receptor, melanotropin**.

corticotropin-induced secreted protein *abbr.*: CISP; *a former name for* thrombospondin-2 (*see* **thrombospondin**).

corticotropin-like intermediate lobe peptide *abbr.*: CLIP; a 22-residue peptide identical in sequence to the C-terminal moiety (residues 18–39) of a molecule of **corticotropin** from the same species. It is obtained from extracts of the pars intermedia of the pituitary of certain animals (rat, pig, dogfish), and has been found also in a carcinoid tumour from a human patient who was secreting ectopic corticotropin; it has not been found in extracts of human pituitary. It is derived from **proopiomelanocortin**. CLIP has B-cell tropin activity and acts to stimulate insulin release from B cells. *See also* **beta-cell tropin**.

corticotropin–lipotropin precursor *see* **proopiomelanocortin**.

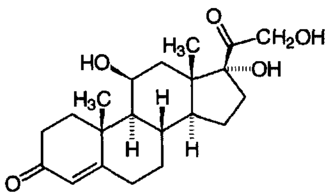
corticotropin receptor one of a number of membrane proteins that bind corticotropin and couple to heterotrimeric G-protein-activated **adenylate cyclase**; they are seven-trans-membrane-domain proteins and are distributed on cells of the adrenal cortex, on cells of the immune system, and in the brain. EC_{50} values for binding are near 10^{-8} M for corticotropin. Example from *Bos taurus*: database code ACTR_BOVIN, 297 amino acids (33.29 kDa). *See also* **melanocortin receptor**.

corticotropin-releasing factor *an alternative name for* corticotropin-releasing hormone (*see* **CRH**).

corticotropin-releasing-factor binding protein a protein that binds and inactivates corticotropin-releasing hormone (**CRH**), to prevent inappropriate pituitary-adrenal stimulation in pregnancy. Example (precursor) from human: database code CRFB_HUMAN, 322 amino acids (36.15 kDa).

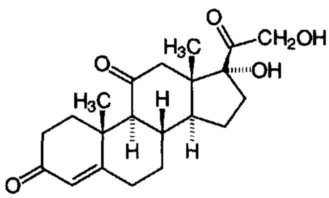
corticotropin-releasing hormone *see* **CRH**.

cortisol *or (recommended name)* **hydrocortisone** 17-hydroxycorticosterone; 11β,17,21-trihydroxy-4-pregnene-3,20-dione; Kendall's compound F; Reichstein's substance M; the most powerful naturally occurring **glucocorticoid** hormone and the most abundant product of the adrenal cortex in humans. It is produced by the cells of the zona reticularis; its best known action is anti-inflammatory. The biosynthesis of cortisol is under the control of **corticotropin**.



cortisol-binding globulin *or* **cortisol-binding protein** *abbr.:* CBG; *an alternative name for* **transcortin**.

cortisone 11-dehydrocortisol; Kendall's compound E; Reichstein's substance F; a prodrug that is biologically inactive but is converted in the liver and, to a lesser extent, in other tissues, to cortisol, a substance with **glucocorticoid** activity. Cortisone was the first glucocorticoid to be isolated and made available for therapeutic use. It is not a primary secretory product of the adrenal cortex, being formed during tissue metabolism from cortisol by 11β-hydroxysteroid dehydrogenase. In fetal plasma, however, cortisone levels are about three times higher than those of cortisol.



cortoic acid any member of a group of acidic metabolites of 17-hydroxycorticosteroids in which the primary alcohol group at C-21 has been oxidized to a carboxylic-acid group. They are the principal acidic metabolites of **cortisol** in humans.

corynecin any of a number of antibiotic compounds produced by the cultivation of *Corynebacterium* KY 4339 in a medium containing C_{12} to C_{14} alkanes as the sole carbon source. Their antibacterial properties are similar to those of **chloramphenicol** but they are less potent.

coryneform **1** club-shaped. **2** describing any Gram-positive, asporogenous, club- or rod-shaped bacterium belonging to the genera *Corynebacterium* or *Brevibacterium*, possibly related to *Streptomyces.* Such bacteria are of biotechnological importance (amino-acid fermentations) and are amenable to genetic manipulation. **3** any coryneform microorganism.

cos *abbr for* **cosine**.

COS cell any member of a cell line, derived from monkey cells, that contains an integrated segment of **SV40** DNA coding for T antigen. Hence these cells will support replication of vector molecules containing the SV40 origin of replication, but no other DNA sequences from that virus.

cosec *abbr. for* cosecant.

cosecant *abbr.*: cosec *or* csc; the reciprocal of the **sine** of an angle.

cosine *abbr.*: cos; a trigonometric function defined, for a given angle of a right-angled triangle, as the ratio of the length of the side adjacent to the angle and the length of the hypotenuse. The cosine of an obtuse angle is numerically equivalent to that of its supplement but of opposite sign.

cosmid a hybrid **cloning vector** used for cloning DNA. Cosmids are derived from plasmids but also contain ***cos* sites** from phage lambda.

***cos* site** the cohesive ends (sticky ends) of certain **phage** DNA molecules. The *cos* site usually referred to is from phage lambda.

costamere a vinculin-containing myofibril attachment site. Costameres form one of the major linkage sites of the skeletal muscle cell, encircling the cell and connecting the Z-discs to the sarcolemma.

co-stimulator *an alternative name for* **interleukin 2**.

cost-selectivity equation a relation between the accuracy gained by editing and the cost in terms of the wasteful hydrolysis of correct products in any enzyme system synthesizing informational macromolecules. It is useful in the rationalization of some observations and proposed editing mechanisms.

cosubstrate an imprecise term used for an organic cofactor that is necessary for the activity of some enzymes. Although not the substrate of principal interest, the cosubstrate acts as an acceptor or donor of atoms or of functional groups and participates stoichiometrically in the reaction. *Compare* **coenzyme**.

COSY *abbr. for* correlated spectroscopy; a technique employed in the first two-dimensional **NMR** experiment, designed to elucidate which resonances of the NMR spectrum are connected by spin–spin J-coupling. J-couplings arise out of the perturbation of the nuclear energy levels of one nuclear spin by another nuclear spin that is propagated through the intervening electronic orbitals. In a one-dimensional NMR spectrum these J-couplings are manifested as a splitting in the Lorentzian lineshape. In 2D-COSY spectra the couplings are indicated by off-diagonal cross-peaks connecting the frequencies of the J-coupled spectra. Ordinarily the J-couplings are only large enough for pairs of ^{1}H nuclei separated by a maximum of two or three bonds to exhibit COSY cross-peaks.

Cot *see* **C_0t**.

cotransduction the **transduction** of two or more identifiable genes in a single event, probably because the transduced element contains more than one genetic locus.

cotranslation *(in recombinant DNA technology)* the synthesis of a hybrid molecule consisting of the desired protein product covalently linked to a normal protein secreted by the organism being used, through translation of a hybrid mRNA molecule generated on the recombinant DNA. —**cotranslate** *vb.*

cotranslational transport the transport of nascent protein chains through the membrane of the cisterna of the endoplasmic reticulum while translation is still proceeding at the ribosome. It is a common phenomenon in higher eukaryotes.

cotransmitter a substance, usually a peptide, that is released from neurons and modifies the actions of the substance regarded as the primary transmitter. For example, vasoactive intestinal peptide is a cotransmitter with acetylcholine, and neuropeptide Y performs a similar role with norepinephrine. Cotransmitters may also have direct effects on target cells.

cotransport the concerted movement of two chemical species across a membrane. *Compare* **antiport**, **symport**.

co-trimoxazole an antibacterial drug that comprises a mixture of **sulfamethoxazole** and **trimethoprim** in a ratio of about 5:1. The combination may have synergistic effects.

Cotton effect an anomalous change in optical rotation with wavelength in the region of an absorption band. As the absorption band is approached there is a change in the rotation to a maximum, which may be either positive or negative, followed by a decrease to a minimum passing through zero at the wavelength of maximum absorption. Similarly there is a change in circular dichroism, either positive or negative, coinciding with the absorption band. The magnitudes and directions of these effects have been used to study macromolecular conformations. [After Aimé Cotton (1859–1951), French physicist.]

cotyledon the leaf-forming part of the embryo of seeds. — **cotyledonous** *adj.*

Couette viscometer an instrument for measuring viscosity, consisting of two concentric cylinders separated by a narrow annulus, which is filled with the sample. One cylinder is fixed and the other is rotated. From the angular velocity at a given torque and the dimensions of the cylinders, the viscosity of the sample can be calculated. The Couette viscometer is designed for use at relatively low shear gradients.

coulomb *symbol*: C; the SI derived unit of electric charge; the quantity of electricity transported in one second by a current of one ampere. Hence, 1 C = 1 A s. [After Charles Augustin de Coulomb (1736–1806), French physicist.]

coulomb force *or* **coulombic force** the electrostatic attraction or repulsion between two charged particles. *See* **Coulomb's law**.

Coulomb's law (of electrostatics) the law stating that the force, F, of attraction or repulsion between two point charges, Q_1 and Q_2, is proportional to the product of the charges and inversely proportional to the square of the distance, d, between them multiplied by the permittivity, ε, of the medium between them. Hence, $F = (Q_1Q_2)/\varepsilon d^2$. A similar law exists for magnetism.

coulometer *or* **coulombmeter** *or* **voltameter** an electrolytic cell for measuring a quantity of electricity passing through a circuit containing the cell by determining the amount of a metal (usually copper or silver) deposited on the cathode from a solution of a salt of the metal.

coulometry *or* **coulombmetry** *or* **voltametry** a method of chemical analysis based on the measurement of a quantity of electricity, in coulombs, in an electrode reaction. —**coulometric** *or* **coulombmetric** *adj.*; **coulometrically** *or* **coulombmetrically** *adv.*

Coulter counter a device for counting particles in a suspension, e.g. blood cells, microorganisms, or spores. Its functioning depends on changes in electric conductivity as the particles pass through a small aperture.

coumarin **1** 1,2-benzopyrone; 2*H*-chromen-2-one; a fluorescent indicator. **2** any of a number of coumarin derivatives. The coumarins are important in the perfume industry since they smell of new-mown hay and grass, though some are no longer used (acernocoumarol and phenindiene) because of adverse allergenic effects. Some are useful in forming fluorescent analytes. Others are antibiotics, e.g. **novobiocin** or mycotoxins (*see* **aflatoxin**).

coumarin anticoagulants *see* **dicoumarol**.

counter any device for enumerating, especially one for detecting and determining particles and photons emitted from the disintegration of a radionuclide. *See*, e.g., **Cherenkov counter**, **Geiger counter**, **scintillation counter**.

countercurrent chromatography a form of **partition chromatography** that eliminates the need for a solid support, in which one of the components of a two-phase solvent system is allowed to flow through the second component in a coiled tube. The technique is claimed to have the advantages of both **countercurrent distribution** and **liquid–liquid chromatography** with the disadvantages of neither, and to be applicable to a broad spectrum of samples ranging from micromolecules to macromolecules, particles, and cells.

countercurrent distribution *or* **countercurrent separation** a method for the separation and purification of (usually organic) substances that depends on the repetitive distribution of solute between two immiscible liquid phases in a series of vessels in which the phases are in contact. The process is continued until homogeneous substances are distributed in various sets of vessels in accordance with their **distribution coefficients** between the two liquid phases. *See also* **Craig apparatus**.

countercurrent transport the movement of a solute in or out of a cell against a concentration gradient by an ATP-independent process. It is due to competition between the solute under consideration and a second solute, with an opposite concentration distribution, for the same membrane carrier protein.

counterelectrophoresis *see* **counterimmunoelectrophoresis**.

counterflow the movement of a substance across a membrane against a concentration gradient.

counterimmunoelectrophoresis *or* **counterelectrophoresis** *other names:* countermigration electrophoresis; immunoelectroosmophoresis; *abbr.:* CIE; a one-dimensional double **electroimmunodiffusion**; a technique wherein diffusion of the reactants towards each other, and hence the formation of precipitin lines, is accelerated by the combination of electrophoresis and electroendosmosis that occurs when an electric potential difference is applied across the gel between the antigen- and antibody-containing wells, with the latter positive to the former.

counterion any ion carrying a charge opposite to that of the charged chemical species with which it is associated; an ion of low M_r with a charge of opposite sign to that of the colloidal ion. *Compare* **co-ion**.

countermigration electrophoresis *an alternative name for* **counterimmunoelectrophoresis.**

counterstain **1** to apply a second stain to a preparation for microscopy. Counterstaining is used to stain in a contrasting way those parts of the preparation that have not been stained by the first stain. **2** any stain so used.

countertranscript *an alternative term for* **antisense RNA**.

countertransport *an alternative term for* **antiport**.

countertrypsin a glycoprotein trypsin inhibitor from mouse plasma and a member of the **fetuin** family.

counting efficiency the ratio of recorded radioactivity events to the number of actual radioactive disintegrations occurring in the same time and in the same sample. It is sometimes expressed as the percentage efficiency.

COUP *other names*: COUP-TF; v-ErbA-related protein; EAR-3; a transcription factor named from one activity, chicken ovalbumin upstream promoter. In association with transcription factor IIB it stimulates initiation of transcription. It is similar to the steroid/thyroid/retinoic nuclear hormone receptors and **seven-up**. Example from human: database code COTF_HUMAN, 423 amino acids (46.10 kDa).

couple **1** a pair of equal and opposite, parallel but not collinear, forces acting on a body, thus producing a turning effect. **2** to join or link (two things) together.

coupled assay a type of enzymic assay of a substrate (or of an enzyme) employed when the reaction in question does not give chromophoric or fluorescent products. A second enzyme reaction that acts on one of the products of the first reaction, and that gives an easily detectable product, is carried out simultaneously with the first reaction. Many coupled assays are based on the formation or disappearance of NADH, which can be easily detected by its absorbance at 340 nm.

coupled oxidation and phosphorylation an intimate relationship between oxidation of substrate and phosphorylation of ADP to ATP such that the two processes cannot proceed independently, as in mitochondria. *See also* **coupling** (def. 3).

coupled reaction either of two reactions, one endergonic and the other exergonic, that are linked energetically, occur simultaneously, and share a common intermediate, such that the overall free energy change for the two reactions is negative.

coupling **1** *(in chemistry)* the covalent linking of one chemical entity to another; *see* **covalent bond**. **2** *(in physics)* **a** an interaction between different parts of a system, e.g. between groups of atoms, electrons, or nuclei. **b** an interaction between two electronic circuits so that power is transferred from one to the other. **3** *(in biochemistry)* the linking of **electron transport** and **oxidative phosphorylation** in isolated mitochondria. Thus, if the **phosphorus:oxygen ratio** is high the mitochondria are said to be tightly coupled. *See also* **uncoupling**.

coupling constant the separation between any two bands of multiple peaks in nuclear magnetic resonance spectroscopy; it is proportional to the magnitude of the **spin–spin coupling** constant.

coupling factor *abbr.*: CF; *archaic term for* either **1** any of the proteins functioning in the coupling of ATP synthesis to electron transport in mitochondria. CF1–CF4 are obsolete names for the α–δ subunits of mitochondrial **H^+-transporting ATP synthase**. **2** any of the proteins required for the coupling of ATP synthesis to the photoinduced electron transport in chloroplasts during photosynthesis. The concept has been replaced by the **chemiosmotic coupling hypothesis**.

coupling membrane any biological membrane in which energy-yielding and energy-requiring biochemical processes occur together.

covalence *or* **covalency** nonionic valence between two atoms in a chemical compound, characterized by their sharing of one or more electrons. —**covalent** *adj.*

covalent bond a chemical **bond** formed between two atoms in a molecule by the sharing of electrons, usually in pairs, by the bonded atoms.

covalent catalysis the catalysis of any reaction in which the substrate is modified by forming a covalent bond with the catalyst (enzyme).

covalent chromatography any technique of **column chromatography** in which the substance of interest binds covalently with groups, e.g. –SH groups, on the support medium and is subsequently displaced by an appropriate reagent.

covalent (closed) circle (of DNA) *or* **covalently closed circular** DNA *or* **closed double-stranded** DNA *abbr.*: cccDNA; any double-stranded DNA molecule that is circular (but not necessarily geometrically circular) and in which both polynucleotide strands are completely continuous. *Compare* **Hershey circle, open circle**.

covalent modification any of a diverse group of processes in which the initially synthesized structures of biopolymers, especially enzymes, proenzymes, or structural macromolecules, are enzymically modified by the breakage of covalent bonds or the addition of new covalently linked groups. It includes **post-translational modification**. The term may sometimes be used in the more restricted sense of reversible interconversion of active and inactive forms of certain metabolic enzymes or other proteins by sets of control enzymes, often by phosphorylation–dephosphorylation.

covariance a measure of the association between two variables. For n pairs of values of two random variables, x and y, this is given by:

$$\text{Cov.}\,(x,y) = (x - \bar{x})(y - \bar{y})/(n - 1)$$

where $\bar{x}$ and $\bar{y}$ are the means of the populations of x and y, respectively. *Compare* **correlation coefficient, regression coefficient, variance**.

covarion any of the codons in a given gene that may concomitantly vary, so resulting in favourable mutations or in mutations leading to amino-acid substitutions that have little or no effect. [From concomitantly variable codon.]

covirus any virus that exists as two or more separate particles all of which must be present together in the host organism for the complete replication cycle of the virus to occur.

covolume the amount by which the apparent volume of a molecule exceeds the sum of the volumes of the constituent atoms. It is usually 13–14 $cm^3\ mol^{-1}$ for protein molecules.

cozymase originally, the heat-stable, diffusible fraction of a crude aqueous extract of brewers' yeast that, if added to the heat-labile, nondiffusible fraction (zymase), would enable the alcoholic fermentation of glucose to occur in a cell-free system. The term was subsequently applied to one particular substance present in that fraction, now known as **nicotinamide adenine dinucleotide**.

C_3 pathway *an alternative name for* **reductive pentose phosphate cycle**.

C_4 pathway *an alternative name for* **Hatch–Slack pathway**.

CPBA *abbr. for* competitive protein-binding assay.

CPE *abbr. for* cytopathic effect.

C peptide **1 a** the inactive polypeptide excised from **proinsulin** during its conversion to insulin. It contains 31 amino-acid residues in the human but shows greater species variability in both length and amino-acid sequence than does insulin. The structure of porcine C-peptide is (including the flanking basic residues):

RREAQNPQAGAVELGGGLGGLQALALEGPPQKR.

It is released into the bloodstream concomitantly with insulin, hence the blood level is useful in evaluation of B-cell function. **b** that segment of the amino-acid sequence of proinsulin lying between the two residues destined to become respectively the C-terminal residue of the B chain of insulin and the N-terminal residue of the A chain. It consists structurally of the above polypeptide plus two flanking pairs of basic residues. C peptide is named from connecting peptide, and perhaps also because, after A (chain) and B (chain), C (peptide) represents alphabetically the third major structural component of proinsulin. **2** *an alternative name for* C fragment (of β-lipotropin) (*see* **lipotronin**).

CpG island a region of 1–2 kb containing a high density of methylated cytosine residues and occurring immediately 5′ to G residues, i.e. in the sequence CpG. CpG islands are frequently found in animal genomes at the 5′ end of genes. In plants the methylated sequence is ...CpNpGp... where N can be any base. Methylation of DNA is a heritable phenomenon that reduces gene expression, probably by increasing the binding of a repressor. CpG islands are a site of high mutational frequency because spontaneous deamination of the methylated cytosine gives thymine, which is not recognized by DNA repair enzymes.

C_3 plant any plant in which fixation of carbon dioxide occurs predominantly by the **reductive pentose phosphate cycle**; i.e. by incorporation initially into the three-carbon compound 3-phosphoglycerate.

C_4 plant any plant in which fixation of carbon dioxide occurs predominantly by the **Hatch–Slack pathway**; i.e. by incorporation initially into the four-carbon compound oxaloacetate. Such plants are found usually in semiarid conditions with high light levels. They include sugar cane, corn (maize), and many weeds.

cpm *or* **c.p.m.** *abbr. for* counts per minute.

Cr *symbol for* chromium.

Crabtree effect the inhibition of respiration by glycolysis. The inhibition is small and occurs only in a few types of cells that possess a high glycolytic capacity, e.g. ascites tumour cells and other neoplastic tissues, renal medulla, leukocytes, and cartilage. *Compare* **Pasteur effect**.

Craig apparatus the most widely used type of apparatus for carrying out **countercurrent distribution** experiments. The earlier version (1944) was constructed of metal and the later version (1949) was of modular design and constructed of glass. [After Lyman Creighton Craig (1906–74).]

crambin a protein obtained from seeds of *Crambe abyssinica*. It comprises 46 amino acids, has a molecular mass of 4.73 kDa (database code CRAM_CRAAB), and is remarkable for its highly ordered crystals and very high resolution. Its 3-D structure is known.

cranin a form of **dystroglycan**.

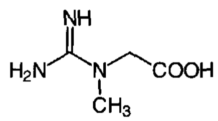
CRE *abbr. for* cyclic AMP response element; a sequence,

GTGACGT[A/G][A/G],

that is present in many viral and cellular promoters. When it binds **CREB protein**, transcription of the genes regulated by such a promoter is turned on.

C-reactive protein *abbr.*: CRP; an acidic, crystallizable, heat-sensitive protein of ≈118 kDa that is detectable in human or monkey blood serum early in the course of various infections or when there is inflammation, tissue damage, or necrosis. One of the so-called **acute phase proteins**, it is normally undetectable. Not an immunoglobulin and of unknown function, it is named for its ability, in the presence of Ca^{2+} ions, to form a precipitate with a pneumococcal somatic C polysaccharide. Example from human (precursor): database code CRP_HUMAN, 224 amino acids (25.04 kDa).

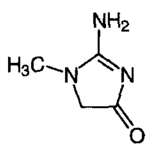
creatine *N*-(aminoiminomethyl)-*N*-methylglycine; an important metabolite in muscle, the precursor of **phosphocreatine**.



creatine kinase EC 2.7.3.2; *systematic name:* ATP:creatine *N*-phosphotransferase; an enzyme that catalyses the formation of **phosphocreatine** from ATP and creatine with release of ADP. The enzyme has M and B subunits, and may exist as the homodimers MM and BB or the heterodimer MB. BB mainly occurs in brain and thyroid, while MM occurs in muscle; MB occurs to a much greater extent in cardiac muscle (30% of total) than skeletal muscle (1% of total) and is thus a useful diagnostic isoenzyme in myocardial infarction, an increased level of the MB isoenzyme occurring between 12 and 36 hours after the infarct. Examples from human; B chain: database code KCRB_HUMAN, 381 amino acids (42.60 kDa); M chain: database code KCRM_HUMAN, 381 amino acids (43.05 kDa); sarcomeric mitochondrial precursor: database code KCRS_HUMAN, 419 amino acids (47.47 kDa); ubiquitous mitochondrial precursor (u-MTCK): database code KCRU_HUMAN, 417 amino acids (46.98 kDa).

creatine phosphate *an alternative name (not recommended) for* **phosphocreatine**.

creatinine 2-amino-1,5-dihydro-1-methyl-4*H*-imidazol-4-one; an end product of creatine metabolism and a normal constituent of urine.



CREB protein *abbr. for* cyclic AMP response element binding protein; a protein that, following phosphorylation by **cyclic AMP-dependent protein kinase** on a single serine residue, binds to the cyclic AMP response element (**CRE**). This then stimulates transcription of genes controlled by regulatory regions that contain the CRE. Example (bovine): database code CREB_BOVIN, 325 amino acids (34.84 kDa).

C region *abbr. for* constant region (of an immunoglobulin).

cretinism a condition of congenital **hypothyroidism** characterized by arrested mental and physical development.

Creutzfeldt–Jakob disease *abbr.*: CJD; a transmissible **spongiform encephalopathy** of humans. It has occurred in three forms: (1) familial (about 15% of cases), (2) iatrogenic or acquired due to inadvertent exposure to CJD-contaminated material, notably human cadaveric growth hormone, and **kuru**, (3) sporadic – the most common form – which is of unknown etiology. They are all associated with the accumulation in affected brains of an abnormal isoform of a glycoprotein. The familial form, which shows an autosomal pattern of disease segregation, is associated with coding mutations in the prion-protein gene. Forms (1) and (3) usually occur in the fifth or sixth decade of life and progress rapidly (within six months) to severe dementia and death typically in one year. Recently a fourth form of CJD has been identified that is believed to be due to consumption of bovine products contaminated with the agent causing **bovine spongiform encephalopathy**. *See also* **prion**. [After Hans Gerhard Creutzfeldt (1885–1964) and Alfons Maria Jakob (1884–1931), German physicians.]

CRF *abbr. for* corticotropin-releasing factor; *see* **CRH**.

CRH *abbr. for* corticotropin-releasing hormone; *other names*: **corticotropin-releasing factor** *or* **corticoliberin** *or* **adrenocorticotropic hormone-releasing factor**; one of a number of peptidic mammalian hypothalamic factors released into the hypophyseal-portal circulation in response to neural and/or chemical stimuli. A 41-residue peptide (M_r 4758.14 human, rat; database code CRF_HUMAN), it controls the secretion by the anterior pituitary gland (i.e. adenohypophysis) of corticotropin, and possibly also of β-lipotropin and pro-γ-melanotropin – all three are peptides formed within the adenohypophysis from proopiomelanocortin. CRH binds to specific seven-transmembrane-domain receptors (e.g. database code CRFR_HUMAN, 444 amino acids (50.72 kDa)) on the **corticotrope** and increases cyclic AMP levels in the cell, stimulating hormone release only in the presence of Ca^{2+}. In addition to increasing the rate of release of corticotropin, it enhances the rate of transcription of mRNA coding for that hormone. *See also* **CRH test**.

CRH test a test that is useful in distinguishing **Cushing's disease** from ectopic corticotropin secretion. If **CRH** is given intravenously, there is an increase in plasma cortisol in normal individuals, while usually in Cushing's disease there is an exaggerated response; in ectopic corticotropin secretion there is usually no cortisol response.

Crick, Francis Harry Compton (1916–), British physicist and molecular biologist famous, *inter alia*, for propounding (with J. D. Watson in 1953) the double-helical structure of DNA, for formulating the adaptor hypothesis of protein synthesis and the central dogma of molecular biology, and for defining the codon; Nobel Laureate in Physiology or Medicine (1962) jointly with J. D. Watson and M. H. F. Wilkins 'for their discoveries concerning the molecular structure of nuclear [*sic*] acid and its significance for information transfer in living material'. *See also* **Franklin**.

Crigler–Najjar syndrome a familial nonhemolytic jaundice and hyperbilirubinemia due to deficiency in hepatic bilirubin glucuronosyltransferase activity. [After John Fielding Crigler (1919–) and Victor Assad Najjar (1914–), US physicians.]

crinophagy a cellular process in which secretion granules fuse with lysosomes. It serves to dispose of excess secretory products in a cell when the stimulus for their discharge is lacking.

crista (*pl.* **cristae**) **1** any of the inward folds of the inner mitochondrial membrane. Their number, extent, and shape differ in mitochondria from different tissues and organisms. They appear to be devices for increasing the surface area of the inner mitochondrial membrane, where the enzymes of electron transport and oxidative phosphorylation are found. Their shape can vary with the respiratory state of the mitochondria. **2** a sensory structure within the ampulla of a semicircular canal of the inner ear.

critical angle *symbol*: *C*; the least angle of incidence at which total internal reflection of a ray of light or other electromagnetic radiation occurs when passing through a medium and meeting the boundary with a medium of lesser density (e.g. of glass with air).

critical electrolyte concentration *or* **critical salt concentration** the concentration of a particular small ion that will displace an ionic molecule (e.g. of a dye or detergent) of like charge from a polyelectrolyte macromolecule. For example, if a polymer P with *z* negative charges is stained with a dye R^+, the dyed polymer, $P^{z-}zR^+$, is formed according to the equation:

$$P^{z-}zM^+ + zR^+ \rightleftharpoons P^{z-}zR^+ + zM^+$$

where M^+ is a small cation. Hence the ratio of dyed to undyed polymer, *D*, is given by:

$$D = [P^{z-}zR^+]/[P^{z-}zM^+] = K[R^+]^z/[M^+]^z$$

where K is a constant. If z is large, there is a very sharp change in D at a critical value of $[R^+]/[M^+]$. Then if $[R^+]$ is held constant and $[M^+]$ varied, an 'all-or-none' change in D is observed at a well-defined $[M^+]$. This concept is used in the **critical electrolyte concentration method** for staining proteoglycans and polynucleotides, and for fractionation of nucleic acids and other polyanions.

critical micellar concentration *or* **critical micelle concentration** *abbr.*: CMC; *symbol*: c_M; the concentration at which surfactant molecules in a lipid–water mixture begin to aggregate amongst themselves to form micelles in the mixture.

critical pressure the minimum pressure capable of liquefying a gas at its critical temperature.

critical temperature **1** the temperature at or above which a partially miscible two-component system exists as one phase and below which it separates into two phases. **2** the temperature above which liquefaction of a gas cannot occur however great the pressure.

Crk *symbol for* a protooncogene related to v-*crk*, the oncogene of avian sarcoma viruses CT10. The name derives from CT10 regulator of kinase, because the protein causes tyrosine phosphorylation but is not itself a tyrosine kinase. Human Crk contains SH2 and SH3 domains (*see* **SH domain**). Database code: Hscrk, 304 amino acids (33.87 kDa).

CRM *abbr. for* (immunologically) cross-reacting material.

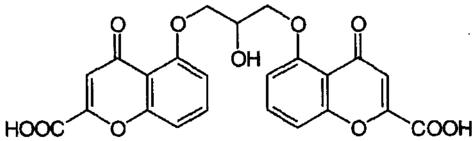
CRM-positive describing a mutant strain of a microorganism that produces a substance that is immunologically cross-reactive with another (specified) substance, e.g. an enzyme or a toxin, produced by a nonmutant (wild-type) strain. Similarly, a **CRM-negative** strain does not produce such a cross-reactive substance.

cRNA *abbr. for* complementary RNA.

cro *symbol for* a regulatory gene in phage lambda, named from 'C-repressor and other things'. Its product, Cro protein, is a repressor that regulates the expression of the *cI* gene. Cro protein is a small globulin, with a **helix-loop-helix** motif; it occurs as a dimer when bound to the operator. Database code RCRO_LAMBD, 66 amino acids (7.36 kDa); 3-D structure known.

Cromer-related blood group a blood group antigen carried by **complement decay accelerating factor**.

cromolyn 1,3-bis(2-carboxychromon-5-yloxy)-2-hydroxypropane; a chromone complex that prevents mast cell degranulation; its mechanism of action is unclear. Commonly prepared as the disodium salt, **disodium chromoglycate**, it is used therapeutically to prevent hypersensitivity reactions, especially asthma.



cross-bridge any connection, seen in electron microcopy, between the contractile elements of muscle, specifically between the head of a myosin molecule and an actin filament.

crossed immunoelectrophoresis *or (sometimes)* **quantitative immunoelectrophoresis** two-dimensional single **electroimmunodiffusion**; a rapid and sensitive technique for separation with high resolution and for quantification of antigenic macromolecules. These are separated first by one-dimensional gel electrophoresis and second by gel electrophoresis orthogonally into a gel containing antibodies to one or more known substances, thereby forming a pattern of loop-shaped precipitation zones between antibodies and corresponding antigens. The height and shape of the loops give information about the quantity and quality of the antigens and of the precipitating antibodies. In **tandem-crossed immunoelectrophoresis** two antigen samples are run at the same time with the application holes placed in such a way that related precipitin peaks formed from the respective samples generate double peaks, permitting direct comparison between them. *Compare* **rocket immunoelectrophoresis**.

crossed immunoisoelectric focusing *or* **crossed immunoelectrofocusing** a two-dimensional separation technique for analysing complex mixtures of macromolecules, e.g. samples of body fluids, that combines the high resolution of **isoelectric focusing** with the selectivity of detection of **immunoelectrophoresis**. Isoelectric focusing in a thin layer of gel (polyacrylamide or, preferably, specially modified agarose) is followed by electrophoresis into an antibody-containing gel orthogonally to the first separation; a pattern of precipitation zones is formed by reaction of antibody with corresponding antigen as in ***rocket immunoelectrophoresis***.

crossed-line immunoelectrophoresis a technique that combines the procedures of **crossed immunoelectrophoresis** and **line immunoelectrophoresis**. Antigens are separated in a one-dimensional run (as in crossed immunoelectrophoresis) and the resulting gel is applied to the edge of a spacer gel in which a strip of gel containing a uniform mixture of the antigens (as used in line immunoelectrophoresis) is embedded parallel to the edge; this combination of the spacer gel and the first gel is then applied to the edge of an antibody-containing gel into which the antigens are introduced by electrophoresis. The resulting sum pattern of crossed immunoelectrophoresis peaks, each standing on individual baselines, permits a direct comparison of crossed and line immunoelectrophoresis patterns and in addition provides a higher power of resolution.

cross-flow filtration a method of operating a filtration device in which retained fluid is circulated over the membrane surface thus preventing undue build up of filtered material on the membrane.

cross-hybridization *or* **cross-hybridisation** the hybridization of a polynucleotide probe to another polynucleotide molecule.

crossing over the reciprocal exchange of segments at corresponding positions along pairs of homologous chromosomes by symmetrical breakage and crosswise rejoining of the chromatids. It results in the **recombination** of alleles. In eukaryotes, crossing over usually accompanies meiosis. Mitotic homologous recombination, while common in yeast, is rare in higher eukaryotes. *See also* **Holliday junction**.

cross-link a side bond between different chains (or parts of a single chain) of a polymer, increasing molecular rigidity. — **cross-linked** *adj.*; **cross-linking** *n.*

cross-matching a serological procedure used to select blood for transfusion. The donor's erythrocytes are mixed with the recipient's serum; agglutination of the erythrocytes indicates incompatibility, demonstrating that the recipient's serum contains antibodies to the donor's erythrocytes.

cross of isocline the dark cross formed in **flow birefringence** experiments between concentric cylinders when the polarizer and analyser are orthogonal. As the system has cylindrical symmetry, there are two regions where the average orientation is parallel to the polarizer direction and two regions, orthogonal to the first two, where the average orientation is parallel to the analyser. In these regions no light is transmitted and a dark cross appears. *See also* **extinction angle**.

crossover **1** *an alternative term for* **crossover chromatid**. **2** the individual resulting from a **crossing over**.

crossover chromatid a **chromatid** resulting from **crossing over** of corresponding segments of the chromatids of homologous chromosomes.

crossover effect *(in pharmacology)* the phenomenon in which a drug shows a positive effect at one concentration and a deleterious effect at another.

crossover method any method for investigating a metabolic (or electron transport) pathway by application of the **crossover theorem**.

crossover nuclease a **deoxyribonuclease** that is involved in recombination.

crossover point the point in a series of reactions in a metabolic (or electron transport) pathway, that, when one reaction is specifically inhibited, marks the boundary between reactants whose concentrations increase and reactants whose concentrations decrease. In particular, it is the point in the respiratory chain that, because of the specific action of an inhibitor, marks the boundary between respiratory catalysts that are more reduced and ones that are more oxidized (at steady-state levels). *See also* **crossover theorem**.

crossover theorem a theorem stating that when a series of reactions in a metabolic (or electron transport) pathway is inhibited at a specific reaction, the concentrations of reactants before the inhibited reaction increase above their steady-state values while the concentrations of the reactants after the inhibited reaction decrease below their steady-state values.

crossover unit *an alternative name for* **map unit**.

cross-react to react with a substance other than that for which the reagent is supposedly specific. *Compare* **cross-reaction**. — **cross-reactive** *adj.*

cross-reacting antibody any antibody that is able to react with an antigen that did not specifically stimulate its production. Such cross-reactions may be weaker than the reaction of the antibody with the antigen that caused its production.

cross-reacting antigen 1 any antigen that is able to react with an antibody produced in response to another antigen. This may be because the two antigens share the same determinants or carry determinants that are sufficiently alike stereochemically to enable the antibody to react with both of them. 2 any antigen that has an identical structure in two strains of microorganism, so that antibody raised against one strain will react with the second strain.

cross-reaction the immunological phenomenon in which an antigen reacts with an antibody raised against a different antigen. *See* **cross-reacting antibody, cross-reacting antigen**.

cross-reactivation the reappearance of activity in the progeny of a lethal mutant virus following mixed infection of a host cell with one or more active viruses. If the active viruses differ from each other in one or more of their genetic loci, the mutant virus is reactivated by genetic exchange leading to the replacement of its damaged nucleic acid.

cross-sensitivity a state of immunological hypersensitivity to one substance produced by priming an animal with another substance that bears **cross-reacting antigen**.

cross-tolerance 1 a state of immunological **tolerance** to one substance produced by priming an animal with another substance that bears **cross-reacting antigen**. 2 a state of physiological tolerance to one drug resulting from chronic administration of another.

croton oil an oil obtained from the seeds of *Croton tiglinum*, a member of the family Euphorbiaceae. It is an irritant, and contains tumour-promoting **phorbol esters**.

crown compound *or* **coronand** any synthetic macrocyclic polydentate nonplanar (generally uncharged) organic compound a molecule of which has three or more ring heteroatoms (usually nitrogen or oxygen) capable of ligation to a metal ion or other cationic entity. The resulting complex is termed a **coronate**. *See also* **crown ether**. *Compare* **cryptand**.

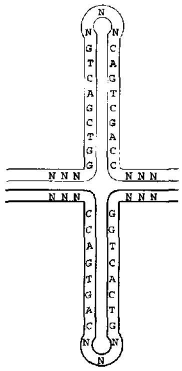
crown ether any of a large subclass of crown compounds that are synthetic macrocyclic polyethers, the molecules of which contain 9–60 atoms in the ring, including 3–20 oxygen atoms. Many crown ethers, especially those containing 5–10 ring oxygen atoms in the molecule, can act as ion-selective complexing agents and ionophores with metal cations; some will solubilize inorganic salts in nonpolar solvents.

crown-gall disease a tumorous disease of many dicotyledonous plants caused by the bacterium *Agrobacterium tumefaciens* and marked by enlargement of the stem near the root crown. The swelling, or **gall**, is produced by the transformation of cells at an infected wound through transfer of the **T-DNA**, part of a large (150–230 kb) plasmid (the **Ti plasmid**) found in oncogenic strains of the bacterium. Transformed tissue is distinguishable by its ability to grow in tissue culture in the absence of plant hormones and by the synthesis of a characteristic **opine**, which is invariably a specific substrate for the particular inducing strain of the bacterium.

CRP *abbr. for* 1 C-reactive protein. 2 cyclic AMP receptor protein; a bacterial **catabolite (gene) activator protein** that complexes with cyclic AMP and binds to specific DNA sites near the promoter to regulate the transcription of several catabolite-sensitive operons. The protein induces a severe bend in DNA and acts as a negative regulator of its own synthesis. It binds to DNA as a dimer. The examples belong to a well-conserved family; there are two motifs. Example from *Escherichia coli*: database code CRP_ECOLI, 210 amino acids (23.61 kDa); 3-D structure of this and many mutants is known. The example from *Shigella flexneri* is identical.

Crt *symbol for* the cerotoyl (i.e. hexacosonyl) group.

cruciform 1 cross-shaped. 2 a DNA conformation that can be generated from long **palindromes**, e.g.



crustacyanin a protein occurring in the carapace of lobsters that binds the carotenoid **astaxanthin**, which provides the blue colour. α-Crustacyanin is a dimer; β-crustacyanin is a 16-mer. There are five types of subunit: A1, A2, A3, C1, and C2; the subunits are **lipocalins**. Example, C1 subunit from European lobster: database code CRC1_HOMGA, 181 amino acids (20.64 kDa).

cruzaine *an alternative name for* **cruzipain**.

cruzipain *or* **cruzaine** a cysteine proteinase (EC 3.4.22.–) that hydrolyses chromogenic peptides at the carboxyl Arg or Lys. It requires at least one more amino acid, preferably Arg, Phe, Val, or Leu, between the terminal Arg or Lys and the amino-blocking group. The purified enzyme digests itself. The enzyme plays a role in development and differentiation of *Trypanosoma cruzi*. Example (precursor): database code CYSP_TRYCR, 467 amino acids (49.78 kDa).

cryo+ *comb. form* indicating low temperature.

cryobiochemistry the study of biochemical substances and processes at subzero temperatures, i.e. below the freezing point of water. The field has expanded because of the development of **cryosolvents**. *See also* **cryoenzymology**.

cryobiology the science that is concerned with the study of the effects of very low temperatures on organisms.

cryoelectron microscopy *see* **electron microscopy**.

cryoenzymology the study of enzymes at low temperatures. By employing temperatures below 0 °C and fluid **cryosolvents**, enzyme reaction rates can be reduced sufficiently for normally transient enzyme–substrate intermediates to be studied.

cryogen any substance used to produce a low temperature; e.g. a freezing mixture.

cryogenic of, or relating to, very low temperatures.

cryogenics the branch of science concerned with the production of very low temperatures and the study of phenomena occurring at these temperatures.

cryoglobulin any **globulin**, especially IgG or IgM, that forms a gel or flocculent precipitate, or spontaneously crystallizes, on cooling of a solution or a sample of serum containing it. *See also* **cold-insoluble globulin**.

cryomicrotome any **microtome** arranged in a **cryostat** and used to prepare thin sections of frozen tissue for microscopic examination.

cryoprecipitate the precipitate of a **cryoglobulin**.

cryopreservation the science of preserving cells, tissues, or organs over long periods with the aid of very low temperatures.

cryoscope *any instrument or device for the determination of* freezing points.

cryoscopy the measurement of freezing points, especially the determination of relative molecular mass or osmotic pressure by measurements of freezing point depressions. —**cryoscopic** *adj.*

cryosolvent any mixture of water and one of a select number of polar organic solvents that is used to maintain proteins in fluid solution at temperatures below 0 °C without denaturation.

cryostat an apparatus used to produce and maintain a constant low temperature in an enclosure or liquid bath.

cryosublimation the process whereby water, or another solvent, is sublimed from a frozen sample and is collected in a cold trap. *Compare* **lyophilization**.

cryotolerant tolerant of low temperatures.

cryptand any synthetic macrooligocyclic organic compound, a molecule of which consists of two or more rings linked together by bridgehead (generally nitrogen) atoms to form a three-dimensional cagelike molecular structure having sufficient space within it to accommodate any one of various cations and thereby form a **cryptate**. *Compare* **crown compound**.

cryptate the adduct of a **cryptand** and a metal ion or other cationic entity. The cation is bound within the molecule by polydentate ligation to heteroatoms in the cryptand, an exceptionally stable compound being formed.

cryptic 1 hidden, secret. 2 not apparent, unrecognized, masked; e.g. of a medical condition. 3 (especially of the colouring of an animal) serving to conceal. 4 of a particle-bound enzyme, that it is not as accessible to substrate as when it has been solubilized.

crypticity *or* **latency** the property of being **cryptic**.

cryptic plasmid a **plasmid** to which no phenotypic traits have been ascribed.

crypto+ *or (before a vowel)* **crypt+** *comb. form* meaning secret, hidden, concealed, unrecognized.

cryptotope a hidden immunological determinant. *Compare* **epitope**.

cryst. *abbr. for* crystalline, or crystallized.

crystal any three-dimensional solid aggregate in which the plane faces intersect at definite angles and in which there is a regular internal structure of the constituent chemical species.

crystal lattice the array of **unit cells** along parallel straight lines that makes up the structure of a crystal.

crystallin any of numerous water-soluble structural proteins found in the lens of the eye. Together the crystallins account for almost 90% of the total lenticular protein. Most vertebrate lenses contain three major classes of crystallins, designated α-, β-, and γ-; in birds and reptiles there is a fourth, designated δ-. ε-Crystallin is homologous with the enzyme **lactate dehydrogenase**. Although there are wide variations between species in the proportions of the various crystallins and in the extent of their post-translational modification or degree of aggregation, they display noteworthy conservation of primary structures. Examples from golden hamster: α-crystallin A chain, database code CRA2_MESAU, 196 amino acids (22.50 kDa); B chain, database code CRAB_MESAU, 175 amino acids (20.07 kDa).

crystalline *abbr.:* cryst.; having the structure or characteristics of a **crystal**; made up of crystals; like a crystal.

crystallize *or* **crystallise** to form, or cause to form, crystals, e.g. by slowly removing solvent from a solution to make a solute form one or more crystals.

crystallized *or* **crystallised** *abbr.:* crys.; (of a substance) having been induced into a crystalline form.

crystallography the study of the geometric forms of crystals. *See also* **X-ray crystallography**. —**crystallographer** *n.*; **crystallographic** *adj.*

crystalloid 1 *an alternative term for* **crystalline**. 2 any solute that can pass through a semipermeable membrane that does not allow the passage of a **colloid**. 3 any of the protein crystals that occur in seeds and other plant storage organs.

Cs *symbol for* cesium.

CS *abbr. for* choriomammotropin.

CSD *abbr. for* cold-shock domain (*see* **cold-shock protein**).

CSF *abbr. for* 1 cerebrospinal fluid. 2 colony stimulating factor.

CSP *abbr. for* cell surface protein; *see* **fibronectin**.

CSPCP *abbr. for* cartilage-specific proteoglycan core.

C_0t *or* **Cot** a measure of DNA kinetic **complexity** used in the renaturation analysis of DNA genomes. It is the product of initial DNA concentration, C_0, and time, t. Highly reiterated sequences will renature at low C_0t values while unique sequences will renature at high C_0t values. $C_0t_{1/2}$ is the value representing half complete renaturation.

CT *abbr. for* computerized tomography (*see* **tomography**).

CTAB *abbr. for* cetyltrimethylammonium bromide.

CTAP-III *abbr. for* connective-tissue activating peptide III; *see* **LA-PF4**.

CTD kinase *see* **[RNA-polymerase]-subunit kinase**.

ctDNA *see* **DNA**.

C terminus *or* **C terminal** that end of any peptide chain at which the 1-carboxy function of a constituent α-amino acid is not attached in peptide linkage to another amino-acid residue. This function may bear an amide group or other substituent in nonpeptide linkage; more commonly it is free (also unprotonated and negatively charged) – the chain end may then also be termed the **carboxy(l) terminus** or **carboxy(l) terminal**. *Compare* **N terminus**. —**C-terminal** *adj.*

C1-tetrahydrofolate synthase a trifunctional enzyme that consists of two major domains: an N-terminal domain, containing N^5,N^{10}-methylenetetrahydrofolate dehydrogenase and N^5,N^{10}-methenyltetrahydrofolate cyclohydrolase (amino acids 1–304 in the example), and a C-terminal domain containing **formate–tetrahydrofolate ligase** domain (amino acids 305–934). It is a homodimer. Example from human: database code C1TC_HUMAN, 934 amino acids (101.31 kDa).

CTF 1 *abbr. for* CAAT box-binding transcription factor *or* nuclear factor 1 (*abbr.:* NF-I); any of a class of **transcription factors** for the eukaryotic RNA polymerase II promoters, **CAAT box** sequences; CTF binds to DNA as a homodimer. Examples from human: CTF1, database code NFI1_HUMAN, 499 amino acids (54.62 kDa); CTF2, database code NFI2_HUMAN, 430 amino acids (47.85 kDa). 2 *abbr. for* competence transcription factor *or* competence protein K; a product of a bacterial regulatory gene required for expression of late competence genes. Example from *Bacillus subtilis*: database code COMK_BACSU, 192 amino acids (22.43 kDa).

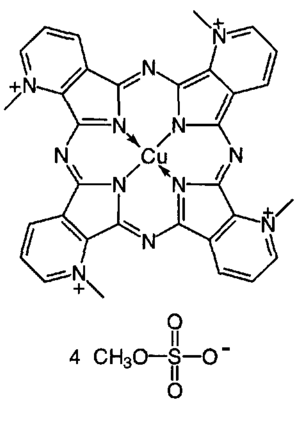
CTGF *abbr. for* connective tissue growth factor.

CTH *abbr. for* ceramide trihexoside (Cer(Hex)$_3$ preferred).

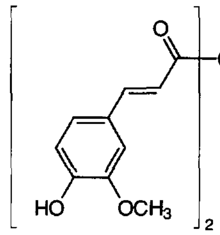
CTP *abbr. for* cytidine 5′-triphosphate.

CTP synthase EC 6.3.4.2; *systematic name:* UTP:ammonia ligase (ADP-forming); *other names:* CTP synthetase; UTP–ammonia ligase. An enzyme responsible for *de novo* synthesis of CTP that in bacteria catalyses the reaction between ATP, UTP, and NH_3 to form CTP, ADP,and orthophosphate. Example from *Escherichia coli*: database code PYRG_ECOLI, 545 amino acids (60.37 kDa). In animals, the amino group is donated by glutamine. The enzyme bears the same EC number and the reaction is analogous to that of the bacterial enzyme, with glutamine replacing ammonium ion, and CTP and glutamate as products. Example from human: database code PYRG_HUMAN, 591 amino acids (66.7 kDa).

CTS *(in clinical biochemistry) abbr. for* catalase.
Cu *symbol for* copper.
CUA a codon in mRNA for L-leucine.
cubic symmetry the symmetry of one of the three **point groups** of a globular oligomeric protein; it may be **tetrahedral**, designated T_n, where n is 12; **octahedral**, designated O_n, where n is 24; or **icosahedral**, with 60 identical subunits.
CUC a codon in mRNA for L-leucine.
cucurbitacin any of a group of complex triterpenoids that are the bitter principles of the cucumber family, Cucurbitaceae.
CUG a codon in mRNA for L-leucine.
culture 1 (a) a collection of cells, tissue fragments, or an organ that is growing or being kept alive in or on a nutrient medium (i.e. **culture medium**); *see also* **cell culture, tissue culture**. (b) any culture medium to which such living material has been added, whether or not it is still alive. 2 the practice or process of making, growing, or maintaining such a culture. 3 to grow, maintain, or produce a culture.
culture medium any nutrient medium that is designed to support the growth or maintenance of a **culture** (def. 1). Culture media are typically prepared artificially and designed for a specific type of cell, tissue, or organ. They usually consist of a soft gel (so-called solid or semisolid medium) or a liquid, but occasionally they are rigid solids.
cUMP *abbr. for* cyclic UMP; *see* **uridine 3′,5′-phosphate**.
cumulative inhibition the inhibition of an enzyme resulting from the action of a number of **effectors**, each of which reduces the activity of the enzyme by a fixed percentage irrespective of the degree of inhibition already extant as a result of the presence of other inhibitors.
cuprein any of a group of copper proteins now known as **superoxide dismutase**.
Cuprolinic Blue *the proprietary name for* quinolinic phthalocyanine (*abbr.:* QPC; no CI number); a new, copper-containing, intensely blue dye used for the visualization of RNA and other polynucleotides. The size and shape of the molecule have been designed to provide an exactly complementary fit between any pair of adjacent isoindole rings in the dye and base pairs in nucleic acids.



cuproprotein any protein containing one or more copper atoms.
curare a poisonous extract from certain tropical South American vines of the genera *Strychnos* and *Chondodendron*, used by Amazon and Orinoco Indians as an arrow poison; it blocks neuromuscular transmission in voluntary muscle and is useful as a muscle relaxant, e.g. in surgery. *See* **(+)-tubocurarine** (the active principle) for the mechanism.
curarize *or* **curarise** to treat with (or as if with) curare; especially to cause muscular relaxation or paralysis. —**curarization** *or* **curarisation** *n.*
curculin a polypeptide present in the fruit of *Curculigo latifolia*, a stimulant herb that grows wild in western Malaysia. It is sweet tasting and has taste-modifying activity. It exists as a homodimer. Database code CURC_CURLA, 114 amino acids (12.37 kDa).
curcumin 1,7-bis(4-hydroxy-3-methoxyphenyl)-1,6-heptadiene-3,5-dione; turmeric yellow; a condiment and dye obtained from dried and powdered rhizomes of *Curcuma domestica (C. longa)*. It provides the characteristic colour of curries, and is also an inhibitor of 5-lipoxygenase and **prostaglandin-endoperoxide synthase** cyclooxygenase activity. *See also* **catechin**.



curdlan *see* **callose**.
curie *abbr.:* Ci *(formerly* C); a non-SI unit of (radio)activity or of radioactive material. Originally (1910) it was defined as the quantity of radon in radioactive equilibrium with one gram of radium. Latterly (1968) it was defined as a unit of activity equal to 3.7×10^{10} disintegrations per second or, less correctly, as the quantity of any radioactive material having such activity. Hence, 1 Ci = 3.7×10^{10} **becquerels**. [After Pierre **Curie** (not after Marie Curie as sometimes stated).]
Curie 1 **Marie Curie** (née Marie Sklodowska, 1867–1934), Polish-born French chemist and physicist, co-discoverer of radium with her husband, P. Curie (*see* def. 2); Nobel Laureate in Physics (1903) jointly with P. Curie 'in recognition of the extraordinary services they have rendered by their joint researches on the radiation phenomena discovered by Professor Henri Becquerel' [prize shared with A. H. Becquerel]; Nobel Laureate in Chemistry (1911) ' in recognition of her services to the advancement of chemistry by the discovery of the elements radium and polonium, by the isolation of radium and the study of the nature and compounds of this remarkable element'. 2 **Pierre Curie** (1859–1906), French physicist and chemist, co-discoverer of radium with his wife, M. Curie (*see* def. 1); Nobel Laureate in Physics (1903) jointly with M. Curie. *See also* **Joliot–Curie**.
curing *(in genetics)* the elimination of a **plasmid** from its host cell.
curtain electrophoresis *see* **continuous-flow electrophoresis**.
cushingoid of, or relating to, **Cushing's disease** or **Cushing's syndrome**, or any of the signs and symptoms associated with these conditions.
Cushing's disease a pituitary-dependent, corticotropin-induced, bilateral adrenal hyperplasia which is one of several conditions embraced by **Cushing's syndrome**. [After Harvey Williams Cushing (1869–1939), US neurosurgeon.]
Cushing's syndrome the group of symptoms and signs resulting from prolonged exposure to inappropriately high plasma **cortisol** concentrations, caused either by primary disease (e.g. tumour) of the adrenal glands or by excessive production of **corticotropin** because of disease (e.g. basophil tumour) of the anterior pituitary. It may also be caused by excessive secretion of cortisol or corticotropin from ectopic sites, or by prolonged therapy with (large doses of) cortisol or other glucocorticoids. The syndrome is characterized by obesity of the trunk, polycythemia, osteoporosis, and glycosuria. The underlying disorder may be corticotropin-dependent or -independent. Of the corticotropin-dependent disorders, **Cushing's disease** is the

most common, accounting for 68% of all types of Cushing's syndrome. The remainder (15% of all types) result from ectopic corticotropin-secreting tumours, that is, non-endocrine tumours such as lung, gut, ovarian, or carcinoid, which in some cases secrete corticotropin. Corticotropin-independent disorders include adrenocortical adenoma (5%) or carcinoma (3%) and nodular adrenal hyperplasia (9%).

cut 1 to sever, to make an incision, to divide (with a sharp instrument). 2 to separate into **fractions**. 3 a double-stranded scission in a polynucleotide duplex. *Compare* **nick**. 4 informal term for a **fraction**, such as one obtained in column chromatography or distillation.

cutaneous of, pertaining to, or in the skin.

cutinase an enzyme that catalyses the hydrolysis of the plant polyester, cutin, to allow penetration by phytopathogenic fungi. Example from *Asochyta rabiei*: database code CUTI_ASCRA, 223 amino acids (23.49 kDa); five motifs.

CUU a codon in mRNA for L-leucine.

cuvette a small transparent vessel, often with parallel sides of specified separation, used to hold a liquid specimen, especially for optical or spectral measurements. *See also* **cell** (def. 2).

C-value the quantity of DNA present in a single haploid genome.

Cya *symbol for* a residue of the α-amino acid cysteic acid (an artefact in peptide sequences).

cyan+ *a variant form (before a vowel) of* **cyano+**.

cyanate 1 the anion, $N{\equiv}C{-}O^-$, derived from cyanic acid ($HOC{\equiv}N$). 2 any salt of cyanic acid. 3 any organic compound containing the monovalent group $-O-C{\equiv}N$. *Compare* **isocyanate**.

cyanelle a cyanobacterium-like structure, considered as a descendant of an ingested cyanobacterium, that exists as an endosymbiont in the biflagellate protist *Cyanophora paradoxa*. The cyanelle has photosynthetic properties resembling those of cyanobacteria but its genome size is about 10% of those reported for cyanobacteria.

cyanhydrin *a former name for* **cyanohydrin**.

cyanide 1 the anion, $N{\equiv}C^-$, derived from hydrocyanic acid (HCN). This ion blocks the electron transport chain by reacting with the ferric form of cytochrome aa_3 and thus acts as an inhibitor of respiration. 2 any salt of hydrocyanic acid. 3 any organic compound containing the monovalent cyano group, $-C{\equiv}N$; such a compound may also be known either as a nitrile or as a carbonitrile, depending respectively on whether its carbon atom is included in or excluded from the numbering of an attached chain.

cyanide-resistant respiration a respiratory pathway, occurring only in plant mitochondria, that is unaffected by **cyanide**. Electron transport from NADH or succinate to O_2 involves a nonheme iron protein as the terminal oxidase; this is insensitive to cyanide. The cyanide-resistant pathway complements the more usual cyanide-sensitive pathway; this contains cytochrome oxidase, which is inhibited by cyanide. Cyanide-resistant respiration is not found in animals.

cyano+ *or (before a vowel)* **cyan+** *comb. form* denoting 1 the cyanide ion or hydrocyanic acid. 2 the –CN group in covalent linkage. 3 (dark) blue.

cyanobacteria *or (formerly)* **blue-green algae** a very heterogeneous group of prokaryotic photosynthetic organisms all of which contain chlorophyll *a*, carotenoids, and phycobiliproteins arranged in thylakoids. They lack chloroplasts, unlike true algae, and differ from photosynthetic bacteria, in, e.g., evolving oxygen during photosynthesis. Many can fix nitrogen.

cyanoborohydride any member of a group of compounds with the structure MBH_3CN, where M is an alkali metal (usually sodium). They are reducing agents similar in action to borohydride but more stable, more lipophilic, and more selective. They are especially useful for the facile reduction of aldehydes and ketones in acid conditions to the corresponding alcohols, for the reductive amination of aldehydes and ketones in the presence of an amine, and for the reductive methylation of amines (including amino acids and proteins) in the presence of formaldehyde.

cyanocobalamin the most usual form of **vitamin B_{12}**, in which the sixth position on the cobalt atom in **cobalamin** is filled by a cyanide group (the cyanide group being an artefact of the isolation procedure).

cyanogen bromide *or* **bromine cyanide** CBrN, or CNBr, or BrCN; a volatile solid (at ordinary temperatures) with two main uses in biochemistry: (1) in the sequencing of proteins, because (in 90% formic acid) it converts methionine residues in peptide linkage to homoserine residues, with simultaneous cleavage of peptide bonds whose carbonyls are contributed by methionine residues; also (in anhydrous heptafluorobutyric acid) it specifically cleaves tryptophanyl peptide bonds; and (2) for activating cross-linked agarose so that almost any molecule containing amino groups can be coupled to it to prepare a specific affinity adsorbent.

cyanogenesis the production or yielding of cyanide ions or hydrocyanic acid.

cyanogenic *or* **cyanogenetic** 1 *or* **cyanophoric** describing any glucoside or other glycoside containing a cyano group that is released as hydrocyanic acid on acid hydrolysis; e.g. amygdalin. Such compounds occur in the kernels of various fruits. 2 describing any plant that synthesizes such a glycoside.

cyanoginosin a cyclic heptapeptide toxic agent responsible for numerous cases of poisoning among South African livestock that consume water contaminated with blooms of cyanobacteria. About seven variants are known.

cyanoglobin an unusual monomeric myoglobin-like hemoprotein found in multicellular cyanobacteria. Example from *Nostoc commune*: database code GLBN_NOSCO, 118 amino acids (12.89 kDa).

cyanohydrin *or (formerly)* **cyanhydrin** any member of a class of α-hydroxynitriles formed by the base-catalysed addition of a cyanide ion to an aldehyde; such compounds are formed also from some reactive ketones.

cyanolabe a pigment sensitive to blue light (445 nm) that is found in some cones of the retina in the eye. It is lacking in subjects with blue-blindness. *Compare* **chlorolabe**, **erythrolabe**.

cyanophage any virus whose host is a cyanobacterium.

cyanophoric *an alternative term for* **cyanogenic** (def. 1).

cyanophycin granule *or* **structured granule** a type of granule, without apparent limiting membrane, that is observed in most, if not all, cyanobacteria. These granules are variable in size and shape, with diameters up to about 500 nm, and contain polypeptides of 25–100 kDa with a 1:1 arginine:aspartic acid ratio. They are believed to act as a nitrogen reserve.

cyanopsin a greenish-blue, light-sensitive, visual pigment occurring in the cone cells of the retinas of freshwater and migratory fishes and some amphibians. It consists of a cone-type opsin combined (as a Schiff's base) with 3,4-didehydro-11-*cis*-retinal. It has an absorption maximum at about 620 nm, and is analogous in function to **iodopsin** in other vertebrates. *See also* **3,4-didehydroretinal**.

cyanosis the bluish appearance of the skin and mucous membranes due to insufficient oxygenation of the blood in the capillaries. —**cyanotic** *adj.*

cyanosome *an alternative name for* a **phycobilisome** in cyanophytes.

cybrid a portmanteau word used to denote a line of cultured mammalian cells that is a cytoplasmic hybrid of two other lines, as shown by inheritance of biochemical or other characteristics.

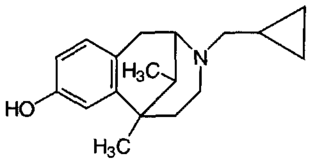
cycacin methylazoxymethanol β-D-glucoside; a toxic substance from the seeds of *Cycas revoluta*, the aglycon of which is carcinogenic.

cycl+ *a variant form (before a vowel) of* **cyclo+**.

cyclamate the anion of cyclamic acid, cyclohexylsulfamic acid. It is used in some countries as a sweetening agent, but is banned in the USA because of its possible carcinogenicity.

cyclase any enzyme that catalyses ring closure, e.g. **adenylate cyclase**.

cyclazocine 3-(cyclopropylmethyl)-1,2,3,4,5,6-hexahydro-6,11-dimethyl-2,6-methano-3-benzazocin-8-ol; a potent benzomorphan narcotic analgesic agent. It is an antagonist at μ-, and an agonist at δ- and especially κ-opioid receptors. Binding to σ-receptors may explain its psychotomimetic actions.



cycle 1 any recurring period of time in which certain events or operations occur, complete themselves, or repeat themselves in a regular sequence. 2 any sequence of changes occurring in a system in which the system is eventually restored to its initial state. 3 *(in biochemistry)* any closed sequence of metabolic reactions in which an end product acts as a reactant in the initiation of the cycle, e.g. the **tricarboxylic-acid cycle**. 4 *(in ecology)* any closed sequence of large-scale processes that describes the nutritional interdependence of animals, plants, and microorganisms, e.g. the **nitrogen cycle**. 5 to process through a cycle or cyclic system; to pass through cycles or to vary in a cyclical manner.

cyclic 1 of, relating to, moving in, or being a cycle. 2 (of an organic chemical compound) containing one (i.e. monocyclic) or more (i.e. polycyclic) rings of atoms. *Compare* **heterocyclic**, **homocyclic**. —**cyclically** *adv.*

cyclic adenosine 5′-diphosphoribose *or* cyclic ADP-ribose *abbr.:* cyclic ADPR *or* cADPR; a metabolite derived from NAD^+ and thought to be responsible for mediating certain of the latter's regulatory effects. It consists of a ribose molecule linked by its C-5 hydroxyl to the *β*-phosphate of ADP and by its C-1 hydroxyl to the adenosine amino group. It is suggested that it is involved in regulating intracellular Ca^{2+} concentrations and in activating the nonskeletal form of the ryanodine receptor Ca^{2+} channel.

cyclical *a variant form of* **cyclic** (def. 1).

cyclic AMP *or* cAMP *abbr. for* adenosine 3′,5′-phosphate. *See also* **adenylate cyclase**.

cyclic AMP-dependent protein kinase a **protein kinase** that is regulated by cyclic AMP (cAMP). There are two types with respect to regulation. In **type I**, the inactive form of the enzyme is composed of two regulatory chains and two catalytic chains; activation by cAMP produces two active catalytic monomers and a regulatory dimer that binds four cAMP molecules. Expression of regulatory chains varies among tissues and is in some cases constitutive and in others inducible. Example (human): database codes: KAP0_HUMAN (α regulatory chain), 381 amino acids (42.93 kDa); KAP1_HUMAN (β regulatory chain), 379 amino acids (43.03 kDa). In **type II**, regulatory chains mediate membrane association by binding to anchoring proteins, including the **MAP-2** kinase and **AKAP**. The inactive form of the enzyme is composed of two regulatory chains and two catalytic chains; activation by cAMP produces two active catalytic monomers and a regulatory dimer that binds four cAMP molecules. Database codes: KAP2_HUMAN (α regulatory chain), 417 amino acids (46.16 kDa); KAP3_HUMAN (β regulatory chain), 403 amino acids (45.34 kDa); catalytic subunits (α, β, γ respectively): database codes KAPA_HUMAN, 350 amino acids (40.41 kDa); KAPB_HUMAN, 350 amino acids (40.44 kDa); KAPG_HUMAN, 350 amino acids (40.24 kDa).

cyclic AMP phosphodiesterase (number EC 3.1.4.17 shared with **cyclic GMP phosphodiesterase**); *recommended name*: 3′,5′-cyclic-nucleotide phosphodiesterase; an enzyme that catalyses the hydrolysis of a nucleoside 3′,5′-cyclic phosphate to a nucleoside 5′-phosphate. The intracellular concentration of cyclic AMP is regulated by low-affinity (type 1) and high-affinity (type 2) enzymes. Examples (yeast) type 1: database code CNA1_YEAST, 369 amino acids (42.00 kDa); type 2: database code CNA2_YEAST, 526 amino acids (60.83 kDa). The enzymes are monomers; type 2 is well conserved.

cyclic AMP receptor any of various receptor molecules that bind cyclic AMP, especially **CRP** (def. 2), a bacterial catabolite (gene) activator protein. Other examples are the signalling receptors of the slime mould, *Dictyostelium discoideum*, which are similar to other eukaryotic G-protein-linked receptors. Receptor 1 is involved in aggregation, and receptor 2 is expressed after aggregation in pre-stalk cells. Receptor 1: database code CAR1_DICDI, 392 amino acids (44.26 kDa); receptor 2: database code CAR2_DICDI, 375 amino acids (43.26 kDa).

cyclic AMP response element *see* **CRE**.

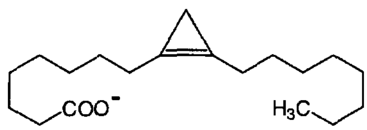
cyclic analysis a procedure that is useful, e.g. in flame spectrophotometry, for minimizing the effects of any mutual interference between two or more components of a sample being analysed. First, the concentration of each such component is estimated. Next, standards for each of the components are prepared containing the other interfering components at their estimated concentrations and the estimations are repeated. Then fresh standards are prepared, using the revised values for the concentrations of the interfering components, and further estimations are made; and so on, until a constant and accurate value for each component is obtained. In practice, usually only two or three such cycles of analysis are necessary.

cyclic CMP *see* **cytidine phosphate**.

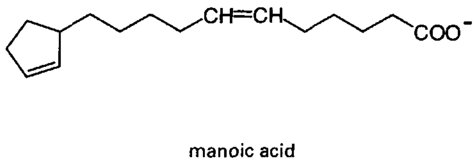
cyclic diguanylic acid bis(3′-5′)cyclic guanylic acid; a potent activator of cellulose synthase in the bacterium *Acetobacter xylinum*.

cyclic electron flow *see* **cyclic photophosphorylation**.

cyclic fatty acid any of a class of fatty acids that contain a carbocyclic unit. They include those with a cyclopropane unit, such as **lactobacillic acid** found in bacterial membranes, those with a cyclopropene unit such as sterculic acid (*see* structure), found in Sterculiaceae, and those with a cyclopentene unit, found in seed fats of Flacourtiaceae, these including manaoic acid (*see* structure). **Mycolic acids** are 2-alkyl-3-hydroxy acids with up to 80 carbons, and contain cyclopropane units. Fatty acids containing a cyclohexane unit are also known.



sterulic acid



manoic acid

cyclic GMP *abbr. for* guanosine 3′,5′-phosphate.

cyclic GMP phosphodiesterase (EC 3.1.4.17, number shared with **cyclic AMP phosphodiesterase**); *recommended name*: 3′,5′-cyclic-nucleotide phosphodiesterase. An enzyme that catalyses the hydrolysis of a nucleoside 3′,5′-cyclic phosphate to a nucleoside 5′-phosphate. It is involved in the regulation of the concentration of cyclic GMP, and thereby involved in the transmission and amplification of the visual signal; *see* **guanosine 3′,5′-phosphate**. Example from mouse rod cells has three subunits: α and β are catalytic subunits (short form of β is β′ from alternative splicing), and γ is a regulatory subunit; α and β are typical cyclic nucleotide phosphodiesterases. α database code CNRA_MOUSE, 858 amino acids (99.40 kDa); β database code CNRB_MOUSE, 856 amino acids (98.39 kDa); γ database code CNRG_MOUSE, 87 amino acids (9.63 kDa).

cyclic IMP *symbol:* cIMP *or* Ino-3′,5′-*P*; inosine 3′,5′-(cyclic)phosphate.

cyclic nucleotide any nucleotide in which the phosphate group is in diester linkage to two positions on the sugar residue, usually 3′,5′ but also sometimes 2′,3′.

cyclic photophosphorylation the processes that occurs in photosystem I in which the high-potential electron in bound ferredoxin can be transferred to cytochrome b_{563}, rather than to $NADP^+$, and then back to the oxidized form of P700 through cytochrome c_{552} and plastocyanin with the generation of one molecule of ATP. *See* **photosystem**.

cyclic symmetry the symmetry of a **point group**, designated C_n, having a geometric arrangement of the subunits such that a rotation of 360°/*n* transposes the structure into itself (i.e. it is indistinguishable from the original one). A dimeric molecule with chemically and spatially identical subunits must necessarily belong to the point group C_2. In general, a molecule with C_n symmetry will be a closed ring containing *n* subunits (e.g. of triangular, square, pentagonal, and hexagonal appearance for molecules containing three, four, five, and six subunits, respectively). If *n* is an odd number, C_n is the only symmetry possible.

cyclic UMP *symbol:* cUMP *or* Urd-3′,5′-*P*; uridine 3′,5′-(cyclic)phosphate.

cyclin any of a group of proteins that are involved in regulation of the eukaryotic **cell-division cycle**. Cyclins periodically rise and fall in concentration in step with the cycle. Individual cyclins activate particular **cyclin-dependent protein kinases** and thereby help to control progression from one stage of the cycle to the next (e.g. p34^{cdc2}/cyclin B regulates the G_2 to M transition in all eukaryotes). All the following examples are from *Saccharomyces cerevisiae*: (1) G_1/S-specific cyclin Cln1; this is essential for control of the cell cycle at the G_1/S (START) transition; it interacts with the Cdc28 protein kinase to form S-phase promoting factor (**SPF**); database code CG11_YEAST, 546 amino acids, (62.11 kDa). (2) G_1/S-specific cyclin Cln2; this has an overlapping function with Cln1. Cln1 and Cln2 mRNAs fluctuate in the cell cycle; database code CG12_YEAST, 545 amino acids (61.62 kDa). (3) G_2/mitotic-specific cyclin C1B1; this interacts with the Cdc2 protein kinase to form MPF. G_2/M cyclins accumulate steadily during G_2 and are abruptly destroyed at mitosis; database code CG21_YEAST, 471 amino acids (54.81 kDa). G_2/mitotic-specific cyclins C1B2–4 are similar. (4) S-phase entry cyclins C1B5 and 6; these are maximally expressed just before the S-phase of the cell cycle; database code CGS5_YEAST, 435 amino acids (50.37 kDa). *See also* ***cdc*** **gene**, **cyclin-dependent protein kinase**, **cyclin regulatory proteins**.

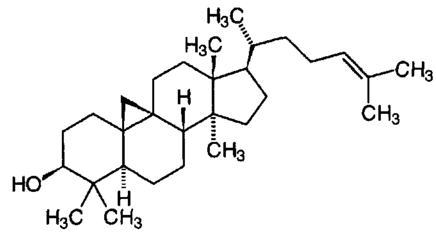
cyclin-dependent protein kinase Cdk; a protein kinase that is only active when complexed with a regulatory protein, a **cyclin**. Different Cdk–cyclin complexes are thought to trigger different steps in the cell-division cycle by phosphorylating specific target proteins. *See also* **cyclin regulatory protein**.

cyclin regulatory protein any protein involved in the regulation of **cyclin** gene expression. An example from *Saccharomyces cerevisiae* is Cdc68. This plays a role in general transcription, with both positive and negative effects on gene expression; it is required for the appropriate synthesis during heat shock, and for continued expression of cyclin genes; database code CC68_YEAST, 1035 amino acids (118.50 kDa). A further example, also from yeast, is the Ser/Thr protein kinase Cdc28 (*see* ***cdc*** **gene**).

cyclitol any cycloalkane containing one hydroxyl group on each of three or more ring atoms. Notable members are the **inositols** and their derivatives.

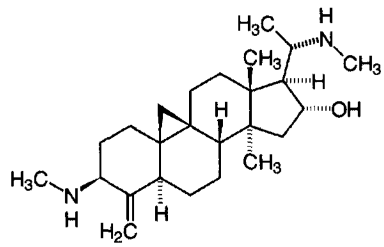
cyclo+ *or (before a vowel)* **cycl+** *comb. form* indicating a circle or ring, especially one in a cyclic organic chemical compound.

cycloartenol the first triterpene intermediate in the biosynthesis of sterols in photosynthetic tissues. It is formed from (*S*)-squalene-2,3-epoxide by a cyclization reaction catalysed by cycloartenol synthase, EC 5.4.99.8. The corresponding intermediate in animal tissues is **lanosterol**.



cyclobutadipyrimidine a type of **pyrimidine dimer** formed within a DNA strand between neighbouring pyrimidine residues.

cyclobuxine a steroidal alkaloid extracted from *Buxus microphylla*. The extracts are used as folk remedies for malaria and venereal diseases. It protects the heart from ischemia and reperfusion damage by inhibiting the release of ATP metabolites during reperfusion.



cyclobuxine D

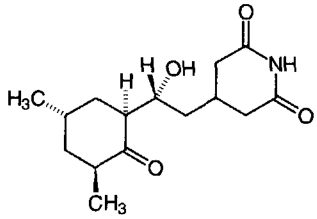
cyclodeaminase *an informal name for* **formiminotetrahydrofolate** cyclodeaminase.

cyclodepsipeptide *see* **depsipeptide**.

cyclodextrin any of a group of cyclic nonreducing **dextrins** formed from starch or glycogen by the extracellular enzyme cyclomaltodextrin glucanotransferase (EC 2.4.1.19), produced by *Bacillus macerans*. They consist of cyclic polymers containing six, seven, or eight α-(1,4)-linked D-glucose residues. These were formerly known, respectively, as Schardinger α-, β-, or γ-dextrins, or α-, β-, or γ-cyclodextrins, but now are more correctly named cyclomaltohexaose, cyclomaltoheptaose, or cyclomaltooctaose. Cyclodextrins have a toroidal molecular structure enclosing a hydrophobic cavity, thereby permitting the formation of inclusion complexes with a wide variety of molecules.

cyclofenil bis-(*p*-acetoxyphenyl)cyclohexylidenemethane; an agent that prevents feedback inhibition of ovulation by inhibiting the binding of estrogen in the brain. The resulting increased secretion of gonadotropin-releasing hormone, gonadotropins, and estrogen promotes ovulation.

cycloheximide *or* **actidione** 3-[2(3,5-dimethyl-2-oxocyclohexyl)-2-hydroxyethyl]glutarimide; an antibiotic produced by some *Streptomyces* spp. It interferes with protein synthesis in eukaryotes by inhibiting peptidyltransferase activity of the 60S ribosomal subunit.



cyclohydrolase *an informal name for* the enzyme **methenyltetrahydrofolate cyclohydrolase**, EC 3.5.4.9.

cyclo-ligase any **ligase** enzyme of the sub-subclass EC 6.3.3 that catalyses the formation of a heterocyclic ring containing a C–N bond; e.g. dethiobiotin synthase, EC 6.3.3.3.

cyclomaltoheptaose *see* **cyclodextrin**.

cyclomaltohexaose *see* **cyclodextrin**.

cyclomaltooctaose *see* **cyclodextrin**.

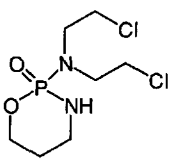
cyclooxygenase *an alternative (informal) name for* **prostaglandin–endoperoxide synthase**.

cyclooxygenation the process that introduces one molecule of dioxygen at C-9 and a second dioxygen at C-15 of a molecule of arachidonic acid, accompanied by formation of a bond between C-8 and C-12 and an endoperoxide bridge across C-9 and C-11.

cyclophilin the protein to which the immunosuppressant **cyclosporin A** binds. Cyclophilin possesses **peptidylprolyl isomerase** activity. The *in vivo* function of this **immunophilin** is not clear.

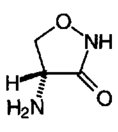
cyclophorase originally, a supposed integrated complex of enzymes that carries out the tricarboxylic-acid cycle, fatty-acid oxidation, oxidative phosphorylation, and terminal electron transport. This system was identified later with **mitochondria**. The cyclophorase complex and the mitochondria are thus the functional and structural sides of the same unit.

cyclophosphamide *N*,*N*-bis(2-chloroethyl)tetrahydro-2*H*-1,3,2-oxazaphosphorin-2-amine 2-oxide; an alkylating agent that is metabolized by the liver cytochrome P450 system to 4-hydroxycyclophosphamide, which is in equilibrium with aldophosphamide. These are transported to tissues, which convert them to the cytotoxic agent, phosphoramine mustard, and acrolein. Cyclophosphamide is useful as an immunosuppressant. *See also* **chlorambucil**.



cyclopropylamine *see* **monoamine oxidase inhibitors**.

D-cycloserine D-4-amino-3-isoxazolidinone; a tuberculostatic antibiotic found in cultures of *Streptomyces orchidaceus*. It is a potent inhibitor of alanine racemase (EC 5.1.1.1) and D-alanine–D-alanine ligase (EC 6.3.2.4; alanylalanine synthetase), two enzymes concerned in the biosynthesis of bacterial cell wall peptidoglycan.



cyclosporin A a cyclic undecapeptide isolated from the fungi *Cylindrocarpon lucidum* and *Tolypocladium inflatum* (formerly *Trichoderma polysporum*). It is a potent immunosuppressive drug that prevents the rejection of allografts, and is used extensively in human transplant operations. Its main target appears to be the T lymphocyte: it contains D-alanine and several nonprotein amino acids, and blocks a calcium-dependent signal from the **T-cell receptor** that normally leads to T-cell activation. When bound to **cyclophilin**, cyclosporin A binds and inactivates the key signalling intermediate **calcineurin**.

cyclotron a machine used to accelerate charged particles of atomic magnitudes to energies of the order of 25 MeV. The particles move in spiral paths under the influence of a uniform vertical magnetic field and are accelerated by a constant frequency electric field.

Cyd *symbol for* a residue of the ribonucleoside **cytidine** (alternative to C).

CydP *symbol for* cytidine phosphate (*see* **cytidine phosphate**).

Cyd2′P *symbol for* cytidine 2′-phosphate (*see* **cytidine phosphate**).

Cyd-2′,3′-P *symbol for* cytidine 2′,3′-phosphate (*see* **cytidine phosphate**).

Cyd3′P *symbol for* cytidine 3′-phosphate (*see* **cytidine phosphate**).

Cyd-3′,5′-P *symbol for* cytidine 3′,5′-phosphate (*see* **cytidine phosphate**).

Cyd5′P *symbol for* cytidine 5′-phosphate (*see* **cytidine phosphate**).

Cyd5′PP *symbol for* cytidine 5′-diphosphate (*see* **cytidine phosphate**).

Cyd5′PPP *symbol for* cytidine 5′-triphosphate (*see* **cytidine phosphate**).

c-yes *see* **YES**.

D-cymarose 2,6-dideoxy-3-*O*-methyl-*ribo*-hexose; the D enantiomer is a constituent of some **cardiac glycosides**.

cyproterone 6-chloro-6-dehydro-17α-hydroxy-1,2α-methyleneprogesterone; a progesterone derivative with anti-androgen activity through partial agonist activity at androgen receptors. It inhibits the secretion of gonadotropins and spermatogenesis in humans.

Cyr61 a growth factor binding protein expressed from G_0/G_1 to mid-G_1 of the cell cycle. It is expressed in high amounts in lung. Example (precursor) from *Mus musculus*: database code CYR6_MOUSE, 379 amino acids (41.66 kDa). *Compare* **connective tissue growth factor**.

Cys *symbol for* a residue of the α-amino acid L-cysteine. When written

$$\begin{array}{c}|\\ \text{Cys}\end{array} \quad \text{or} \quad \begin{array}{c}\text{Cys}\\ |\end{array},$$

it represents L-half-cystine.

cyst **1** any saclike structure, generally of a pathological nature, comprising a central cavity lined with epithelium and containing fluid or semisolid material. **2** the capsule enclosing a dormant larva or spore. —**cystic** *adj.*

cystamine β,β'-diaminodiethyl disulfide; 2,2′-dithiobis(ethylamine); 2,2′-dithiobisethanamine; $H_2N–CH_2–CH_2–S–S–CH_2–CH_2–NH_2$; the decarboxylation product of cystine and an oxidation product of **cysteamine**.

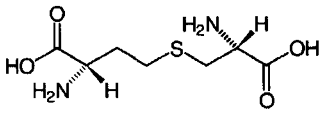
cystate **1** the **cystine** anion; the dianion, $^{-}OOC–CH(NH_2)–CH_2–S–S–CH_2–CH(NH_2)–COO^{-}$, derived from cystine. **2** any salt containing the cystine anion. **3** any ester of cystine. *Distinguish from* **cysteate** and **cysteinate**.

γ-cystathionase *see* **cystathionine γ-lyase**.

cystathionine *trivial name for* *S*-(β-amino-β-carboxyethyl)-homocysteine; (R*)-2-amino-4-[(S*)-2-amino-2-carboxyethylthio]butanoic acid; $HOOC–CH(NH_2)–CH_2–S–CH_2–CH_2–CH(NH_2)–COOH$; a di($\alpha$-amino acid) and a mixed thioether with two chiral centres. L-Cystathionine

$$(\textit{symbol}: \begin{array}{c}\text{Ala}\\ |\\ \text{Hcy}\end{array}),$$

S-(L-alanin-3-yl)-L-homocysteine; (*S*)-2-amino-4-[(*R*)-2-amino-2-carboxyethylthio)butanoic acid – in which both moieties are in the L-configuration, does not occur in proteins but is an intermediate in the transfer of the sulfur atom from methionine, via homocysteine, to cysteine in mammals. (*Note*: The pair of (R^*,R^*)-enantiomers, in which one moiety is in the D-configuration and the other in the L-configuration, is sometimes known as **allocystathionine**.)



cystathionine γ-lyase EC 4.4.1.1; *systematic name*: L-cystathionine cysteine-lyase (deaminating); *other names*: homoserine deaminase; homoserine dehydratase; γ-cystathionase; cystine desulfhydrase; cysteine desulfhydrase; cystathionase; a multifunctional pyridoxal-phosphate enzyme that catalyses the formation of L-cysteine with release of 2-oxobutanoate and ammonium ion from L-cystathionine and water. It also acts on homoserine and cysteine to remove NH_3, forming oxobutanoate from homoserine or pyruvate and H_2S from cysteine. The enzyme requires pyridoxal-phosphate as coenzyme. Example from human: database code CGL_HUMAN, 361 amino acids (39.49 kDa). It carries out the final reaction in the pathway for the synthesis of cysteine from methionine. Example from *Saccharomyces cerevisiae*: database code CYS3_YEAST, 393 amino acids (42.41 kDa).

cystathionine β-synthase EC 4.2.1.22; *systematic name:* L-serine hydro-lyase (adding homocysteine); *other names:* serine sulfhydrase; β-thionase; methylcysteine synthase. An enzyme that catalyses the reaction between L-serine and L-homocysteine to form cystathionine and water. This is the first step in the synthesis of cysteine from serine. Pyridoxal-phosphate is a cofactor. Example from *Saccharomyces cerevisiae*: database code STR4_YEAST, 507 amino acids (55.96 kDa).

cystathionine γ-synthase EC 4.2.99.9; *recommended name:* *O*-succinylhomoserine (thiol)-lyase. An enzyme in the pathway for the synthesis of methionine in plants and bacteria. It catalyses the formation of cystathionine from *O*-succinyl-L-homoserine and L-cysteine to form succinate. Pyridoxal-phosphate is a cofactor. The enzyme can utilize hydrogen sulfide or methanethiol as a thiol substrate. Example from *Escherichia coli*: database code METB_ECOLI, 386 amino acids (41.50 kDa).

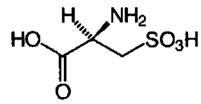
cystathioninuria a condition of humans in which there is excessive excretion of L-cystathionine in the urine. It may be caused by an inherited deficiency of cystathionine γ-lyase.

cystatin any of a group of proteins, present in tissues and body fluids, that inhibit cysteine proteinases. The cystatins constitute a single evolutionary protein superfamily, and are classified into three different families. Family 1 proteins contain about 100 amino-acid residues (M_r 11 000–12 000) and lack disulfide bonds. Family 2 proteins have about 120 residues (M_r 13 000–14 000) with two intrachain disulfide bonds. Family 3 proteins (*also called* **kininogens**) contain three cystatin-like domains, each of which has two disulfide bonds at positions homologous to those in family 2. Examples: cystatin A (*also called* stefin A), database code CYTA_HUMAN, 98 amino acids (10.99 kDa); type 2 cystatin C (*also called* **neuroendocrine basic polypeptide** *or* **post-γ-globulin**): precursor database code CYTC_HUMAN, 146 amino acids (15.78 kDa); this is expressed in highest levels in the epididymis, vas deferens, brain, thymus, and ovary, and is found in sera from patients with Icelandic cerebrovascular type amyloidosis and related diseases.

cysteamine thioethanolamine; β-mercaptoethylamine; 2-aminoethanethiol; $HS–CH_2–CH_2–NH_2$; the decarboxylation product of cysteine. It readily oxidizes in air to **cystamine**. Pantetheine, a component of coenzyme A, is the amide formed between cysteamine and pantothenic acid.

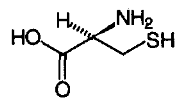
cysteate **1** *the common name for* cysteate(1–); cysteic-acid monoanion; hydrogen cysteate. (*Note:* In theory, the term denotes any ion or mixture of ions formed from **cysteic acid** and having a net charge of minus one, although the species $^-O_3S–CH_2–CH(NH_3^+)–COO^-$ generally predominates in practice.) **2** *the systematic name for* cysteate(2–); cysteate-acid dianion. **3** any salt containing an anion of cysteic acid. **4** any ester of cysteic acid. *Compare* **cystate**, **cysteinate**.

cysteic acid *the trivial name for* 3-sulfoalanine; α-amino-β-sulfopropionic acid; 2-amino-3-sulfopropanoic acid; $HO_3S–CH_2–CH(NH_2)–COOH$; a chiral α-amino acid and an oxidation product of cysteine or cystine. Residues of L-cysteic acid (*symbol*: Cya), (*R*)-2-amino-3-sulfopropanoic acid, do not occur normally in proteins although they are found in wool from the outer part of the fleece of sheep following exposure to sunlight. They are commonly encountered during sequencing of peptides and proteins as the end product of oxidative cleavage of the disulfide bonds of cystine residues and concomitant oxidation of cysteine residues.



cysteinate **1** cysteinate(1–); cysteine monoanion; the anion, $HS–CH_2–CH(NH_2)–COO^-$, derived from **cysteine**. **2** cysteinate(2–); cysteine dianion; the anion, $^-S–CH_2–CH(NH_2)–COO^-$, derived from cysteine. **3** any salt containing an anion of cysteine. **4** any ester of cysteine. *Compare* **cystate**, **cysteate**.

cysteine *the trivial name for* β-mercaptoalanine; α-amino-β-thiolpropionic acid, 2-amino-3-mercaptopropanoic acid; $HS–CH_2–CH(NH_2)–COOH$; a chiral α-amino acid. In neutral or alkaline solution it is readily oxidized by air to **cystine**. L-Cysteine (*symbol*: C *or* Cys), (*R*)-2-amino-3-mercaptopropanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: UGC or UGU. In mammals it is a nonessential dietary amino acid and is glucogenic. D-Cysteine (*symbol*: D-Cys *or* DCys), (*S*)-2-amino-3-mercaptopropanoic acid, occurs naturally in firefly **luciferin**. (*Note*: application of the **sequence rule** to the enantiomers of cysteine results in their stereochemical designation as either *R* or *S* being the converse of that possessed by most chiral α-amino acids found in proteins.)



L-cysteine

cysteine (acid) the protonated form of a **cysteine** residue, i.e. with an undissociated thiol group.

cysteine desulfhydrase *see* **cystathionine γ-lyase**.

cysteine endopeptidase *or (formerly)* **thiol proteinase** any enzyme of the sub-subclass EC 3.4.22, which consists of proteinases characterized by having a cysteine residue at the active centre and by being irreversibly inhibited by sulfhydryl reagents such as iodoacetate. The group includes cathepsins B, H, and L (*see* **cathepsin B**, **cathepsin H**, and **cathepsin L**), **clostripain**, **ficain**, and **papain**. *See also* **apoptosis**.

cysteine-type carboxypeptidase EC 3.4.18.1; *other names*: lysosomal carboxypeptidase B; cathepsin B_2; cathepsin IV;

acid carboxypeptidase; an enzyme that catalyses the liberation of a C-terminal amino-acid residue from a protein. It is an enzyme with broad specificity, but lacks action on C-terminal proline.

cysteinyl the acyl group, $HS-CH_2-CH(NH_2)-CO-$, derived from **cysteine**. (*Note:* Although this name is formed atypically, its use is recommended to avoid confusion with **cysteyl**.) *Compare* **cystyl**, **half-cystyl**.

cystein-S-yl the alkylthio group, $HOOC-CH(NH_2)-CH_2-S-$, derived from **cysteine**.

cysteyl 1 the acyl group, $HO_3S-CH_2-CH(NH_2)-CO-$, derived from **cysteic acid**. 2 *an alternative name for* **cysteinyl** (use now discouraged).

cystic fibrosis a heterogeneous metabolic disorder affecting all exocrine glands and probably other tissues. It is the most common lethal genetic disorder of Caucasians, although the disease is also found in other races. There is an abnormality in the secretions of the exocrine glands, which have higher than normal concentrations of Na^+, Cl^-, nucleotides, Ca^{2+}, etc.; a number of abnormal protein factors have been described in the exocrine glands of patients. Many cases are due to an abnormality of **cystic fibrosis transmembrane conductance regulator**.

cystic fibrosis transmembrane conductance regulator *abbr.:* CFTR; a 168 kDa integral membrane protein encoded by the cystic fibrosis gene and with a structural similarity to membrane-associated ATP-dependent transporter protein (*see* **ABC transporters**), one of a family of proteins involved in the active transport of ions and small molecules across membranes. CFTR belongs to the **MDR protein** subfamily. The CFTR protein appears to be involved in the functioning of a chloride ion channel. A mutation in this protein has been identified as being responsible for 70% of the cystic fibrosis chromosomes in Northern Europe. Example from human: database code CFTR_HUMAN, 1480 amino acids (167.92 kDa).

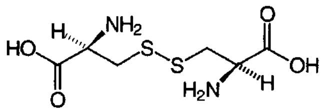
cystine *the trivial name for* dicysteine; β,β'-dithiodialanine; α,α'-diamino-β,β'-dithiobis(propionic acid); (R^*,R'^*)-3,3′-dithiobis(2-aminopropanoic acid);

$$HOOC-CH(NH_2)-CH_2-S-S-CH_2-CH(NH_2)-COOH;$$

a di(α-amino acid) with two chiral centres, formed by the oxidation of **cysteine**. L-Cystine,

$$[\textit{symbol}: \begin{matrix}\text{Cys}\\ |\\ \text{Cys}\end{matrix}],$$

(R,R')-3,3′-dithiobis(2-aminopropanoic acid), is found in peptide linkage in proteins, having been formed by post-translational oxidation of cysteine. In mammals it is a nonessential dietary amino acid and is glucogenic. As it is somewhat insoluble in neutral aqueous solutions, it sometimes forms urinary stones. *meso*-Cystine, (R,S')-3,3′-dithiobis(2-aminopropanoic acid), is the internally compensated achiral diastereoisomer, in which D- and L-half-molecules are combined. *See also note at* **cysteine**.



L-cystine

cystine desulfhydrase *see* **cystathionine γ-lyase**.

cystine knot a structure in a protein resulting from three disulfide bonds formed between six cystein residues that are positioned in a way that results in formation of a knot-like topology. It was originally found in transforming growth factor-β and platelet-derived growth factor.

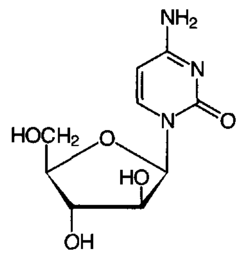
cystinosis a recessively inherited human metabolic disorder characterized by a high intracellular content of free (nonprotein) cystine, apparently compartmentalized within lysosomes, and resulting in deposition of cystine crystals in the cornea, conjunctiva, bone marrow, lymph nodes, leukocytes, and internal organs. —**cystinotic** *adj.*

cystinuria a heritable disorder of cystine and diamino acid (arginine, lysine, and ornithine) transport in humans affecting the epithelial cells of the renal tubules and the gastrointestinal tract. It is characterized by the formation of cystine stones in the urinary tract and elevated levels of cystine and diamino acids in the urine.

cystyl the diacyl group, $-OC-CH(NH_2)-CH_2-S-S-CH_2-CH(NH_2)-CO-$, derived from **cystine**. *Compare* **cysteinyl**, **half-cystyl**.

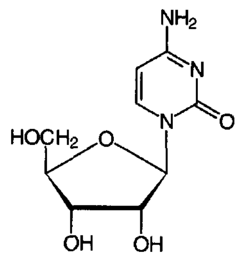
Cyt *symbol for* a residue of the pyrimidine base cytosine (alternative to C).

cytarabine β-cytosine arabinoside; 1-β-D-arabinofuranosylcytosine; *abbr.* ara-C; a cytotoxic drug used as an antineoplastic and antiviral agent. It is effective in the treatment of acute myeloblastic leukemia. It is converted metabolically to 1-β-D-arabinofuranosylcytidine 5′-triphosphate (*abbr.*: *ara*-CTP), a potent inhibitor of DNA synthesis.



+cyte *comb. form* indicating a **cell** (def. 1).

cytidine *symbol*: C *or* Cyd; cytosine riboside; 1-β-D-ribofuranosyl cytosine; a widely distributed **nucleoside**.



cytidine 2′,3′-(cyclic)phosphate *see* **cytidine phosphate**.

cytidine 3′,5′-(cyclic)phosphate *or* **cytidine 3′,5′-cyclophosphate** *see* **cytidine phosphate**.

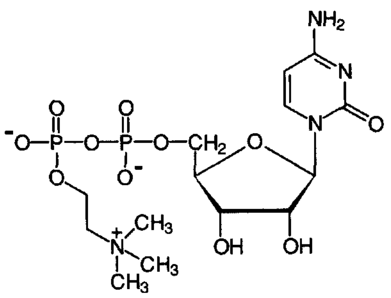
cytidine deaminase EC 3.5.4.5; *systematic name:* cytidine aminohydrolase. An enzyme that catalyses the reaction:

$$\text{cytidine} + H_2O = \text{uridine} + NH_3.$$

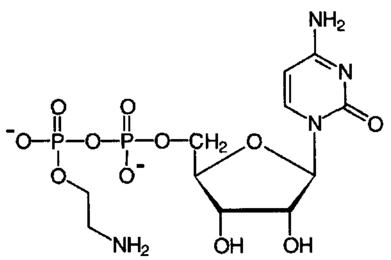
It is a homodimer, with bound Zn^{2+}. Specificity is poor, and deoxycytidine is also a substrate. Example from *Escherichia coli*: database code CDD_ECOLI, 294 amino acids (31.50 kDa).

cytidine 5′-diphosphate *symbol*: Cyd5′*PP or* ppC; *the recommended name for* cytidine diphosphate (*abbr.*: CDP), 5′-diphosphocytidine, 5′-cytidylyl phosphate, cytidine 5′-(trihydrogen diphosphate), a universally distributed nucleotide that occurs both in the free state and as a component of certain **nucleotide coenzymes**. It is a metabolic precursor of cytidine 5′-triphosphate, CTP. It is a product of nucleoside triphosphatase, EC 3.6.1.15, and is also formed in a few cases when CTP is involved in phosphotransferase reactions (e.g. dolichol kinase, EC 2.7.1.108); it is converted to deoxycytidine 5′-diphosphate by **ribonucleoside-diphosphate reductase** (EC 1.17.4.1).

cytidine(5′)diphosphocholine *or* **cytidine diphosphate choline**; intermediate in phospholipid synthesis, especially of phosphatidylcholines and sphingomyelins.



cytidine(5′)diphosphoethanolamine *or* **cytidine diphosphate ethanolamine**; intermediate in the synthesis of phosphatidylethanolamines.



cytidine monophosphate *abbr*.: CMP; an alternative name for any cytidine phosphate, but in particular for cytidine 5′-phosphate, especially when its distinction from cytidine (5′-)-diphosphate and cytidine (5′-)triphosphate requires emphasis.

cytidine phosphate *symbol*: Cyd*P*; cytidine monophosphate (*abbr*.: CMP); any phosphoric monoester or diester of cytidine. There are three monoesters: cytidine 2′-phosphate, cytidine 3′-phosphate, and **cytidine 5′-phosphate**, and two possible diesters: cytidine 2′,3′-phosphate and cytidine 3′,5′-phosphate, although cytidine 5′-phosphate is the ester commonly denoted (the locant being omitted if no ambiguity may arise). Cytidine 2′,3′-phosphate (*symbol*: Cyd-2′,3′-*P*), also named cytidine 2′,3′-(cyclic)phosphate *or* cyclic cytidine 2′,3′-monophosphate (*abbr*.: 2′,3′-cyclic CMP), is formed as an intermediate during the alkaline hydrolysis of ribonucleic acid. This diester then readily hydrolyses to a mixture of the two monoesters cytidine 2′-phosphate (*symbol*: Cyd2′*P*), *also named* cytidine 2′-monophosphate (*abbr*.: 2′CMP) *or* 2′-cytidylic acid *or* cytidylic acid a, and cytidine 3′-phosphate (*symbol*: Cyd3′*P*), also named cytidine 3′-monophosphate (*abbr*.: 3′CMP) *or* 3′-cytidylic acid *or* cytidylic acid b. Cytidine 3′,5′-phosphate (*symbol*: Cyd-3′,5′-*P*), also named cytidine 3′,5′-(cyclic)phosphate *or* cyclic cytidine 3′,5′-monophosphate (*abbr*.: 3′,5′-cyclic CMP), is a product of cytidylate cyclase, EC 4.6.1.6.

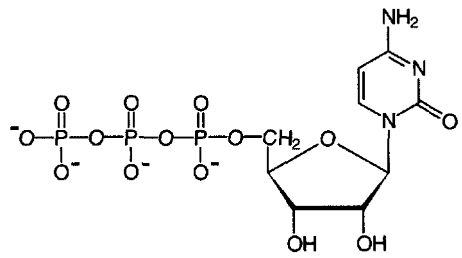
cytidine 2′-phosphate *see* **cytidine phosphate**.

cytidine 2′3′-phosphate *see* **cytidine phosphate**.

cytidine 3′-phosphate *see* **cytidine phosphate**.

cytidine 5′-phosphate *or* **5′-cytidylic acid** *or* **5′-phosphocytidine** *or* **5′-O-phosphonocytidine**; *symbol*: Cyd5′*P*; *alternative recommended names for* cytidine 5′-monophosphate (*abbr*.: 5′CMP), cytidine 5′-(dihydrogen phosphate), cytidine (mono)-ribonucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation). It is formed by pyrophosphatase-catalysed cleavage of (inorganic) diphosphate from CTP, and also as a product of a variety of reactions, e.g. that in which phosphatidylcholine is formed from CDPcholine and diacylglycerol (diacylglycerol cholinephosphotransferase, EC 2.7.8.2).

cytidine 5′-triphosphate *symbol*: Cyd5′*PPP or* pppC; *recommended name for* cytidine triphosphate (*abbr*.: CTP), 5′-triphosphocytidine, 5′-cytidylyl diphosphate, cytidine 5′-(tetrahydrogen triphosphate), an important coenzyme. It is formed by **CTP synthase**. It participates as coenzyme in reactions in which a cytidylyl group is transferred e.g. in lipid metabolism, to choline phosphate (EC 2.7.7.15, to form CDP-choline) and similar reactions, or in carbohydrate metabolism to form CDPglucose (EC 2.7.7.33), or in the phosphotransferase dolichol kinase (EC 2.7.1.108).



cytidyl any chemical group formed by the loss of a 2′-, a 3′-, or a 5′-hydroxyl group from the ribose moiety of cytidine. *Compare* **cytidylyl**.

cytidylate 1 either the mono- or the di-(phosphoric) anion of **cytidylic acid**. 2 any mixture of cytidylic acid and its anions. 3 any salt or ester of cytidylic acid.

cytidylic acid *the trivial name for* any phosphoric monoester of cytidine. The position of the phosphoric residue on the ribose moiety of a given ester may be specified by a prefixed locant. *See* **cytidine phosphate**. However, 5′-cytidylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5′-Cytidylic acid is also an alternative recommended name for **cytidine 5′-phosphate**.

cytidylyl the cytidine[mono]phospho group, the acyl group derived from **cytidylic acid**. *Compare* **cytidyl**.

cytidylylation the process of introducing a cytidylyl group into a substance.

cyto+ *comb. form* indicating a **cell** (def. 1).

cytochalasin any of a group of structurally related fungal metabolites (designated A, B, C, D, E, F, etc.) that share a number of unusual and characteristic effects on cultured mammalian cells. The most notable are inhibition of cytoplasmic cleavage following nuclear division, induction of nuclear extrusion, and inhibition of cell motility. They bind to the growing plus ends of actin filaments, preventing further addition of actin molecules, and thus affect functions that involve assembly and disassembly of actin filaments. The basic structure is a benzyl-substituted hydroisoindolone nucleus to which is fused a small macrolide-like ring. [From Greek *cytos*, cell; *chalasis*, relaxation.]

cytochemical bioassay a technique for the assay of various polypeptide hormones, based on function rather than structure. It involves measuring changes in the biochemical activity of a specific target cell, appropriate to the physiological effect of the hormone in question, by quantitative cytochemical methods. It is claimed to be more sensitive and more specific than equivalent radioimmunoassays.

cytochemistry the chemistry of living cells, especially the cellular localization of biochemical substances and processes; microscopical biochemistry. —**cytochemical** *adj.*

cytochrome any member of a class of hemoproteins whose characteristic mode of action involves the transfer of electrons in association with a reversible change in oxidation state (between Fe(II) and Fe(III), and sometimes Fe(IV)) of the heme iron. The name embraces all intracellular hemoproteins except hemoglobin, myoglobin, peroxidases, catalase, tryptophan 2,3-dioxygenase, nitrite and sulfite reductases, and heme-thiolate proteins. There are four major established subclasses; well-established members of each subclass are generally distinguished by consecutive numerical subscripts; remaining members are designated by a suffixed number based upon the wavelength (in nanometres) of the α-band (e.g. cytochrome c_1, cytochrome c_{555}); *see* **cytochrome absorption bands**. (1) In **cytochromes *a*** the heme prosthetic group contains a formyl side-chain, i.e. heme *a*. (2) **Cytochromes *b*** contain protoheme or a related heme (without a formyl group) as prosthetic group, not covalently bound to the protein. The group includes **cytochrome *o***, **cytochrome P-450**, and **helicorubin**, as well as **ubiquinol-cytochrome *c* reductase**, **cytochrome *b***, **cytochrome b_2**, **cytochrome b_5**, **cytochrome b_{245}**, **cytochrome b_{599}**, **cytochrome b_6**, and **cytochrome b_6–*f*** complex. (3) In **cytochromes *c*** there are covalent linkages, not necessarily thioether linkages, between the heme side-chains and the protein (*see also* **cytochrome *c***, **cytochrome c_1**). (4) **Cytochromes *d*** contain a tetrapyrrollic chelate of iron as prosthetic group, in which the degree of conjugation of double bonds is less than in porphyrin. *See also* **cytochrome *f***.

cytochrome absorption bands three principal absorption bands in the absorption spectrum of many ferrohemoproteins, being the α-band, β-band, and γ-band (*or* **Soret band**), resulting from the absorption of radiation by the heme groups. They are useful in the characterization and study of these proteins. Characteristic absorption maxima for cytochromes *a*, *b*, *c*, and c_1 are for the α-band: 600, 563, 550, and 554 nm respectively; for the β-band: (none for *a*) 532 (*b*), 531 (*c*), and 524 (c_1) nm; and for the γ-band 439, 429, 415, and 418 nm.

cytochrome *b* a component of the eukaryotic mitochondrial **respiratory chain** complex III. It is an integral membrane protein with two hemes, b_{562} and b_{566}. Example (human): database code CYB_HUMAN, 380 amino acids (42.68 kDa).

cytochrome b_2 EC 1.1.2.3; an enzyme (*recommended name*: L-lactate dehydrogenase (cytochrome)) that catalyses the oxidation of L-lactate by 2 ferricytochrome *c* to form pyruvate and 2 ferrocytochrome *c*. Flavin mononucleotide (FMN) and protoheme IX are cofactors. Cytochrome b_2, is a mitochondrial matrix protein induced by lactate during respiratory adaptation (in yeast). Database code CYB2_YEAST (precursor), 591 amino acids (65.46 kDa); the mature enzyme is a homotetramer; 3-D structure is known.

cytochrome b_5 an electron carrier for several oxygenases, present in animal microsomes and in erythrocytes. It is reduced by **cytochrome b_5 reductase**. Two forms are synthesized by alternative splicing. Example (human): database code CYB5_HUMAN, 133 amino acids (15.18 kDa).

cytochrome b_5 reductase EC 1.6.2.2; a flavoprotein (FAD) enzyme that catalyses the reaction:

$$\text{NADH} + 2\ \text{ferricytochrome}\ b_5 = \text{NAD}^+ + 2\ \text{ferrocytochrome}\ b_5.$$

cytochrome b_6 *or* **cytochrome b_{563}** closely related to cytochrome *b*, that is part of the *bc* complex functioning between photosystems II and I of photosynthesis. Example (barley, wheat): database code CYB6_WHEAT, 215 amino acids (24.14 kDa).

cytochrome b_6–*f* complex a complex, occurring in chloroplasts, of cytochrome b_6, the 17 kDa polypeptide (*petD* gene product), cytochrome *f*, and the **Rieske protein**.

cytochrome b_{245} a component of the phagocyte oxidase complex. The protein is a dimer (human) of α and β subunits. Examples: α, database code C24A_HUMAN, 194 amino acids (20.80 kDa); β, database code C24B_HUMAN, 569 amino acids (65.13 kDa).

cytochrome b_{599} a cytochrome of photosystem II of photosynthesis. It exists as a dimer of α and β chains. Examples (pea): α chain, database code PSBE_PEA, 82 amino acids (9.27 kDa); β chain, database code (identical in *Oenothera bertiana*), PSBF_PEA, 39 amino acids (4.42 kDa). *See* **photosystem**.

cytochrome *c* a soluble mitochondrial matrix protein that associates with the inner mitochondrial membrane and functions to transfer electrons between **respiratory chain** complexes III and IV. Its structure has been extensively studied and the sequence from many (>60) species determined. Phylogenetic studies show that about 26 residues are completely invariant, especially those with side chains packing against the heme. The apoprotein is nucleus-encoded, synthesized on free ribosomes, and transported into the mitochondrial matrix, for which the action of heme lyase (holocytochrome-*c* synthase) is essential. Example (human and chimpanzee): database code CYTC_HUMAN, 104 amino acids (11.60 kDa).

cytochrome c_1 an inner mitochondrial membrane integral protein that is a component of **respiratory chain** complex III. Comprising 241 amino acids (bovine heart), and with a molecular mass of 27.94 kDa, it transfers electrons directly to cytochrome *c*. Under some conditions it co-purifies with the 'hinge protein' (9.2 kDa) and a small (7.2 kDa) subunit of unknown function.

cytochrome *c* oxidase *or* **cytochrome oxidase** *or* **cytochrome aa_3** EC 1.9.3.1; a copper-containing oxidoreductase enzyme that catalyses the oxidation of cytochrome *c* by dioxygen with the production of water. The reaction is:

$$4\ \text{ferrocytochrome}\ c + O_2 = 4\ \text{ferricytochrome}\ c + 2\ H_2O.$$

It is an integral mitochondrial membrane protein comprising a complex of 13 subunits; subunits I, II, and II are encoded in the mitochondrion and the remaining ten are nucleus encoded. Electrons originating in cytochrome *c* are transferred via heme *a* and Cu(a) to the binuclear centre formed by heme a_3 and Cu(b). Subunit I binds the two hemes and Cu(b): database code COX1_HUMAN, 513 amino acids (56.97 kDa). Subunit II binds Cu(a) and cytochrome *c*: database code COX2_HUMAN, 227 amino acids (25.53 kDa).

cytochrome *c* reductase *see* **NADH dehydrogenase**.

cytochrome *f* a cytochrome, designated after the Latin *frons* (leaf), that occurs in high concentration in leaves (up to 25% of the concentration of chlorophyll). It is a typical *c*-type cytochrome, similar to c_1, and is a component of thylakoid membrane, transferring electrons from photosystem II to I. Example, apocytochrome *f* precursor (tobacco): database code CYF_TOBAC, 320 amino acids (35.21 kDa). *See also* **cytochrome b_6**, **cytochrome b_6–*f* complex**.

cytochrome *m* *an alternative name for* **cytochrome P450**.

cytochrome *o* any *b*-type prokaryotic cytochrome containing protoheme as the prosthetic group that, unlike a typical cytochrome *b*, serves as a terminal electron acceptor (cytochrome oxidase) and is autoxidizable by dioxygen. Use of the term is discouraged.

cytochrome P450 *or (sometimes)* **cytochrome *m*** *abbr*.: (of protein) CYP, (of gene) *CYP*; any member of a group of *b*-type cytochromes that are apparently unique in having a sulfur atom ligated to the iron and that, when reduced, form carbon monoxide complexes – the latter are pigments with a major absorption peak (Soret band) near 450 nm hence the name. These cytochromes are enzymes, including unspecific monooxygenase (EC 1.14.14.1), camphor 5-monooxygenase (EC 1.15.15.1), alkane 1-monooxygenase (EC 1.14.15.3), steroid 11β-monooxygenase (EC 1.14.15.4), cholesterol monooxygenase (side-chain-cleaving) (EC 1.14.15.6), prostacyclin synthase (EC 5.3.99.4), and thromboxane synthase (EC 5.3.99.5). They catalyse the reaction:

$$RH + \text{reduced flavoprotein} + O_2 = ROH + \text{oxidized flavoprotein} + H_2O.$$

They occur in most animal cells and organelles, and in plants and microorganisms, and their chief function is thought to be the monooxygenation of lipophilic substances (RH can be steroid, fatty acid, or xenobiotics), as in the case of cytochrome $P450_{SCC}$, cholesterol monooxygenase (side-chain-cleaving). In mammals, as liver microsomal enzymes, many are inducible by drug and other treatment e.g. the major phenobarbital-inducible cytochrome in rabbit liver microsomes, known as $P450_{LM2}$. The name **heme-thiolate** (or Brit. **haem-thiolate**) **protein** has been proposed on the grounds of a thiolate ligand at the heme being responsible for the group's unusual spectral and catalytic properties.

A classification of this extensive family is being evolved in a new systematic nomenclature. At present, there are 27 families, of which 10 are found in all mammals. Each family consists of a group of proteins that exhibit 40% or greater sequence identity, and the family is described by an Arabic number. Within these families, subgroups of proteins having greater that 55% identity are classified into subfamilies, designated by a capital letter, and individual proteins within subfamilies are arbitrarily assigned numbers. Under this nomenclature, $P450_{LM2}$ is known as CYP2B4, or P4502B4, and the gene as *CYP2B4*. Example, P450 IA1, TCCD inducible, also P450 type 6, P450-C (human): database code CP11_HUMAN, 512 amino acids (58.10 kDa).

cytocuprein *an alternative name for* **superoxide dismutase**.

cytoflav(e) *or* **cytoflavin** *a historical name for* the yellow pigment(s) discovered in heart muscle and other tissues. In some cases these pigments were subsequently shown to be riboflavin or a flavin-containing enzyme, especially **yellow enzyme**.

cytofluorometry a technique for the detection, localization, and estimation of substances in or attached to cells, using naturally fluorescent substances or those to which a fluorescent label has been attached. The technique has many applications, including specific labelling of cells using fluorescent antibodies, using fluorescent **probes** to report intracellular events or membrane dynamics, or following the fate of a protein with a fluorescent label after injection into a cell.

cytogenetics the branch of genetics concerned with correlating inherited traits with cytological features, especially the appearance, structure, and behaviour of chromosomes. —**cytogenetic** *adj.*

Cytohelicase *the proprietary name for* a commercial enzyme preparation consisting predominantly of a 1,3-β-D-glucanase. It is useful for the isolation of protoplasts because it lyses cell walls.

cytokeratin *see* **keratin**.

cytokine any of a varied group of proteins that are released by mammalian cells and have **autocrine** or **paracrine** activity. They elicit from the target cell a variety of responses depending on the cytokine and target cell. Cytokine actions include control of cell proliferation and differentiation, regulation of immune responses, hemopoiesis, and inflammatory responses. Cytokines produced by lymphocytes are known also as **lymphokines**; those produced by monocytes are also called **monokines**. Other types of cytokine include **chemokines**, growth factors, colony stimulating factors, transforming growth factors, interferons, and tumour necrosis factors. *See* **angiogenin, colony stimulating factor, epidermal growth factor, erythropoietin, fibroblast growth factor, heparin-binding EGF-like growth factor, hepatocyte growth factor, interferon, insulin-like growth factor, interleukin, leukemia inhibitory factor, macrophage inflammatory protein, nerve growth factor, oncostatin M, platelet-derived endothelial growth factor, platelet-derived growth factor, RANTES, stem cell factor, transforming growth factor, tumour necrosis factor, vascular endothelial growth factor**.

cytokine receptor common γ chain a glycoprotein subunit common to the receptors for a variety of interleukins (IL-2, IL-4, IL-7, and IL-13). It is a type I **membrane protein**. Example (precursor) from *Mus musculus*: database code CYRG_MOUSE, 369 amino acids (42.19 kDa).

cytokinesis **1** the changes that occur in the cytoplasm of a cell during nuclear division. **2** the division of the cytoplasm of a parent cell into daughter cells after nuclear division. *Compare* **karyokinesis**.

cytokinin *or* **phytokinin** any of a group of plant hormones that have a stimulatory effect on the division of plant cells and a retarding effect on leaf senescence. They are purine derivatives.

cytolemma *an alternative name for* **cell membrane**.

cytology the branch of science dealing with the origin, structure, function, biochemistry, and life history of cells. It includes the (microscopic) examination of cells, particularly in the detection and diagnosis of disease.

cytolysin any protein, especially an antibody, that can induce **cytolysis**.

cytolysis the breakdown of cells, especially by destruction of their outer membranes; the lysis of cells. —**cytolytic** *adj.*

cytometer **1** a **cell counter**. **2** a device for measuring cells.

cytometry the counting and/or measuring of cells in a fluid suspension.

cyton *or* **cytone** the body of a cell, especially a nerve cell, as distinct from its processes.

cytopathic damaging to cells.

cytopathic effect the morphological changes that cells may show in response to toxic agents or viruses.

cytopempsis the intracellular phase of **transcytosis**, whereby droplets, molecules, or particles are transported across a polarized cell from one surface to another.

cytophilic having an affinity for cells.

cytophilic antibody any antibody that binds to the surface of cells by means other than by combination of the antibody-combining site with antigenic determinants on the cell surface. Antibody thus fixed to a cell is still able to bind antigens in the vicinity; i.e. the cells become able specifically to absorb antigen.

cytophotometry a technique that employs a combination of microscopy and spectrophotometry for the detection, localization, and determination of substances in cells. It may be done on stained or unstained preparations.

cytoplasm all the **protoplasm** of a living cell excluding the nucleus and the **plasma membrane** but including the other intracellular organelles and structures. Some authors currently also exclude mitochondria, chloroplasts, or other structures containing independently replicating DNA. The term is not synonymous with **cytoplasmic fraction**. *See also* **cytosol**. —**cytoplasmic** *adj.*

cytoplasmic droplet a large (1–8 μm) piece of membrane-bound cytoplasm, containing the usual cytoplasmic organelles (but no nucleus) and one or more large lipid droplets, found in skimmed freshly secreted goat milk. The droplets are capable of triglyceride biosynthesis.

cytoplasmic fraction the material obtained after the removal of unbroken cells, cell debris, and nuclei from a **homogenate** of eukaryotic cells or tissue. It may be further resolved into a **mitochondrial fraction**, a **microsomal fraction**, and a soluble fraction. The term is not synonymous with **cytoplasm**. *See also* **cytosol**.

cytoplasmic gene any extranuclear gene located in certain eukaryotic cellular organelles, e.g. mitochondria and chloroplasts.

cytoplasmic inheritance the inheritance of characters or traits whose determinants are not located on a chromosome. The determinants may be, e.g., viral, mitochondrial, or on plasmids.

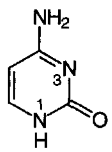
cytoplasmic matrix *an alternative name for* **groundplasm**.

cytoplasmic membrane **1** any of the membranous structures within the **cytoplasm** of eukaryotes, such as the rough and smooth **endoplasmic reticulum** and the **Golgi apparatus**. *See also* **microsomal fraction**, **microsome**. **2** *an alternative term for* **cell membrane** (especially in relation to bacteria).

cytoplast 1 an anuclear eukaryotic cell resulting, e.g., from treatment with cytochalasin B (*see* **cytochalasin**). **2** the apparent single structural and functional unit of the eukaryotic cell, consisting of the **microtrabecular lattice** linking the subcellular organelles and structural fibres.

cytosegresome *an alternative name for* **autophagic vacuole**.

cytosine *symbol:* Cyt; 4-amino-2-hydroxypyrimidine; 4-amino-2(1*H*)-pyrimidinone; a pyrimidine derivative that is one of the five main bases found in nucleic acids; it occurs widely in **cytidine** derivatives.



β-cytosine arabinoside *see* **cytarabine**.

+cytosis *suffix* indicating an increase in the number of cells (in an indicated tissue or organ).

cytoskeleton any of the various filamentous elements within the cytoplasm of eukaryotic cells that remain after treatment of the cells with mild detergent to remove membrane constituents and soluble components of the cytoplasm. The term embraces **intermediate filaments**, **microfilaments**, **microtubules**, and the **microtrabecular lattice**. The various elements of the cytoskeleton not only serve in the maintenance of cellular shape but also have roles in other cellular functions, including cellular movement, cell division, endocytosis, and movement of organelles. (*Note:* The term was coined to describe hypothetical structural cytoplasmic mosaics envisaged as providing physical support for, and thus spatial coordination of, enzymes effecting sequences of reactions, and as imposing restriction of mobility of the cell. Subsequently the cytoskeleton was regarded also as a possible subcellular site of action of various hormones.) —**cytoskeletal** *adj.*

cytosol the part of the **cytoplasm** that does not appear to contain membranous or particulate subcellular components. The term is not synonymous with **soluble fraction**. —**cytosolic** *adj.*

cytosol aminopeptidase *another name for* **leucyl aminopeptidase**.

cytosome any cytoplasmic structure bounded by a single **unit membrane** and of dubious identity.

cytostatic any agent that suppresses cell growth and multiplication.

cytotactin *another name for* **tenascin**.

cytotaxigen any substance that acts indirectly to induce **cytotaxis** by inducing **cytotaxin** formation. For instance, the fixing of complement by antigen–antibody complexes causes the liberation of cytotaxins, and hence cytotaxis. —**cytotaxigenic** *adj.*

cytotaxin *or (sometimes)* **chemotaxin** any chemotactic substance acting directly to induce **cytotaxis**. *See also* **cytotaxigen**.

cytotaxis 1 positive or negative **chemotaxis** of motile cells in response to diffusible substances (cytotaxins) emitted by other cells. **2** the rearrangement of cells leading to the ordering and arranging of new structures. —**cytotactic** *adj.*

cytotoxic causing cell death.

cytotoxicity the attribute of being cytotoxic.

cytotoxicity test any test of cell viability after exposure of cells to (potentially) toxic agents, e.g. antibody and complement. Loss of viability may be demonstrated by a **dye exclusion test**.

cytotoxic T-cell any of a subset of T lymphocytes, bearing the CD8 (human) or Ly2 (mouse) surface marker, that have the ability to kill infected cells. This requires cooperation from T-helper cells. Each cytotoxic T-cell specifically recognizes one antigen, which must be associated with a syngeneic class I MHC molecule.

cytotropism the mutual attraction of two or more cells. —**cytotropic** *adj.*

cytovillin *another name for* **ezrin**.

Dd

d *symbol for* **1** deci+ (SI prefix denoting 10^{-1} times). **2** a deuteron. **3** day.

d+ **1** *abbr. for* deoxy+ (def. 1); used as a prefix especially to form the symbol dRib for deoxyribose. **2** deoxy+ (def. 2); used as a prefix to form symbols for 2′-deoxyribonucleosides (e.g. dAdo *or* dA for deoxyadenosine) and abbreviations for 2′-deoxyribonucleotides (e.g. dGMP for deoxyguanosine monophosphate).

d *symbol for* **1** diameter, distance, or thickness. **2** relative density.

d- *abbr. for* dextro- used *(formerly) as symbol* denoting dextrorotatory; (+)- should now be used. *See* **optical isomerism**.

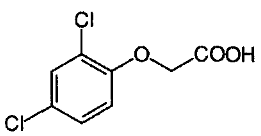
D *symbol for* **1** a residue of the α-amino acid L-aspartic acid. **2** a residue of an incompletely specified base in a nucleic-acid sequence that may be adenine, guanine, or either thymine (in DNA) or uracil (in RNA). **3** a residue of the ribonucleoside (5,6-)dihydrouridine. **4** debye. **5** deuterium (use deprecated).

D600 gallopamil, 5-methoxyverapamil, α-(3-{[2-(3,4-dimethoxyphenyl)ethyl] methylamino}propyl)-3,4,5-trimethoxy-α-(1-methylethyl) benzeneacetonitrile; a drug used in the laboratory to block transport of calcium ions in biological membranes; it is more potent than **verapamil**.

2,4-D *abbr. for* 2,4-dichlorophenoxyacetic acid, a synthetic **auxin** used as a herbicide and to increase latex output from old rubber trees. When sprayed onto foliage at appropriate concentrations it kills broad-leaved plants but not grasses.

D *symbol for* **1** diffusion coefficient. **2** (decadic) attenuance. **3** absorbed dose. **3** (bond) dissociation energy. **4** dilution rate.

$\boldsymbol{D_i}$ *symbol for* **1** electric polarization. **2** internal transmission density (*formerly called* optical density).



D- *prefix* denoting one of two possible **absolute configurations** around a chiral element in certain classes of compounds. *See* **D/L convention**.

da *symbol for* deca+ (the SI prefix denoting 10 times).

dA a residue of the deoxynucleoside deoxyadenosine (alternative to dAdo).

Da *symbol for* dalton.

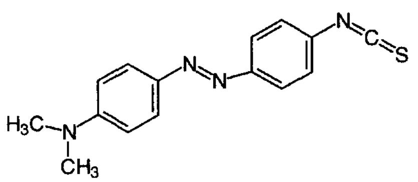
DA *abbr. (sometimes) for* dopamine.

Dab *see* $\mathbf{A_2bu}$.

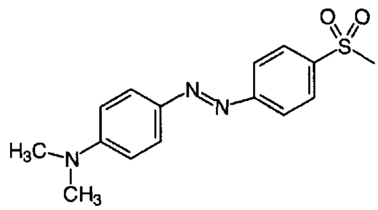
DAB *abbr. for* diaminobutyric acid.

DABA *abbr. for* diaminobenzoic acid.

DABITC *abbr. for* dimethylaminobenzene isothiocyanate, 4-*N,N*-dimethylaminoazobenzene 4′-isothiocyanate; a sensitive and versatile chromophoric reagent used in sequencing polypeptides by the **Edman degradation**.



dabsyl *abbr. for* dimethylaminobenzenesulfonyl, 4-dimethylaminobenzene-4′-sulfonyl; a group used as a chromophoric label for free amino groups in amino acids, peptides, and proteins. It is introduced by reaction with dabsyl chloride, 4-dimethylaminobenzene-4′-sulfonyl chloride.



dAdo *symbol for* a residue of the deoxynucleoside deoxyadenosine; 2′-deoxyribosyladenine; adenine 2′-deoxyriboside (alternative to dA).

dAdo5′*P* *symbol. for* deoxyadenosine 5′-phosphate.

DAF *abbr. for* complement decay-accelerating factor.

DAG *abbr. for* diacylglycerol.

DAHP *abbr. for* 3-deoxy-D-*arabino*-(2-)heptulosonate 7-phosphate.

dal *(sometimes) symbol for* dalton (Da is usual).

Dale, (Sir) Henry Hallett (1875–1968), British physiologist and pharmacologist distinguished for his discovery of acetylcholine and for his fundamental physiological studies of the actions of acetylcholine, epinephrine, and histamine (*see also* **Schultz–Dale reaction**); Nobel Laureate in Physiology or Medicine (1936) jointly with O. Loewi ‘for their discoveries relating to chemical transmission of nerve impulses’.

Dale's principle the principle that a given neuron is likely to release the same active substance from all its active terminals. Sometimes aggrandized into **Dale's law** – one neuron, one neurotransmitter – its only subsequent modification has been the recognition that a neuron may release more than one active substance. *See* **cotransmitter**.

dalton *symbol*: Da; a non-SI unit of atomic and molecular mass, equal to one-twelfth of the mass of the nuclide carbon-12. It is identical with the **unified atomic mass unit**. It is widely used in biochemistry. Sometimes it is used incorrectly as a supposed unit of the dimensionless terms **relative atomic mass** and **relative molecular mass**. [After John Dalton (1766–1844), British chemist and physicist.]

Dalton's law of partial pressures a statement that the total pressure of a mixture of gases is equal to the sum of the partial pressure of the constituent gases.

Dalziel coefficient any of the coefficients of an equation expressing the initial rate, ν, of a sequential two-substrate enzymic reaction of the form:

$$\nu = EAB\,/\,[AB\phi_0 + B\phi_1 + A\phi_2 + \phi_{12}]$$

where E is the enzyme concentration, A and B are the substrate concentrations, and ϕ_0, ϕ_1, ϕ_2, and ϕ_{12} are the Dalziel coefficients. The method can be extended to the evaluation of the eight Dalziel coefficients needed to describe the rate of a three-substrate reaction. [After Keith Dalziel (1921–94), British biochemist.] *Compare* **Michaelis kinetics**.

dam a gene of *Escherichia coli* encoding DNA-adenine methylase. *See* **DNA methylase**.

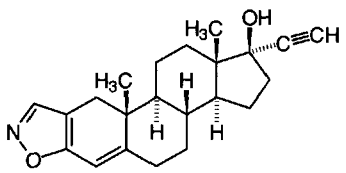
Dam, Henrik Carl Peter (1895–1976), Danish biochemist noted for his discovery in green plants of an antihemorrhagic principle which he named vitamin K (later termed vitamin K_1 or phylloquinone); Nobel Laureate in Physiology or Medicine

(1943) ' for his discovery of vitamin K' [prize shared with E. A. **Doisy**].

damascenine 3-methoxy-2(methylamino) benzoic acid methyl ester; a **protoalkaloid** from the seeds of *Nigella damascena* (love-in-a-mist) and the odoriferous principle of the oil of nigella.

dAMP *abbr. for* deoxyadenosine monophosphate, the common name for 2′-deoxyadenosine 5′-phosphate; 2′-deoxy-5′-adenylic acid; adenine 2′-deoxyriboside 5′-phosphate.

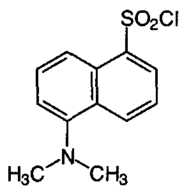
danazol 17α-pregna-2,4-dien-20-ynol[2,3-*d*]isoxazol-17β-ol; a derivative of ethisterone that is an inhibitor of anterior pituitary function. It inhibits secretion of gonadotropins and synthesis of sex steroids and acts as a ligand for their receptors.



Dane particle *see* **hepatitis**.

Danielli–Davson model *see* **Davson–Danielli model**.

dansyl *symbol*: Dns; *abbr. for* 5-dimethylaminonaphthalene-1-sulfonyl, a group used in **extrinsic fluorescence** studies of proteins. *See also* **dansylate**.



dimethylaminonaphthalene sulfonyl chloride

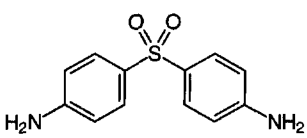
dansylate to derivatize with a dansyl group by reacting with dansyl chloride, 5(dimethylamino)naphth-1-ylsulfonyl chloride. The method is used to acylate free amino groups in protein end-group analysis. The dansyl amino acids, isolated after hydrolysis of the protein, are highly fluorescent and amounts as small as 1 nmol may be detected. —**dansylated** *adj.*; **dansylation** *n.*

Danysz's effect *or* **Danysz's phenomenon** the variation in the toxicity of toxin–antitoxin mixtures dependent on whether toxin is added to equivalence in one lot (nontoxic mixture produced) or in smaller lots over a time period (toxic mixture produced). When added in small lots the toxin reacts with more than one equivalent of antitoxin (antibody) so that insufficient antibody is left to neutralize all the subsequent lots of toxin. [After its discoverer, Jean Danysz (b. 1860), Polish pathologist working in Paris.]

DAP *abbr. for* diaminopimelate.

DAP pathway *see* **diaminopimelic-acid pathway**.

dapsone 4,4′-diaminodiphenyl sulfone; bis(4-aminophenyl)sulfone; an antibacterial and antiprotozoal substance used especially against *Mycobacterium leprae*, the causative agent of leprosy, and as an adjunct in the treatment of malaria. It acts as a competitive antagonist of *p*-aminobenzoic acid and an inhibitor of folate synthesis.



dark adaptation the process by which the rods of the retina gradually become fully responsive to dim light when no longer exposed to bright light. —**dark-adapted** *adj.*

dark band *see* **light band**.

dark current the electric current observed in any photoelectric device in the absence of incident radiation. It is caused by thermal effects.

dark-field microscope *or* **dark-ground microscope** a form of light microscope in which the specimen is illuminated by the apex of a hollow cone of light. On diverging, the light does not enter the aperture of the objective lens, so that unstained or transparent specimens appear as bright objects on a black background.

dark reactions of photosynthesis traditional term for the reactions of the **reductive pentose phosphate cycle** that utilize the ATP and NADPH produced by the light-driven reactions of photosynthesis. The term misleadingly indicates that these reactions occur in the dark. This is true only in the sense that, over very short time intervals, they can occur in the absence of light provided ATP and NADPH are present. However, they are subject to regulatory mechanisms that are stimulated by light, and inhibited by the dark. In growing plants, these reactions occur during daylight hours. The reactions of photosynthesis are used to convert CO_2 into hexoses and other organic compounds by the reactions of the reductive pentose phosphate cycle.

dark reactivation *or* **dark repair** any light-independent enzymic mechanism for the repair of damaged DNA molecules. *Compare* **photoreactivation**.

DATD *abbr. for* diallyltartardiamide, a cross-linking agent.

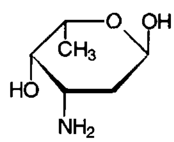
dative bond *an alternative term for* **dipolar bond**.

dATP *abbr. for* deoxyadenosine triphosphate, the common name for 2′-deoxyadenosine 5′-triphosphate, 2′-deoxy-5′-adenylyl diphosphate, adenine 2′-deoxyriboside 5′-triphosphate.

daughter *(as modifier)* **1** denoting a cell (or cell nucleus) produced by the fission of the original (mother) cell or nucleus. **2** denoting either of the two strands of DNA that originate from the parental duplex DNA molecule at replication. **3** denoting a nuclide produced by the nuclear disintegration (either spontaneous or induced) of another nuclide.

daunomycin *or* **daunorubicin** an anthracycline antibiotic and antineoplastic isolated from fermentation broths of *Streptomyces peucetius* widely used in the treatment of cancer. It is a glycoside formed from the tetracyclic aglycon daunomycinone ($C_{21}H_{18}O_8$) and the aminohexose **daunosamine**. It inhibits DNA replication by intercalation into duplex DNA and also inhibits RNA transcription. *Compare* **doxorubicin**.

daunosamine 3-amino-2,3,6-trideoxy-L-*lyxo*-hexose; the aminodeoxy sugar component of **adriamycin**, **daunomycin**, **doxorubicin**, and other antibiotics.



α-L anomer

Dausset, Jean Baptiste Gabriel Joachim (1916–), French physician and hematoimmunologist noted for first establishing the existence in humans of the major histocompatability complex; Nobel Laureate in Physiology or Medicine (1980) jointly with B. Benacerraf and G. D. Snell 'for their discoveries concerning genetically determined structures on the cell surface that regulate immunological reactions'.

Davson–Danielli model *or* *(sometimes)* **Danielli–Davson**

model a structural model of the **plasma membrane** consisting, as originally proposed, of a lipid bilayer covered on both surfaces by a film of adsorbed globular proteins. Essentially it was a lamellar model: two sheets of protein separated by a sheet of lipid. The lipid layer had a hydrophobic core and more hydrophilic surfaces. To account for the permeability properties of the plasma membrane the model was subsequently modified by the proposal that pores allow polar solutes to penetrate the lipid layer, and that these pores are lined by protein molecules, providing hydrophilic tubes through the membrane. The model has been totally superseded by more recent concepts. *See* **cell membrane.** [After Hugh Davson (1909–96), British biochemist and physiologist and J. F. Danielli (1911–84), British biochemist.]

day *symbol*: d; a non-SI unit of time equal to 24 hours or 86 400 s.

dbcAMP *abbr. for* dibutyryl cyclic AMP; N^6,$O^{2'}$-dibutyryladenosine 3′,5′-(cyclic)phosphate; a fat-soluble analogue of cyclic AMP, **adenosine 3′,5′-phosphate**.

DBL a gene, isolated and named from a human diffuse B cell lymphoma, that encodes a protein of unknown function and is expressed tissue-specifically in brain, adrenals, and gonads. The protein has been found to catalyse guanine nucleotide exchange *in vitro* on the CDC42Hs protein, a human cell-division-cycle protein. Database code DLB_HUMAN, 925 amino acids (107.66 kDa).

DBM *abbr. for* 2-diazobenzyloxymethyl, $-CH_2-O-CH_2-C_6H_4-N_2^+$; a group that introduces reactive diazonium groups into cellulose or paper forming **DBM-cellulose** or **DBT-paper** in **blot transfer** of proteins or nucleic acids. DBM is formed by diazotization of 2-aminobenzyloxymethyl (ABM) groups previously coupled to the support.

DBMIB *abbr. for* 2,5-dibromo-3-methyl-6-isopropyl-*p*-benzoquinone; a competitor of **plastoquinone** in photosynthetic electron flow.

DBP *see* **vitamin D binding protein.**

d.c. *or* **DC** *abbr. for* **1** direct current. **2** a voltage of constant polarity.

dC *symbol for* a residue of the deoxynucleoside deoxycytidine (alternative to dCyd).

DCC *or* **DCCD** *or* **DCCI** *abbr. for* *N,N′*-dicyclohexylcarbodiimide, a coupling agent useful in peptide synthesis and other coupling reactions. *See also* **carbodiimide**.

DCC a human gene, named from 'deleted in colonic carcinoma', that is considered to encode a tumour suppressor, characterized as a cell adhesion molecule of the immunoglobulin superfamily. Database code HSDCCG, 1447 amino acids (158.28 kDa).

D cell *or (formerly)* **A_1 cell** *or* **α_1 cell** a type of endocrine cell of the **apud** series, found in gut mucosa and the pancreatic islets. It was formerly thought to produce gastrin, but is now believed to produce **somatostatin**.

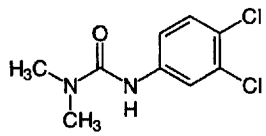
dCMP *or* ***dCyd5′P*** *abbr. for* deoxycytidine monophosphate, the common name for 2′-deoxycytidine 5′-phosphate; 2′-deoxy-5′-cytidylic acid; cytosine 2′-deoxyriboside 5′-phosphate.

dCMP hydroxymethylase *systematic name*: 5,10-methylenetetrahydrofolate:deoxycytidylate 5-hydroxymethyl transferase. EC 2.1.2.8; *other names*: deoxycytidylate hydroxymethylase; deoxycytidylate hydroxymethyltransferase; an enzyme that catalyses the reaction:

$$\text{5,10-methylenetetrahydrofolate} + H_2O + \text{deoxycytidylate} = \text{tetrahydrofolate} + \text{5-hydroxymethyldeoxycytidylate.}$$

It is a gene product of T-even bacteriophage; the DNA of these phages contains glycosylated 5-hydroxymethylcytosine residues in place of cytosine residues. Example from T4 phage: database code DCHM_BPT4, 246 amino acids (28.46 kDa); there is similarity to *Escherichia coli* **thymidylate synthase**.

DCMU *abbr. for* 3-(3,4-dichlorophenyl)-1,1-dimethylurea; a herbicide that blocks electron transport on the reducing side of photosystem II. One proprietary name is Diuron.



DCMU

Dco *symbol for* the decanoyl group, $CH_3-[CH_2]_8-CO-$.

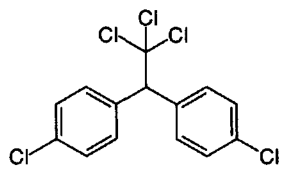
dCTP *abbr. for* deoxycytidine triphosphate, the common name for 2′-deoxycytidine 5′-triphosphate, 2′-deoxy-5′-cytidylyl diphosphate, cytosine 2′-deoxyriboside 5′-triphosphate.

dCyd *symbol for* a residue of the deoxynucleoside deoxycytidine; 2′-deoxyribosylcytosine; cytosine 2′-deoxyriboside (alternative to dC).

dCyd5′P *symbol for* deoxycytidine 5′-phosphate.

dd+ *abbr. for* dideoxy+ (def. 2); **1** used as a *prefix* to the appropriate three-letter or one-letter symbols for ribonucleosides in order to form symbols for 2′,3′-dideoxyribonucleosides (e.g. ddNuc *or* ddN, *symbol for* any (unknown or unspecified) 2′,3′-dideoxyribonucleoside; ddAdo *or* ddA, *symbol for* dideoxyadenosine). **2** used similarly to form abbreviations for 2′,3′-dideoxyribonucleoside 5′-triphosphates (e.g. ddNTP, *abbr. for* any (unknown or unspecified) 2′,3′-dideoxyribonucleoside 5′-triphosphate; ddCTP, *abbr. for* dideoxycytidine triphosphate).

DDT *abbr. and common name for* dichlorodiphenyltrichloroethane; 1,1,1-trichloro-2,2-bis(*p*-chlorophenyl)ethane; a contact insecticide that has proved particularly valuable in controlling insect disease vectors, especially mosquitoes (*Anopheles* spp.). However, it is toxic to higher animals as well. It has a very long persistence of activity from residual deposits which has caused it to be banned in many countries.



de+ *prefix (in organic chemical nomenclature)* denoting replacement of a specified group or atom in a specified chemical compound by a hydrogen atom. An exception is **deoxy+** when applied to hydroxy compounds, which denotes replacement of a hydroxyl group by a hydrogen atom. *See also* **dehydro+, des+**.

deacylase *informal name for* any enzyme that catalyses a deacylation reaction; it may be a hydrolase or a transferase.

deacylation the act or process of removing an acyl group from a chemical compound.

DEAD box a highly conserved motif in a family of putative RNA **helicases**, named after the single-letter code for the amino-acid sequence involved, i.e. Asp-Glu-Ala-Asp. The first DEAD-box protein that was shown to have RNA-helicase activity was the eukaryotic translational initiation factor eIF4-A. Other proteins have since been included in the family, based on sequence similarities; some have been shown to have ATP-dependent RNA-helicase activity, while in others this remains to be experimentally confirmed. *See also* **DEAH family, Appendix E**.

dead-end complex *an alternative name for* **abortive complex**.

DEAE *abbr. for* diethylaminoethyl, $(C_2H_5)_2N-CH_2-CH_2-$, a weakly basic group widely used as a substituent in hydroxylated chromatographic support matrices, e.g. cellulose, Sephadex, to form anion exchangers. *Compare* **QAE**.

de-aerate to remove air from a liquid or a slurry, commonly by heating and/or reduction of pressure.

DEAH family yeast proteins that are structurally similar to **DEAD-box** proteins, but larger. They differ fom DEAD-box proteins in parts of the conserved region, especially in that they have a conserved DEAH sequence rather than DEAD. Some of them are known to be involved in RNA splicing, e.g. in *trans*-esterification steps and spliceosome release.

deamidase *name sometimes used for* an enzyme that catalyses the hydrolytic cleavage of the C–N bond in a carboxamide, but that is correctly termed an **amidase**.

deamidation the process of removing an amino group from an amide, either by hydrolysis or by a transfer reaction.

deaminase any of the enzymes that catalyse the nonoxidative removal of amino groups, with production of ammonia. They are usually aminohydrolase enzymes (sub-subclass EC 3.5.4), which hydrolytically deaminate amino-substituted cyclic amidines including various (free or combined) pterins, purines, or pyrimidines. Certain hydro-lyase (sub-subclass EC 4.2.1) and hydrolytic or nonhydrolytic ammonia-lyase (sub-subclass EC 4.3.1) enzymes also are commonly called deaminases.

deamination the process of removing an amino group from an organic compound. *Compare* **transamination**.

deaminooxytocin *or* **demoxytocin** a synthetic analogue of **oxytocin**, in which the N-terminal half-cystine residue has been replaced by a residue of 3-mercaptopropanoic acid. It is more potent than the parent compound.

de-ashing a procedure in which interfering inorganic (i.e., ash-forming) components are removed from samples of carbohydrates being analysed by liquid chromatography by passing them first through anion- and cation-exchange resins.

death domain a region of limited homology, consisting of about 80 residues close to the intracellular C terminus of certain cell-membrane receptors (e.g. the **TNF receptor**) that is essential for the receptors to generate a signal leading to apoptosis.

deblock to remove a **blocking group**. —**deblocking** *n.*

debrancher deficiency *an alternative name for* type III **glycogen** (storage) **disease**.

debranching enzyme any enzyme that catalyses the cleavage of branch points in particular branched biopolymers, e.g. **isoamylase**, **oligo-1,6-glucosidase** and **pullulanase**.

debye *symbol*: D; a non-SI unit of molecular dipole moment: 1D ≈ 3.335 64 × 10^{-30} C m.

Debye, Petrus ('Peter') Josephus Wilhelmus (1884–1966), Dutch-born US physicist and physical chemist; Nobel Laureate in Chemistry (1936) 'for his contributions to our knowledge of molecular structure through his investigations on dipole moments and on the diffraction of X-rays and electrons in gases'.

Debye–Hückel theory a theory used to calculate the electric potential at a point in a solution in terms of the concentrations and charges of the ions and the properties of the solvent. It yields the logarithm of the mean rational activity coefficient:

$$\lg f^{+/-} = (A|z_1z_2|\, I^{1/2})/(1 + BaI^{1/2}),$$

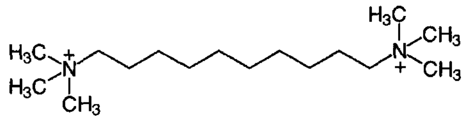
where the constants A and B involve the temperature and the dielectric constant of the solvent, a is an adjustable 'ion-size' parameter, I is the **ionic strength**, and $|z_1z_2|$ is the absolute value of the valence product of the electrolyte under consideration. [After P. J. W. **Debye** and Eric Hückel (1896–1980), German physicist.]

dec+ *see* **deca+** (def. 2).

Dec *symbol for* the decyl group, $CH_3–[CH_2]_8–CH_2–$.

deca+ 1 *symbol*: da; the SI prefix denoting 10 times. **2** *or (before a vowel)* **dec+** *comb. form* denoting ten, tenfold, or ten times; formerly sometimes written as deka+ *or* dek+.

decamethonium N,N,N,N^1,N^1,N^1-hexamethyl-1,10-decanediaminium a nicotinic receptor ligand that causes sustained activation of the receptors, with consequent blockade of synaptic transmission (termed 'depolarizing block'). *See also* **suxamethonium**.



decanoate 1 the anion, $CH_3–[CH_2]_8–COO^–$, derived from decanoic acid (formerly capric acid). **2** any mixture of decanoic acid and the anion. **3** any salt or ester of decanoic acid.

decanoyl *symbol*: Dco; the univalent acyl group, $CH_3–[CH_2]_8–CO–$, derived from decanoic acid. *See also* **caproyl** (def. 1).

decant to pour off a supernatant liquid from a precipitate or other sediment. —**decantation** *n.*

decap to (enzymically) remove a **cap** (def. 1) from the 5′ end of a capped mRNA molecule. —**decapped** *adj.*

decapsidate to remove the capsid from a virus. —**decapsidation** *n.*

decarboxylase any of the enzymes of the sub-subclass EC 4.1.1, carboxy-lyases; they catalyse the removal of carboxyl groups from carboxylic acids.

decarboxylation the act or process of removing the carboxyl group from a carboxylic acid as carbon dioxide. The reaction may be enzyme-catalysed by a **decarboxylase** or, in some instances, particularly with 2-oxo acids, it may be spontaneous. —**decarboxylate** *vb.*; **decarboxylated** *adj.*

decay 1 a decrease in activity, force, quantity, or other physical attribute, especially with a characteristic time constant. **2** (a) **nuclear decay**: the spontaneous change of one nuclide into another, its daughter, with a different proton number or nucleon number. (b) **radioactive decay**: nuclear decay in which particulate or electromagnetic radiation (or both) is emitted, or in which spontaneous fission of the nucleus occurs. *See also* **disintegration**. **3** the decline of an atom or molecule from an excited state. **4** the biochemical decomposition of plant and animal remains, due chiefly to the activities of microorganisms.

decay accelerating factor *see* **complement decay-accelerating factor**.

decay constant *an alternative term for* **disintegration constant**.

deci+ 1 *symbol*: d; the SI prefix denoting 10^{-1} times. **2** *comb. form* denoting one tenth, one in ten.

decimolar *(informal)* one-tenth molar; i.e. denoting a solution of concentration 0.1 mol dm^{-3}.

decinormal *(informal; not now recommended)* one-tenth normal; i.e. denoting a solution of concentration 0.1 equivalents per litre (*see* **normal** (def. 2)).

Declomycin *proprietary name for* **demeclocycline**. *See also* **tetracycline**.

decoding *(informal) an alternative term for* **translation** (def. 1).

decontaminant an agent for removing or neutralizing a contaminant.

decontamination the act or process of removing or neutralizing any toxic or potentially toxic materials, or of reducing their concentrations to nonhazardous levels. The term is used especially in regard to microbes, toxic chemicals, carcinogens, and radioactive substances. *Compare* **contamination**.

decorin *or* **bone proteoglycan II precursor** a collagen-binding protein of the extracellular matrix, so called because it 'decorates' collagen fibrils. The core protein has a single **glycosaminoglycan** chain, together with other oligosaccharide substituents. Example from human: database code HSRNAFIB, 376 amino acids (43.10 kDa).

decose any **aldose** having a chain of ten carbon atoms in the molecule.

dCTP deaminase EC 3.5.4.13; *systematic name:* dCTP aminohydrolase; *other name*: deoxycytidine triphosphate deaminase.

An enzyme that hydrolyses the amino group of dCTP to form dUTP and NH_4^+. Example from *Escherichia coli*: database code DCD_ECOLI, 193 amino acids (21.23 kDa).

deculose any **ketose** having a chain of ten carbon atoms in the molecule.

decyl *symbol*: Dec; the alkyl group, $CH_3–[CH_2]_8–CH_2–$, derived from decane.

***n*-decyl-*β*-D-glucopyranoside** a nonionic detergent, useful for solubilizing membrane proteins; CMC 2–3 mM.

***n*-decyl-*β*-D-maltopyranoside** a nonionic detergent, useful for solubilizing membrane proteins; CMC 1.6 mM.

dedifferentiation the loss of the characteristics of a specialized cell and regression to a simpler state. —**dedifferentiate** *vb.*

de Duve, Christian René Marie Joseph (1917–), Belgian biochemist and cell biologist distinguished for his discoveries of the lysosome and the peroxisome and for studies of their properties in health and disease; Nobel Laureate in Physiology or Medicine (1974) jointly with A. Claude and G. E. Palade 'for their discoveries concerning the structural and functional organization of the cell'.

defective virus a virus that is unable to reproduce itself whilst infecting a cell without the presence of a **helper virus**.

defensin any of a family of small cationic peptides, containing six cysteines in disulfide linkage, that have broad-spectrum antibiotic action and contribute to host defence against microorganisms. They are abundant in phagocytes and small intestinal mucosa of humans and other mammals and in the hemolymph of insects. Defensins adopt multimeric pore-forming complexes in membranes, thereby rendering the membrane permeable. They are distinguished by a predominantly **beta-sheet** structure. Examples, β-defensin 1 from *Bos taurus*: database code BD01_BOVIN, 38 amino acids (4.28 kDa); hemolymph defensin from *Mytilus galloprovincialis*: database code DEFI_MYTGA, 38 amino acids (4.27 kDa).

deficiency disease any disease resulting from the deficiency of one or more essential nutrients, e.g. a vitamin, an essential amino acid, or an essential mineral.

deficient lacking some essential component; inadequate in amount. The term can be used, e.g., for a deficient (culture) medium, deficient diet, or deficient strain (of an organism).

defined medium any culture medium in which the concentrations of all the constituents, including trace substances, are quantitatively known.

deformylase *see* **formylmethionine deformylase**.

degeneracy 1 *(in a code)* the existence of two or more synonyms specifying a particular item or function; *see also* **degenerate code**. 2 *(in a quantized system)* the existence of two or more linearly independent **eigenfunctions** with the same **eigenvalue**, especially the condition of an atom or molecule. 3 *(of an atom or molecule)* the condition in which the energy levels of different quantum states have the same energy level. —**degenerate** *adj.*

degenerate code a code in which more than one symbol represents the same entity. One such is the **genetic code**, wherein a specific amino acid may be encoded by two or more nucleotide triplets, or **codons** present in a nucleic acid or polynucleotide.

degeneration the process whereby a cell, tissue, or organ changes to a less specialized or functionally less active form.

degradation *(in chemistry)* 1 the gradual stepwise and deliberate conversion of a molecule into smaller chemical entities, commonly to elucidate its chemical structure. 2 any undesired breakdown of a molecule or material with impairment or loss of its characteristic properties. 3 depolymerization; decomposition. 4 *(in biochemistry)* catabolism.

degranulation 1 any process by which granular cells lose their granules. 2 the stripping of polysomes from rough endoplasmic reticulum *in vitro*.

degree *see* **angle**, **temperature**.

degree Celsius *symbol*: °C; the SI derived unit of **Celsius temperature**. It denotes the same interval of temperature as the **kelvin** on a scale with different reference points.

degree of polymerization the number of **monomeric units** in the molecule of a polymer.

degrees of freedom 1 *(in statistics)* the number of unrestricted independent variables constituting a statistic; thus in a sample of *n* values of *x* there are *n* deviations $(x_i - \bar{x})$, where $\bar{x}$ is the estimate of the mean, used in calculating the sample **variance**, s^2, but they are not all independent, since there is a linear relation connecting them, namely:

$$\sum_{i=1}^{n} (x - \bar{x}) = 0$$

hence only $(n - 1)$ of the quantities are independent, and there are $(n - 1)$ degrees of freedom in the calculation of s^2. 2 *(in physics)* the minimum number of parameters necessary to the description of the state of a system. 3 the number of independent equations by which the energy of a system may be expressed. 4 *(in chemistry)* the number of independently variable intensive properties that must be specified in order to describe completely the state of the system. *See* **phase rule**.

de Hevesy *see* **Hevesy**.

dehydrase *obsolete term for* **dehydratase** or **dehydrogenase**.

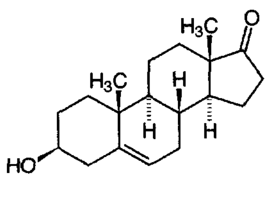
dehydratase *or (formerly)* **dehydrase** any hydro-lyase enzyme, of sub-subclass EC 4.2.1, that catalyses the (reversible) breakage of a carbon–oxygen bond leading to the formation of an unsaturated product and the elimination of water; e.g. citrate dehydratase, EC 4.2.1.4. When named after the alternative, i.e. unsaturated, substrate, the enzyme is termed a **hydratase**; e.g. aconitate hydratase, EC 4.2.1.3. *See also* **carbonate dehydratase**.

dehydration 1 the removal of combined, or adsorbed, water from a chemical compound; *see also* **anhydrous**. 2 the process or result of losing water, especially the abnormal loss of water from an organism. 3 the replacement of water by solvents.

dehydro+ *prefix (in chemical nomenclature)* denoting the loss of two hydrogen atoms. In common usage it is often used in place of **didehydro+**; e.g. 7,8-didehydrocholesterol is often termed dehydrocholesterol. *Distinguish from* **anhydro+**.

2-dehydro-3-deoxyphosphooctonate aldolase *another name for* **3-deoxy-D-*manno*-octulosonic acid 8-phosphate synthase**.

dehydroepiandrosterone *abbr.*: DHEA; androstenolone, *trans*-dehydroandrosterone (*abbr.*: DHAS), 3*β*-hydroxyandrost-5-en-17-one; an intermediate in androgen and estrogen biosynthesis, formed from 17*α*-hydroxypregnenolone in testis, ovary, and adrenal cortex by the action of 17*α*-hydroxyprogesterone aldolase, EC 4.1.2.30. It is the immediate precursor of androstenedione, which is formed from it by the action of 3*β*-hydroxy-Δ^5-steroid dehydrogenase, EC 1.1.1.145 and it can be converted to androst-5-en-17-one 3*β*-sulfate (DHEA-S), in which form it is measured in plasma, there being about 1000 times more DHEA-S than DHEA in plasma.



dehydrogenase any donor:acceptor **oxidoreductase** enzyme, of the class EC 1, when named after the alternative, i.e. more reduced, substrate and when dioxygen is not the acceptor.

dehydrogenate to oxidize an organic compound by the removal of one or more hydrogen atoms. —**dehydrogenation** *n.*

3-dehydroquinate dehydratase EC 4.2.1.10; *systematic name*: 3-dehydroquinate hydro-lyase; *other name*: 3-dehydroquinase; an enzyme that catalyses the reaction:

3-dehydroquinate = 3-dehydroshikimate + H_2O.

The reaction forms part of the **shikimate pathway**. For example, *see* **aro1**. *See also* **pentafunctional arom polypeptide**.

3-dehydroquinate synthase EC 4.6.1.3; an enzyme that catalyses the hydrolysis of 7-phospho-3-deoxy-*arabino*-heptulosonate to 3-dehydroquinate and orthophosphate. *See* **pentafunctional arom polypeptide; shikimate pathway**.

(3-)dehydroretinal *an alternative name for* **3,4-didehydroretinal** (not recommended, except for nutritional usage).

(3-)dehydroretinoic acid *an alternative name for* **3,4-didehydroretinoic acid** (not recommended, except for nutritional usage).

(3-)dehydroretinol *an alternative name for* **3,4-didehydroretinol** (not recommended except for nutritional usage).

deionize *or* **deionise** to remove all ions from a solution. —**deionization** *or* **deionisation** *n*. *See also* **desalt**.

deionizer *or* **deioniser** a device used to remove ions from solutions, usually by the use of **ion exchangers**.

Deisenhofer, Johann (1943–), German-born US biochemist; Nobel Laureate in Chemistry (1988) jointly with R. Huber and H. Michel 'for the determination of the three-dimensional structure of a photosynthetic reaction centre'.

deka+ *or (before a vowel)* **dek+** *former spelling of* **deca+** (def. 2).

delayed early gene *see* **early gene**.

delayed hypersensitivity *or* **type IV hypersensitivity** *(in immunology)* a **hypersensitivity** reaction that arises more than a day after encounter with the antigen and is mediated by antigen-sensitized $CD4^+$ T cells that release **cytokines**, attracting macrophages to the site and activating them. This type of hypersensitivity is seen in skin contact reactions and in response to some chronic pathogens such as those that cause leprosy, tuberculosis, and schistosomiasis. *Compare* **immediate hypersensitivity**.

Delbrück, Max (1906–81), German-born US physicist, molecular geneticist, and sensory physiologist; notable in particular for his development of the plaque technique for studying viruses and for his research on the interaction of bacteriophages with bacteria; Nobel Laureate in Physiology or Medicine (1969) jointly with A. D. Hershey and S. E. Luria 'for their discoveries concerning the replication mechanism and the genetic structure of viruses'.

deletion the loss of a segment of genetic material from a chromosome. A deletion can vary in size from a single nucleotide residue to a segment containing a number of genes. A **terminal deletion** occurs at one end of the chromosome, whereas an **intercalary deletion** occurs elsewhere along the chromosome.

delipidation the removal of lipid from a sample.

deliquescent *(of certain salts)* tending to absorb water from the surrounding atmosphere at ordinary temperature and pressure and then to dissolve in the water so taken up. *Compare* **hygroscopic**. —**deliquesce** *vb.*; **deliquescence** *n.*

delocalization *or* **delocalisation** a quantum-mechanical concept implying that the unpaired electron density in a molecule is spread over several atoms. Analogously, delocalization of the spin of an unpaired electron can occur.

delta *symbol*: δ (lower case) *or* Δ (upper case); the fourth letter of the Greek alphabet. For uses *see* **Appendix A**.

deltahedron (*pl.* **deltahedra**) any polyhedron whose faces are all equilateral triangles. Some viruses disaggregate to form particles that are deltahedra. —**deltahedral** *adj.*

deltakephalin (D-Thr^2)-Leu-enkephalin-Thr; a highly potent and specific synthetic hexapeptide δ-opiate receptor agonist; structure Tyr-D-Thr-Gly-Phe-Leu-Thr. *See also* **enkephalin**.

delta-sleep-inducing peptide *or* **sleep peptide** a nonapeptide, Trp-Ala-Gly-Gly-Asp-Ala-Ser-Gly-Glu, found in the plasma of rats and rabbits, that shows a several-fold increase in concentration during sleep. It rapidly induces slow-wave (i.e. delta) sleep on injection into awake animals.

dematin a protein involved in the formation of actin bundles, abundantly expressed in brain, heart, skeletal muscle, kidney, and lung, whose activity is abolished upon phosphorylation by cAMP-dependent protein kinase. It is a member of the **villin** family. Example from human: database code DEMA_HUMAN, 383 amino acids (43.07 kDa).

demeclocycline 6-demethyl-7-chlorotetracycline; *see* **tetracycline**.

demethylation any reaction involving the removal of one or more methyl groups from a molecule, whether by oxidation or transfer to an acceptor molecule.

demineralization *or* **demineralisation** **1** the process of removing inorganic salts, e.g. from water or other liquid; deionization. **2** excessive loss of mineral salts from the tissues, especially bone, as occurs in certain diseases.

demoxytocin *an alternative name for* **deaminooxytocin**.

denaturation **1** *(of a protein)* a process in which the three-dimensional shape of a molecule is changed from its **native state** without rupture of peptide bonds. It is sometimes taken to include disulfide bond rupture or chemical modification of certain groups in the protein if these processes are also accompanied by changes in its overall three-dimensional structure. Denaturation is frequently irreversible and accompanied by loss of solubility (especially at the isoelectric point) and/or of biological activity. **2** *(of a nucleic acid)* a process whereby a molecule is converted from a firm, two-stranded, helical structure to a flexible, single-stranded structure. *Compare* **renaturation**.

denaturation loop a non-base-paired interstrand loop formed in the more thermally unstable regions (i.e., of high A+T content) of a duplex DNA molecule during **melting** (def. 3).

denaturation mapping the identification of regions of low thermal (or alkali) stability (i.e., of high A+T content) in a duplex DNA molecule, by trapping the partly melted structure and blocking renaturation, e.g. with formaldehyde (which preferentially couples with the amino groups of the single-stranded regions), and subsequently examining the specimen by electron microscopy.

dendrite a freely branching protoplasmic process of a nerve cell that transmits nerve impulses to the body of the cell and that communicates functionally with dendrites or bodies of other nerve cells.

dendrogram any branching treelike diagram that illustrates the relationships between organisms or objects.

Denhardt's solution a buffer, containing Ficoll, polyvinylpyrrolidine, and bovine serum albumin, that is used as a blocking reagent for **hybridization** with nitrocellulose filters. [After David Tilton Denhardt (1939–), US biochemist.]

denitrification the reduction, especially by denitrifying bacteria, of simple inorganic nitrogen compounds, such as nitrates and nitrites, to gaseous products such as nitric oxide, nitrous oxide, and/or dinitrogen. *See also* **nitrate reduction**. *Compare* **nitrification, nitrogen fixation** (def. 1).

denitrifying bacteria soil bacteria that, in the absence of dioxygen, reduce nitrates and nitrites to gaseous products. *Compare* **nitrifying bacteria**.

de novo *Latin* anew, from the beginning; e.g. *de novo* synthesis.

densimeter *an alternative name for* **densitometer** (def. 2).

densitometer **1** *or* **photodensitometer** an apparatus for measuring the extent to which a material absorbs light (*see* **absorptiometer** (def. 1)) or reflects light (*see* **reflectometer**). It is useful for scanning chromatograms or electrophoretograms, for determining the opacity of solutions or suspensions, or for measuring the blackening of photographic films. **2** *or* **densimeter** an instrument for measuring the mass density or relative density of liquids or gases. —**densitometry** *n.*; **densitometric** *adj.*

density **1 a** *or* **mass density** *symbol*: ρ; the mass of a substance per unit volume. **b** *or* **linear density** *symbol*: ρ_l; the mass of a substance per unit length. **c** *or* **surface density** *symbol*: ρ_A *or* ρ_S; the mass of a substance per unit area. *Compare* **relative density**. **2** the distribution of a scalar physical quantity per unit area of space (e.g. radiant energy density) or per unit volume (e.g. charge density). **3** the distribution of a vector physical quantity per unit area of space (e.g. electric current density, magnetic flux density). **4 optical density**. *See* **absorbance**.

density-gradient centrifugation *see* **isopycnic centrifugation**.

density-gradient sedimentation equilibrium *see* **isopycnic centrifugation**.

density-marker bead any small coloured bead of accurately known mass density that is used for calibrating density gradients. Generally a set of several such beads, covering a range of densities, is used.

dentate having teeth or toothlike projections. *See also* **multidentate**.

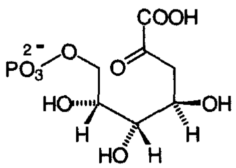
denticity the number of distinct but linked atoms of a specified ligand binding to the central atom in a coordination entity. *Compare* **ligancy**.

dentine *or* **dentin** the hard, dense, calcified tissue of a tooth lying between the enamel and the pulp. It is 75% mineral, mainly hydroxyapatite.

deoxy+ *or (formerly)* **desoxy+** *prefix (in organic chemical nomenclature)* **1** denoting the replacement of a hydroxyl group by a hydrogen atom; *see also* **de+**. **2** (*abbr*.: d+) indicating the presence of one or more residues of 2-deoxy-D-ribose. *Compare* **ribo+** (def. 2). **3** used especially with hemoglobin, to signify absence or removal of dioxygen. *Compare* **oxy+**.

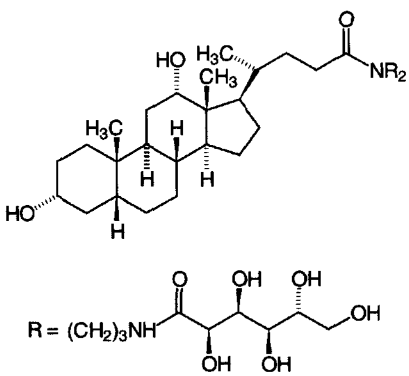
5′-deoxy-5′-adenosylcobalamin *or* **adenosylcobalamin** *see* **cobamide coenzyme**.

3-deoxy-D-*arabino*-(2-)heptulosonate 7-phosphate *or (less formally)* **3-deoxy-D-*arabino*-heptulosonate 7-phosphate**; *abbr*.: DAHP; an early intermediate in aromatic amino acid biosynthesis by the **shikimate pathway** in autotrophic organisms.

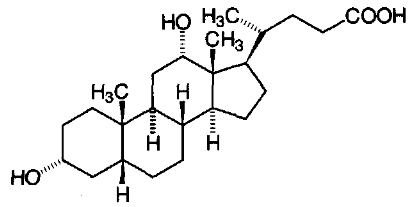


α-pyranose form

deoxy-BigCHAP *N*,*N*-*bis*-(3-D-gluconamidopropyl)-(7-deoxy)-cholamide; a nonionic detergent analogue of **CHAPS**, **CHAPSO**, and **BigCHAP**; aggregation number 8–16; CMC 1.1–1.4 mM.

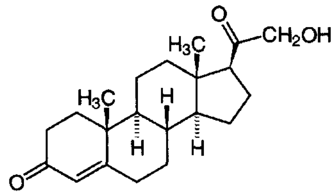


deoxycholic acid 3α,12α-dihydroxy-5β-cholan-24-oic acid; one of the **bile acids** occurring as its conjugate with glycine or taurine in the bile of most mammals, including dog, goat, human, ox, sheep, and rabbit. The sodium salt, sodium deoxycholate, is used as a detergent. Aggregation number 3–12; CMC 2–6 mM.



deoxycholic acid

deoxycorticosterone *abbr*.: DOC; Kendall's desoxy compound B, Reichstein's substance Q, 21-hydroxyprogesterone, 21-hydroxypregn-4-ene-3,20-dione; a **mineralocorticoid** hormone with little glucocorticoid activity, synthesized by the adrenal gland. For clinical purposes it may be administered as its ester, deoxycorticosterone acetate (*abbr*.: doca *or* DOCA).



deoxycytidine deaminase *see* **cytidine deaminase**.

deoxycytidylate deaminase *an alternative name for* dCMP deaminase.

3-deoxy-D-*manno*-octulosonic acid 8-phosphate synthase EC 4.1.2.16; *recommended name*: 2-dehydro-3-deoxyphosphooctonate aldolase; *other name*: phospho-2-keto-3-deoxyoctonate (*abbr*.: phosphoKDO) aldolase; an enzyme that catalyses the formation of 3-deoxy-D-*manno*-octulosonic acid 8-phosphate from D-arabinose 5-phosphate and phospho*enol*pyruvate with release of orthophosphate. It is the first step in the biosynthesis of the bacterial lipopolysaccharide, lipid A, and is required for cell growth. Example from *Escherichia coli*: database code KDSA_ECOLI, 284 amino acids (30.80 kDa).

deoxynojirimycin an antibiotic produced by *Bacillus* species, the reduced form of **nojirimycin**; it inhibits formation of N-linked complex oligosaccharides in culture.

deoxynucleoside *an alternative name for* **deoxyribonucleoside**.

deoxynucleotide *an alternative name for* **deoxyribonucleotide**.

deoxynucleotidyltransferase EC 2.7.7.31; *recommended name*: DNA nucleotidylexotransferase; *other names*: terminal addition enzyme; terminal transferase; terminal deoxyribonucleotidyltransferase. It is a Mg^{2+}-dependent polymerase enzyme that extends the 3′ end of a DNA strand one nucleotide at a time, independent of a template. It catalyses the reaction (if *n* nucleotides are added in all):

$$n\text{ (deoxynucleoside triphosphate)} + \text{(deoxynucleotide)}_m = n\text{ (pyrophosphate)} + \text{(deoxynucleotide)}_{m+n}.$$

Example (bovine): database code TDT_BOVIN, 520 amino acids (59.60 kDa).

deoxypentose any monosaccharide in which one alcoholic hydroxyl group of a pentose has been replaced by a hydrogen atom.

deoxypentose nucleic acid *former term for* **deoxyribonucleic acid**.

deoxypolymeric tailing extension of the 3′ end of a strand of DNA, or of each strand of (a fragment of) duplex DNA, by a tail comprising a homooligomer or homopolymer of a deoxyribonucleotide. It is effected by the enzyme **deoxynucleotidyl**

transferase. The process is useful for forming cohesive termini to DNA fragments in the preparation of **recombinant DNA**. *See* **cohesive end**.

deoxyribonuclease *abbr*.: DNase (*or sometimes* DNAse *or* DNAase); any enzyme within subclass EC 3.1, esterases, that catalyses the hydrolytic cleavage of phosphodiester linkages in DNA. If such an enzyme catalyses endonucleolysis (i.e. hydrolysis of nonterminal phosphodiester linkages) it is termed an **endodeoxyribonuclease**; if it catalyses exonucleolysis (i.e. hydrolysis of terminal linkages) it is termed an **exodeoxyribonuclease**. *See also* **AP endonuclease**, **crossover point**, **deoxyribonuclease I**, **deoxyribonuclease II**, **deoxyribonuclease III**, **deoxyribonuclease IV (phage T4-induced)**, **deoxyribonuclease V**, **deoxyribonuclease X**, **deoxyribonuclease Kl**, **deoxyribonuclease (pyrimidine dimer)**, **endonuclease**, **exodeoxyribonuclease I**, **exodeoxyribonuclease V**, **exodeoxyribonuclease VII**, **exodeoxyribonuclease (phage SP3-induced)**, ***myt5***, **restriction enzyme**.

deoxyribonuclease I EC 3.1.21.1; *other names*: pancreatic DNase; DNase; thymonuclease. An enzyme that catalyses the endonucleolytic cleavage of DNA to 5′-phosphodinucleotide and 5′-phosphooligonucleotide end products. It shows a preference for double-stranded DNA. Example from *Bos taurus*: database code DRN1_BOVIN, 260 amino acids (29.06 kDa). *See also* **streptodornase**.

deoxyribonuclease II EC 3.1.22.1; *other names*: DNase II; pancreatic DNase II. An enzyme that catalyses endonucleolytic cleavage of DNA to 3′-phosphomononucleotide and 3′-phosphooligonucleotide end products.

deoxyribonuclease III EC 3.1.11.2; *other name*: exonuclease III. An enzyme that catalyses exonucleolytic degradation of double-stranded DNA progressively in the 3′- to 5′-direction, releasing 5′-phosphomononucleotides. Example endodeoxyribonuclease III from *Escherichia coli*: database code EX3_ECOLI, 268 amino acids (30.91 kDa); this enzyme is also the major **AP endonuclease** (i.e. an endonuclease) in *E. coli*.

deoxyribonuclease IV (phage T4-induced) EC 3.1.21.2; *other names*: endodeoxyribonuclease IV (phage T4-induced); endonuclease IV. An enzyme that catalyses endonucleolytic cleavage of DNA to 5′-phosphooligonucleotide end products. Example from *Bacillus subtilis*: database code END4_BACSU, 297 amino acids (33.07 kDa).

deoxyribonuclease V EC 3.1.22.3; *other name*: endodeoxyribonuclease V. An enzyme that catalyses endonucleolytic cleavage at apurinic or apyrimidinic sites within DNA to yield products with a 3′-phosphate. It is involved in DNA repair. The enzyme from *Escherichia coli* consists of three subunits, RECB, RECC, and RECD, also known as the β chain, γ chain, and α chain, respectively. Example, α chain: database code EX5A_ECOLI, 608 amino acids (66.96 kDa).

deoxyribonuclease X EC 3.1.22.5; an enzyme that catalyses endonucleolytic cleavage of supercoiled plasmid DNA to produce linear DNA duplexes.

deoxyribonuclease K1 *or* ***Aspergillus*** **deoxyribonuclease K1** EC 3.1.22.2; *other name*: *Aspergillus* DNase K1. An enzyme that catalyses endonucleolytic cleavage of DNA to 3′-phosphomononucleotide and 3′-phosphooligonucleotide end products.

deoxyribonuclease (pyrimidine dimer) EC 3.1.25.1; *other name*: endodeoxyribonuclease (pyrimidine dimer). An enzyme involved in DNA repair; it can induce single-strand breaks in DNA on the 5′-side of pyrimidine dimers in the strand containing the lesion, after formation of the dimers as a result of, e.g., ultraviolet radiation.

deoxyribonucleate 1 *abbr*.: DNA; any anionic form of deoxyribonucleic acid. 2 any salt of deoxyribonucleic acid.

deoxyribonucleic acid *former names*: desoxyribonucleic acid *or* thymonucleic acid; *also* deoxypentose nucleic acid *or* deoxyribose nucleic acid *or* desoxyribose nucleic acid *or* thymus nucleic acid. *See* **DNA**.

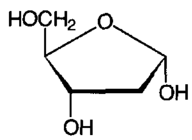
deoxyribonucleoprotein *abbr*.: DNP; any conjugated protein that contains DNA as the nonprotein moiety.

deoxyribonucleoside *or* **deoxynucleoside** any **nucleoside** (def. 1) in which the glycose moiety is 2-deoxy-β-D-ribofuranose.

deoxyribonucleoside monophosphate kinase any of several enzymes within sub-subclass EC 2.7.4 that catalyse the transfer of a phosphoric residue from a nucleoside triphosphate to a deoxyribonucleoside monophosphate, e.g. from ATP to a deoxyribonucleoside phosphate to form ADP and a deoxyribonucleoside diphosphate.

deoxyribonucleotide *or* **deoxynucleotide** a deoxyribonucleoside in ester linkage to phosphate, commonly at the 5′ position of its deoxyribose moiety. *See* **nucleotide**.

deoxyribose 1 *symbol*: dRib; *the trivial name for* 2-deoxyribose; 2-deoxy-*erythro*-pentose. The D enantiomer, as 2-deoxy-β-D-ribofuranose, forms the glycose moiety of all deoxyribonucleosides, deoxyribonucleotides, and deoxyribonucleic acids. 2 3-deoxyribose (*see* **cordycepose**).



2-deoxy-α-D-ribofuranose

deoxyribose nucleic acid *former name for* **deoxyribonucleic acid**.

deoxyriboside any glycoside or glycosylamine in which deoxyribose forms the glycose moiety, especially a deoxyribonucleoside.

deoxyribosome *see* **nu body**.

deoxyribosylthymine *an alternative name for* thymidine.

deoxyribotide any deoxyriboside esterified with phosphate at the 5′ position on its ribose moiety, especially a deoxynucleotide.

deoxyribovirus *an alternative name for* **DNA virus**.

deoxysaccharide *or* **deoxysugar** any saccharide in which one alcoholic hydroxyl group has been replaced by a hydrogen atom.

depactin a protein isolated from starfish eggs, able to depolymerize F-actin and inhibit the extent of actin polymerization. It binds to actin monomers from filaments and in solution. Example from *Asterias amurensis*: database code DEPA_ASTAM, 150 amino acids (17.21 kDa).

dependent form of glycogen synthetase the phosphorylated form of the enzyme glycogen synthetase that is 'dependent' for its activity on the presence of glucose 6-phosphate, a positive effector.

dependent variable any variable in a mathematical statement or equation whose value depends on the values taken by the related independent variables.

dephospho-[reductase kinase] kinase EC 2.7.1.110; *other name*: reductase kinase kinase; an enzyme that catalyses the phosphorylation of dephospho-[[3-hydroxy-3-methylglutaryl-CoA reductase (NADPH)] kinase] by ATP with release of ADP. It is involved in the regulation of **hydroxymethylglutaryl-CoA reductase** (HMGCoA reductase). By phosphorylating reductase kinase, it activates that enzyme to phosphorylate HMGCoA reductase, thereby inactivating the latter.

dephosphorylation the process of removing one or more phosphoric (ester or anhydride) residues from a molecule.

depolarization *or* **depolarisation** 1 a reduction in the electric potential measured across a cell membrane, especially that brought about in a nerve or muscle cell by an electrical stimulus or a drug. 2 the decrease of the angle of polarization of polarized light so that it appears altered when viewed through an analyser. *See also* **fluorescence depolarization**.

depolarization ratio *or* **depolarisation ratio** *(in light scattering)* the ratio of the horizontally to the vertically polarized components of the light scattered by scattering particles when illuminated by unpolarized light. If the scattering particles are isotropic the depolarization ratio will be zero; if they are anisotropic it will not be zero.

deproteinization *or* **deproteinisation** the process of removing protein from a biological sample, either by precipitation of protein from solution or by hydrolysis of protein with proteolytic enzymes.

depside any of a family of natural or artificial compounds derived from two or more molecules of various trihydroxybenzoic acids by esterification of a carboxyl group on one molecule with a hydroxyl group on another molecule (which may or may not be of the same structure as the first). By analogy with peptides, depsides derived from two, three, etc., or several trihydroxybenzoic acid units may be termed **didepside**, **tridepside**, etc., or **polydepside**, respectively. Depsides occur widely in lichens and also as components of some tannins.

depsidone any compound derived from two phenolcarboxylic acid molecules (which may or may not be the same molecules) that are joined together by both an ester and an ether link. Such substances are found in lichens.

depsipeptide any linear or macrocyclic compound containing (alternating) amino-acid and hydroxy-acid residues, linked by amide, *N*-methylamide, and ester bonds. Among the naturally occurring **cyclodepsipeptides** (sometimes alternatively termed peptolides) are the antibiotics **valinomycin**, the **enniatins**, and the monamycins.

depurinate to remove purine residues from a polynucleotide. —**depurinated** *adj.*; **depurination** *n.*

depyrimidinate to remove pyrimidine residues from a polynucleotide. —**depyrimidinated** *adj.*; **depyrimidination** *n.*

derepression the phenomenon in which a repressor is prevented from interacting with the operator component of an **operon** and an increase in the synthesis of a gene product thereby occurs. With an inducible enzyme, the inducer derepresses the operon. Derepression may also result from mutations, either in the **regulator(y) gene**, thus blocking repressor synthesis, or in the operator gene, thus rendering it insensitive to the repressor.

deRib *(sometimes) symbol for* 2-deoxy-D-ribose (dRib is the correct symbol).

derivative 1 any compound that may, at least theoretically, be formed from another compound to which it is structurally related. **2** *see* **derivative of a function**.

derivative of a function *(in mathematics)* the limit of the ratio of an increment of a dependent variable to the corresponding increment of an associated independent variable as the latter increment tends to zero.

derivatize *or* **derivatise** to convert a chemical compound into a derivative. —**derivatization** *or* **derivatisation** *n.*

dermaseptin any of a series of antimicrobial peptides derived from amphibian skin. Dermaseptins s1–5 constitute a family of cationic lysine-rich amphipathic antifungal peptides of 28–34 residues. Presumed to protect the naked frog skin from infections, they have wider application, being the first vertebrate peptides to show lethal effects against the filamentous fungi that can cause infections in immunodeficiency syndrome, or during the use of immunosuppressive agents. *Compare* **magainin**.

dermatan sulfate *or* **chondroitin sulfate B** any of a group of glycosaminoglycans with repeats consisting of β1→4-linked L-iduronyl-(β1→3)-*N*-acetyl-D-galactosamine 4-sulfate units. They are important components of ground substance or intercellular cement of skin and some connective tissues (*see* **matrix** (def. 3)). They usually occur attached to proteins as proteoglycans, the mode of linkage being the same as for **chondroitin sulfate**. A hereditary deficiency in the ability to degrade dermatan sulfate characterizes certain types of **mucopolysaccharidosis**.

dermis *or* **corium** the thick layer of tissue forming part of the skin and lying beneath the epidermis. It consists of loose connective tissue in which are blood capillaries, smooth-muscle fibres, sweat glands and sebaceous glands with their ducts, hair follicles, and sensory nerve endings. —**dermal, dermic** *adj.*

dermorphin any of a group of heptapeptide amides of the general structure H-Tyr-DAla-Phe-Gly-Tyr-Xaa-Ser-NH_2, isolated from the skin of frogs of the genus *Phyllomedusa* and some other amphibians; they show high and long-lasting opiate-like activity.

DES *abbr. for* diethylstilbestrol.

des+ *prefix* **1** (in the trivial name of a polycyclic compound, e.g. a steroid) indicating the removal of a terminal ring, with addition of a hydrogen atom at each junction arm with the adjacent ring; it should be followed by the italic capital letter designating the terminal ring in question, e.g. des-*A*-androstane. **2** (in the trivial name of a polypeptide) indicating the removal of an amino-acid residue, terminal or otherwise; it should be followed by the name or symbol for the amino acid in question and an arabic number indicating its position in the normal polypeptide, e.g. des-7-proline-oxytocin, or des-Pro^7-oxytocin. **3** indicating the removal of a specified atom or group from a molecule of an indicated substance; *see* **desamido+**, **desthio+**; *see also* **descarboxy-clotting factor**. **4** *(formerly) a variant of* **de+** (still preferred in some languages, e.g. French, German).

desalinate to remove salt, especially from seawater to render it suitable for drinking or irrigation. *Compare* **desalt**. —**desalination** *n.*

desalt to remove small, usually inorganic ions from a sample. Methods that may be used include electrodialysis, electrophoresis, gel filtration, and ion-exchange chromatography. *See also* **deionize**. *Compare* **desalinate**.

desamidation the hydrolysis of one or more carboxamide groups in a molecule to carboxyl groups.

desamido+ *prefix* indicating **desamidation** of a molecule, especially of a peptide.

desaturase *general name for* any enzyme catalysing a desaturation reaction. The name specifically often refers to any of several fatty-acyl-CoA desaturases, enzymes coming within sub-subclass EC 1.14.99; in animals these are associated with the endoplasmic reticulum, require dioxygen, and insert double bonds into saturated and *Z*(i.e. *cis*)-unsaturated fatty acids. They also occur in higher plants, protozoa, and fungi. These enzymes are specific for position in the fatty-acyl chain, and are termed Δ^5-desaturase, Δ^9-desaturase, and so on, according to the distance from the carboxyl carbon atom to the first carbon atom of the double bond. In mammals, there is a lack of any desaturase that inserts a double bond nearer to the terminal methyl group of the fatty acid than nine carbon atoms, hence the need for *n*-3 and *n*-6 precursor fatty acids (the **essential fatty acids**). *See also* **linoleic family, linolenic family**.

desaturation 1 any process or reaction in which an organic compound becomes **unsaturated** (def. 2), e.g. by removal of two hydrogen atoms, or a hydrogen atom and a hydroxyl group, from adjacent carbon atoms. **2** any process in which ligands are removed from a macromolecule so that all the binding sites for that ligand are no longer occupied. —**desaturate** *vb.*; **desaturated** *adj.*

descarboxy-clotting factor any of the abnormal blood clotting factors containing glutamic acid residues instead of γ-carboxyglutamic-acid residues. They are formed in animals deficient in vitamin K or during administration of vitamin K antagonists. *Compare* **PIVKA**.

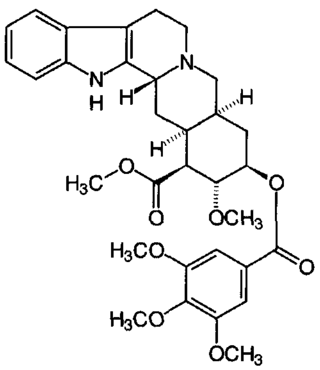
Descartes, René (1596–1650), French philosopher and mathematician. *See also* **Cartesian**.

descending chromatography any chromatographic technique in which a liquid mobile phase runs downwards through the supporting matrix.

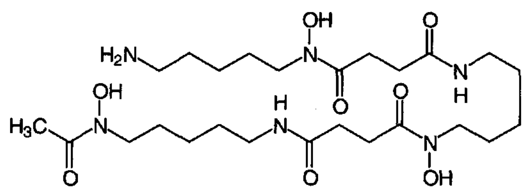
descriptive name name of an organic substance that is more descriptive of action or function and often more convenient than its **trivial name**, **semisystematic name**, **systematic name**, or other type of name permitted by the rules of chemical nomenclature.

desensitization *or* **desensitisation 1** a spontaneous decline in response resulting from continuous application of agonist, or to repeated applications or doses; such attenuation may result within minutes of prior stimulation, or may take hours to develop. It is recommended that this term be used where **fade** or **tachyphylaxis** are considered to involve the receptor itself, or to be a direct consequence of receptor activation. **Homologous desensitization** results from prior stimulation with the receptor's own agonist, whereas **heterologous desensitization** results from the activity of other agonists acting at other receptors. On the longer time scales, desensitization may result from receptor internalization in **coated pits**, and over shorter time scales from modification of the receptor in a way that uncouples it from signal transduction systems, e.g. by phosphorylation (*see* **adrenergic receptor kinase**). **2** the modification of an enzyme, either by mutation or chemically, so that it no longer responds to **effectors** but retains its catalytic activity. **3** the suppression of an established (immunological) hypersensitivity by injection of antigen, usually in very small doses. In delayed hypersensitivity the antigen reacts with sensitized cells, so neutralizing them. In immediate hypersensitivity the antigen either causes blocking antibody production (in atopy) or forms nontoxic complexes with existing antibodies (in anaphylaxis). —**desensitize** *vb.*; **desensitized** *adj.*

deserpidine desmethoxyreserpine, 11-desmethoxy-*O*-(3,4,5-trimethoxybenzoyl)reserpic acid methyl ester; an alkaloid derived from roots of *Rauwolfia canescens* with pharmacological actions similar to the parent methoxy compound, **reserpine**.



desferrioxamine a siderophore of the hydroxamate type derived from *Streptomyces pilosus* that forms a chelate complex specifically with Fe(III) ions, yielding **ferrioxamine**. It is used parenterally to treat acute iron poisoning and to reverse iron and aluminium overload. Its natural function is to transport iron into the microbial cell and/or to make iron available for the synthesis of heme.



desiccant a substance such as $CaCl_2$ or P_2O_5 that has a high affinity for water and is used to remove moisture from a gas or another substance; a drying agent.

desiccate to remove (most of) the water from a substance or material containing moisture, especially with the aid of a **desiccant**; to dry. —**desiccation** *n.*

desiccator an apparatus in which to effect desiccation.

desmin *or (formerly)* **skeletin** a 50–55 kDa protein found in the Z disk of skeletal and cardiac muscle cells and in **intermediate filaments** in smooth muscle and non-muscle cells.

desmo+ *or (before a vowel)* **desm+** *comb. form* denoting ligament or bond.

desmofibrin a stabilized insoluble fibrin polymer formed from soluble fibrin polymer (*see* **fibrin**) in the final stage of **blood coagulation** by the action of factor XIII (fibrin-stabilizing factor) (*see* **blood coagulation**).

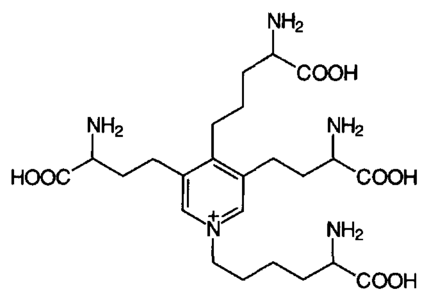
desmoglein a type I **membrane protein** of the mature desmosomal junction, involved in the interaction of plaque proteins and intermediate filaments mediating cell–cell adhesion. It contains **cadherin**-like repeats. Example (desmoglein 1 precursor) from human: database code DSG1_HUMAN, 1049 amino acids (113.59 kDa).

desmolase *former term for* any enzyme that catalyses the formation or rupture of a carbon–carbon bond.

desmoplakin any of several intracellular desmosome proteins that are probably involved in attaching intermediate filaments (e.g. keratin filaments in epithelial cells, desmin filaments in heart cells) to the desmosome. Example, desmoplakin I from human foreskin: database code HUMDPI, 2832 amino acids (327.35 kDa). *See also* **plakoglobin** (for desmoplakin III).

desmopressin 1-(3-mercaptopropanoic acid)-8-D-arginine vasopressin, desamino-[8-D-arginine]vasopressin; a synthetic analogue of **vasopressin** with greater antidiuretic activity and less pressor activity than the parent compound.

desmosine 4-(4-amino-4-carboxybutyl)-1-(5-amino-5-carboxypentyl)-3,5-bis(3-amino-3-carboxypropyl)pyridinium; a tetra(α-amino acid) isolated from hydrolysates of the fibrous protein **elastin**. Desmosine and **isodesmosine** covalently crosslink two or more polypeptide strands in this protein, the postulated mode of formation of both compounds being condensation of two residues of **allysine** in one polypeptide chain with one allysine and one lysine residue in a second polypeptide chain.

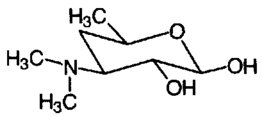


desmosome *or* **macula adherens** a patchlike intercellular junction found in vertebrate tissue. It consists of parallel zones of two cell membranes, separated by an interspace of 25–35 nm and having dense fibrillar plaques in the subjacent cytoplasm. The interspace is continuous with the intercellular space and contains a dense, central plaque of material rich in protein. These proteins are known as linker proteins and consist of cell–cell adhesion molecules of the **cadherin** type. Desmosomes are important in cell-to-cell adhesion and are particularly numerous in stratified squamous epithelium that is subject to mechanical stress. A similar type of intercellular junction, the **zona adherens**, sometimes also referred to as a desmosome, encircles the cell like a belt. —**desmosomal** *adj.*

desmotubule *see* **plasmodesma**.

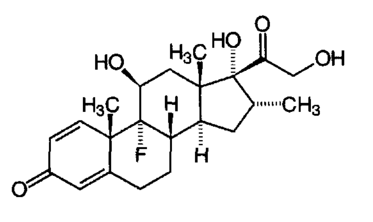
desorb to remove adsorbed substances from a surface. —**desorption** *n.*; **desorbed** *adj.*

desosamine 3,4,6-trideoxy-3-(dimethylamino)-D-*xylo*-hexose; an aminodeoxy sugar component of several **macrolide** antibiotics, including **erythromycins** A, B, and C.



β-D anomer

desoxy+ *an obsolete variant of* **deoxy+**.
destain to remove dye in excess of that required to stain specifically the material in question, e.g. in staining proteins in an electrophoretogram. —**destaining** *n.*
destainer any apparatus used for removing excess dye electrophoretically from stained gels.
desthio+ *prefix* indicating the replacement of a sulfur atom by two hydrogen atoms.
destrin an **actin binding protein** similar to **cofilin** with wide tissue distribution. It severs actin filaments and binds to actin monomers. Example from pig: database code DEST_PIG, 165 amino acids (18.48 kDa).
Desulfovibrio a genus of Gram-negative, obligately anaerobic, asporogenous, chemotrophic bacteria that obtain energy by anaerobic respiration using reducible inorganic sulfur compounds.
detergent 1 a cleansing agent, especially a soap or (usually synthetic) soaplike substance that, when used in aqueous solution, cleanses by reducing surface tension. 2 having the properties of a detergent. *See also* **BigCHAP**, **Brij**, **C_xE_y**, **cetyltrimethylammonium bromide**, **CHAPS**, **CHAPSO**, **cholic acid**, ***n*-decyl-β-D-glucopyranoside**, ***n*-decyl-β-D-maltopyranoside**, **deoxy-BigCHAP**, **deoxycholic acid**, **digitonin**, **dodecylglucopyranoside**, **dodecylmaltoside**, ***n*-dodecyl-*N*, *N*-dimethylglycine**, **Genapol**, **glycocholate**, **glycodeoxycholate**, **heptyl glucoside**, **heptyl thioglucoside**, **hexyl glucoside**, **lauryldimethylamine oxide**, **sodium dodecylsulfate**, **Lubrol**, **Mega-8**, **nonyl glucoside**, **NP-40**, **octyl glucoside**, **octyl thioglucoside**, **taurocholate**, **taurodeoxycholate**, **Triton**, **Tween**, **Zwittergent**.
detergent gel electrophoresis *an alternative name for* **SDS-polyacrylamide gel electrophoresis**.
deteriosome a microvesicle that is found in eukaryotic cells and is rich in phospholipase A_2 (*see* **phospholipase**). It is believed to represent a stage in membrane degradation.
detoxication *or (sometimes)* **detoxification**; the act or process of counteracting or rendering innocuous a poisonous substance. In an organism this involves various enzymic reactions by which harmful compounds, whether produced in the organism or introduced into it, are converted into (normally) less harmful substances and into more readily excretable products. Enzymic detoxication is brought about by chemical modification (frequently by oxidation) and/or conjugation with a normal metabolite. The enzymic process does not always result in a less toxic product and the term is sometimes restricted to those instances in which a less toxic product actually occurs, as distinct from **intoxication**. Enzymic detoxication may be divided into two phases. Phase 1 concerns metabolic transformation by a wide variety of oxidations, reductions, hydrolyses, etc., usually resulting in the introduction of functional groups that increase the polarity of the molecule and act as centres for phase 2 reactions. Phase 2 involves conjugation, in which the foreign compound itself or the compound resulting from phase 1 reactions is combined with endogenous substances, such as acetate, glucuronate, sulfate, or an amino acid, or is alkylated by introduction of methyl, or other groups, to form the more water-soluble ultimate excretory metabolite. *See also* **lethal synthesis**. —**detoxicate** *vb.*
detoxification 1 the removal of poison. 2 *(sometimes)* **detoxication**.
detoxify 1 to remove a poison or the effect of a poison. 2 to bring about **detoxication**.
deuterate to combine or label with **deuterium**. —**deuteration** *n.*; **deuterated** *adj.*
deuterium *symbol*: ^2H *or (deprecated)* D; a stable isotope of hydrogen, relative atomic mass 2.014, sometimes known as heavy hydrogen. Its relative abundance in natural hydrogen is 0.015 atom percent. It was much used as a tracer in studies of the chemistry and biochemistry of compounds containing nonlabile hydrogen before the ready availability of tritium and carbon-14; its use continues in tracer experiments with detection by NMR spectroscopy, and where the use of radioactive compounds is contra-indicated. *See also* **heavy water**, **protium**, **tritium**.
deuterium (discharge) lamp a **hydrogen (discharge) lamp** filled with dideuterium in place of dihydrogen; this substitution increases the intensity 3- to 5-fold for the same power consumption and extends the upper limit of the continuum further into the visible range.
deutero+ *or (before a vowel)* **deuter+** *comb. form* 1 denoting that an indicated chemical compound or class of compound contains deuterium. 2 denoting second or secondary.
deuteron 1 *symbol*: $^2H^+$; the cation derived from an atom of deuterium. *Compare* **hydron**. 2 *symbol*: d; a particle of nucleon number two, having a charge equal and opposite to that of an **electron** and having a mass of 2.0136 dalton.
develop 1 to process an exposed photographic film with a **developer** in order to render the image visible. 2 to render visible the colourless substances on a chromatogram or electrophoretogram by treatment with a stain or chromogenic reagent. 3 to cause a solvent to flow through a chromatographic support in order to separate the components of an applied mixture. 4 *(of a cell, organ, or organism)* to undergo or cause to undergo a series of orderly changes whereby a mature state is attained. —**development** *n.*
developer 1 a solution of a reducing agent that renders visible the latent image on an exposed photographic film or paper by reducing to metallic silver those areas of silver halide that had been exposed to light. 2 a solution of a stain or chromogenic reagent used to render visible the colourless substances on a chromatogram or electrophoretogram.
Dewar flask *or* **Dewar** a double-walled vessel or flask, usually of silvered glass but sometimes of metal, in which the space between the walls is evacuated to prevent conduction and convection of heat between the inner and outer walls. It is used to maintain the contents at temperatures either higher or lower than that of the environment. [After its inventor Sir James Dewar (1842–1923), British chemist and physicist.]
dexamethasone 9*α*-fluoro-16*α*-methylprednisolone; a synthetic **glucocorticoid** with negligible mineralocorticoid activity that has actions similar to adrenal corticosteroids.



dextr+ *see* **dextro+**.
dextran any glucan of very high relative molecular mass and consisting of linear chains of *α*(1→6)-linked D-glucose residues, often with *α*(1→2)- or *α*(1→3)-branches. Native dextran, produced by a number of species of bacteria of the family Lactobacilliaceae, is a polydisperse mixture of components

of M_r <10^3 to >10^7. Specific fractions of (partially hydrolysed) native dextran that have a narrow and well-defined size distribution may be designated by a numeric suffix equal to 10^{-3} × mass-average molar mass; e.g. Dextran T 500. Dextrans used in medicine are acid-degraded native dextrans.

dextranosome a **lysosome** loaded with dextran. It has higher than normal density.

dextransucrase EC 2.4.1.5; *systematic name*: sucrose:1,6-α-D-glucan 6-α-D-glucosyltransferase; *other name*: sucrose 6-glucosyltransferase. An extracellular bacterial enzyme that catalyses the reaction:

sucrose + (1,6-α-D-glucosyl)$_n$ = D-fructose + (1,6-α-D-glucosyl)$_{n+1}$.

Streptococcus mutans dextransucrases are of several types: dextransucrase S produces water-soluble glucans (α(1→6)-linked glucose); dextransucrase I produces water-insoluble glucans (α(1→3)-linked glucose and some α(1→6)-linkages); and dextransucrase SI produces both types; the sequences are related. Example dextransucrase SI precursor from *S. mutans*: database code GTFC_STRMU, 1375 amino-acid residues (152.85 kDa); the first 24 residues are the signal. *See also* **sucrase** (def. 3).

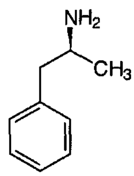
dextrin any of the poly-D-glucosides of intermediate chain length that are formed during the hydrolytic degradation of starch or glycogen by enzymes, acid, or heat. *See also* **limit dextrin**, **Schardinger dextrin**.

α-dextrin endo-1,6-α-glucosidase EC 3.2.1.41; *other names*: pullulanase; pullulan 6-glucanohydrolase; limit dextrinase; debranching enzyme; amylopectin 6-glucanohydrolase; the starch-debranching enzyme, which hydrolyses (1→6)-α-D-glucosidic linkages in pullulan and starch to form maltotriose. Example from *Klebsiella pneumoniae*: database code PULA_KLEPN, 1090 amino acids (117.97 kDa).

dextro- *symbol*: *d*-; a prefix, no longer used, to denote a compound that is **dextrorotatory**; the dextrorotatory enantiomer of compound A is now designated as (+)-A.

dextro+ *or (before a vowel)* **dextr+** *comb. form* denoting on or towards the right. *Compare* **levo+**.

dextroamphetamine (*S*)-methylbenzeneethanamine; the potent (–)-stereoisomer of **amphetamine**. One proprietary name is Dexamphetamine.



dextrorotatory *or (US)* **dextrorotary** **1** *symbol*: (+)- *or (formerly) d*-; describing a chemical compound that rotates the plane of polarization of a transmitted beam of plane-polarized light to the right, i.e. in a clockwise direction as viewed by an observer looking toward the light source. **2** describing any rightwards or clockwise rotation. *Compare* **levorotatory**. —**dextrorotation** *n.*

dextrorphan *see* **levorphanol**.

dextrose *traditional name (still used in pharmacy)* for D-glucose, D-(+)-glucopyranose, the dextrorotatory component of **invert sugar**. *Compare* **levulose**.

DFP *abbr. for* diisopropyl fluorophosphate.

dG *symbol for* a residue of the deoxynucleoside deoxyguanosine (alternative to dGuo).

dGMP *abbr. for* deoxyguanosine monophosphate, the common name for 2′-deoxyguanosine 5′-phosphate; 2′-deoxy-5′-guanylic acid; guanine 2′-deoxyriboside 5′-phosphate.

dGuo *symbol for* a residue of the deoxynucleoside deoxyguanosine; 2′-deoxyribosylguanine; guanine 2′-deoxyriboside (alternative to dG).

dGuo5′*P* *symbol for* deoxyguanosine 5′-phosphate.

dGTP *abbr. for* deoxyguanosine triphosphate, the common name for 2′-deoxyguanosine 5′-triphosphate, 2′-deoxy-5′-guanylyl diphosphate, guanine 2′-deoxyriboside 5′-triphosphate.

DH+ *abbr. for* dehydro+ *or* dihydro+ *or* dihydroxy+, sometimes prefixed to chemical names or alternatives. Its use is deprecated.

DHAS *abbr. for trans*-dehydroandrosterone, i.e. **dehydroepiandrosterone**.

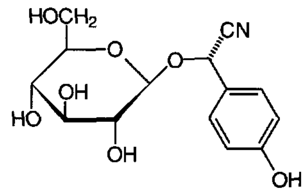
DHEA *abbr. for* dehydroepiandrosterone.

DHFR *abbr. for* dihydrofolate reductase.

DHPR *abbr. for* dihydropteridine reductase.

DHU arm *abbr. for* dihydrouridine arm of transfer RNA.

dhurrin L-*p*-hydroxymandelonitrile-β-D-glucopyranoside; a glucoside from the fruit of young *Sorghum vulgare*. *See also* **amygdalin**.



di+ *comb. form* **1** *(in chemical nomenclature) (distinguish from* **bis+** (def. 2)) **a** indicating the presence in a molecule of two identical unsubstituted groups, e.g. diethylsulfide, 1,3-dihydroxyacetone. **b** indicating the presence in a molecule of two identical inorganic oxoacid residues in anhydride linkage, e.g. adenosine 5′-diphosphate. **2** *or* **bis+** (def. 1) denoting two, twofold, twice, doubled.

diabetes any clinical condition characterized by thirst and the passing of large volumes of dilute urine, especially **diabetes mellitus**. *See also* **diabetes insipidus**. —**diabetic** *adj., n.*

diabetes-associated peptide *abbr.*: DAP; *another name for* **amylin**.

diabetes insipidus a disease characterized by thirst and the excretion of large volumes of hypotonic (dilute) urine, caused by any lesion, nervous or endocrine, that interferes with the normal secretion of **vasopressin** by the posterior pituitary gland. *See also* **nephrogenic diabetes insipidus**.

diabetes mellitus *or* (commonly) **diabetes** a disorder in which the level of blood glucose is persistently above the normal range. Two main forms of the disease are generally recognized. **Type 1 diabetes mellitus**, the so-called juvenile type, often manifests itself in children and young adults. There is a marked failure to release **insulin** from the B cells of the **islets of Langerhans** in the pancreas, and frequently an almost complete absence of insulin in the B cells. The only effective treatment is administration of insulin, hence the frequent designation of this form as **insulin-dependent diabetes mellitus** (*abbr.*: IDDM). Type 1 diabetes mellitus is thought to arise in many patients from autoimmune attack on the B cells. Without insulin a severe **ketoacidosis** rapidly develops, with tissue wasting resulting from breakdown of muscle protein and hydrolysis of triacylglycerols in fat depots. **Type 2 diabetes mellitus** or **maturity onset diabetes** tends to manifest itself in adults, especially if obese, though the distinction between types 1 and 2 is blurred. In type 2 there may often be secretion of significant amounts of insulin, but for reasons not well understood these are not fully effective. There is by definition a failure to lower blood glucose adequately after a **glucose tolerance test**; patients of this type seldom develop severe ketoacidosis. **Tolbutamide** and related sulfonylureas can be used to stimulate insulin release in type 2. In both types of diabetes mellitus there may develop serious complications, including neuropathy, retinopathy, atherosclerosis, peripheral vascular disease, and nephropathy. Type 2 diabetes mellitus is also referred to as **non-insulin-dependent diabetes**.

diabetes mutation a mutation first identified in mice that spontaneously develop diabetic symptoms. It appears to be a mutation in the **leptin receptor** gene.

diabetic a sufferer from diabetes.

diabetogenic causing or being caused by **diabetes**.

diabetogenic hormone *former term for* **somatotropin**.

diabolic a term suggested to categorize enzymes that appear to serve a complex or multiple metabolic function. *See also* **amphibolic**.

diacetylchitobiose 4-*O*-(2-acetamido-2-deoxy-β-D-glucopyranosyl)-2-acetamido-2-deoxy-D-glucose; the repeating unit of **chitin**. *See also* **chitobiose**.

diacetylmorphine *other names*: diamorphine, heroin; a semisynthetic narcotic analgesic derived from **morphine** by acetylation and used either as the free base or its hydrochloride. It has more potent analgesic activity than the parent compound but otherwise its actions are similar. Noted for its dependence-inducing actions and potential for abuse, it is a controlled substance under the US Code of Federal Regulations; import and manufacture of diacetylmorphine and its salts are forbidden in the USA and other countries. *See also* **codeine**.

diacylglycerol *or (formerly)* **diglyceride** any (1,2- or 1,3-) diester of glycerol and two fatty acids; the latter may be identical or different.

diacylglycerol *O*-acyltransferase EC 2.3.1.20; *other name*: diglyceride acyltransferase; an enzyme that catalyses the synthesis of triacylglycerol from acyl-CoA and 1,2-diacylglycerol with release of CoA.

diacylglycerol cholinephosphotransferase EC 2.7.8.2; *other names*: phosphorylcholine–glyceride transferase; alkylacylglycerol cholinephosphotransferase; 1-alkyl-2-acetylglycerol cholinephosphotransferase; an enzyme of the pathway for the *de novo* synthesis of phosphatidylcholines. It catalyses the formation of a phosphatidylcholine from CDPcholine and 1,2-diacylglycerol with release of CMP. Example from yeast: database code CPT1_YEAST, 407 amino acids (46.25 kDa).

diacylglycerol kinase *abbr*.: DGK; EC 2.7.1.107; *systematic name*: ATP:1,2-diacylglycerol 3-phosphotransferase; *other name*: diglyceride kinase. An enzyme that catalyses the reaction between ATP and 1,2-diacylglycerol to form ADP and 1,2-diacyl-*sn*-glycerol 3-phosphate (phosphatidic acid). It plays an important part in the response of cells to agonists that stimulate the **phosphatidylinositol cycle**, being involved in the resynthesis of inositolphospholipids. The enzyme is stimulated by Ca^{2+} and phosphorylated by protein kinase C; it is an **EF-hand** protein. Example from human: database code KDGL_HUMAN: 735 amino acids (82.49 kDa).

diacytosis the process of transporting material across a cell, especially an epithelial cell, by **pinocytosis** at one surface, movement of the **pinocytic vesicle** across the cell, and **exocytosis** of its contents at another part of the cell surface.

diafiltration the process of separating microsolutes, e.g. salts, from a solution of larger molecules (or of exchanging them for different microsolutes) by **ultrafiltration** with continuous addition of solvent (or of a solution of the new microsolutes). This rapidly removes the original microsolutes from the solution, whose volume remains constant.

diagonal chromatography a two-dimensional chromatographic technique used to determine the sensitivity of a constituent of a mixture to some (photo)chemical process, e.g. oxidation. The sample is chromatographed in one direction, the process carried out *in situ*, and the specimen is then rechromatographed at right angles. Compounds unmodified by the treatment all lie on a diagonal line across the chromatogram.

diagonal electrophoresis a two-dimensional electrophoretic technique for determining the position of disulfide bonds in a polypeptide. The polypeptide is partially hydrolysed and the resulting smaller peptides separated by paper electrophoresis in one direction. After treatment on the paper with performic acid vapour, they are subjected to a second electrophoresis step at right angles. Those peptides not falling on a diagonal line on the paper contain cysteic acid residues formed by the oxidation of disulfide groups.

diakinesis the final phase of prophase I in **meiosis**.

dialdose any monosaccharide derivative containing two (potential) aldehydic carbonyl groups. *Compare* **aldose**, **diketose**.

diallyltartardiamide *abbr*.: DATD; *N*,*N'*-diallyl-L-tartramide, $[CH(OH){-}CO{-}NH{-}CH_2{-}CH{=}CH_2]_2$; a compound sometimes used as a cross-linking agent in the preparation of polyacrylamide gels for electrophoresis or related separative procedures. Its structure contains a 1,2-diol grouping, which is susceptible to **periodate oxidation**. Such gels may thereby be solubilized after use to facilitate recovery of separated substances.

dialysable *or (esp. US)* **dialyzable** capable of being purified or separated by **dialysis**. The term has been used also, variously, to describe material that either can or cannot traverse a dialysis membrane. Some authorities prefer the term **diffusible** for material that can traverse a dialysis membrane. *See also* **dialysate**.

dialysate *or (esp. US)* **dialyzate** the solution resulting from **dialysis** that is free from colloidal material, i.e. free from material unable to pass through a dialysis membrane; the opposite of **retentate**. In the past the term has been used confusingly to describe both the material retained by the membrane and the material that has passed through a membrane. Some authorities prefer the term **diffusate** for the resulting colloid-free material and the term **analysis residue** for the retentate. *See also* **dialysable**.

dialyser *or (esp. US)* **dialyzer** an apparatus consisting of two or more compartments separated by **dialysis membrane(s)**, in which to carry out **dialysis**.

dialysis a process in which solute molecules are exchanged between two liquids through a membrane in response to differences in chemical potentials between the liquids. It is used especially for separating macromolecules (colloids) in solution from smaller dissolved substances (crystalloids) by selective **diffusion** through a **semipermeable membrane**, which allows the passage of all components of the system except the macromolecules. *See also* **diafiltration**, **diasolysis**, **electrodecantation**, **electrodialysis**, **equilibrium dialysis**, **reverse dialysis**, **thin-film dialysis**, **ultrafiltration**. [From Greek *dialusis*, a separation or dissolution.] —**dialyse** *or (esp. US)* **dialyze** *vb*.; **dialytic** *adj*.; **dialytically** *adv*.

dialysis bag the significant component of a simple **dialyser**, commonly made from a convenient length of **dialysis tubing** by knotting or tying off one or both ends. When filled with the colloidal solution to be dialysed it is immersed or suspended in an appropriate colloid-free aqueous medium.

dialysis cell any small device for use as a **dialyser**.

dialysis membrane any semipermeable membrane suitable for **dialysis**, often made of cellulose acetate, cellulose nitrate, or regenerated cellulose. It may be fashioned into flat sheets or into continuous tubing. Most dialysis membranes are permeable to molecules of up to 12–14 kDa.

dialysis residue *an alternative name for* **retentate**.

dialysis tubing seamless tubular **dialysis membrane**.

diamagnetic pair a diamagnetic dimer produced by spin coupling of two paramagnetic (molecular or atomic) species.

diamagnetism the property displayed by substances that have a very small negative magnetic susceptibility, due to the changes induced in the orbital motions of atomic electrons by an applied external magnetic field; such changes generate a magnetic field in opposition to the applied field. Whereas such a phenomenon occurs in all substances it is often masked by the opposing and stronger effects of **ferromagnetism** or **paramagnetism**. —**diamagnetic** *adj.*

diamine any compound containing two amino groups or substituted amino groups.

diamine oxidase *see* **amine oxidase (copper-containing)**.

diamino+ *prefix (in chemical nomenclature)* indicating the presence of two (unsubstituted or substituted) amino groups.

diamino acid any organic acid, usually an alkanoic acid, that contains two (unsubstituted or substituted) amino groups.

di(α-amino acid) any compound whose molecules may be derived from two (identical or differing) molecules of α-aminoalkanoic acid however they are linked. The term includes (but is not limited to) any dipeptide.

diaminobenzoic acid *abbr.*: DABA; 3,5-diaminobenzoic acid; a compound that gives fluorescent products when heated in mineral acid solution with aldehydes. It is used for the microfluorimetric determination of DNA (as it does not react with RNA) and in the analysis of sialic acid.

diaminobutyric acid *symbol*: A_2bu; *abbr.*: Dab; α,γ-diaminobutyric acid; 2,4-diaminobutanoic acid; a diamino acid not occurring in proteins. The L enantiomer is sometimes found replacing *meso*-diaminopimelic acid in peptidoglycan of bacterial cell walls and several residues per molecule are present in the peptide antibiotics of the polymyxin group (together with one residue of the D enantiomer in polymyxin B_1). It also occurs in higher plants as a nonprotein amino acid.

diaminocaproic acid *see* **lysine**.

2,7-diaminofluorene a dye used in the chromogenic assay of peroxidases. It replaces benzidine, which was formerly used but is now known to be carcinogenic.

diaminopimelic acid *symbol*: A_2pm *or* Dpm; *abbr.*: DAP; α,ε-diaminopimelic acid, 2,6-diaminoheptanedioic acid; a α,ε-diaminodicarboxylic acid with two chiral centres. *meso*-Diaminopimelic acid is the stereoisomer occurring in peptidoglycan in the cell walls of *Escherichia coli* and all other Gram-negative bacteria. *See also* **diaminopimelic-acid pathway**.

diaminopimelic-acid pathway *or* **DAP pathway** a metabolic pathway for the synthesis of L-lysine from L-aspartic acid via α,ε-diaminopimelate. Found in bacteria, some lower fungi, and green plants, it involves an initial condensation between pyruvate and aspart-4-al, yielding, following dehydration, 2,3-dihydropicolinate. This substance is then reduced; ring opening occurs followed by succinylation to form α-(succinylamino)-ε-oxo-L-aminopimelic acid. Transamination, deacylation, and epimerization yield the corresponding *meso*-α,ε-diaminopimelic acid; then decarboxylation completes the overall synthesis of L-lysine. *Compare* **aminoadipic pathway**.

diamorphine *see* **diacetylmorphine**.

diaphorase *name formerly applied indiscriminately to* any enzyme capable of catalysing the oxidation of either NADH or NADPH in the presence of an electron acceptor such as a dye (e.g. Methylene Blue, or 2,6-dichlorophenolindophenol), ferricyanide, a quinone, or a cytochrome but not dioxygen. *Also formerly known as* **coenzyme factor**.

diaphragm 1 *(in chemistry)* a semipermeable membrane, or porous plate, used to separate two solutions in **dialysis** or **osmosis**. 2 *(in anatomy)* any separating membrane, especially the thin, domed fibromuscular partition separating the thorax from the abdomen. Flattening of the diaphragm, accompanied by expansion of the rib cage, causes air to be sucked into the lungs. 3 *(in optics)* any device, sometimes adjustable (e.g. iris diaphragm), that can limit or vary the aperture of a lens or optical system. 4 any other thin dividing membrane.

diarylpropane peroxidase *see* **ligninase**.

diasolysis a membrane permeation process for the separation of organophilic from hydrophilic solutes, akin to **dialysis** but differing in that separation depends primarily on the relative solubilities of the solutes in the membrane rather than on their molecular dimensions. Such a process may play a part in the selective passage of solutes through membranes of living cells.

diastase 1 *obsolete name for* α-amylase (EC 3.2.1.1). 2 the crude mixture of amylases obtained commercially as a yellowish white amorphous powder from malt. [From Greek *diastasis*, a separation.] —**diastatic** *adj.*

diastatic activity the enzymic activity of α-amylase.

diastereoisomer *or* **diastereomer** any **stereoisomer** that is not **enantiomeric**; it may be either chiral or achiral. The term includes ***cis–trans* isomer**, **configurational isomer**, **conformational isomer**, and **epimer**. —**diastereoisomeric** *or* **diastereomeric** *adj.*; **diastereoisomerism** *or* **diastereomerism** *n.*

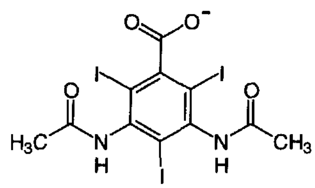
diastereotopic 1 chemically-like ligands in constitutionally equivalent locations that are not symmetry related (cannot be interchanged by rotation about an axis (C_n) or alternating axis (S_n) of symmetry) are diastereotopic; the two ligands are in a stereochemically different, nonmirror-image environment. Separate replacement of each ligand by a different achiral ligand yields two products that are in a diastereoisomeric relationship. Example: replacement of the H_R hydrogen of the methylene group in (*R*)–CH_3–CH_2–CHBr–CH_3 with chlorine yields (2*R*,3*R*)-2-bromo-3-chlorobutane; the similar replacement of the H_S hydrogen yields the diastereoisomeric (2*R*,3*S*)-2-bromo-3-chlorobutane. A biochemically important example is that both of the methylene groups of citric acid have diastereotopic hydrogens. 2 the two faces of a double bond or of a planar cyclic ring system that are not related by any symmetry operation (axis, plane, centre, or alternating axis) are diastereotopic; the two faces show stereochemically different, nonmirror-image environments. Separate addition of an achiral reagent to each face yields two diastereoisomeric products. Example: hydrogenation of the C=O bond in (3*S*)–CH_3–CH_2–$CH(CH_3)$–CO–COOH (in absence of any chiral influence such as a chiral catalyst) yields the two diastereoisomers, (2*R*,3*S*)-2-hydroxy-3-methylpentanoic acid and (2*S*,3*S*)-2-hydroxy-3-methylpentanoic acid. A biochemical example is the two diastereotopic faces, *A* and *B*, of the (dihydro)pyridine ring of nicotinamide-adenine coenzymes (*see* **nicotinamide-adenine dinucleotide (reduced)**). *Compare* **enantiotopic**.

diastole 1 the passive dilatation of the chambers of the heart that occurs between the rhythmical contractions. 2 the rhythmical expansion of a pulsating vacuole. *Compare* **systole**. —**diastolic** *adj.*

diatomaceous earth *or* **kieselguhr** a friable, whitish material consisting mostly of silica, derived from diatomite, a rock consisting largely of the remains of diatoms. It is used as a filter aid and as an absorbent in column chromatography. *See also* **Berkefeld filter**.

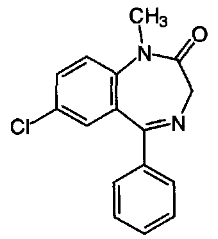
diatomic *(of a molecular entity)* composed of only two atoms. —**diatomicity** *n.*

diatrizoate the anion 3,5-diacetamido-2,4,6-triiodobenzoate. Its sodium salt forms solutions of low viscosity and high relative density. A mixture with Ficoll provides a solution having the optimal density and osmolarity for the rapid, one-step isolation of lymphocytes from small volumes of blood by centrifugation. It is also used as a radiopaque material.



diauxy the adaptation of microorganisms to culture media containing two different carbohydates. Growth occurs in two phases separated by a period of less rapid or zero growth. In the first phase, the organism utilizes the carbohydrate for which it possesses constitutive enzymes. During the interlude the organism synthesizes the required induced enzymes for the metabolism of the other carbohydrate, which it then proceeds to exploit in the second growth phase. —**diauxic** *adj.*

diazepam 7-chloro-1,3-dihydro-1-methyl-5-phenyl-2*H*-1,4-benzodiazepin-2-one; one of the most widely used **benzodiazepine** drugs, noted for its CNS depressant effects. Used as an anti-anxiety–hypnotic agent at subsedative doses, its actions are similar to those of chlordiazepoxide, but it has greater potency. One proprietary name is Valium.



diazepam binding inhibitor a 10 kDa 86-residue polypeptide, that acts as an endogenous ligand for a mitochondrial receptor (formerly regarded as a peripheral benzodiazepine binding site) in steroidogenic cells (i.e. adrenocortical, glial, and Leydig cells) and regulates stimulation of steroidogenesis by tropic hormones. It also binds to the benzodiazepine binding site on the $GABA_A$ receptor and modulates glucose-dependent insulin secretion and synthesis of acyl-CoA esters.

diazo compound any organic compound, other than an **azo compound**, that contains the bivalent **azo** group –N=N– (or $=N_2$) in covalent linkage. It may be (1) any aliphatic compound of general formula $R–CH=N_2$, in which both nitrogen atoms are linked to the same carbon atom by replacing two hydrogen atoms of the parent molecule, e.g. diazomethane, $CH_2=N_2$; or (2) any aromatic compound of general formula Ar–N=N–X in which one nitrogen atom is linked to a carbon atom of the aromatic ring by replacing one hydrogen atom of the parent molecule, and the other nitrogen atom is linked covalently to a group X through an atom other than carbon (except when X = CN), e.g. benzenediazohydroxide, $C_6H_5–N=N–OH$.

diazonium compound any aromatic organic compound of general formula $Ar–N_2^+$, containing the univalent cationic **diazonium group**, $–N^+\equiv N$, linked to a carbon atom of an aromatic ring system by replacing one hydrogen atom of the parent molecule. The positive charge is balanced by an anion to form a **diazonium salt**, e.g. benzenediazonium chloride, $C_6H_5–N_2^+Cl^-$.

diazotize *or* **diazotise** to cause a primary aromatic amine to form a **diazonium compound** by reaction with nitrous acid in the presence of a strong acid, according to the equation:

$$Ar\text{-}NH_2 + HONO + H^+ = Ar\text{-}N_2^+ + 2H_2O.$$

—**diazotized** *adj.*; **diazotization** *n.*

dibasic cleavage site a site within a proprotein structure at which endopeptidases act specifically to cleave the protein to one or more active polypeptides. It comprises a connected pair of basic amino-acid residues (Arg–Arg, Arg–Lys, Lys–Arg, or Lys–Lys). The cleavage is effected by a family of cellular endoproteases. These have catalytic domains related to **subtilisin** and go under the general name mammalian **prohormone convertases** (PCs). Examples are ***kex2***, first described in yeast, PC2 in human insulinoma, PC3, and **furin**. Dibasic cleavage sites occur e.g. in **proinsulin**, **proopiomelanocortin**, and **chromogranins**.

dibenzazepine any of a group of compounds with a tricyclic structure related to phenothiazines that prevent the neuronal reuptake of norepinephrine or 5-hydroxytryptamine from the synaptic cleft, and are useful as drugs in the treatment of depression, e.g. imipramine, clomipramine.

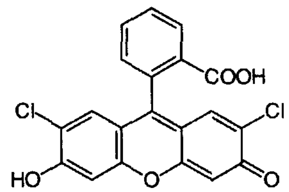
dibenzocycloheptene any of a group of agents that prevent neuronal reuptake of norepinephrine and 5-hydroxytryptamine from the synaptic cleft and are useful as drugs in the treatment of depression, e.g. amitriptyline.

dicarboxylate transport protein *former name for* **oxoglutarate/malate carrier protein.**

dicarboxylic acid any organic compound containing two carboxyl groups.

dicaryon *a variant spelling of* **dikaryon.**

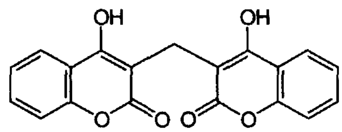
2′,7′-dichlorofluorescein a dye useful as a detection reagent for lipids, yielding fluorescent spots on chromatograms under UV light.



dichroic ratio the ratio of the absorbance of light polarized perpendicularly to the axis of a nonisotropic sample (e.g. the fibre axis of a protein or nucleic-acid molecule or one axis of a crystal) to the absorbance of light polarized parallel to the sample axis. If the dichroic ratio is greater than unity, the vibrational transition observed is said to exhibit **perpendicular dichroism**; if less than unity, it is said to show **parallel dichroism.**

dichroism 1 the directional effect observed in the absorption of electromagnetic radiation resulting from the relative orientations of the absorbing chromophore and the direction of the electric vector in the polarized electromagnetic radiation. Thus the absorption of such radiation may be restricted only to atoms or groups that vibrate in a specific direction or to a component of the radiation that is polarized in a specific direction. *See* **circular dichroism, dichroic ratio, electric dichroism. 2** concentration- and path-length-dependent changes in the frequencies of electromagnetic radiation absorbed by solutions of certain substances. —**dichroic** *adj.*

dicoumarol *or* **dicumerol** *or (esp. US)* **dicumarol** (*formerly* bishydroxycoumarin); 3,3′-methylenebis(4-hydroxycoumarin), 3,3′-methylenebis(4-hydroxy-2*H*-1-benzopyran-2-one); a substance originally isolated from spoiled sweet clover but now synthetic, that prevents the normal formation of prothrombin and other blood coagulation factors. It interferes with the action of vitamin K in γ-carboxylation of certain glutamic residues in the precursors of the coagulation proteins. It is used therapeutically as an anticoagulant, and is also used as a rodenticide. **Warfarin** is one of the **coumarins.**



dicoumarol

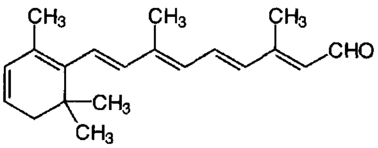
dictyosome 1 *or* **golgiosome** *an alternative name for* **Golgi apparatus**, especially in cells of invertebrates and plants; there are usually several scattered dictyosomes per cell. **2** a collar-like structure that may form an anchor for the base of the flagellum in uniflagellate fungi.

Dictyostelium discoideum a cellular **slime mould**. It exists in the vegetative form as unicellular, haploid **myxamoebae**, 8–12 µm maximum dimension, using bacteria as food. In the absence of bacteria, chemotactic aggregation of myxamoebae occurs, mediated by myxamoebal secretion of acrasin (i.e., cyclic AMP), to form an elongated, finger-like, multicellular **pseudoplasmodium**, 1–4 mm maximum dimension. Aggregation is assisted by the synthesis of a multivalent carbohydrate-binding lectin, **discoidin**, found in the outer membrane of starved cells. The pseudoplasmodium differentiates to form a single, unbranched stalk, the **sorocarp**, which carries a spheroidal or ovoid mass of yellow spores, each commonly 5–10 µm × 3–6 µm with a thick cellulose-containing wall. Germination of spores only occurs in the presence of an adequate supply of amino acids.

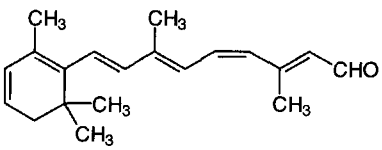
dicyclohexylcarbodiimide *see* **DCC**. *See also* **carbodiimide**.

didehydro+ *prefix (in chemical nomenclature)* denoting the loss of two hydrogen atoms. *See also* **dehydro+**.

3,4-didehydroretinal *recommended trivial name for* (3-)dehydroretinal; (3-)dehydroretinaldehyde; retinal$_2$; vitamin A_2 aldehyde; retinene$_2$; (2*E*,4*E*,6*E*,8*E*)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1,3-dien-1-yl)nona-2,4,6,8-tetraen-1-al; the didehydro derivative of **retinal** (def. 2). This all-*E*-isomer, 3,4-didehydro-all-*trans*-retinal, often termed all-*trans*-retinal$_2$, plays a role in the visual process in freshwater fish and some amphibians analogous to that of all-*trans*-retinal in other vertebrates; *see* **retinal** (def. 2). Its isomerization product, the (2*E*,4*Z*,6*E*,8*E*)-stereoisomer, generally termed 11-*cis*-retinal$_2$, combines (as a **Schiff base**) with an opsin to form (in cone cells) **cyanopsin** or (in rod cells) **porphyropsin**. *Note*: The numbering system of the systematic name is different from that of the trivial name, the latter following the rules used for carotenoids.

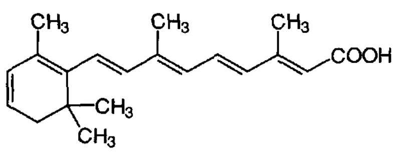


(all-*E*) isomer

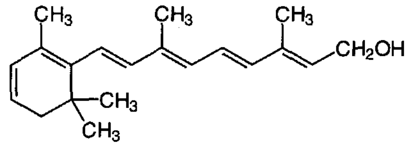


(2*E*,4*Z*,6*E*,8*E*) isomer

3,4-didehydroretinoic acid *recommended trivial name for* (3-)dehydroretinoic acid; vitamin A_2 acid; (2*E*,4*E*,6*E*,8*E*)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1,3-dien-1-yl)nona-2,4,6,8-tetraenoic acid; the didehydro derivative of **retinoic acid**.

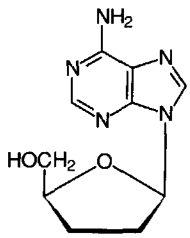


3,4-didehydroretinol *recommended trivial name for* (3-)dehydroretinol; retinol$_2$; vitamin A_2; vitamin A_2 alcohol; (2*E*,4*E*,6*E*,8*E*)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1,3-dien-1-yl)nona-2,4,6,8-tetraen-1-ol; the didehydro derivative of **retinol**, being characterized by a second double bond in the terminal ring of the molecule. It is the predominant form of vitamin A in freshwater fishes. *See also* **vitamin A**.

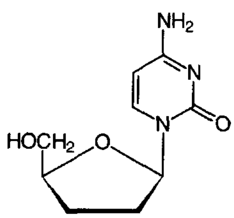


dideoxy+ *prefix (in organic chemical nomenclature)* **1** denoting the replacement of two hydroxyl groups by two hydrogen atoms. **2** (*abbr*.: dd+) indicating the presence in a nucleoside of a residue of 2,3-dideoxy-D-ribose.

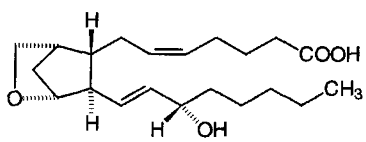
dideoxyadenosine *symbol*: ddAdo *or* ddA; *abbr*.: DDA; 2′,3′-dideoxyadenosine; an adenosine analogue with antiviral activity.



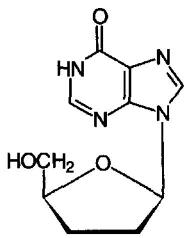
dideoxycytidine *symbol*: ddCyd *or* ddC; *abbr*.: DDC; 2′,3′-dideoxycytidine; a cytidine analogue with antiviral activity.



9,11-dideoxy-9α,11α-epoxymethanoprostaglandin F$_{2α}$ U46619; a stable **thromboxane** A_2 mimetic.



dideoxyinosine *symbol*: ddIno *or* ddI; *abbr*.: DDI; 2′,3′-dideoxyinosine; an **inosine** analogue with antiviral activity.

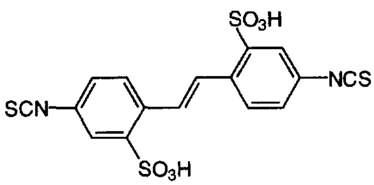


dideoxyinosine

dideoxy method *or* **dideoxy(nucleotide) sequencing analysis** *an alternative name for* **chain-termination method** (for sequencing DNA molecules).

dideoxynucleoside triphosphate *abbr*.: ddNTP; any artificial nucleoside triphosphate in which both of the hydroxyl groups on C-2′ and C-3′ of the pentose moiety have been replaced by hydrogen atoms. They are used in sequencing DNA molecules by the **chain-termination method**.

DIDS *abbr. for* 4,4′-diisothiocyano-2,2′-disulfonic acid stilbene; a powerful anion-transport inhibitor



didymium glass a glass with a bluish-pink coloration due to the inclusion of a mixture of rare-earth elements, chiefly neodymium and praseodymium. It has intense, narrow, light-absorption bands, and is used for calibrating spectrometers. An example is a glass known as Corning 5120, which has absorption bands at 441, 528.7, 585, 684.8, 743.5, 754, 808, 883, and 1067 nm.

dielectric 1 a substance or medium that can sustain an electric field but does not conduct electricity; an insulator. **2** of, relating to, or being a dielectric.

dielectric absorption the strong absorption of microwave electric fields shown by polar liquids, particularly water. It is especially troublesome in **electron spin resonance spectroscopy**.

dielectric constant *an obsolete term for* **relative permittivity**.

dielectric dispersion variation of the **relative permittivity** with the frequency of the applied electric field.

dielectric increment the change in the **relative permittivity** of a solution with the change in the concentration of the solute. For low concentrations of solute in polar solvents, the dielectric increment, $\delta D/\delta c$, is equal to $(D - D_0)/c$, where D and D_0 are the relative permittivities of the solution and solvent, respectively, and c is the solute concentration.

dielectrophoresis *the translational motion of neutral* suspended particles relative to that of the suspending medium when the suspension is subjected to a nonuniform electric field. The movement is the result of polarization induced in the particles, the degree of which is dependent on the ratio of the relative permittivities (*formerly* dielectric constants) of particles and medium. In **positive dielectrophoresis** (that most commonly observed), movement of the particles is towards the region of highest field intensity, irrespective of its sign (*compare* **electrophoresis**). **Negative dielectrophoresis** may be observed in certain frequencies when an alternating field is used, the particles then being attracted less than the medium to the region of most intense field; at such frequencies the polarizability of the particles is not greater than that of the medium. Dielectrophoresis is used, e.g., in the concentration of certain biological products, the characterization of single-celled organisms or cellular organelles by their characteristic yield spectrum (i.e. yield vs. field frequency), and the determination of the diffusion coefficients of proteins in solution.

diene any **alkadiene** or substituted alkadiene.

+diene(+) *or (before a vowel)* **+dien+** *(in chemical nomenclature) infix or suffix* in systematic names denoting the presence in a molecular structure of an unsaturated aliphatic hydrocarbon chain containing two (conjugated or unconjugated) double bonds. The position of each double bond may be indicated by a locant for its lowest-numbered carbon atom; e.g. cyclohexa-1,3-diene, eicosa-8,11-dienoic acid.

dienoic indicating a compound having an aliphatic chain with two double bonds and a carboxyl group.

dienoyl *general name for* any acyl group derived from an unsaturated fatty acid containing two (conjugated or unconjugated) double bonds.

2,4-dienoyl-CoA reductase (NADPH) EC 1.3.1.34; *systematic name*: *trans*-2,3-didehydroacyl-CoA:NADP$^+$ 4-oxidoreductase; *other name*: 4-enoyl-CoA reductase (NADPH). An enzyme of the pathway for oxidation of unsaturated fatty acids that catalyses the reduction by NADPH of *trans,trans*-2,3,4,5-tetradehydroacyl-CoA to *trans*-2,3-didehydroacyl-CoA, a normal substrate of β oxidation. For the bacterial enzyme, this is a direct product of the enzyme, but for the mammalian enzyme the product is *trans*-3,4-didehydroacyl-CoA, which must undergo isomerization to *trans*-2,3-didehydroacyl-CoA. Example from rat liver: database code RATDCR, 335 amino acids (36.16 kDa). *See also* **unsaturated fatty acid oxidation**.

diesterase *see* **phosphodiesterase**.

dietary deficiency a nutritional deficiency due to inadequate intake of one or more specific and necessary constituents of the diet, even though the diet may in other respects be adequate quantitatively.

diethyl ether *see* **ether**.

diethylstilbestrol *or (esp. UK)* **diethylstiboestrol** 4,4′-(1,2-diethyl-1,2-ethenediyl)bisphenol; a nonsteroidal compound that possesses the activity of an **estrogen**. It is used therapeutically to replace natural hormones, and was formerly used as a growth promoter in livestock (this use is now prohibited in many countries, including the UK and USA). It was originally used for the treatment of prostatic cancer.

difference spectrophotometry a spectrophotometric method for investigating the effects of potential perturbants on a chemical substance or system in solution. Two samples are prepared containing identical solutions except that one contains a potential perturbant, e.g. a protein denaturant. The absorbances of the two samples at each frequency are subtracted one from another generating a **difference spectrum** and thereby highlighting any small differences between the spectra of the normal and perturbed systems.

differential centrifugation centrifugation of a suspension in a succession of increasing gravitational fields so that particles of differing sizes and/or relative densities may be separated and collected.

differential counting counting of the pulses produced in a **scintillation counter** that fall within two or more differing energy ranges selected for with a pulse-height analyser. The technique is used for counting the disintegrations due to two or more radionuclides in the same sample.

differential labelling a method of determining the part of a molecule that reacts in a particular way by blocking one part with a specific reagent and investigating whether a particular reaction will still occur. It is used, e.g., for investigating the combining sites of antibodies.

differential medium any medium used for growing micro-organisms that reveals specific properties of the organism(s) grown and hence aids in identification of the organism(s). *See also* **selective medium**.

differential scanning calorimetry *abbr.*: DSC; a form of **differential thermal analysis** in which the temperatures of both the sample and the reference substance are maintained at either the same or a constant difference; the amount of heat that must flow into the sample to maintain the temperature differential during a transition is then measured.

differential thermal analysis *abbr.*: DTA; a method for analysing temperature-induced transitions in a sample. The sample and an inert reference material are heated or cooled at the same rate and the difference in temperature between them is monitored. This difference is zero or constant until a thermally induced transition occurs in the sample, when the temperature difference changes. The direction of the change shows whether the transition is endo- or exothermic.

differential titration the titration of a group, e.g. an acid group, before and after another group in the same molecule has reacted with some other reagent. It is used to determine the extent, if any, of **linkage** between the functions of the groups in question. *See* **titrate**.

differentiating-stimulating factor *see* **leukemia inhibitory factor**.

differentiation the process whereby relatively unspecialized cells, e.g. embryonic or regenerative cells, acquire specialized structural and/or functional features that characterize the cells, tissues, or organs of the mature organism or some other relatively stable phase of the organism's life history.

differentiation antigen any cell-surface antigen whose expression varies during successive developmental stages of a particular cell type.

diffraction the bending of light, or other waves, by an obstacle or aperture, into the region of the geometrical shadow of the object; any redistribution in space of the intensity of waves resulting from the presence of an object in their path causing variations of either their amplitude or their phase. The effect is particularly marked when the size of the object is of the same order as that of the wavelength of the waves. Diffraction accounts for the phenomenon observed when light passes through an aperture and falls on a screen; this takes the form of patterns of light and dark bands (with monochromatic light) or coloured bands (with heterochromatic light) near the edges of the beam and extending into the shadow. Diffraction is also responsible for the propagation of radio waves over the surface of the globe and of sound waves around solid objects. *See* **diffraction grating**, **electron diffraction**, **X-ray diffraction**.

diffraction grating a glass or quartz plate on the surface of which there is a large number of equidistant parallel black lines (500 or more per millimetre). When light or other electromagnetic radiation passes through such a grating it is diffracted in a manner dependent on the wavelength of the radiation. Other plane surfaces, e.g. metals, may be used.

diffusate a product of a process of diffusion, especially the material that enters into or traverses a barrier of restricted or selective permeability. In **dialysis** the term is sometimes preferred to **dialysate**. *Compare* **nondiffusible material**.

diffusible **1** capable of diffusing. **2** *(sometimes) a preferred alternative term for* **dialysable** (i.e. capable of passing through a semipermeable membrane used in dialysis).

diffusion **1** the net flow of molecules from a region of high concentration, or high chemical potential, to one of low concentration, or low chemical potential, due only to random thermal motion. *See also* **facilitated diffusion**. **2** the irregular reflection or transmission of light or other electromagnetic radiation; scattering. **3** the act or process of dispersing or of being dispersed widely. —**diffuse** *vb., adj.*

diffusion cell *or* **diffusion chamber** **1** a cell or chamber with porous walls that allow the passage of dissolved substances but not of biological cells. It is used in studying materials released by cells. **2** a device for measuring the diffusion of macromolecules and consisting of two chambers separated by a porous membrane of nearly parallel channels. The channels are large compared with the size of the macromolecules but small enough that the bulk flow of liquid through them is negligible.

diffusion coefficient *or* **diffusion constant** *symbol*: D; a coefficient of proportionality between flux and concentration gradient for a solute diffusing in a system of constant temperature and pressure in accordance with **Fick's first diffusion law** and **Fick's second diffusion law**. For an ideal solute it is a constant, but in practice usually varies with concentration. The cgs unit for D is $cm^2 s^{-1}$ (*see also* **Fick unit**) and the SI unit is $m^2 s^{-1}$.

diffusion-controlled *or* **diffusion-limited** describing a chemical reaction (or other process) whose velocity is dependent on the rate at which one or all of the reactants (or participants) can diffuse to the site at which the reaction (or process) occurs, as to the active site of an enzyme catalysing the reaction.

diffusion gradient any gradient e.g. in the concentration of a dissolved solute that forms if the solute is capable of diffusing from a region of higher concentration to a region of lower concentration until its concentration in the solution is uniform.

diffusion-limited *see* **diffusion-controlled**.

diffusion potential an electric potential difference across a membrane due to the diffusion through the membrane of more cationic charges than anionic charges, or vice versa.

diffusive characterized by diffusion. —**diffusively** *adv.*

DIFP *see* **DFP**.

digest **1** to subject material to enzymic or chemical breakdown. **2** to subject food to **digestion** (def. 1). **3** to break down nitrogenous compounds by hot sulfuric acid to convert them to ammonium salts. **4** to treat biological materials with powerful oxidants to facilitate determination of trace elements present. **5** the mixture of compounds obtained from the enzymic or chemical degradation of a macromolecule or other substance. —**digester** *n.* an apparatus that facilitates digestion procedures.

digestion **1** the whole of the physical, chemical, and biochemical processes carried out by living organisms to break down ingested nutrients into components that may be easily absorbed and directed into metabolism. **2** the process of breaking down complex substances or materials into smaller parts either enzymically or chemically, especially in the course of their analysis.

digestive enzyme any hydrolytic enzyme that brings about or assists in **digestion** (def. 1).

digestive juice any of the secretions entering the digestive tract that contain digestive enzymes. They include saliva, **gastric juice**, **intestinal juice**, and **pancreatic juice**.

digestive system the digestive tract together with related organs, such as the pancreas and liver.

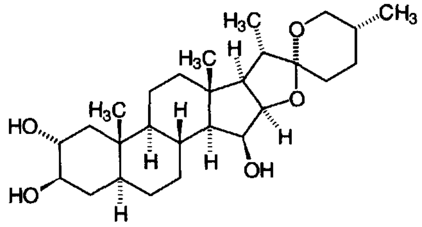
digestive tract *or* **alimentary tract** the tubelike passage in higher animals, extending from the mouth to the anus, in which the food is digested, from which absorption of nutrients occurs, and which serves to eliminate waste products.

diginose the D enantiomer, 2,6-dideoxy-3-*O*-methyl-D-*lyxo*-hexose, is a component of some **cardiac glycosides**

digitalis **1** any plant of the genus *Digitalis*. **2** a preparation of powdered dried leaves of the purple foxglove (*Digitalis purpurea*). It contains the cardiac glycosides **digitoxin**, **gitoxin** (a glycoside of 3,14,16-trihydroxycard-20(22)-enolide), and **gitaloxin** (16-formylgitoxin), and is used therapeutically to increase the strength of the heart's contraction and to slow auricular–ventricular conduction.

digitalose the D enantiomer, 3-methyl-D-fucose, 6-deoxy-3-*O*-methyl-D-galactose, is a component of some **cardiac glycosides**, e.g. digitoxin.

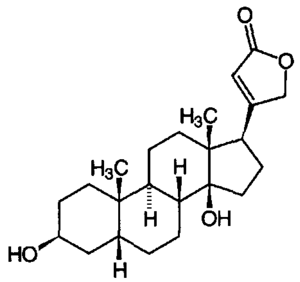
digitogenin (25*R*)-5α-spirostan-2α,3β,15β-triol; the aglycon of the glycosides (saponins) present in **digitonin**.



digitogenin

digitonin a mixture of at least four steroid saponins obtained from the seeds of the purple foxglove, *Digitalis purpurea*. It is useful in the fractional precipitation and analysis of cholesterol and other steroids and as a nonionic detergent in the solubilization of membrane proteins. Aggregation number 60. Its aglycon is **digitogenin**.

digitoxigenin 3β,14-dihydroxy-5β-card-20(22)-enolide; the aglycon of **digitoxin**.

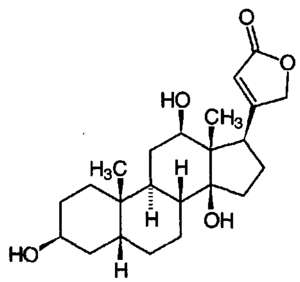


digitoxin a secondary glycoside, or mixture of glycosides comprised mainly of digitoxin, that is extracted from *Digitalis purpurea*. It has actions similar to, but longer-lasting than, **digoxin**. Its aglycon is digitoxigenin. *See also* **cardiac glycoside**.

digitoxose the D enantiomer, 2,6-dideoxy-D-*ribo*-hexose, is a component of some **cardiac glycosides**.

diglyceride *former name for* **diacylglycerol** (it is still quite commonly used, as in 'diglyceride fraction', but its use is discouraged for such purposes as diglyceride is properly a generic term for any compound having two glyceryl residues, e.g. **phosphatidylglycerol**).

digoxigenin $3\beta,12\beta$,14-trihydroxy-5β-card-20(22)-enolide; the aglycon of **digoxin**.



digoxin a secondary glycoside extracted from the Austrian (or woolly) foxglove, *Digitalis lanata*. It acts on the heart to potentiate contractility and reduce conductivity mainly through inhibition of Na^+,K^+-ATPase, hence its use in congestive heart failure and atrial dysrhythmia. Its aglycon is digoxigenin. *See also* **cardiac glycoside**.

dihedral having or contained by two planes or plane faces.

dihedral angle the inclination of two planes that meet at an edge.

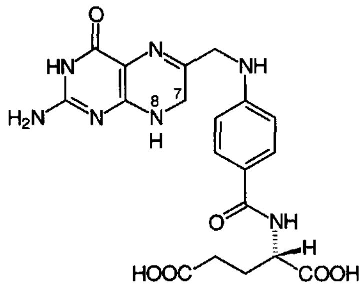
dihedral symmetry the symmetry of a **point group**, designated D_n, containing an n-fold axis with n twofold axes perpendicular to it.

dihomo-(6,9,12)-linolenate dihomogammalinolenate, dihomo-γ-linolenate, (*all-Z*)-eicosa-8,11,14-trienoate; an intermediate in the synthesis of arachidonate from linoleate, formed by Δ^6 desaturation and chain elongation. It acts as a precursor for the PG_1 series of **prostaglandins**. *See also* ***eicosatrienoic acid***.

dihydric describing a chemical compound containing two hydroxyl groups per molecule. It is used especially of **alcohols**. *Compare* **dihydroxy+**.

dihydro+ *prefix (in chemical nomenclature)* indicating the presence of two additional hydrogen atoms.

dihydrofolate *abbr*.: H_2folate (DHF not recommended); **1** 7,8-dihydrofolate, 7,8-dihydropteroylglutamate; *the trivial name for* the 7,8-dihydro derivative of **folate** (def. 1). **2** any **folate** (def. 2) that is a dihydro compound, especially any of the respective 7,8-dihydro intermediates in the conversion of nonreduced folates to their corresponding folate coenzymes (i.e. tetrahydrofolates) by the enzyme dihydrofolate reductase. *See also* **antifolate, tetrahydrofolate**.



dihydrofolate reductase *abbr*.: DHFR; EC 1.5.1.3; *systematic name*: 5,6,7,8-tetrahydrofolate:$NADP^+$ oxidoreductase; *other name*: tetrahydrofolate dehydrogenase. An enzyme that catalyses the interconversion of 5,6,7,8-tetrahydrofolate and 7,8-dihydrofolate, and also slowly interconverts 7,8-dihydrofolate and folate; both reactions are linked to the reduction/oxidation of $NADP^+$/NADPH. It is inhibited by **aminopterin**, amethopterin (*also termed* **methotrexate**), and **trimethoprim**. Example from human: database code NRL_1DRF, 186 amino acids (21.30 kDa); 3-D structure known; four motifs.

dihydrofolate synthase EC 6.3.2.12; *other name*: dihydrofolate synthetase; an enzyme of the pathway for the *de novo* synthesis of dihydrofolate. It catalyses the formation of dihydrofolate from ATP, dihydropterate, and L-glutamate with release of ADP and orthophosphate. Example from *Escherichia coli*: database code FOLC_ECOLI, 422 amino acids (45.39 kDa).

dihydrogen *systematic name for* the diatomic compound H_2, commonly termed (molecular) hydrogen.

dihydrolipoamide *S*-acetyltransferase EC 2.3.1.12; *other names*: lipoate acetyltransferase; thioltransacetylase A; an enzyme that catalyses the formation of acetyl-CoA and dihydrolipoamide from *S*-acetyldihydrolipoamide and CoA. A lipoyl group is required as a coenzyme, the enzyme having three covalently linked lipoyl groups. This is the E_2 component of the **pyruvate dehydrogenase complex**. Example: dihydro-